Stefano Caselli
Stefano Gatti *Editors*

Structured Finance

Techniques, Products and Market

Second Edition

Springer

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Foreword by Gianfranco Bisagni

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ISBN 978-3-319-54123-5 ISBN 978-3-319-54124-2 (eBook)
DOI 10.1007/978-3-319-54124-2

Library of Congress Control Number: 2017950267

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2nd edition: © Springer International Publishing AG 2017
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Printed on acid-free paper

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Foreword

Structured finance is an important part of the evolution that has taken place in the banking industry in the past decades. In the “old world”, banks had a simpler job: basically, they handled liquidity, borrowing and lending cash from/to people or companies, and they provided services to support trade. In that world, the most important skill that the banks needed to have was the ability to assess credit risk.

Today the scenario has evolved significantly. While assessing credit risk is still a “must”, it is not enough. The world is changing at an unprecedented speed and the business of our clients has become more and more complex. Technological evolution, demographic changes, and the new economic order which is reshaping global demand, the urgency to achieve critical mass to exploit economies of scale and to intercept efficient global supply chain: these are only a few examples of the challenges that companies have to face every day. For banks, meeting these needs requires a significant effort in improving the offer. For example, if today clients are becoming more and more global, the ability to serve the client everywhere is a precondition to many other services. But also being able to help clients grow and to offer hedging solutions to properly protect customers from new risks is another important prerequisite. It is a matter of symbiotic business sustainability, both for customers and banks. As a result, bankers had to build new expertise in products and structures, as well as new skills in market and operational risks.

Today, the financial industry is affected by several factors, among which:

1. Increasing pressure from the regulators: the new guidelines are forcing banks to build larger capital buffers and stronger liquidity;
2. Stricter rules on compliance: sanctions and Know Your Customer (KYC) regulations are affecting the day-by-day activity of banks;
3. The digital revolution: it is affecting the profitability of the banks and forces them to invest in IT platforms substantially;
4. New entrants: Fintech companies are jeopardizing banks’ own territory.

This process is irreversible and banks have to adapt to it by becoming ever more strategic partners to their clients. Structured finance fits perfectly in this scenario and it is certainly one of the major components in the offering of a large corporate and investment bank. It is an excellent opportunity for banks to demonstrate that their know-how can generate value for clients.

Structured finance is a complex business that has to be approached with a solid professional behavior, always looking at the long-term risks and their consequences. In fact, one of the causes of financial crises, especially the most recent ones, was a misuse of structured finance. The possibility of repackaging assets, and together with them their risks, in apparently safer different structures, could lead companies and investors to enter in much riskier scenarios. However, if used wisely, structured finance is a powerful tool that banks can use to support clients' growth and success, in particular by offering unique and tailor-made solutions to customers. The use or misuse is part of the ethical and sustainable evolution of the banking industry.

As mentioned above, structured finance is a complicated business, both from technical and regulatory points of view. Its complexity goes beyond the pure technicality of the product. It requires a mix of hard and soft skills. On the one hand, hard skills can easily be grasped with formal training. Banks have invested and are investing significant resources to train their staff, both on technical and compliance topics. On the other hand, soft skills are much more difficult to develop. It requires an understanding of several variables that only a proper experience in the field can provide.

Certainly banks are leveraging on past experiences and are shaping their business models in order to achieve more stability and sustainability. Regulators are playing a role in this evolution, but what should drive banks future set up and product offer will be the long-term client relation and the perfect alignment of interests. Risks will never disappear, but structured finance will help to manage them more effectively. Bankers will have to exercise their competence by offering their client sustainable and ethical solutions.

Milan, Italy
May 2017

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Preface

Structured finance is a business area that encompasses a wide range of transactions. In this work, the authors opted to include securitization, project finance and PPP transactions, leasing (as a deal representative of asset-based finance,) and acquisition finance activities conducted by utilizing a deal design based on a strong debt component (essentially LBOs in all their contractual variations). Given the importance of the management of nonperforming loans (NPL) in the European Banking system, a chapter of the book is dedicated to the analysis of the securitization of this asset class. This perimeter of analysis does not lend itself to meticulous theoretical or empirical debate. The evidence which emerges from observation of the managerial practices of international and domestic intermediaries that compete in this business (which are described in this work) substantially confirm this choice.

Though defining the boundaries of structured finance is not particularly problematic, a comprehensive study on business practices of European financial intermediaries in this business area does not exist. In actual fact, neither in national nor international literature can systematic studies be found which deal with both positioning of actors on the European market as well as the choice of organizational structures at the basis of services offered. This was the primary motivation for drafting the present study in its second edition after the first one published in 2005.

The second reason that prompted the authors to prepare the second edition of this book lies in the dramatic transformation of the strategy that various European banks implemented in the past 10 years, reacting to pressure followed to the 2008 financial crisis, the 2011–2012 Eurozone Sovereign debt crisis and the subsequent changes imposed by the supervisory authorities. These stimuli translated into the need for a more serious commitment to supporting financial policies of client companies by offering a complete range of services.

The shift from a logic of corporate lending to one of “full service”, typical of corporate and investment banking, calls for change in two respects:

1. the range of financial services proposed;
2. the choice of organizational structure adopted to supply these services.

As for the first point, structured finance transactions share many basic principles with medium to long-term lending. In fact, in the past European banks have been participants in project finance, acquisition finance, and securitization. The novelty lies in the current need for national and pan-European intermediaries to serve as direct interlocutors with customers in such transactions. In practice, this materializes in the need to integrate the offer of funding (which already exists) with consulting on modeling and assembling the transaction in question. Winning credibility in higher value added advisory and arranging activities has become the central objective for banks that wish to implement credible corporate and investment banking strategies.

As regards the choice of organizational structure, it is only natural to expect that adopting a higher profile in financial consultancy to support credit deals requires banks to rethink the way they interface with customers. Designing managerial practices to handle customer relations and, in this context, creating organizational roles for contact and transactions become critical elements in guaranteeing the effectiveness of the business proposal.

Within this framework of changes in transactional and commercial choices, structured finance can be seen as an attractive business area for European banks, most certainly in view of potential market size. In this regard, at least three examples can be given.

The first involves project finance. In a context of European infrastructure needs estimated to be about 2 trillion euro in the period 2015–2020 and with scarcity of public resources available to fill in the infrastructure gap, private money and Public—Private Partnerships (PPPs) become a viable solution to convey money of investors to this infrastructure asset class. Not only do banks play a direct role but they can and must become a catalyst of institutional investors' pool of resources via partnerships and innovative structured finance solutions.

The second example refers to securitization. The market shows as of yet untapped potential in the segment of corporate originators and even more so in the application of the technique for an active management of NPLs. The dialectic role that even medium-sized banks can play in this segment should not be underestimated.

The final example refers to LBOs in their many contractual variations. Such transactions underscore structural problems involved in family succession and governance transfer in most European countries. The development of the private equity industry, which has recently begun to reabsorb the recessive trends of the post crisis period, is apt to proliferate the possibilities of credit intervention in replacement phases in the future.

The evidence presented above provides a more accurate framework for the objectives of this research project. In fact, for each structured transaction, this work aims to do the following:

1. Analyze the key criticalities which emerge in the relationship between bank and customer. Critical success factors are then identified which enable banks to compete as credible service providers in this business area.
2. Examine the advantages which can be gained by originators/sponsors from each type of structured finance.
3. Quantify the current dimensions of the market and identify areas for development which as yet have not been completely exhausted by the competitive game.

In addition, a transversal objective in the study of transactions and business areas is to illustrate the macrostructural profiles that characterize business units tasked with advisery services and funding of structured transactions. This is particularly useful in research with an empirical bent so as to verify the existence of best practice benchmarks.

Lastly, from a methodological viewpoint this study has been conducted on two levels. First, considering the scarcity of literature dealing with the field of analysis in its entirety in any methodical way, every contributor reviews the best available literature on each single structured finance transaction. Second, to analyze its present and potential dimensions, an in-depth, empirical investigation is carried out to arrive at an accurate quantification of the current size of the European market. Wherever possible, qualitative and quantitative indications are also provided on the potential for development of this market.

Milan, Italy Stefano Caselli
Milan, Italy Stefano Gatti
September 2017

Contents

Chapter 1
Characteristics and Common Features of Structured Finance Transactions

Stefano Caselli and Stefano Gatti

Abstract This chapter gives a brief introduction to the characteristics that various structured finance transactions analyzed in the following chapters have in common. It is worthwhile to consider the entire set of such transactions in order to provide readers with a general framework and to focus attention directly on aspects which characterize each one. After having described the basis of structured transactions, points of divergence with respect to usual corporate lending techniques are presented, highlighting the advantages that can be achieved from realizing a transaction following structured finance logic.

1.1 Introduction

This chapter gives a brief introduction to the characteristics that various structured finance transactions analyzed in the following chapters have in common. It is worthwhile to consider the entire set of such transactions in order to provide readers with a general framework and to focus attention directly on aspects which characterize each one.

After having described the basis of structured transactions, points of divergence with respect to usual corporate lending techniques are presented, highlighting the advantages that can be achieved from realizing a transaction following structured finance logic.

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S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_1

1.2 Typical Features of Structured Finance Transactions

Using the logic behind arranging financing on a structured basis, a transaction can be included in the business area of structured finance when the following conditions hold true:

1. The recipient of the funds raised is a separate entity from the party or parties sponsoring the transaction. This separation is achieved by creating vehicle companies (SPVs, Special Purpose Vehicles, or SPCs, Special Purpose Companies—the terms are synonymous) designated to take on the initiative and to secure cash receipts and payments which result.
2. Consequent to the previous point, since the initiative to be financed is undertaken by a legal entity set up for this specific purpose, all economic consequences generated by the initiative in question are attributed to this SPV. Financers, therefore, grant financing to the vehicle and not to the parties (sponsors or originators) who founded this company.
3. Since the SPV is the recipient of the financing, and considering that this vehicle has its own net worth, the assets instrumental to managing the project are separated from the remaining assets of the parties that created the vehicle. Hence, along with the cash flow from the initiative, the SPV's assets become collateral for creditors.

The three conditions cited above explain why structured finance transactions are also called "off-balance sheet financing".

The presence of a separate vehicle company which is interested in obtaining financing for the realization of a specific initiative detached from other projects underway, implies that loan repayment is guaranteed primarily by the generation of cash by the assets tied up in the initiative in question. The net worth of the sponsors is, in theory, irrelevant in assessing the financial sustainability of the loans. This is due to the fact that creditors are dealing with no-recourse financing or limited recourse financing, in very specific cases, on the assets of parties that set up the vehicle company.

The use of ad hoc vehicles which encapsulate projects or asset portfolios finds a very wide range of applications; examples are plentiful.

1. In securitization, the Special Purpose Company (or securitization vehicle) issues bonds on the market against real or financial assets segregated in that same vehicle. The only source of reimbursement on capital and interest is the ability of the pool of assets to generate cash in equal measure. The cash flow profile is improved by means of internal and external credit enhancement techniques, which are discussed in Chaps. 2 and 7 (with the focus on Securitization of NPLs).
2. In project finance transactions, industrial projects are segregated in an SPV. Financing is then awarded to this company, which is secured through a series of contractual agreements with key counterparts (contractors, purchasers, suppliers,

operator agents, etc.) for the purpose of improving volatility profiles of free cash flows, as will be seen in Chaps. 3 and 6 (with a specific attention to PPPs).
3. In leasing transactions (Chap. 4)—in particular big ticket deals, i.e. those involving complex, large-scale assets (airplanes, ships, large real estate projects)—the preference is to draw up the contract with an ad hoc legal entity as counterpart to allow better correspondence between cash outflow from payments on installments and inflow generated by the financed asset.
4. In leveraged buyouts (Chap. 5) setting up a vehicle company facilitates capital budgeting of the initiative. Taking into account cash flow deriving from the target company (and only from that company), an assessment is made of the sustainability of the emerging financial structure based on the total liabilities of the target and the Newco (the vehicle utilized for the acquisition) with respect to the latter's equity.

1.3 The Advantages of Assembling a Financing Transaction in Structured Form

The advantages to be had through off-balance sheet forms of financing can be ascertained by analyzing the differences between on-balance financing logic (or corporate financing) and that based on the creation of ad hoc companies.

As explained in the preceding section, the goal in encapsulating an initiative or a pool of assets in an ad hoc organization is to isolate the fate of these assets in relation to those of the sponsor or sponsors of the transaction. This isolation works both ways. An initiative with poor prospects, even where default is a possibility, does not impact the performance or the survival of the company, due to the principle of limited shareholder responsibility set down in the regulatory framework of many countries. On the other hand, a project's worth should, at least in theory, remain untainted by business dealings that could negatively affect its shareholders. In this sense, a project's creditors continue to claim rights to the assets and cash flows of the initiative, even if its shareholders go bankrupt.[1]

The clear separation between the initiative and the sponsoring party also means that the two can have very different creditworthiness. One extreme may be strong sponsors and weak initiatives segregated in a vehicle. The other extreme (more commonly found in practice) could be cases where sponsors have rather low creditworthiness but nonetheless are able to make the initiative hinge on a vehicle company which, appropriately secured by credit enhancement mechanisms, can obtain a higher credit rating than its originators.

[1]We say in theory because in practice one of the events of default included in the loan agreement is precisely the default of one or more sponsors. If this occurred, it would also have repercussions on the initiative and thus on its creditors.

The first economic benefit of structured transactions lies in the cost of funding of new financial resources for the initiative. If the benefits of a reduced cost of funding are greater than the cost of credit enhancement (of whatever kind: a purchasing contract or a tranche-based bond issue, a pledge to pay penalties signed by a counterpart of the SPV, insurance coverage), realizing the initiative on a structured basis is advantageous for sponsors.

The second advantage in separating the initiative from the sponsor(s) lies in maintaining financial flexibility of this company or companies. In fact, financing is granted to a legal entity separate from the sponsor, which therefore does not tap into the latter's credit lines. A specially secured initiative is proposed to the pool of financers, for which a specific return/risk combination is offered in the face of new credit lines. Prior credit lines are not drawn on, nor are there any induced effects on the cost of already existing funding for the sponsor.

Chapter 2
The Alchemy of Securitization. Evolution and Perspectives

Paolo Comuzzi, Roberto Tasca and Simona Zambelli

Abstract The term securitization is used to represent the process whereby assets are pooled together, with their cash flows, and converted into negotiable securities to be placed into the market (Tasca and Zambelli 2005) or to be used as collateral in refinancing transactions with Central Banks or with market participants. Securitization is aimed at transforming illiquid assets into securities. These securities are backed or secured by the original underlying assets and are generally defined as Asset Baked Securities (ABS). Theoretically, any financial assets producing cash flows (receivables, residential and commercial mortgages, credit card receivables, and other consumer and commercial loans) can be securitized (see, e.g., Buchanan 2016, Buchanan 2017; Kara et al. 2016; Malekan 2014; Lipson 2012). The main purpose of the chapter is to analyze the basic characteristics and the market structure of traditional securitization, especially with reference to the Italian market.

Simona Zambelli wrote Sects. 2.1, 2.2 and 2.3; Roberto Tasca wrote Sects. 2.4 and 2.6; Paolo Comuzzi wrote Sects. 2.5 and 2.7. The conclusive paragraph has been written jointly by the authors.

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S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_2

2.1 Introduction and Literature

As an aftermath of the global financial crisis, securitization has been often accused of involving high level of asymmetric information and moral hazard, lack of proper monitoring and insufficient disclosure, as well excessive risk transferred to retail investors without proper regulation (see, e.g., Beltran et al. 2017).[1] Securitization has also been connected to an increase in insider trading activity (Ryan et al. 2016). After the 2007–2009 crisis, securitization has been accused of being one of the causes that contributed to exacerbating the global financial instability (see, e.g., Neuhauser 2015). Therefore, a number of stringent laws and regulatory proposals have been introduced worldwide to regulate securitization more heavily, in the hope of guaranteeing a higher financial stability and a greater investor protection (see, e.g., Beltran et al. 2017; Fligstein and Roehrkasse 2016; Kravitt et al. 2015; Albertazzi et al. 2015; Neuhauser 2015; Cherasi and Rochet 2014; Acharya and Richardson 2009; Beck 2014; Lützenkirchen et al. 2013; Juhas 2013; Solomon 2012).[2]

Unlike US banks, Italian banks showed a lower exposure to toxic assets and a higher involvement in a less risky form of securitization, called traditional securitization. An empirical study on the securitization activity of Italian banks confirmed the positive effects of traditional securitization (Albertazzi et al. 2011) compared with the riskier synthetic securitization, which was more often adopted in the US (Egly et al. 2015).

The main purpose of this chapter is to analyze the basic characteristics and the market structure of traditional securitization, especially with reference to the Italian market. In particular, this chapter intends to answer the following questions:

1. What are the characteristics of traditional securitization?
2. How is the traditional transaction structured?
3. What is the role of financial intermediaries within the securitization process, especially in Italy?
4. What are the main characteristics of the Italian securitization activity?

To answer the above questions, we will first explain the basic components of a securitization transaction, describing the typical structure and the main players involved. Secondly, we will analyze the Italian securitization market, emphasizing its peculiarities through an international comparative analysis.

Generally speaking, the purpose of securitization is to transform illiquid assets into securities. For the purpose of this chapter, the term securitization is used to represent the process whereby assets are pooled together, with their cash flows, and converted into negotiable securities to be placed into the market (Tasca and

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[1]For the lack of regulation, the securitization activity has also been labeled as shadow banking (see, e.g., Gennaioli et al. 2013).

[2]For current updates about rules and regulations on securitization in Italy see, e.g., the following website: http://www.securitisation-services.com/it/normative/index.php.

Zambelli 2005) or to be used as collateral in refinancing transactions with Central Banks or with market participants. These securities are backed or secured by the original underlying assets and are generally defined as Asset Baked Securities (ABS).[3] Theoretically, any financial assets producing cash flows (receivables, residential and commercial mortgages, credit card receivables, and other consumer and commercial loans) can be securitized (see, e.g., Buchanan 2016, 2017; Kara et al. 2016; Malekan 2014; Lipson 2012).[4]

This chapter is organized as follows. Section 2.2 describes the basic structure of the typical securitization transaction (traditional securitization). Section 2.3 discusses the securitization process. Section 2.4 emphasizes the particular role of financial intermediaries within this process, and Sect. 2.5 discusses regulatory issues on securitization. Section 2.6 highlights the role of financial intermediaries within the securitization process. Section 2.7 presents current and future trends of the securitization market, from an international perspective. The last section provides concluding remarks.[5]

2.2 The Traditional Securitization: Definition and Deal Process

Securitization is a financial tool aimed at transforming a pool of assets into marketable securities, which are secured by the cash flow stream related to the underlying assets (Asset Backed Securities—ABS).

It is realized through a transfer of assets by a company (Originator) to a separate entity (Special Purpose Vehicle—SPV), which then issues securities, in the form of

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[3]An ABS represents a security backed by specific assets. This means that principal and interest repayment rely directly on the capability of the underlying assets to generate the expected cash flows. In the US it is common to distinguish between:

- Asset backed securities (ABS), which represent securities backed by specific assets (auto loans, credit card receivables, student loans, equipment leases). This definition does not include mortgages loans or corporate bond loans;
- Mortgage backed security (MBS), which are securities backed by specific mortgage loans.

Outside the US, the definition of ABS may include deals backed by mortgages loans. For the purpose of this chapter, we will use the term of ABS to indicate all classes of securitized instruments. See: Bhattacharya and Fabozzi (1997), Saunders and Cornett (2004), Burton et al. (2003), Spotorno (2003).

[4]For a recent overview of the literature on securitization and its real effects see, e.g., Berg et al. (2015), Buchanan (2016, 2017), Lu et al. (2013), Albertazzi et al. (2011). See also: Battacharya and Fabozzi (1997), Fabozzi (1999), Colagrande et al. (1999), Bontempi and Scagliarini (1999), Artale et al. (2000), Damilano (2000), De Angeli and Oriani (2000), Rumi et al. (2001), Porzio et al. (2001), Galletti and Guerrieri (2002), Ferro Luzzi (2000), Gualtieri (2000), La Torre (2000), (Caneva 2001), Navone et al. (2002).

[5]Sections 2.2, 2.3, 2.4, 2.6 represent a revised version of Tasca and Zambelli (2005).

debt instruments, to be placed into the market through a private or public offering (see, e.g., Kara et al. 2016) or to be used as collateral in refinancing transactions.

The concept of asset securitization was introduced in the US financial system in the 1970s, when the Government National Mortgage Association issued securities backed by a pool of loans, represented by residential mortgages.[6]

Two main types of securitization transactions exist:

1. **Cash flow based (CFB) securitization**. The transaction is structured as a sale of assets by a company (Originator) to a special entity (Special Purpose Vehicle, SPV), which then issues securities backed by the underlying assets. The CFB securitization is also defined as *Funded Securitization*, because the Originator can raise money through the asset sale, diversifying its financing sources (for details see, e.g., Lipson 2012).[7]
2. **Synthetic securitization**. It is a transaction through which the credit risk or a right to access to stream of future cash flows, associated with a reference pool of assets, is transferred to a separate entity (SPV). It is not a sale of assets, so the Originator does not receive any cash flow. The SPV in this case is not the owner of a pool of assets, but only the entity that carries the associated credit risk. It is realized through the use of derivatives (total return swaps and credit derivatives, or CDO; for more details see, e.g., Berg et al. 2015).

For the purpose of this chapter, we will focus on the first type of transaction, which has been subject to a higher level of regulation and has showed a much lower level of risk.[8] For an overview of synthetic securitization transaction, see: Caputo et al. (2001). In order to analyze the basic structure of a securitization transaction, let us consider the following example. The Originator is a bank, willing to raise money by liquidating a specific pool of loans through securitization.

As Fig. 2.1 shows, we can identify two key elements of the securitization transaction:

1. Asset sale[9];
2. Issuance of Asset Backed Securities.

[6]See: Saunders and Cornett (2004), Burton et al. (2003), Spotorno (2003).

[7]Recently, another form of securitization became important: the whole business securitization in relation to leveraged buyout acquisitions (SBO deals). For more details on SBO deals, see Buszko et al. (2013). For more details on Leveraged Buyout deals and their legal structure, see, e.g., Zambelli (2008, 2010).

[8]For a recent overview of the literature on securitization and its real effects see, e.g., Berg et al. (2015), Buchanan (2016, 2017), Lu et al. (2013), Albertazzi et al. (2011). For a recent overview of the risks connected to securitization see, e.g., Chen et al. (2017), Le et al. (2016), Laurent et al. (2016), Cifuentes and Pagnoncelli (2014), Wu et al. (2011).

[9]The example represented in Fig. 2.1 considers a transfer of credits and receivables between the Originator and the SPV.

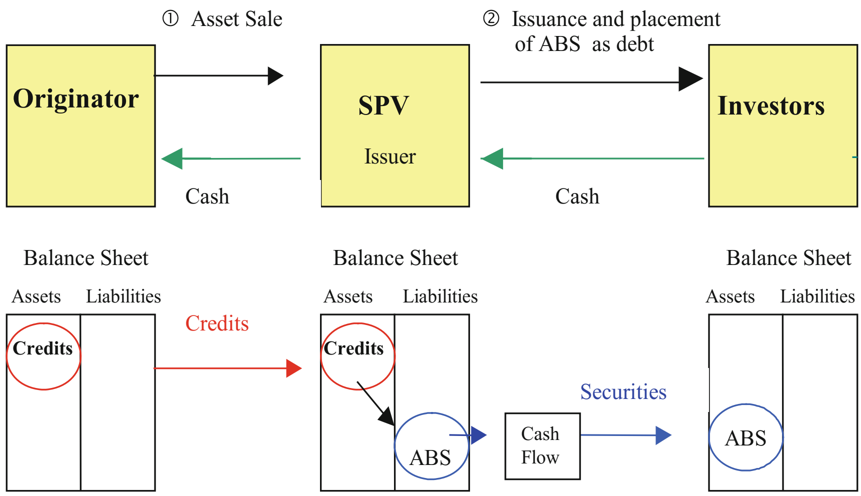

Fig. 2.1 Transactions involved and relative funds flow. *Source* Tasca and Zambelli (2005, p. 8)

– **Asset Sale**. The first element is represented by a true sale of assets between two parties:

(a) One party is the seller of the pool of assets and is known as the "Originator". In our example it is represented by a bank;
(b) The other party is an independent entity, established for the purpose of buying the assets and transforming them into negotiable securities to be placed into the capital market. This entity serves only as securitization vehicle and so it is often defined as "Special Purpose Vehicle" (SPV) or "Special Purpose Company" (SPC). It may take the organizational form of corporation or limited partnership.

– **Issuance of Asset Backed Securities**. In order to finance the asset purchase, the SPV issues securities (usually debt obligation instruments), which are backed by the acquired assets (Asset Backed Securities—ABS). The cash flows originated by the acquired pool of assets are then used to pay the principal and interest on the securities sold to the final investors (holders of ASB securities).[10]

As a result of the securitization:

1. The Originator can use its illiquid assets to raise funding to use immediately for its business activity, without waiting for the maturities of each credits;
2. The underlying securities, issued by the SPV, are backed by a portfolio of assets, which allows a better segregation of risks and the optimization of asset-liability matching risk exposure;

[10]The payments collection related to the securitized portfolio is managed by a third party, the Servicer, which usually is represented by the Originator itself.

3. The issuance of Asset Backed Securities contributes to satisfy different investors' risk appetites and to develop primary financial markets, allowing a transfer of certain performance risks to the final investors.

The risks carried by the investors depend mainly on the quality of the performance of the underlying assets, rather than the creditworthiness of the issuer or the Originator. A careful evaluation of the assets' characteristics and track record is then essential before performing any securitization transaction. The quality of the assets in fact will affect:

1. The creditworthiness of the related ABS, which is usually represented by a rating assigned by specialized agencies[11];
2. The type and the amount of credit enhancement tools, which might be necessary to lower some of risks associated with the asset pool expected performances and improve their rating.

A securitization differs from a traditional equity or debt financing for at least two reasons. First, it is not a loan. It implies an asset segregation from the Originator to the SPV. Second, the buyers of Asset Backed Securities rely primarily on the cash flows generated by the underlying pool of assets, rather than the cash flows generated by the business activity of the issuer.[12]

2.2.1 The Role of the True Sale of Assets to the Special Purpose Vehicle (SPV)[13]

Securitization is realized through a true sale of assets by the Originator to a separate company (SPV), which issues securities backed by those assets. The true sale mechanism allows a company to isolate a group of financial assets, separating the risk of the firm as a whole from the risk associated with the securitized assets.[14] Second, the SPV represents a critical actor within the securitization process: it servers as a vehicle to accomplish a securitization transaction.

In order to understand the crucial role played by the SPV, let us consider the following scenario. Imagine that the Originator could directly issue securities

[11]For a critical overview on the real efficacy of rating agency to actually forecast the default probability of a issuer, see, e.g. Lützenkirchen et al. (2013), Peicuti (2013).

[12]In a traditional financing, instead, the return to investors depends mainly by the capacity of the company to generate sufficient cash flows to repay its debt obligation.

[13]This Section is based on Tasca and Zambelli (2005).

[14]The expected return to investors depends mainly on the risk associated with the cash flows guaranteed by the securitized assets, rather than the default risk of the Originator.

backed by a pool of assets, without selling it to an intermediate vehicle. In this scenario, the investors interested in buying the Asset Backed Securities would carry both the default risk connected to the entire business activity of the Originator and the risk related to the securities. In reality, in a securitization transaction investors are willing to assume only the risk related to the performance of the pool of assets they are investing in. In order to protect final investors against the bankruptcy risk of the Originator, it is crucial to isolate the securitized assets from its business activity and its creditors. To guarantee this asset isolation, it is necessary to structure the transaction as a "true sale" of assets between the Originator and a third independent entity, the SPV, a purpose company which is established exclusively for the accomplishment of the financing.

The SPV involvement provides an investor with greater protection against the default risk of the Originator.

The SPV is a separate company which is intended to generate a perfect legal segregation of the assets from the originator. In principle, once a pool of assets is transferred to a special independent vehicle, it is no longer available to the Originator or to its creditors. The assets "isolated" from the Originator and the cash flows arising from them can only be used by the SPV to make payments to the holders of the Asset Backed Securities. The vehicle can only hold specific assets and issue in turn securities backed by these assets. The SPV is not allowed to begin other business activities and to assume new obligations. This is why the vehicle is also called a "bankruptcy remote entity".[15]

With reference to the asset isolation effect, it is important to highlight a crucial difference between the Italian and the US regulation system.

In the US the assets sold by a borrower before falling into bankruptcy do not become part of the bankruptcy procedure. Consequently, the Originator is not allowed to reclaim the transferred assets and so, in case of default, its creditors cannot call on them to satisfy their claims.[16]

By contrast, according to the Italian Securitization Law, the risk of reclaiming the sold assets is eliminated only if the sale occurred more than one year before the bankruptcy event (Fig. 2.2). Secondly, it is necessary to demonstrate that the assets have been sold to a fair price. If the above conditions are not satisfied, it is legally possible for the Originator (or its creditors) to reclaim the assets transferred to the SPV.[17]

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[15] JCR-VIS Credit Rating Company Limited (2003), Bond Market Association (2002).

[16] In the US there is no bankruptcy code provision regulating the legal status of securitized assets. Securitized assets are then considered as legally sold and are excluded from the eventual bankruptcy procedure of the Originator. For more information see: Nomura Fixed Income and Research (2002).

[17] In case of bankruptcy of the SPV, the period in which it is possible to reclaim the sold assets is six months. See: Spotorno (2003).

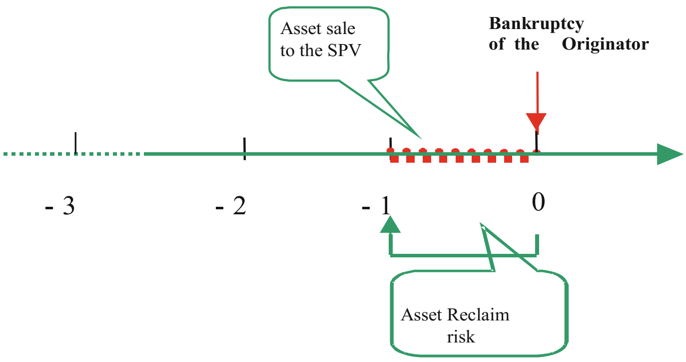


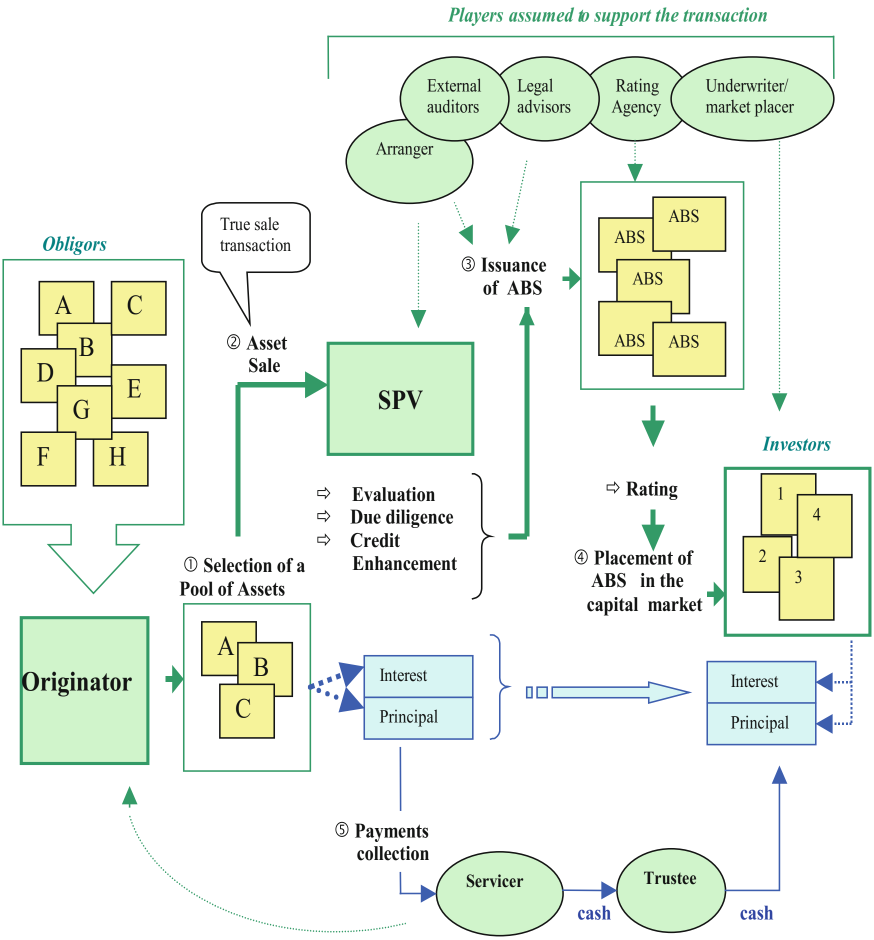
Fig. 2.2 Asset Reclaim Risk, according to the Italian Securitization Law (L. 130/99). *Source* Tasca and Zambelli (2005, p. 11)

2.3 The Alchemy of Securitization: The Process from Assets to Securities

In order to understand how it is possible to transform illiquid assets into marketable securities, let us describe the main steps that are required for accomplishing a typical securitization transaction.[18] As Fig. 2.3 shows, a securitization process involves the following phases.

- **Selection of a pool of assets**. In the first place, the Originator has to identify a pool of assets with similar characteristics. Theoretically, any asset producing cash flows (receivables, residential and commercial mortgages, credit card receivables, and other consumer and commercial loans) can be securitized, including non-performing loans, as we will emphasize in the course of the analysis of the Italian securitization market.
- **Asset sale/True sale transaction**. In the second place, it is necessary to guarantee the isolation of the pool of assets from the Originator. As noticed, this is achieved by structuring a true sale of the pool of assets by the Originator to an external entity (SPV), which has no business other than acquiring assets and issuing securities backed by these assets.
- **Definition of the capital and the legal structure of the transaction**. Arrangers and advisors (legal and agents) define the capital structure of the transaction, define the flow of funds (waterfall) and draft the set of legal contracts to govern them.
- **Issuance of asset backed securities**. To finance the acquisition of the assets, the SPV issues securities to be sold in the marketplace to investors. These debt

[18]See, e.g., Saunders and Cornett (2004), Burton et al. (2003), Spotorno (2003), Leixner (http://pages.stern.nyu.edu), JCR-VIS Credit Rating Company Limited (2003), Nomura Fixed Income and Research (2002), Bond Market Association (2002).



Example

Let as imagine that the loan originator is a financial institution, such as a bank. It wants to raise money by liquidating a pool of mortgage loans. A SPV is established for this purpose and a securitization transaction is then structured following several steps.

① A pool of assets is selected by the Originator (bank). The bank decides to sell a group of commercial and residential loans.

② A true sale transaction follows. The bank sells the selected pool of loans to a SPV.

③ After a detailed evaluation of the assets and a credit enhancement procedure, the SPV then issues securities backed by the acquired underlying assets (ABS). To make them more attractable, a rating is associated with the asset baked securities.

④ A placement in the market place follows. Underwriters and market placer will help the SPV to sell the securities into the market. Typical final investors are: mutual funds, banks, insurance companies and pension funds.

⑤ The servicer (usually represented by the originator) then collects the monthly payments (in the form of interest and principal) associated with the acquired assets and then forward these cash flows to the Trustee. In the end, the Trustee forwards these payments to the final investors.

Fig. 2.3 Basic securitization process. *Source* Tasca and Zambelli (2005, p. 13)

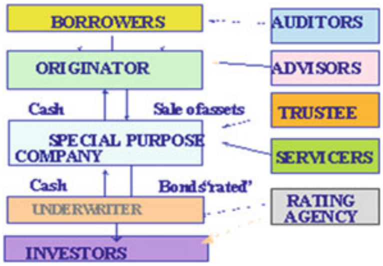


Fig. 2.4 Players involved in the securitization process. *Source* Tasca and Zambelli (2005, p. 17)

securities are secured by the underlying assets acquired by the vehicle (Asset Backed Securities—ABS).

– **Market placement**. The SPV sells these securities on the capital market, through a private placement or public offering, with the help of underwriters. Usually, the ABSs are purchased by banks, insurance companies, pension funds and other institutional investors.
– **Payment of the asset purchase**. In the end, the funds raised by the SPV from the market placement are used to pay the pool of assets originally acquired by the vehicle.

As a result of the securitization process, funds will flow from the investors to the issuer (SPV) and from the issuer to the Originator.[19]

The securitization process is simplified in Fig. 2.3 and the actors involved are summarized in Fig. 2.4. The process summarized in Fig. 2.3 is quite basic. In reality, it actually involves more steps and additional players, especially financial intermediaries and legal firms, to support the accomplishments of the entire operation.

For example, in order to ensure marketability to the ABSs and to make them more appealing, a credit rating from specialized agencies can be a requirement for the issued securities. The rating identifies the creditworthiness, in term of the default risk, of the issuance and could affect the cost of the entire operation, paid by the Originator.[20]

Some credit enhancement strategies can be useful to improve the credit rating of the marketable securities and to limit some of the risks that are transferred to investors. These procedures aim at creating specific mechanisms to mitigate potential losses.

Credit enhancement can be either internally determined within the transaction structure (internal enhancement) or externally provided by a third party (external enhancement).

[19]Saunders and Cornett (2004), Burton et al. (2003), Spotorno (2003), Leixner (http://pages.stern.nyu.edu), JCR-VIS Credit Rating Company Limited (2003), Bond Market Association (2002).

[20]For more information: JCR-VIS Credit Rating Company Limited (2003), Nomura Fixed Income and Research (2002).

Typical examples of internal credit enhancement provisions are the following:

1. Overcollateralization. In this case, the fair value of the underlying assets acquired by the SPV is greater than the total face value of the issued securities. Excess cash flows will then be used to cover potential losses or to cash flows temporary shortfalls;
2. Spread accounts. The spread is represented by the positive difference between: the yield generated by the underlying assets and the yield associated to the related securities, issued by the SPV. This spread is retained by the SPV in order to absorb potential losses;
3. Reserve Funds. A cash reserve fund might be created in order to cover potential cash flows shortfalls;
4. Senior/subordinated debt structure. With this provision, the SPV sells different types of securities (senior, subordinated) with different risk/return/maturity characteristics. In particular, the securities have different payment priority. Senior securities are characterized by a lower risk, higher rating and lower return. Conversely, junior securities are more risky. As a consequence, they are associated with: lower rating and higher expected return. In the worst-case scenario, senior securities give the holder the right to receive the related payments before the subordinated securities ones. Consequently, cash flows will be used to pay the senior securities and eventually, only if sufficient capital is left, to satisfy the subordinated securities.

External credit enhancement examples are the following:

1. Letter of credits by a bank or insurance company, to guarantee the security issuance;
2. Insurance contracts;
3. Special guarantees from a third party.

As Fig. 2.3 shows, among the parties that are involved into the securitization process there are: the Servicer and the Trustee.

The Servicer is responsible for the management of the portfolio and the collection of receivables and other payments on the assets acquired by the SPV. Often the management of the portfolio is delegated the Servicer to a Special Servicer, an entity whose capabilities and track record are specifically focused on the management of a particular asset type.

The Trustee is an independent third party (usually a bank) assumed to monitor the entire collection process and to make payments to the security-holders. Its aim is to protect the investors' interests, monitoring the regular payment-reports prepared by the Servicer.

Often the Originator acts as a Servicer. This occurs when the Originator is represented by a financial institution. In this situation, the obligors continue to make payments to the Originator, which will forward the cash flows to the Trustee. Then, the Trustee will forward the cash flows collected by the Servicer to the final investors.

2.4 The Main Players Involved in a Securitization Transaction

The securitization process requires that:

1. The asset seller (Originator) shall be a company satisfying the requisites provided by law;
2. The sums paid by the assigned debtors shall be exclusively dedicated to the satisfaction of the debt service and principal payment of the securities issued by the special purpose vehicle and to the payment of the transaction-costs.

Credit derivatives, wholesale securitization and synthetic securitization transactions are not included into the above legal definition.[21]

As it is shown in Fig. 2.4, the securitization process involves many players, with different roles: borrowers, loan Originator, special purpose company, rating agency, credit enhancer, underwriter and investors. In particular, the main actors of securitization can be summarized as follows:

1. The **Asset Seller** (Originator);
2. The **Special Purpose Vehicle** (“SPV”): the SPV is the entity specifically established to undertake particular securitization transactions, which would hold the legal rights over the assets transferred by the Originator. It can take the form of limited liability company (s.r.l.). A disposition of the Central Bank Governor also requires the SPV to be recorded into a special register (according to the provisions of the Legislative Decree 58/98 art. 107) and to satisfy minimum capital levels, depending on the volume of transactions managed[22];
3. The **Arranger**: the financial institution (bank or other) which agrees to structure a securitization transaction. The Arranger is responsible, alone or through a syndicate structure, for the placement of the ABS;
4. The **Servicer**: entity which is responsible for the day-to-day collection. In many case the Originator also performs the role of the Servicer, based on a servicing agreement;
5. The **Trustee**: institution (bank or other) which administers the securitization transaction, manages the inflow and outflow of moneys and does all acts needed for protecting the investors’ rights.
6. **Other actors**: collection account bank, deposit account bank, cash manager and paying agent, representative of the noteholders, liquidity facility provider, corporate services and administrative provider, hedging counterparts. These actors play an important role in order to implement the collection, deposit, management and hedging of the cash-flows and risks related to the securitization process.

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[21] See Caputo et al. (2001).

[22] By law, the SPV needs be exclusively involved with one or more securitization processes.

Moreover, other agents may be involved into a securitization process:

- **Rating agencies**: institutions which assign credit rating to debt obligations after analyzing the issuer and the transaction characteristics;
- **Legal consultants** for the deal structuring;
- **Auditors**: institutions dedicated to the due diligence of the credit portfolio.

Other financial institutions might be involved in the securitization process, in order to guarantee collateral services, such as the hedging against the interest rate risk.

2.5 Key Regulatory Issues on Securitization

The issuance of a dedicated Regulation is essential when establishing a framework for securitization in any jurisdiction. In every jurisdiction the key elements of securitization are regulated within specific set of regulatory standards: asset and cash flows segregation from the originator default risk, limited recourse rights towards SPV assets against any third party claim, limited enforcement rights against the sale of the securitized portfolio, asset sale de-recognition in the originator's books.

But regulation plays a relevant role in enhancing the development of the securitization market also at multi-jurisdiction level in leveling the playing field for originators and investors.

As a response to the global financial crisis, policy makers and Regulators around the world started introducing a more and more stringent framework for securitization in order to contribute to make the financial industry more resilient and to facilitate the sustainability of the development of the securitization market. The regulatory effort is still in progress but the key elements are related to the evidence that securitizations can be structured in many different ways and in different jurisdictions but the capital treatment of the exposures toward risks embedded in a securitization must be considered on the basis of its economic substance than its legal form. The stress on economic substance versus legal form is one of the milestones now achieved.

Another pillar stone of the regulatory effort is to enhance the level of transparency and disclosure, both on the securitized underlying portfolio level and on the securitization legal and capital structure level. Simple, transparent and comparable (STC) securitization criteria has become a mantra for regulators.

Simplicity refers to the homogeneity of the underlying asset pool and the capital structure of the transaction.

Transparency aims to providing a sufficient level of disclosure on the portfolio, the structure of the transaction, the parties involved in it and in promoting the understanding of the risks embedded in the structure at closing and during the life of the transaction.

Comparability intends to facilitate asset class analysis and performance valuation within different jurisdictions.

STC criteria are intended to facilitate any party involved in a securitization transaction (originators, investors, agents, accountants and regulators), in the assessment of the risks and the returns involved in a particular transaction and to enable them to make more easy comparisons across different deals within an asset class and across portfolios originated in different jurisdictions.

STC criteria help investors in accessing to a greater and reliable set of information about the underlying asset characteristics and performance and the legal and capital structure of the transaction. STC criteria compliance will be rewarded with a preferential regulatory capital requirements standards.

2.6 The Role of Financial Intermediaries Within the Securitization Process

Financial intermediaries play a crucial role within the securitization process, which includes different steps.

The first step includes the identification of homogeneous financial assets, which can be securitized, according to the provisions specified by the art. 1 of Law 130/99 and the rating criteria used by specialized agencies. Informally, the Originator and the Arranger might contact rating agencies in order to receive general advices on the feasibility of the operation and on the entity of credit enhancements. The aim of credit enhancement is to enhance the security or the rating of the securitized instruments. Credit enhancement can be internal (subordination, over-collateralization, yield spread, excess spread, reserve funds), or external (third party guarantee, letter of credit, cash collateral account, collateral invested account).

The second step of the securitization process refers to the asset-evaluation implemented by rating agencies, on the basis of investigation and analysis of the transaction and the issuer's characteristics. In particular, rating agencies tend to evaluate:

- Assets characteristics;
- Credit tracking;
- Payment methods and timing;
- Diversification of the asset portfolio;
- Default rate;
- Recovery Timing.

According to the above assets characteristics, the chosen credit rating agency defines together with the Arranger the particular financial structure of the securities that will be issued into the market and backed by the assets. Credit rating agencies usually identify: collateral guarantees, "priority of payments", and those credit events that will imply an anticipated reimbursement (trigger events).

If the analysis implemented by credit rating agencies is positive, a pre-sale report is then written. In the same time, external auditors implement a due diligence of the asset portfolio.

Once the pre-sale report has been completed, legal firms implement the deal structuring.

The basis set of legal contracts that are governing a securitization transaction can be divided into two main blocks: the asset transfer block and the issuance block. In the transfer block there are the following contracts:

- Transfer Agreement
- Corporate and Administrative Services Agreement
- Servicing and Sub-Servicing Agreement.

In the issuance block there are the following contracts:

- Term and Conditions of the Notes;
- Intercreditor Agreement;
- Master Definition Agreement;
- Cash Management Agreement e del Collateral Management Agreement;
- Trust deed;
- Deed of Pledge;
- Subscription Agreement;
- Prospectus.

After the deal structuring phase, it follows the marketing activity, including a road show aimed at presenting the transaction characteristics to institutional investors (usually assumed as target investors).

The fifth phase implies the issuance and the placement of the asset backed securities into the primary market. If the securities are sold to professional investors, it is important to design an Offering Circular, according to the provision of art. 2, Law 130/99. If the securities are sold to private investors, it is necessary to elaborate a particular information prospect, according to the Legislative Decree 58/98.[23] The underwriter works together with SPV and credit enhancer entity in order to arrange the placement of the securities into the primary market, usually offering to the Originator a service of acquiring those securities that remain unsold.

The next step of the securitization process is the acquisition of the securities by investors, represented exclusively by institutional investors. Once the placement of the securities into the primary market has been completed, the resulting net inflow is transferred by the SPV to the Originator. In this way, the debt obligation related to the asset sale initiated by the Originator is honored.

After the market placement, it is important to monitor and manage the cash flows payments that are related to the asset backed securities (servicing activity). Usually the Originator is responsible for the management of the portfolio and the collection

[23] So far, the issuances of ABS in Italy have been directed to qualified investors and consequently no information prospects have been elaborated.

of the cash flows arising from the portfolio. Sometimes, a separate Special Servicer might be employed for the management of the entire portfolio and collection process.

The Servicer must ensure that the SPV keeps the cash flows derived by the asset backed securities separated from the other SPV's assets. This "management and accountant isolation" represents a condition required by Law.

In order to better protect investor's interests, a Servicer must be enrolled into a special register held by the Regulation Authorities and also must have a sufficient own equity capital and has the duty to inform the monitoring Authority whenever irregularities in the payments-process occur.

2.6.1 Strategic Business Areas of Activities

As anticipated, financial intermediaries play a crucial role within the securitization process and can be distinguished according to the different areas of activity:

1. asset selling and underwriting (primary market);
2. cash flow collection and asset management (secondary market);
3. market making (secondary market).

1. The Arranger is the institution responsible for the asset placement into the primary market. The success and the costs of the whole transaction depend mainly on the placement skills of the Arranger. The choice of the Arranger is then very important and usually takes into account different elements: international experience; track record; business contacts (with investors and rating agencies); reputation; placement and underwriting capacity. There is a trade-off between reputation and cost of the deal of the Arranger: the greater is the reputation of the Arranger and the higher is the commission requested by him.
2. The Servicer is the institution responsible for the cash-flows collection. If it is a financial institution, the Originator act as Servicer. The Servicer activity is usually played by the financial institutions which originated the securitization process, or by a subsidiary (captive organizational structure). Independent Servicer institutions play a marginal role (but increasing) within the Italian securitization market. Regarding the securitization process initiated by big-size corporate, the Servicer activity is usually done by a separate financial institution, who takes care of the all collection activity. Small-medium size firms, on the other hands, usually sign a contract of sub-service with a financial institution, acting as a primary Servicer.[24]
3. The marketing maker activity is currently limited to the transactions among qualified investors, especially institutional investors (gross market).

[24]In the US market, the servicer activity is usually played by auditing companies and not by financial institutions.

2.7 The Securitization Industry: Current and Future Trends

The purpose of this section is to provide a picture of the recent trends in the international securitization industry and its level of vitality in term of issuance and outstanding volumes and in term of underlying pools diversification. The analysis of these elements are fundamental when assessing the maturity achieved any domestic jurisdiction, its peculiarities, its growth potential and its sustainability in the medium time horizon.

During the financial crisis Central banks support to the securitization primary and secondary market has played a fundamental role in maintaining the market alive and in providing and adequate level of support and confidence for a new start. Investor confidence suddenly faded away when Lehman collapsed, but the US Federal Reserve stepped in granting the market a huge ABS purchase support program that prevented the collapse of the US and International securitization industry.

ECB followed the same path during the Eurozone confidence crisis arising from the default of Greece. ECB first move was starting accepting ABS as collateral in its repo financing transaction, allowing European banks to regain access to interbank financing. The second move ECB has set in place is the ABS secondary market purchase program.

The first ECB measure was intended to prevent European ABS collapse while the second is intended to facilitate the restart of the market and to restore investor confidence.

The relevance of the Central banks interventions in the securitization industry testify how relevant is the ABS market regular functioning for the international banking industry in order to provide economies with and adequate level of financing to sustain growth.

The figures in Table 2.1 evidence the recovery of the US ABS market vis a vis of the European market, volumes in the US primary market show a good level of recovery, while European resilience in the level of issuance is supported by ECB.

In term of market diversification Europe is much more developed than its peers (see Table 2.2) but the differences can be explained by the different features of the financial industry for SME and Corporate financing in Europe and in the US and Australia.

In terms of ABS volumes outstanding, see Table 2.3 for European consolidated data, the figures show that some asset type are more resilient that others during the period, CLO/CDO and CMBS outstanding volumes shrank dramatically while RMBS decline path is subsequent the economic crisis that hit most part of the European domestic economies.

In Table 2.4 UK, Italy and the Nederland are the most relevant market in Europe in term of ABS issuance (in the figures Covered Bonds are not considered).

Table 2.1 Securitisation historical issuance (€ Billions)

	EU	US	AU
2007	594.9	2080.5	34.3
2008	818.7	934.9	6.6
2009	423.9	1385.3	9.7
2010	378.0	1203.7	15.5
2011	376.8	1056.6	20.4
2012	257.8	1579.2	14.8
2013	180.8	1515.1	22.4
2014	217.0	1131.5	22.1
2015	216.4	1620.7	19.9
2016	237.6	1792.9	9.6

Source Bloomberg, Bank of America-Merrill Lynch, Citigroup, Dealogic, Deutsche, JP Morgan, Macquarie, RBS, Thomson Reuters, Unicredit, AFME & SIFMA

Table 2.2 Issuance by Collateral (€ Billions)

	EU		US		AU	
	2016	2015	2016	2015	2016	2015
ABS	70.6	66.4	162.0	176.5	2.2	4.2
CDO/CLO	21.2	14.2	25.4	72.9		
CMBS	3.7	6.0	1529.5	1297.0	0.3	0.2
RMBS	119.4	102.0	76.0	89.0	7.1	15.5
SME	19.8	27.1				
WBS/PFI	2.9	0.8				
Total	237.6	216.4	1792.9	1635.4	9.6	19.9

Source Bloomberg, Bank of America-Merrill Lynch, Citigroup, Dealogic, Deutsche, JP Morgan, Macquarie, RBS, Thomson Reuters, Unicredit, AFME & SIFMA

Table 2.3 Outstandings by Collateral in Europe (€ Billions)

	2016	2015	2014	2013	2012	2011	2010
ABS	209.4	201.0	199.9	208.3	205.9	204.4	201.4
CDO/CLO	125.7	105.0	118.7	145.5	187.5	212.5	255.5
CMBS	67.6	79.3	86.0	100.7	117.5	134.0	154.5
RMBS	712.9	752.4	855.8	883.8	1008.7	1228.9	1325.4
SME	89.6	94.6	106.9	121.4	154.9	174.8	162.0
WBS/PFI	68.7	69.6	73.9	72.8	70.0	67.6	70.5
Total	1274.0	1,301.9	1,441.1	1532.7	1744.5	2022.2	2169.2

Sources Bloomberg (US & Europe), Fannie Mae (US), Federal Reserve (US), Freddie Mac (US), Ginnie Mae (US), Macquarie, Thomson Reuters (US), AFME & SIFMA

Table 2.4 Issuance by Country of Collateral (€ Billions)

	2016	2015	2014	2013	2012	2011	2010	2009
Belgium	3.6	1.2	4.1	2.0	15.4	19.0	14.1	27.4
Denmark			0.0	0.8			1.5	
France	20.3	17.0	50.6	9.9	14.9	16.4	9.2	6.9
Germany	17.5	45.1	18.3	21.9	10.2	12.9	13.4	17.5
Greece	1.3		0.2		2.0	6.4	1.0	22.5
Ireland	4.6	0.7	2.1	1.0	1.2		6.6	25.1
Italy	41.3	33.7	19.7	27.0	62.8	51.3	16.4	69.3
Netherlands	33.4	21.4	25.2	39.6	48.7	85.6	137.6	44.2
Portugal	1.3	4.9	2.9	3.3	1.4	10.6	16.9	10.5
Spain	34.0	26.3	27.2	27.5	18.6	61.7	55.4	64.9
UK	55.1	45.9	49.1	33.5	78.6	100.4	101.0	88.7
Other EU	1.4		0.8	1.6	1.3	0.6	0.1	
Other Europe	2.0	2.7	1.1	3.1	1.7	2.8	1.5	1.8
PanEurope	21.7	15.0	15.4	9.2	0.4	3.1	2.6	21.4
Multinational		1.2	0.2	0.4	0.5	6.0	0.8	23.7
European Total	237.6	216.4	217.0	180.8	257.8	376.8	378.0	423.9

Source Bloomberg, Bank of America-Merrill Lynch, Citigroup, Dealogic, Deutsche, JP Morgan, Macquarie, RBS, Thomson Reuters, Unicredit, AFME & SIFMA

2.8 Summary and Conclusions

In this chapter we analyzed the securitization process. The securitization is a process through which illiquid assets are transformed into negotiable securities to be placed in the market.

The Originator pools together a set of assets with homogeneous characteristics, in order to sell them to a particular special purpose company (SPV), which buys the rights to receive the related future expected payments. Thanks to the asset sale to a SPV, the transaction becomes unaffected by the bankruptcy of the Originator. To finance the purchase of the assets, the SPV then issues asset backed securities (ABS), with the help of investment underwriters.

Many steps are necessary to accomplish securitization transactions. The credits are first transferred by the Originator to a SPV, and then transformed into marketable securities to be sold to final investors. The payments on the receivables are then collected by a servicing institution (Servicer), usually represented by the Originator. The funds are forwarded to a third party (Trustee) which forwards them to the security holders.

As a response to the global financial crisis, policy makers and Regulators around the world started introducing a more and more stringent framework for securitization in order to contribute to make the financial industry more resilient and to facilitate the sustainability of the development of the securitization market. The regulatory effort is still in progress.

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Chapter 3
Project Finance

Stefano Gatti

Abstract Project finance is the funding solution that financial markets have developed to convey private capital to infrastructure investments. In this chapter, we first analyze the key characteristics of this structured finance transaction, pointing out the main ways it differs from corporate finance and the benefits it offers to sponsors and lenders. We then look at the market trends at a global and European level, indicating how the market has evolved in the past few years. As a final point, we focus on how capital markets providing debt to infrastructure have changed in response to the financial crisis and the drop in interest rates. What we'll discover is that the project finance market, once firmly under control of banks, is becoming an area of cooperation between banks and long term institutional investors. The entrance of new actors in the market of structured finance for infrastructure poses critical questions for regulators, who have to strike the right balance between the need to push private investment in infrastructure and to maintain financial market stability.

3.1 Introduction

This chapter provides an analysis of project finance transactions, centering on the key characteristics of this financial technique and the recent evolution of the market for infrastructure financing in Europe. We also look at trends in the European debt capital market for infrastructure and the emergence of new institutional investors in this asset class. Our objective is to show that the traditional approach to structuring a project finance deal (based on a strict bilateral relationship between a group of industrial sponsors and a group of syndicated bank lenders) has become more complex as a result of macroeconomic factors following the two financial crises of 2008 and 2011–2012. Financing infrastructure is no longer an exclusive area of banking activity; now a much larger number of actors play a part. The different

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© Springer International Publishing AG 2017

S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_3

utility functions of the various parties involved makes project finance deal structuring one of the most complex processes in the business of structured finance.

This chapter is organized as follows. Section 3.2 outlines the typical features of a project finance deal and compares it to traditional corporate financing (Sect. 3.2.1), highlighting potential benefits for sponsors and lenders in applying this structured finance technique (Sects. 3.2.2, 3.2.3. respectively). Sections 3.2.4, 3.2.5 present the network of contracts which form the basis of a project finance deal, and how this contributes to risk analysis and risk allocation. Section 3.3 analyzes the sectors where the technique is mainly applied. First we review the topic from a historical perspective and then we move on to the European experience. Section 3.4 explores the evolution of debt capital markets in Europe from two different viewpoints: the emergence of project bond issues and the new role played by long term institutional investors as possible partners for arranging banks. The main conclusion that emerges is that this business, traditionally in the hands of a small group of banks, is becoming an area of structured finance where the competences accumulated by banks in the past can be shared with the availability of long term financing that institutional investors are willing to commit to infrastructure. Section 3.5 draws up the main conclusions.

3.2 Distinctive Features of the Transaction

A rather common definition of project finance describes it as financing (by banks or via bond issues) that as a priority doesn't depend on the reliability and creditworthiness of the sponsors. Sponsors are the parties who propose the business idea to launch the project and who finance it with equity contributions. Approval doesn't even depend on the value or solidity of assets the sponsors themselves are willing to make available to financers. Instead, what really counts is the project's ability to repay the contracted debt and remunerate the capital invested at a level that's consistent with the degree of risk inherent in the initiative. We could also say that project finance is the structured financing of a specific economic unit (the SPV or Special Purpose Vehicle) which the sponsors create by means of equity. It's the cash flow from the SPV which the lender considers as the source of loan reimbursement, whereas project assets only represent collateral.

Below are, in essence, the distinctive features of a project finance transaction.

1. The debtor is a project company-SPV set up on an ad hoc basis, which is financially and legally independent from the sponsors.
2. Financers have only limited recourse (or in some cases no recourse) to the sponsors from the inception of the project. Specifically, the sponsors' involvement in the deal is limited in terms of: time (generally during the set-up to start-up period); amount (they can be called on for equity injections if certain economic-financial tests prove unsatisfactory); and quality (managing the system efficiently and ensuring certain performance levels).

3. Project risks are allocated appropriately among all the parties who participate in the transaction.
4. Cash flow generated by the enterprise must be sufficient to cover operating costs and service the debt, to repay the principal plus interest. As these cost items are priority, only after they are covered is any residue used to pay dividends to the sponsors.
5. Collateral is allowed by the sponsors to lenders as security written on assets belonging to the SPV.

3.2.1 A Comparison Between Project Finance and Corporate Finance

As the five points above indicate, the distinguishing feature of project financing is that this financial technique is based on the analysis of a stand-alone infrastructure project and on its ability, as a going concern, to generate a sufficient amount of cash flow. The no- or limited-recourse clause in the lending agreement obviously makes it less essential to carefully assess project sponsors.

Real-life experience shows that evaluations made by financers are much more complex and in depth than a mere certification of the soundness of the sponsors and the project they intend to launch. In fact, a project-financed deal must first and foremost show that it has a balanced, viable allocation of risks among the various parties concerned (see Sect. 3.2.5), regardless of whether they are from manufacturing, transport, telecommunications or any other sector. Apart from the risk in terms of the sponsors' economic and financial soundness, in project financing it is vital that financers also make a detailed analysis and evaluation of all other parties involved (Sect. 3.2.4): plant and equipment suppliers, constructors, managers, suppliers of raw materials, purchasers or users of the end product or service. From a bank's standpoint, project finance is above all seen as a question of credit, so the long term solvability of all parties gravitating around the Special Purpose Vehicle is of prime importance.[1]

Yet all of this in no way detracts from the essential differences between project financing and corporate financing. In the latter case, the lender has to assess a project that will have a direct impact on a company's financial statements and so

[1]Intuitively, the financial soundness of the parties involved in a project finance initiative should have a significant effect on the pricing of the transaction. However, there is very little evidence of this. Dailami and Hauswald (2007) found a strong negative correlation between the existence of a take-or-pay agreement (namely, long-term unconditional purchase contracts signed by a large buyer/offtaker) with sound counterparts and spreads set above leading rates for the issue of project bonds. Furthermore, the authors demonstrated a strong correlation between project bond pricing and the rating trend, and the product or service purchaser's financial situation. Similar results are shown by Bonetti et al. (2010) and Corielli et al. (2010) who demonstrate that when there are sponsors who also play the role of key counterparties of the SPV, this has beneficial effects on the loan spread and the Debt/Equity ratio of the initiative.

Table 3.1 Main differences between project finance and corporate finance. *Source* Gatti (2012)

	Corporate finance	Project finance
Collateral for financing	Assets of the borrower	Project assets
Effect of financial flexibility	Reduces the borrower's financial flexibility	Non-existent or very reduced as regards the sponsor's flexibility
Accounting treatment	On-balance sheet	Off-balance sheet (the only effect is the cash outlay to subscribe the equity in the SPV or disbursement of subordinated loans)
Main variables considered when granting credit	Customer relations Financial soundness Profitability	Future cash flows
Sustainable leverage	Depends on the effects on the borrower's balance sheet	Depends on cash flow generated by the project (leverage is usually greater)

requires a combined project-company evaluation. This will certainly include an appraisal of the project in its own right, but also the balance sheet, the economic and financial situation of the company that will carry out the project, its past history and existence of longstanding customer relations (and, therefore, the company's business turnover with the bank).

This implies that corporate financing is based partly (perhaps even entirely) on being able to tap much broader asset base than those relating specifically to the individual project: if the latter fails, the financer can always count on the company's other assets. This is not the case with project finance, where financing is granted on a no-recourse or limited-recourse basis. These clauses (especially the first) mean that banks have no access to the sponsors' assets if the project should fail. Even if there is limited recourse instead of no recourse, the situation doesn't change. While it is true that no absolute division exists between sponsors and the project, the conditions under which banks could have recourse to collateral given by the vehicle company's sponsors are just possibilities. In any event, full enforcement of the guarantees would not enable full recovery of the sums that were loaned.

Table 3.1 summarizes the main differences between project as opposed to corporate logic behind granting financing.

3.2.2 Why Use Project Finance: Benefits for Sponsors

In order to correctly assemble project-based financing, the arranging bank must first understand the sponsors' reasons for wanting to completely separate the destiny of the project from that of the respective companies.

The reason is not necessarily always to cut costs. Normally project financing actually costs more than a regular medium term industrial loan, a corporate bond issue or equipment leasing. Apart from the spread paid above interbank market base rates, project finance costs increase considerably as a result of technical and legal consultancy fees, which are fundamental to the successful outcome of the transaction.[2]

This means that the reasons for recourse to project finance lie elsewhere. Usually when a venture is significantly larger than the borrower's current activities, it's funded off-balance sheet by means of structured financing. In a structured transaction, in fact, the outcome of the venture doesn't affect the company or companies that want to launch the project, as the project itself is segregated in a separate company, for which several legal systems recognize limited shareholder-sponsor liability. The advantages of isolating the project from the company work both ways: a project with poor prospects doesn't affect the company's performance nor its survival; conversely, default of the company shouldn't affect the project's performance or the possible claims of creditors who are financing it (Brealey et al. 1996; Shah and Thakor 1987; Leland 2007; Esty 2002; Gatti 2012).

Therefore, off-balance sheet financing implies recognizing an important fact: if a venture is more creditworthy than the originating company, then realizing the project by isolating it in a vehicle company provides certain benefits that could otherwise not be realized. Naturally, the superiority of the new enterprise compared to the merits of the originator means that risks must be identified in advance and covered by contractual credit enhancement-type instruments or insurance, as we've already discussed in Point 3 of Sect. 3.2. Only by doing so can the enterprise's creditworthiness be rated even higher than that of the sponsor company or companies. This in turn reduces the cost of obtaining financial resources for funding, in addition, of course, to increasing the degree of financial leverage utilized.

In essence, recourse to an off-balance sheet deal to realize and finance a project potentially enables sponsors to enjoy several other advantages, such as:

1. cost reductions as regards funding for the sponsor;
2. the sponsor's financial flexibility remains intact;
3. the sponsor is buffered against any negative impact from the project.

The first is achieved in relation to the structuring cost for the initiative (which is very high, especially if the deal is extremely complex), if this cost is less than the savings on funding costs, owing to the fact that the venture can earn a higher credit rating than the sponsor. So a cost benefit is achieved if the structuring cost for the transaction and risk identification and allocation is less than the lower cost of resources raised to realize the initiative.

[2]Megginson and Kleimeier (2000) empirically show a modest difference between pricing (measuring the spread above interbank base rates) for project loans and syndicated corporate loans. This, according to the authors, is because a project finance transaction isn't necessarily riskier than a corporate-financed project if the underlying contracts constitute a valid mitigant for the financers' credit risk. In any case, the cost of structuring the deal in project finance form, if measured in terms of upfront fees paid, is relevant. Esty (2004) indicates an average of 5–10% of the total investment cost is paid for advisory fees to the different professionals who participate in the transaction.

As regards the second advantage—financial flexibility—we should mention that if the project is particularly large (that is, if it represents a substantial part of the sponsor's assets and consequently it's financial structure), realizing the venture would erode current credit lines, increasing debt and precluding possible future initiatives. Incidentally, we also need to remember that an increase in indebtedness could also produce immediate side-effects on the future cost of any new funding. By using an off-balance sheet structured deal we can avoid this effect: the funding concerns an ad hoc legal entity involving limited or no recourse to the sponsor/originator.[3]

The third is an "insurance-type" benefit against the possible default of the initiative. Depending on how the project's risks have been foreseen and allocated, if it fails the sponsor company or companies will still survive. By the same token, a further advantage of this kind is that if a company finances a project in-house (namely, if it includes the project in its own financial statements), banks can use the total company assets as collateral, not just project assets (i.e. co-insurance effect). Instead, as mentioned in Point 5 of Sect. 3.2, in project finance the only collateral is the project's assets, whereas the sponsor's assets remain unencumbered.

3.2.3 Why Use Project Finance: Benefits for Lenders

The separate incorporation of a project in an SPV is also seen favourably by lenders. It is true that creditors lose the advantage of the co-insurance effect of the more traditional corporate finance/on-balance sheet solutions, where in case of project default, creditors can fall back on the overall portfolio of corporate assets. Nevertheless, separate incorporation has its own advantages (see Novo 2012).

The first is that setting up an SPV completely isolates the initiative's cash flow from that of the originating companies/project sponsors, giving creditors full control over the vehicle's performance. The second is that financing an SPV allows creditors to concentrate their analysis on the performance of a single project, reducing information asymmetries that could arise in a corporate finance setting. The third reason, following on from the previous one, is that precisely because lenders finance one single initiative, they are in the position to dictate managerial behavior with the inclusion of an articulated set of covenants in the credit agreement that limit the discretionary power of sponsors and make monitoring easier than in normal corporate finance settings.

[3]The Special Purpose Vehicle's indebtedness has a real effect on sponsors when accounts are consolidated. The consolidated financial statements will actually show an increase in indebtedness as a result of the debt of the subsidiary vehicle company. If, however, financing granted by the creditor is on a company basis, not a consolidated basis, then the off-balance sheet financing fully achieves its aim.

3.2.4 Structuring a Project Finance Deal: The Contractual Network

As mentioned in Sect. 3.2.1, project finance is based on the incorporation of an SPV that becomes the center of a complex network of contracts (see Dailami and Hauswald 2007; Bonetti et al. 2010; Corielli et al. 2010).

An example is provided in Fig. 3.1.

Figure 3.1 shows all the key contracts that typically are set up in an infrastructure project finance. The project shown in the figure (Via A11), finalized at the end of 2014, involved the construction of a new motorway connecting the harbour of Zeebrugge with the E 40 to Ghent and E 34 to Antwerp in Belgium.

The right-hand side of Fig. 3.1 (Contract 1) shows the relationship between the SPV and the Public Authority (the Flemish Region). This is the entity that grants approval to the SPV to carry out the project with authorizations and permits. Sometimes, the private sector is mandated to operate the business based on a long-term concession agreement [like in the Via A11 case a DBFM (Design, Build, Finance and Manage)] (Valila 2005).

The upper part of Fig. 3.1 shows the relationship between the SPV and its sponsors. Project sponsors provide equity to the SPV based on an Equity Contribution Agreement (Contract 2). Funds are allocated in proportion to the selected debt/equity ratio for the deal on a pro-rata basis. In addition to pure equity, project sponsors can also offer subordinated debt (Contract 2a) to the SPV in order to optimize their payoff, particularly in the first years of operations (Yescombe 2002; Gatti 2012).

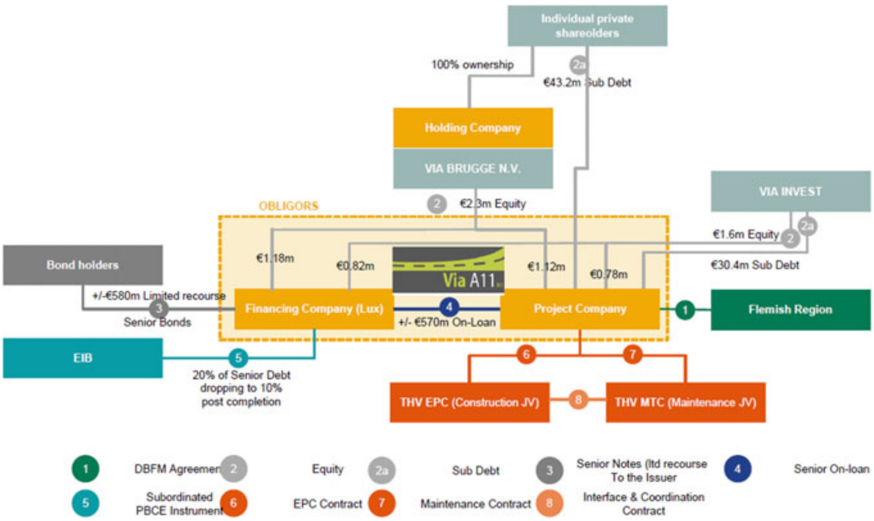

Fig. 3.1 Typical contractual structure of a project finance deal. *Source* Vecchi et al. (2015)

The left-hand side of Fig. 3.1 (Contracts 3–5) shows the debt funding received by the SPV. In this case, funding was sourced in debt capital markets via a bond issue backed by a 20% guarantee by the European Investment Bank (European Commission 2011). Proceeds from the bond sale were transferred to the SPV through a conduit vehicle domiciled in Luxembourg (Contract 4). As shown in Sect. 3.4.1, bond issues are becoming a viable alternative to traditional project finance bank loans. However, the latter still retain a large percentage stake in the overall debt market for infrastructure financing.

When bank loans are used, the typical structure is that of a syndicated loan. This loan is based on a multi-tranche facility consisting of a base facility to finance construction, start up costs and capitalized interests and fees, a value added tax facility in countries where VAT is in place, and a stand-by facility covering financial needs once the base facility has been exhausted. The loan package also includes a working capital facility for day-to-day needs once construction is complete (Gatti 2012; Finnerty 2007).

The bottom right-hand side of Fig. 3.1 shows the key project contracts underpinning the deal (Corielli et al. 2010; Novo 2012). Although any number of contracts can be drafted in a project finance transaction, four of them are essential for the soundness of the venture. Construction Contracts and EPC-Engineering, Procurement and Construction Contracts (Contract 6) are closed on a turnkey basis to make plant and equipment available to the SPV, usually at predefined prices, and respecting established delivery times and standards of performance. These contractual features are useful to shift the construction risk from the SPV to the contractor (Blanc-Brude et al. 2006).

Operation and Maintenance (O&M) Agreements (Contract 7) are designed to provide the SPV with efficient and effective plant maintenance, compliant with predefined service-level agreements, to avoid operational risk to the SPV.

Additional key project contracts include purchasing agreements and selling agreements. Raw material supply agreements are stipulated with relevant suppliers who guarantee input to the SPV at predefined quantities, quality, and prices on a put-or-pay basis. This means that the supplier unconditionally guarantees the needed input or pays liquidated damages to the Vehicle, insulating it from the risk of lack of raw materials during the operational phase. Selling agreements, often known as take-or-pay or off-taking agreements, enable the SPV to sell part or all of its output to a third party (offtaker) who accepts to unconditionally buy the output, again at predefined prices and for a given period of time. In this way, market risk is shifted to a third party.

It is important to note that in most project finance transactions, project sponsors are also contractual counterparties of the SPV. This is perfectly natural if we consider that the primary interest of sponsors is to appropriate the highest share of cash flows generated by the project. Combining the shareholder role and contractual counterparty role is a way to prevent opportunistic behavior of project sponsors and to limit agency costs (Brealey et al. 1996; Corielli et al. 2010).

Figure 3.1 can also be viewed from the financial/capital budgeting standpoint. Each contract between the SPV and its counterparties generates cash inflows and

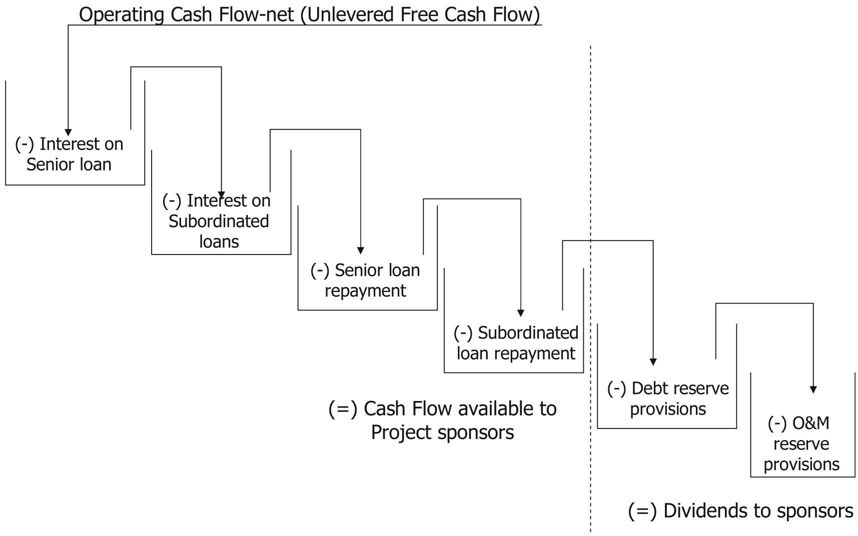

Fig. 3.2 The cash flow waterfall of a typical project finance transaction. *Source* Gatti (2012)

outflows that contribute to the overall ability of the vehicle to repay operational costs, debt service and dividends to sponsors. A reinterpretation of Fig. 3.1 in light of cash flow dynamics is shown in Fig. 3.2.

3.2.5 Structuring a Project Finance Deal: Risk Analysis and Risk Management

In structured finance transactions, risk analysis and risk management are the keys to the success of the deal, much more than in a traditional corporate finance setting. The reason for this is that the SPV manages only one single investment, and if this investment is excessively risky, creditors could face situations where the cash flows produced by the new initiative are not sufficient to repay debt service. In corporate finance, instead, even if a project experiences troubles, other assets of the firm can contribute to generating cash flows earmarked to repay creditors.

Sponsors, banks and their technical, legal and financial advisers carry out a thorough due diligence of the risks of the project as a preliminary step to any capital budgeting exercise. Due diligence goes through the following key steps:

1. risk identification and risk analysis;
2. risk transfer and allocation to the actors best suited to ensure coverage against these risks;
3. residual risk management.

Step 1 sees sponsors and lenders together carefully mapping out all the possible risks that could arise during the life of the project and assessing their probability of occurrence and severity (Gatti et al. 2007; Borgonovo et al. 2009). Risks may arise either during the construction phase, when the project is not yet able to generate cash, or during the operating phase. Here is a possible classification of risks (Gatti 2012):

- precompletion risks
- postcompletion risks
- risks found in both the pre- and postcompletion phases.

Regarding precompletion risks, the most common stem from bad activity planning, or entail technology risk and construction risk (in the form of delayed completion, cost overruns or completion with performance deficiency). Postcompletion risks are associated with the supply of input, the performance of the plant as compared to minimum project performance standards, and the sale of the product or service. Risks found in both phases involve financial risks (interest rate risk, currency risk, inflation risk), regulatory risk related to permits and authorizations, political and country risk (Hainz and Kleimeier 2012) and legal risk linked to contract enforceability and creditor rights protection (Subramanian and Tung 2016).

Once risks are mapped and analyzed, Steps 2, 3 serve to identify the strategies that the SPV can use in order to mitigate the impact of risks on project cash flows. Three different options are available to sponsors and lenders:

1. to transfer the risk by allocating it to one of the key counterparties
2. to transfer the risk to professional insurers
3. to retain the risk.

Risk transfer by means of project contracts is the most often used risk management strategy in project finance. Going back to Sect. 3.2.4, the key contracts signed by the SPV (EPC, Supply, Purchase, O&M Agreements) allocate rights and obligations to this Vehicle and its respective counterparties. These contracts can be used as risk mitigation techniques if the counterparty in the best position to control and manage a given risk is made liable for the negative effects on cash flows that would result. If the risk does materialize, some form of indemnification must be paid to the SPV. The related damage-payment due incentivizes the counterparty to respect the original agreement in order to avoid the negative fallout determined if the risk in question should emerge. If a risk arises and it has been allocated (transferred) to a third party, this same party will bear the cost of the risk without affecting the SPV or its lenders.

The second option available for risk management is risk insurance. This alternative must be used for risks that cannot be controlled and managed by any one of the SPV counterparties. In this case, insurers cover the SPV risks against the payment of an insurance premium. These companies can do so because they

	Risks found in both the pre- and post-completion phases							Pre-Completion Phase Risks		Post-Completion Phase Risks		
	Rate Risk	Interest Rate Risk	Inflation Risk	Environmental risk	Regulatory Risk	Political Risk	Country Risk	Technological, Planning, or Design Risk	Construction Risk	Operational Risk	Supply Risk	Demand Risk
Special Purpose Vehicle	Currency matching							Sponsors' guarantees to lenders				
Contractor					Limited to obtaining building permits			Included in the construction agreement	Fixed price turnkey agreement	Turnkey agreement (first test)		
Technology Supplier								Penalties to be paid				
Operator										Penalty payments and removal of operator (later tests)		
Buyers			Establishing pre-agreed inflation adjustments									Take or pay
Suppliers			Establishing pre-agreed inflation adjustments								Put or pay agreement or through put agreement	
Export Credit Agencies (ECAs)						Credit insurance programs	Credit insurance programs					
Banks	Derivative products and coverage instruments	Derivative products									Endorsement credit to back supplier's loans	Endorsement credit to back buyer's loans
Insurance Companies				Insurance policies			Insurance policies		Insurance policies	Insurance policies		
Independent Engineering Firms								Assessments on technological validity		Certification of later testing		

Fig. 3.3 The risk matrix of a project finance deal. *Source* Gatti (2012)

manage large risk portfolios where the joint probability of emergence of all the risks in the portfolio at the same time is very low.[4]

The final option to control the risk is to retain it and to try to limit its effects on SPV operations by means of well-designed internal risk procedures. Risk retention is a common practice in already-existing corporations (Carter et al. 2006; Froot et al. 1993; Minton and Shrand 1999; Nocco and Stulz 2006). A firm might decide to deliberately retain risks because it considers risk allocation to third parties too expensive, or the cost of insurance policies excessive compared to the effects determined by the risks in question. Risk retention, as a residual risk management policy, is more effective for existing corporations than for SPVs. This is because in standard corporate finance, operational risk can be diversified across the whole portfolio of real assets managed by an existing firm. Instead, operational risk for a SPV is idiosyncratic, as it refers to a single project. For this reason, the unallocated portion of risk plays a key role in the credit spread and debt/equity ratio level (Corielli et al. 2010).

The output of the risk analysis and risk allocation process is the risk matrix of the project finance transaction (Fig. 3.3). This matrix is easy to read. Rows indicate the contribution of each party involved in the deal in eliminating project risks; columns show whether each of these risks has been identified and adequately covered. The matrix can be used by lenders to map risks and assess whether they have been assigned to at least one of the SPV's counterparties. The principle is rather intuitive: the greater the number of intersections between actors and risks where the SPV does not appear, the lower the project risk that the SPV's lenders have to face.

[4]Regarding insurance coverage, lenders require the SPV to undersign insurance policies against a number of risks identified by an insurance broker as a condition precedent for any debt drawdown. The different insurance products available on the market (Gatti 2012) are coordinated and linked to the project's contractual structure via an OPIC (Owner Controlled Insurance Programme) (Cottone and De Rocco 2016).

3.3 Project Finance: How It Is Applied and When It Is Most Often Used

This section provides a market analysis of project-financed initiatives. First we'll look at how the market has developed over the years, describing long-term trends and discussing the sectors where the technique is most often used. Following this, we'll take a close look at the European situation, again to identify the most attractive sectors for adopting project finance from the standpoint of private investors.

3.3.1 Past Development of Project Finance and Market Sectors

Modern examples of project finance date back to the Thirties in the United States when the technique was used to fund oil exploration and successive well drilling in Texas. Financing was granted based on the producer's ability to repay capital and interest using revenue from the sale of crude, often counter-guaranteed by long term supply contracts. Then in the Seventies project finance was also adopted in Europe, again in the oil sector: in fact it became the financial technique of choice used for funding oil extraction off the British coast. Again in the Seventies, when the power production market was regulated in the USA (by PURPA, the Public Utility Regulatory Policy Act of 1978), Congress encouraged energy production from alternative sources by requiring that utility companies purchase all the power output from qualified producers (IPPs, Independent Power Producers). From then on, project finance was applied more and more often to create power production plants using both traditional and alternative or renewable sources.

So from a historical standpoint, project finance was first launched in well-defined sectors marked by two salient factors:

1. the presence of a captive market, thanks to long-term contracts to secure pre-determined revenues, involving large, financially sound buyers (so-called offtakers);
2. the absence of high-level technological risks during plant construction.

In these sectors right from the early days, the role of sponsor was covered by large international constructors/developers and multinationals in the oil sector.

Instead the growth trend in the following years moved along two lines. During the first expansion, project finance was exported to developing countries. This was promoted by developers themselves who, faced with a progressive reduction in market opportunities at home, proposed the technique to the governments in these nations as a means to rapidly create basic infrastructures and ensure a greater contribution from private capital, guaranteed by Export Credit Agencies in their own countries.

But the second change in the project finance market took place in industrialized countries themselves, where the technique had initially been tested out in more traditional sectors. This expansion came about from using project finance as a means to realize the following types of off-balance sheet initiatives.

1. Projects that had a lower or less effective market risk coverage: examples are sectors in which there was no single large buyer, like toll road infrastructures, leisure facilities, telecoms or urban parking solutions.
2. Projects in which the government intervened to promote the realization of public works. In many cases, these initiatives were incapable of repaying investment costs, operating expenses and debt servicing by means of market tariffs; as such they had to be subsidized to a greater or lesser degree by recourse to public funds. In certain countries in Europe, first and foremost in the UK, the use of the project finance technique to realize public works reached considerable proportions in the wake of the PPP (Public-Private Partnership) program, known as PFI (Private Finance Initiative). In other European countries, recent years have seen an intensification of this phenomenon and nowadays most countries have in place special regulations involving the procurement of infrastructure projects via partnerships with the private sector (Gatti 2012; EPEC 2016).

The reasons given above can be summarized in the following figure (Fig. 3.4).

As we note, Segment II constitutes the benchmark for project finance initiatives. The two arrows pointing towards Segment I and Segment III indicate market trends that followed the initial use of project finance technique. Also, it is clear that while Segment IV is possible, it represents an unfavorable combination of factors for project finance applications. In fact, faced with high uncertainty, a very inflexible contract structure and high financial leverage, it would be difficult for management

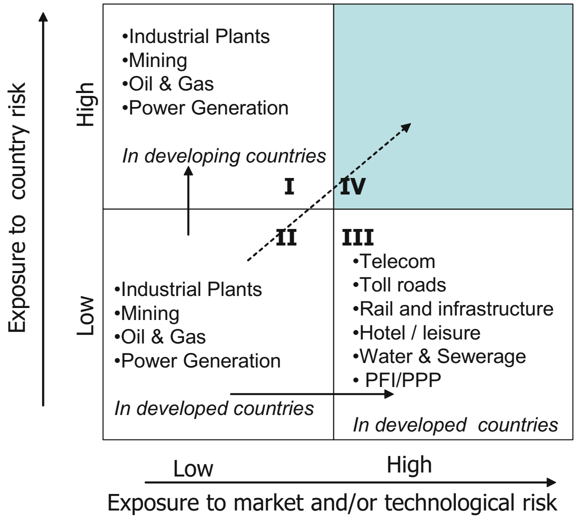

Fig. 3.4 Project finance market trend by market and degree of underlying risk. *Source* The author, based on Esty (2002)

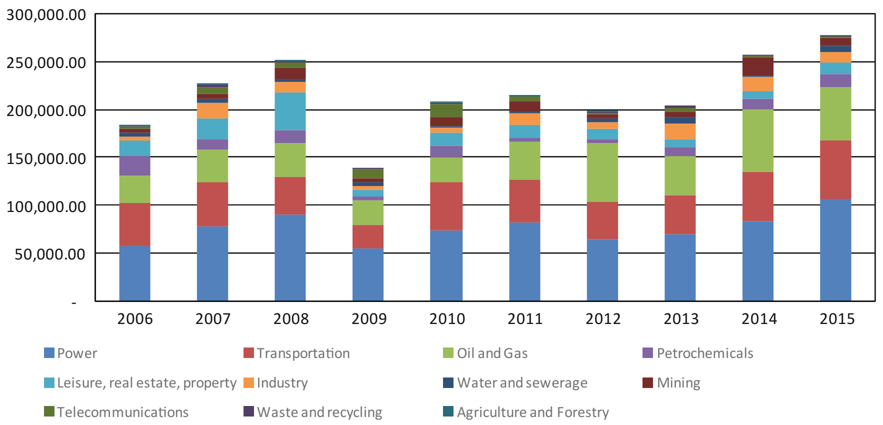

Fig. 3.5 Global Project Finance—Breakdown by Sector 2006–2015 (in millions of dollars) *Source* Thomson Deal Intelligence

to respond or adapt rapidly to change. In such circumstances, recourse to a corporate finance approach is undoubtedly advisable.[5]

An analysis of data at a global level confirms these considerations. Figure 3.5 plots the evolution of the global project finance market between 2008 and 2015. As this figure clearly shows, three sectors (power, transportation and oil & gas) represent more than 75% of the global market; the others are less relevant and stable across the period.

After a major drop in 2008, following the Lehman crisis, the market has rebounded quite rapidly. Indeed, at the end of 2015, it has gone beyond complete recovery, with amounts higher than the pre-crisis period. Compared to other more traditional asset classes like listed stocks and bonds, infrastructure financing has proven to be a resilient asset class, even able to overcome stressed macroeconomic scenarios.

The matrix in Fig. 3.4 is also insightful from another standpoint. In fact a distinction is made between sectors in which project finance is a viable option depending on the initiative's capacity to sustain the attendant investments and costs from the cash flow it generates. In particular, Segments II and I encompass sectors where the product can be sold at market prices based on long-term supply contracts (take-or-pay agreements or offtake agreements). However, projects in Segment III usually have problems in setting a market price capable of generating sufficient returns for sponsors (except for the hotel & leisure business). These are goods that have a pronounced spillover effect (for instance, water management or urban and social development in depressed areas), or that are associated with a population's needs, the cost of which has a significant impact on less well-off segments (health

[5]Recent examples of investments in telecom towers in some Sub-Saharan countries by infrastructure private equity funds are clear examples of projects belonging to Segment IV (Blas 2014).

and personal care, for example). In these cases, total privatization of the sector would mean that the service offered once the project is realized would not be accessible to certain segments of the population. So recourse has to be made to public financing by means of capital grants, which mitigate investment costs for private sponsors, consequently lowering the price or tariff levels imposed on end-users.

Based on the above, project finance deals can be divided into fully self-financed initiatives (project finance interpreted in a strict sense) and partially self-financed initiatives. As regards the first category, the basis for evaluation is the soundness of the contractual architecture and counterparts; for the second category, apart from the factors indicated, bankability will be determined by the level of the public grant. However, it must be noted that the involvement of a public body in project finance could have possible negative effects for the private party, exposing it to the risk of renegotiation if the project is not strategic for the country or when public finance is extremely limited (Deloitte 2016; Fitch 2015).

3.3.2 The European Situation[6]

In this section, we focus our analysis on the trends in the European market for project and infrastructure financing. The timeframe, covering the period from 2008 to 2015, allows us to examine the effects of severe macroeconomic stress which occurred in Europe on two occasions: the post Lehman crisis of 2008–2009 and the Sovereign Debt Crisis of 2011–1012.

Figure 3.6 shows the trends in the European project finance market in terms of project value, along with the percentage of the projects represented by PFIs or PPPs.

From a quantitative standpoint, the European project finance market shows two big drops in value in 2009 and 2012 as a response to the macroeconomic recession. However, in line with what we've seen at a global level, the trend has been positive since 2013. From about 59 billion euro, the market has reached a record peak of 70.3 billion, surpassing the 2008 value of 66.3 billion euro. The most representative sectors, in terms of average percentage of the total over the whole period, are road

[6]Data referred to in this section are taken from Dealogic's ProjectWare database and include both new initiatives and the refinancing of previously syndicated transactions on a project finance basis. Recourse to this type of data must be weighed carefully because of the limitations of the databases themselves. More precisely, transactions recorded in ProjectWare don't cover the entire universe of project finance transactions assembled in any one year or in a certain country, given that input to the databases is provided by advisers or arrangers themselves. Normally this means data concerning the majority of smaller projects assembled at a local level (and probably not even syndicated) are not captured as they are structured directly by the promoting bank. This limitation becomes even more critical in more detailed surveys pertaining to an individual country or geographical area.

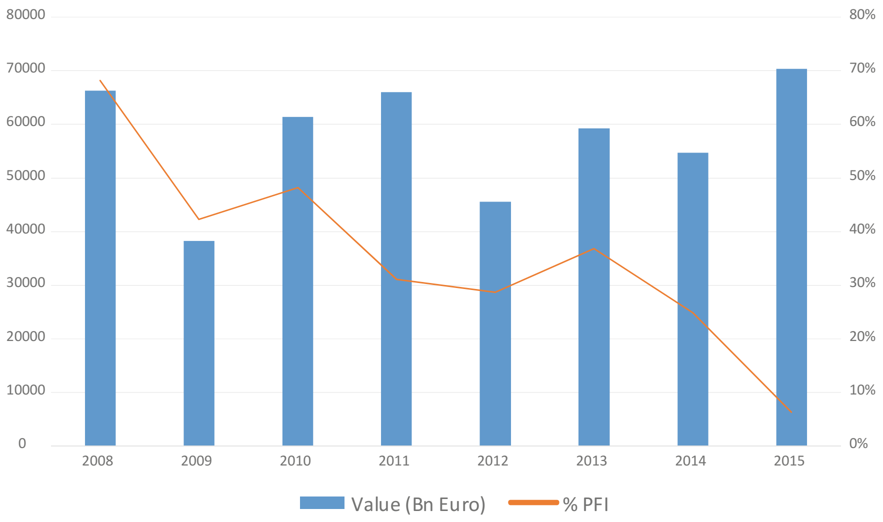

Fig. 3.6 European project finance initiatives 2008–2015 (in millions of euro). *Source* Dealogic ProjectWare

projects (13.5%), renewables (12%), power and energy (9%), rail and infrastructure (8.3%), and gas pipelines (8.2%). Details are shown in Table 3.2.

An analysis of the data clearly indicates that sectors listed in Segment III of Table 3.1 (in particular rail, roads and PPP/PFIs) dominate in terms of amount financed compared with more traditional sectors, such as energy. This confirms a progressive move toward the use of project finance from Segment II to Segment III, as highlighted previously.

With reference to the energy sector, it must be noted that renewables are taking over a significant stake of the project finance market in Europe. The increased attention of policymakers to the Kyoto protocol and the recent Paris meetings regarding global warming will probably boost this percentage even more in the near future. Utilities and oil and gas companies are also upping their investments in renewables in response to more intense attention of institutional investors to global warming and sustainability issues, which are changing the asset allocation of their holdings (Blackrock 2016).

Additional information provided by Fig. 3.6 is the declining pattern of PFI/PPP projects. From a percentage of over 65% in 2008, the value dropped to 25% at the end of 2014. This trend reflects growing budget constraints that European countries are facing given the high public debt/GDP ratios and the fiscal compact rules imposed by the European Union. These restrictions make it difficult to fill in the estimated two trillion euro infrastructure gap by 2020 and justify recent steps taken by the European Commission to relaunch infrastructure investment with the support of the European Investment Bank. Two initiatives stand out in this respect: the 2020

Table 3.2 Amount (in millions of euro) and percentage breakdown by sector of project-financed transactions in Europe—2008–2015. *Source* Dealogic—ProjectWare

	2008	%	2009	%	2010	%	2011	%	2012	%
Airport	151.04	0.228	5433.79	14.195	1774.62	2.891	3131.15	4.743	5903.41	12.967
Bridge			24	0.063	16	0.026				
Defence	3858.12	5.817	83	0.217			1550.49	2.348		
Education	1876.29	2.829	1900.91	4.966	4254.64	6.93	918.832	1.392	845.781	1.858
Gas Distribution	120	0.181	630	1.646	3395.33	5.531	2642.39	4.002		
Gas pipeline	8585.86	12.946	1599	4.177	5985	9.749	10,477.48	15.87	2799.14	6.148
Gasfield exploration and development	13,148.12	19.825	2019.07	5.274	1358.21	2.212	4257.31	6.448	4079.18	8.96
Government buildings	57.373	0.087	254.753	0.665	973.232	1.585	268.023	0.406	895.756	1.968
Mining	455.19	0.686	6	0.016	375.605	0.612	549.004	0.832	832.267	1.828
Oil pipeline	787.87	1.188			30	0.049				
Oil Refinery/LNG and LPG Plants	3548.71	5.351	986.429	2.577	168	0.274	440	0.666	64.879	0.143
Oilfield exploration and development	1551.62	2.34	176.05	0.46	3542.83	5.771	550.934	0.834	2662.62	5.848
Other downstream	555	0.837							482.633	1.06
Other Infrastructure Projects	1211.06	1.826	674.998	1.763	161.142	0.262	2110.74	3.197	222.186	0.488
Other upstream									30	0.066
Petrochemical/Chemical Plant	459	0.692	1263.1	3.3	1985.82	3.235	4386.10	6.644	50	0.11
Port	435.5	0.657			469.307	0.764	2989.21	4.528	2084.53	4.579
Power	5395.98	8.136	2223.5	5.808	6508.67	10.602	2915.20	4.416	4516.96	9.921
Prison	419.75	0.633	99.215	0.259	372.229	0.606	430.906	0.653		

(continued)

Table 3.2 (continued)

	2008	%	2009	%	2010	%	2011	%	2012	%
Rail-Infrastructure	975	1.47	2873.7	7.507	5448.3	8.875	9227.22	13.976	5860.23	12.872
Renewable fuel	9310.88	14.039	5810.24	15.178	8075.11	13.154	10,385.55	15.731	3998.89	8.783
Road	9698.94	14.624	9617.11	25.123	8968.26	14.608	6365.53	9.642	6398.39	14.054
Telecom	138.9	0.209	190	0.496	2624.62	4.275	421.23	0.638	425.898	0.935
Tunnel	1597.9	2.409							25.15	0.055
Urban railway/LRT/MRT	212.92	0.321	211.94	0.554	3073.89	5.007	614.724	0.931	927.7	2.038
Waste	1148.94	1.732	1232.11	3.219	996.589	1.623	787.677	1.193	562.229	1.235
Water and sewerage	620.7	0.936	971.397	2.538	833.391	1.358	600.772	0.91	1859.79	4.085
Total	66,320.65	100	38,280.31	100	61,390.80	100	66,020.47	100	45,527.61	100

	2013	%	2014	%	2015	%	Total	%
Airport	5042.59	8.507	4935.19	9.015	3806.73	5.41	30,178.52	6.5%
Bridge							40	0.0%
Defence							5491.61	1.2%
Education	1472.13	2.483	475.075	0.868	3448.06	4.9	15,191.71	3.3%
Gas Distribution	5860.13	9.886	4024.21	7.351	2895.37	4.115	19,567.44	4.2%
Gas pipeline	3740.84	6.311	3368.53	6.154	1464.91	2.082	38,020.76	8.2%
Gasfield exploration and development					2749.90	3.908	27,611.79	6.0%
Government buildings	503.436	0.849	658.528	1.203	323.661	0.46	3934.76	0.9%
Mining	538.263	0.908	2186.36	3.994	691.047	0.982	5633.74	1.2%
Oil pipeline	373.985	0.631			1616.53	2.297	2808.39	0.6%
Oil Refinery/LNG and LPG Plants	1750.32	2.953	1259.84	2.301	2841.79	4.039	11,059.96	2.4%
Oilfield exploration and development	7565.91	12.763	14,485.22	26.461	3656.55	5.197	34,191.73	7.4%
Other downstream	574.973	0.97			1022.84	1.454	2635.45	0.6%
Other Infrastructure Projects	491.115	0.828	932.56	1.704	89	0.126	5892.80	1.3%

(continued)

Table 3.2 (continued)

	2013	%	2014	%	2015	%	Total	%
Other upstream							30	0.0%
Petrochemical/Chemical Plant	200	0.337	748.504	1.367	1845.14	2.622	10,937.66	2.4%
Port	1089.52	1.838	1830.56	3.344	2285.90	3.249	11,184.53	2.4%
Power	4726.42	7.973	4880.52	8.916	10,243.20	14.558	41,410.44	9.0%
Prison	643.81	1.086	190.28	0.348			2156.19	0.5%
Rail-Infrastructure	3796.62	6.405	3469.68	6.338	6743.44	9.584	38,394.17	8.3%
Renewable fuel	2542.33	4.289	3973.36	7.258	10,300.23	14.639	54,396.59	11.8%
Road	10,233.76	17.264	5032.92	9.194	5850.31	8.315	62,165.22	13.5%
Telecom	5008.60	8.449	390.2	0.713	2939.74	4.178	12,139.18	2.6%
Tunnel			140.352	0.256	240	0.341	2003.40	0.4%
Urban railway/LRT/MRT			1245.33	2.275	2602.27	3.698	8888.77	1.9%
Waste	2187.92	3.691	280.464	0.512	166.6	0.237	7362.54	1.6%
Water and sewerage	936.369	1.58	234.03	0.428	2538.61	3.608	8595.06	1.9%
Total	59,279.04	100	54,741.71	100	70,361.81	100	4,61,922.40	

PBI-Project Bond Initiative and the EFSI-European Fund for Strategic Investments (Gatti 2016).

A look at the geographical breakdown of project finance deals can be useful to understand which countries in Europe are more active in using the technique. Table 3.3 indicates that a small number of nations in the period 2008–2015 represent about 70% of the market. United Kingdom (23%), Spain (16%), Russia (10%), France (10%) and Italy (7%) stand out as big adopters of project finance. The UK is always cited as the country where PPPs are used in large measure to finance new infrastructure projects or to refurbish existing ones. It is also the country that is particularly active in trying to attract pension savings to infrastructure with the creation of a two billion pound national pension platform (Plimmer and Pickard 2014). In the UK, institutional investors have demonstrated clear interest in entering the infrastructure finance market.

Italy and Spain rank high due to the boost given by tax incentives in the business of renewables. Although these incentives have been progressively shrinking, a large portion of project finance initiatives has benefitted from favourable conditions proposed to private parties in the period in question.

Finally, Russia has recently enacted a reform of PPP legislation with the objective of privatizing the infrastructure sector, in particular transportation projects (highways and airports).

3.4 Recent Trends in Infrastructure Investing

For many years, the traditional capital structure used for infrastructure financing was based on a simple combination of multiple-tranche syndicated bank loans and equity provided by corporate sponsors and developers. This business practice has undergone a remarkable transformation in the past few years. The acceleration of the change was particularly evident starting from the second part of 2000s when an increased risk appetite among institutional investors coupled with a favourable environment of low interest rates spurred a flow of financing from investors other than banks and industrial sponsors.

This renewed interest in infrastructure was equally sizeable in terms of equity and debt. On the debt side, the role played by monoline insurers contributed to attract capital flows into capital market debt instruments (at least until the 2007–2008 crisis exploded, triggering a series of downgrades). The default of Lehman Brothers in September 2008 partially curbed this appetite. However, data seem to indicate that an upward trend is emerging again, with institutional investors still looking for yield on long-term assets with a clear and stable pattern of cash flows.

In this section, we concentrate our analysis on the evolution of debt instruments used for financing infrastructure. For information regarding how the ownership of SPVs is changing thanks to the greater contribution of institutional investors in providing equity to infrastructure, see Probitas Partners (2016) and Gatti (2016). In Sect. 3.4.1, we examine the project bond market, exploring how bonds can

Table 3.3 Amount (in millions of euro) and percentage breakdown by country of project-financed transactions in Europe—2008–2015. *Source* Dealogic—ProjectWare

	2008	%	2009	%	2010	%	2011	%	2012	%
Armenia										
Austria										
Azerbaijan	70.731	0.107			1750.70	2.852	67.503	0.102		
Belgium	1531.60	2.309	426.47	1.114	2689.88	4.382	1186.75	1.798	300.731	0.661
Bulgaria							48	0.073	223.977	0.492
Croatia	413.6	0.624								
Czech Republic			44.814	0.117	366.261	0.597	334.67	0.507		
Cyprus	21.893	0.033								
Denmark			348.61	0.911	161.602	0.263	760.936	1.153		
Estonia					221	0.36				
Finland			31	0.081	100	0.163	735.099	1.113	1,328.00	2.917
France	3117.64	4.701	474.171	1.239	4471.49	7.284	12,054.47	18.259	9,233.11	20.28
Georgia							178.386	0.27		
Germany	2526.10	3.809	3824.35	9.99	2688.84	4.38	1556.84	2.358	3,259.76	7.16
Greece	1036.85	1.563	186.68	0.488	611	0.995				
Hungary	520	0.784	200	0.522	63	0.103				
Iceland										
Ireland	98.9	0.149	156.805	0.41	480.5	0.783	11	0.017	120.805	0.265
Italy	2275.37	3.431	4250.00	11.102	3450.61	5.621	5,577.55	8.448	2,451.50	5.385
Kazakhstan	6974.13	10.56					153.184	0.232	754.631	1.658
Latvia					150	0.244				
Luxembourg										
Macedonia					100	0.163				

(continued)

Table 3.3 (continued)

	2008	%	2009	%	2010	%	2011	%	2012	%
Malta										
Netherlands	3377.86	5.093	422.26	1.103	1456.15	2.372	346.6	0.525	1,211.22	2.66
Norway			176.05	0.46	314.923	0.513	2,368.99	3.588	2,352.86	5.168
Poland	1777.87	2.681	1610.00	4.206	248.575	0.405	503.932	0.763	436.464	0.959
Portugal	5541.18	8.355	2159.72	5.642	4302.77	7.009	1,384.40	2.097	99.141	0.218
Romania			308	0.805					24.446	0.054
Russian Federation	13,108.47	19.765	3727.72	9.738	11,614.40	18.919	13,567.96	20.551	1,040.81	2.286
Slovak Republic	185.189	0.279	1165.06	3.043					14.9	0.033
Slovenia	24	0.036					850	1.287	30	0.066
Spain	12,575.96	18.92	11,056.44	28.83	18,014.72	29.344	11,439.72	17.328	2,807.30	6.166
Sweden	127.111	0.192							163.158	0.358
Switzerland			99.215	0.259			854.141	1.294		
Tajikistan										
Turkmenistan							2,812.84	4.261		
Ukraine									290	0.637
United Kingdom	8242.05	12.428	7612.94	19.887	8134.39	13.25	9,227.50	13.977	15,305.63	33.618
Uzbekistan	2774.14	4.183							4,079.18	8.96
Total	66,320.65	100	38,280.31	100	61,390.80	100	66,020.47	100	45,527.61	100

	2013	%	2014	%	2015	%	Total	%
Armenia					311.178	0.442	311.18	0.07%
Austria	60	0.101	75	0.137			135.00	0.03%
Azerbaijan					5109.31	7.261	6998.25	1.52%
Belgium	3157.97	5.327	2463.37	4.5	3267.28	4.644	15,024.04	3.25%
Bulgaria	112.028	0.189	91.797	0.168	288.073	0.409	763.88	0.17%

(continued)

Table 3.3 (continued)

	2013	%	2014	%	2015	%	Total	%
Croatia	279	0.471					692.60	0.15%
Czech Republic	1,998.56	3.371	1,231.24	2.249	352.616	0.501	4,328.17	0.94%
Cyprus							21.89	0.00%
Denmark	149.179	0.252	766.621	1.4	311.623	0.443	2,498.57	0.54%
Estonia							221.00	0.05%
Finland	1,650.00	2.783	3,160.00	5.773	639	0.908	7,643.10	1.65%
France	3,030.65	5.113	3,415.47	6.239	10,588.11	15.048	46,385.11	10.04%
Georgia					894.094	1.271	1,072.48	0.23%
Germany	2,902.39	4.896	5,881.08	10.743	1,341.77	1.907	23,981.12	5.19%
Greece	3,112.91	5.251	172	0.314	68.723	0.098	5,188.16	1.12%
Hungary	312.5	0.527	1,367.00	2.497	1,130.00	1.606	3,592.50	0.78%
Iceland					61.863	0.088	61.86	0.01%
Ireland	165.343	0.279	868.701	1.587	500.708	0.712	2,402.76	0.52%
Italy	5,173.65	8.728	4,582.61	8.371	2,630.49	3.739	30,391.78	6.58%
Kazakhstan	412.8	0.696	62.366	0.114	392.736	0.558	8,749.85	1.89%
Latvia							150.00	0.03%
Luxembourg	749.232	1.264					749.23	0.16%
Macedonia							100.00	0.02%
Malta					450	0.64	450.00	0.10%
Netherlands	2,162.10	3.647	2,157.96	3.942	2,541.12	3.612	13,675.27	2.96%
Norway	911.895	1.538	8,228.37	15.031	1,802.03	2.561	16,155.12	3.50%
Poland	887.432	1.497	1,075.79	1.965	694.939	0.988	7,235.01	1.57%
Portugal	78.893	0.133	73.632	0.135	339.56	0.483	13,979.30	3.03%
Romania	327.766	0.553	264.411	0.483			924.62	0.20%

(continued)

Table 3.3 (continued)

	2013	%	2014	%	2015	%	Total	%
Russian Federation	48.4	0.082	2,562.69	4.681	1,133.01	1.61	46,803.46	10.13%
Slovak Republic	2,742.70	4.627	1,780.00	3.252	450	0.64	6,337.85	1.37%
Slovenia							904.00	0.20%
Spain	6,714.81	11.327	2,440.56	4.458	7,967.76	11.324	73,017.26	15.81%
Sweden			142.829	0.261	5,483.90	7.794	5,917.00	1.28%
Switzerland							953.36	0.21%
Tajikistan			56.084	0.102			56.08	0.01%
Turkmenistan							2,812.84	0.61%
Ukraine							290.00	0.06%
United Kingdom	22,138.82	37.347	11,822.13	21.596	21,611.92	30.715	1,04,095.39	22.54%
Uzbekistan							6,853.33	1.48%
Total	59,279.04	100	54,741.71	100	70,361.81	100	4,61,922.40	100.00%

represent a supplemental source of financing for infrastructure in addition, or as an alternative, to syndicated loans. In Sect. 3.4.2, we show how institutional investors have approached the infrastructure financing market and the options available to convey financial resources to project finance debt.

3.4.1 *Infrastructure and Debt Capital Markets: The Use of Project Bonds*

Still today, limited or non-recourse syndicated loans are the most used source of financing for infrastructure projects. Between 2007 and 2015, project finance loans represented between 5 and 24% of the global syndicated loans market, equaling an amount ranging from $140 billion and $280 billion (see Fig. 3.7). A much lower percentage was represented by funding on the debt capital markets. In the same period, project bonds (i.e. bonds issued by SPVs in project finance transactions) bounced between $8.5 billion and $50 billion. Funding on debt capital markets is still limited to a marginal fraction of the total debt funding for these kinds of initiatives. Furthermore, after peaking in 2013, the use of project bonds decreased until 2015, at which time project bonds represented 13% of the total syndicated loans market.

The breakdown by geographical area and sector shows a clear concentration of bonds issued by SPVs in some sectors (infrastructure, power and social infrastructure) and a polarization in USA/Canada, UK and Western Europe. Clearly, this indicates that the appetite of long-term institutional investors for this kind of debt instrument is higher in more developed financial markets. However, developing economies have recently started using such instruments too, as reported in Gatti (2014).

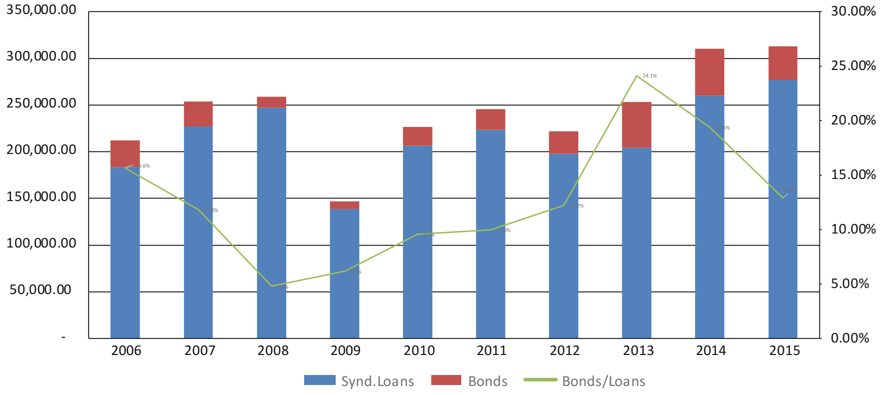

Fig. 3.7 Trends of global project finance loans and bonds (2006–2015)—Data in millions of dollars. *Source* Thomson One Banker, Project Finance International

Fig. 3.8 Trends of European project finance loans and bonds (2006–2015)—Data in millions of dollars. *Source* Thomson One Banker, Project Finance International

With specific reference to Europe, Fig. 3.8 shows that between 2008 and 2012 the market for project bonds was almost nil. Then came the action taken by the European Commission to relaunch infrastructure investments via the credit enhancement scheme supported by the European Investment Bank through the 2020 Project Bond Initiative. The market has since taken over, with percentages fluctuating between 10 and 23% in the period 2013–2015. Anecdotal evidence reported in Sect. 3.4.2. seems to indicate that institutional investors are allocating more resources to fixed income instruments like project bonds. This is in response to the launch of Quantitative Easing by the European Central Bank and the subsequent drop of yields on government securities and investment grade corporate bonds.

The current state of affairs with regard to debt instruments in Europe indicates that the market is increasingly considering bonds and loans as complementary sources of funding for infrastructure investments. This is due to two possible reasons:

1. Their contractual design is different, and combining the two can generate funding mixes that benefit infrastructure projects.
2. Project bonds have the potential to be an asset class of particular interest for long term investors other than banks, like pension funds, life insurance companies or foundations.

Several characteristics of the contractual design of project bonds make them more attractive to long term investors vis-à-vis syndicated loans. First, compared to such loans, bonds are more standardized capital market instruments. All else being equal, this feature enhances the liquidity of the instrument, provided that the issue size is sufficiently large. What's more, larger issues may also be included in bond indices and this could make them even more enticing for bond market investors. Second, project bonds are attractive instruments for long term investors, as they can be issued with longer maturities than the tenors of syndicated loans that banks normally accept (Gatti 2016). Finally, if well structured, a project bond issue can benefit from a comparatively lower cost of funding and from less stringent covenants than a traditional syndicated loan.

Although project bonds present comparative advantages over syndicated loans (EPEC 2010), applying bonds to infrastructure financing requires careful consideration of some drawbacks that make them less attractive for institutional investors. Here are a few (Gatti 2014):

1. Project bonds are less suitable than syndicated loans if they are used during the construction phase of an infrastructure project. The reason is that credit risk for bondholders is too high (or, put another way, the rating of the project is too low) and too difficult to assess (Moody's 2010). Instead, project bonds are more useful for initiatives that have already passed the construction phase, or as a refinancing technique for syndicated loans granted to the project during the construction phase. When bonds are issued after the commercial operating date, potential synergies emerge with bank loans, allowing banks to shorten the maturity of loans or to use mini-perm structures.[7]
2. Standardisation: Institutional investors are more familiar with the traditional contractual design of a bond that pays coupons at regular intervals and then remits the entire amount of capital as a bullet repayment at the end of life of the bond. The consequence of standardisation is that bond proceeds are cashed in by the issuer as a lump sum at the beginning of the construction phase. But since the construction of the infrastructure can take a long time, the SPV is forced to keep cash in liquid form with a high opportunity cost (negative carry). This doesn't happen with syndicated loans, which are disbursed following a milestone schedule (Gatti 2012).
3. Refinancing risk: Regardless of whether project bonds are utilized at the inception of the initiative or at the beginning of the operational phase, the project will be subject to refinancing risk. If mini-perm syndicated loans are used during the construction phase, the risk for banks is that general debt market conditions may worsen between the loan disbursement and the time when bonds are issued. If instead the bond option is applied only during the operational phase of the infrastructure, and its economic life is particularly long, the bond could have a final maturity that is actually shorter than the life of the project, which triggers once again a refinancing risk for project sponsors.

[7]Mini-perm loans are typically characterized by a bullet payment for the total or partial amount of the principal. Such a loan would be used to finance the construction phase but must be repaid only after a short period of time during the construction phase, forcing the SPV to refinance the loan, exposing it to refinancing risk.

4. Rating: The availability of a large investor base for project bonds depends on the rating these instruments receive from relevant agencies. It's true that available evidence demonstrates that project finance loans are no more risky than traditional corporate loans (Standard and Poor's 2009). But it's undeniable that if a bond does not receive an investment-grade rating, the public of potential investors remains confined to those who specialise in high yield instruments. Furthermore, by-laws of many institutional investors include an explicit ban on investing in sub-investment grade instruments.

3.4.2 New Forms of Debt Infrastructure Investments

In addition to the use of bonds to finance infrastructure projects, banks have also modified their approach to lending as a consequence of a massive deleveraging and a modified regulatory environment (particularly the enactment of Basel III rules regarding the Net Stable Funding Ratio, NSFR). The increased interest by institutional investors in long term infrastructure investments has also contributed to the shift toward an originate-to-distribute model, and away from the more traditional originate-to-keep model. This trend has been particularly evident in Europe.

The originate-to-distribute model sees banks cooperating with institutional investors in channelling debt funds to infrastructure (Cei 2016). Although the market is still in its early stage of development and information is very limited, the practice seems to point to four alternative structures that enable institutional investors to approach long term infrastructure investments:

1. The partnership/co-investment model
2. The securitisation model
3. The debt fund model
4. The direct origination of loans by institutional investors.

In the partnership/co-investment model, an institutional investor finances infrastructure loans originated by a Mandated Lead Arranger (MLA) Bank. The fund provision is regulated by a set of eligibility criteria and the MLA Bank retains a pre-agreed percentage of each loan in its portfolio. With this co-investment, an institutional investor can build a portfolio of infrastructure loans, relying on the servicing of the loans in the portfolio provided by the originating bank. The bank in turn may well extend the partnership to a number of institutional investors.[8]

In the securitization model, a bank that is active in the project finance market sets up an SPV with the purpose of pooling a number of project finance loans; these

[8]An examples of the partnership model is the relationship between the French Bank Natixis and Belgian Insurance Company Ageas. A similar structure characterizes the partnership between Crédit Agricole and Crédit Agricole Assurances (Gatti 2014).

become collateral for an asset-backed bond issue (Buscaino et al. 2012). In this way, a portfolio of infrastructure loans can be tailored to the specific needs of institutional investors. The basic difference between this model and the partnership one is that in the latter the institutional investor becomes one of the direct lenders to the SPV, while the securitisation model involves participation in a pool of loans originated by the bank. (In other words, the institutional investor finances infrastructure projects only indirectly.)

The debt fund model is probably the easiest way to approach the infrastructure market for institutional investors without a specific, dedicated team to invest in infrastructure assets, or for those who lack past experience in investing in this asset class. In the debt fund model, an institutional investor provides funding to a resource pool (the fund) managed by an asset manager. The strategic asset allocation is defined from the inception of the project, which allows institutional investors to select the fund that best suits their investment needs. The success of the debt fund model is contingent on a strong deal flow.

Compared to partnerships or securitisations, debt funds are based on fixed and pre-agreed investment criteria, while the other two alternatives have a greater possibility to adapt the financial structures to investors' needs.

The market for debt funds is still very young and information available on salient trends is very scarce. This said, however, Probitas Partners (2016) report that infrastructure debt funds are attracting increasing interest. Almost negligible in 2011 in terms of total fundraising (less than 1%), they reached a remarkable 8% in 2016.[9]

The last alternative model of infrastructure debt investing is the direct origination of loans. This trend is confined to the most sophisticated investors who have decided to invest in internal skills development to create internal teams specialised in infrastructure investment. In Europe, Allianz Global Investors started originating infrastructure loans in 2012; Legal & General in the UK originated commercial real estate loans for student accommodation facilities. M&G (Prudential) has been active in student housing loan projects and hospital facilities.

3.5 Conclusions

In this chapter, we have analysed project and infrastructure finance. We have shown the differences with respect to corporate finance, and the benefits that the deal gives to project sponsors and lenders. We have also reviewed the network of contracts

[9]Examples of the debt fund model are the European Infrastructure Debt Platform created by Blackrock with a focus on Germany, France, the UK and Benelux countries, the Senior European Loan Fund originated by Natixis AM, the senior debt fund dedicated to property lending launched by M&G (Prudential), and the recent debt fund dedicated to European Infrastructure launched by Allianz Global Investors.

underpinning a project finance deal and the risk management process that lies at the core of deal structuring.

In terms of market trends, the chapter has traced the evolution of the project finance starting in the early Nineties, when initial experiments in this technique centered on the power sector (especially for cogeneration projects). Over time project finance has spread to other industries, often because of the public sector's need for a more substantial involvement of private capital in partially self-financed projects. Today project financing is applied in a vast array of sectors. This has also brought about a change in the essential features of the transaction, especially as regards risks remaining with the SPV and, consequently, its financers. In our opinion, current market trends favor projects in which market risk is not completely excluded, but instead left (at least in part) as a risk for the vehicle company. Intermediaries should take this factor into account when setting adequate pricing, given the risk a particular transaction involves.

Another important conclusion concerns the trend of supplying funds for infrastructure financing. Although banks still retain an sizeable percentage of lending, capital markets and institutional investors are playing a bigger role in providing funds for infrastructure development. The big question that still remains unanswered is whether or not this trend is simply due to extraordinary conditions in capital markets (too low interest rates for too a long time) or if we are facing a structural reallocation of resources from more traditional asset classes to alternative investments. If the latter is the case, two challenges open before us. On one hand, institutional investors must make up for their lack of competences vis-à-vis banks in analyzing the risk profile of infrastructure deals. The use of alternative strategies discussed in Sect. 3.4.2 can be a constructive pattern to followed. The second challenge involves regulators. Basel III and Solvency II have set down precise regulations for coping with credit risk that arises for traditional intermediaries who invest in alternative asset classes. However, a big portion of the funds in question comes from non-regulated (or loosely regulated) investors. Striking the right balance between the need to attract funds to fill the infrastructure gap and the need for financial stability will be one of the key topics to grapple with in the years to come.

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Chapter 4
Structured Leasing Transactions

Stefano Caselli

Abstract The aim of this chapter is to investigate and highlight the typical features of the transactions and market covering a family of instruments known as structured leasing deals, namely, complex, intricate deals assembled ad hoc, with an organizational and contractual framework that is grafted onto a leasing transaction. To achieve the above, this chapter has been structured in four parts. The first part tackles the issue of defining structured leasing deals by examining their characteristics within the broader context of the structured finance and leasing market. The second part, instead, focuses on the market set-up, highlighting basic trends, its size and internal organization, and the types of financial intermediary operating within it. The third part looks at the leasing transaction from a tax standpoint, examining its distinctive features and possible room for maneuver, which is fundamental when assembling a structured transaction. At this juncture it would appear significant to conduct a sensitivity analysis to measure the effects on cost of capital produced by changes in the basic components of the leasing contract and tax implications for the lessee. Lastly, the fourth part examines the issue of classifying and analyzing structured leasing transactions in order to provide an initial classification scheme, illustrating their characteristics from a standpoint of the transaction's basic set-up, tax and financial architecture and main results achieved.

4.1 Introduction

The aim of this chapter is to investigate and highlight the typical features of the transactions and market covering a family of instruments known as structured leasing deals, namely, complex, intricate deals assembled ad hoc, with an organizational and contractual framework that is grafted onto a leasing transaction.

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S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_4

Given the tenor of the definition, the first task is to establish the boundary between conventional and structured leasing transactions in a clear and correct manner. This would seem to be very complicated given both the novel nature of these instruments and the objective difficulty of establishing distinctions that may then prove arbitrary and excessively theoretical. In this sense the decisive factor is to examine what options and practices have been adopted by financial intermediaries at a global market level and, above all, what the most advanced organizational-contractual forms are. This, in order to survey on a supply-and-demand basis the needs behind the demand for structured transactions and the characteristics of products proposed by operators in the sector.

To achieve the above, this chapter has been structured in four parts.

The first part tackles the issue of defining structured leasing deals by examining their characteristics within the broader context of the structured finance and leasing market, both in terms of indications coming from the international financial market and the limited scientific references available. The second part, instead, focuses on the market set-up, highlighting basic trends, its size and internal organization, and the types of financial intermediary operating within it. This necessarily means looking at the international scenario, both because this is a new issue and, above all, because of the very nature of the transactions themselves, which in terms of size and intrinsic features are such that it would not be easy to refer to one single, specific domestic market.

The third part looks at the leasing transaction from a tax standpoint, examining its distinctive features and possible room for maneuver, which is fundamental when assembling a structured transaction. At this juncture it would appear significant to conduct a sensitivity analysis to measure the effects on cost of capital produced by changes in the basic components of the leasing contract and tax implications for the lessee. Lastly, the fourth part examines the issue of classifying and analyzing structured leasing transactions in order to provide an initial classification scheme, illustrating their characteristics from a standpoint of the transaction's basic set-up, tax and financial architecture and main results achieved.

4.2 Basic Characteristics of Structured Leasing Transactions

A detailed definition of structured leasing transactions is hampered by a series of factors that limit possibilities to outline the significant characteristics in a sufficiently clear manner. These factors are the combined result of a number of situations, above all linked to the initial development phase of structured leasing and low diffusion of the necessary operational know-how required to manage this instrument. As for the first point, structured leasing is essentially a new type of transaction, confirmed by the fact that there were very few references to it or analyses in specialized publications until early in 2000. Initial considerations on this subject—

have only appeared recently, together with the first accounts of transactions assembled by specialized financial intermediaries.[1] As a result structured leasing was recognized as a new transaction and an independent form of leasing within the structured finance and asset finance sector. On the second point, a certain reticence of operators who have acquired significant expertise in the sector seems to be justified by their need to safeguard know-how believed to constitute an essential competitive edge and by the considerable level of financial resources involved. With specific reference to the Italian financial market, it should also be mentioned that structured leasing is in its infancy and only slowly making inroads, as can be seen from continuing recourse to the conventional form of leasing, above all for extremely large transactions.

Examining transactions about which there is news, above all those that have taken place in more progressive financial markets as regards the use of financial leasing (USA, UK and, in part, France and Germany), certain essential elements can be pinpointed that enable reasonably accurate identification of structured leasing transactions, distinguishing them from conventional leasing and, in part, from project finance leasing, with which they can often overlap.

While there is still no precise, accepted definition at international level, structured leasing is intended as those transactions that, to differing degrees, develop an organized synergy between funding policy, asset and risk management of the underlying asset and tax-based finance. The aim being to manage the need for complex financing in a creative manner, as opposed to just raising funds by mobilizing a party's wealth by means of recourse to the leasing instrument (see Fig. 4.1). It follows that the very essence of such transactions is the existence of an asset and the simultaneous and indispensable presence of the above three distinctive elements, which can be combined in different manners depending on the specific need profile. This very possibility to exploit a different interaction between the factors concerned makes structured leasing a highly versatile tool. So versatile that its use can range from transactions in the corporate finance context—with a view to financing the company—to those in high-profile private banking.

In analytical terms, a structured leasing deal formally comprises a financial leasing contract concerning various types of underlying asset. In this sense, at first sight structured leasing would not seem to be an original, distinct type of financial transaction but instead a type of financing lease referring to a very clear category of assets.[2] The structure of the transaction calls for a succession of installments over

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[1]Works published on the subject of structured leasing transactions appear to be few and far between, and so reference has to be made to studies carried out in the research environment and activities in closely associated operational areas. For this reason there is one type of material that covers the subject directly and another, broader category of material that, when referring to conventional and project leasing, deals with the issue of "structuring a leasing transaction". For a broad overview see Ascari and Albisetti (1996) and the more recent update based on the paper written by Pinto and Pacheco (2014).

[2]For a review of the basic features of leasing transactions see the Italian handbook written by Carretta and De Laurentis (1998) as well Baker and English (2009).

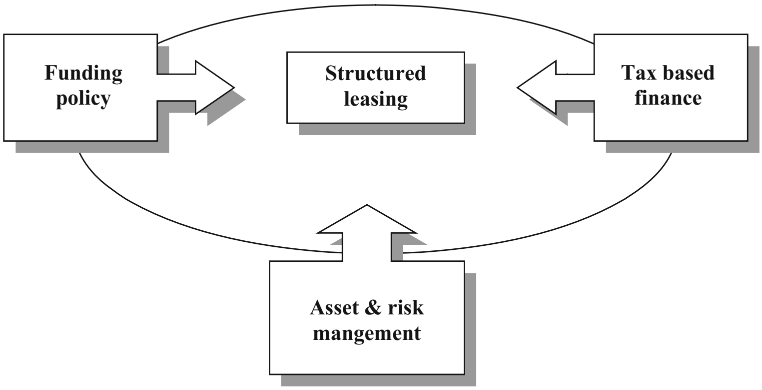

Fig. 4.1 Distinctive elements of structured leasing transactions. *Source* The author

the entire duration of the operation and a final one-time payment made as a call option on the underlying asset. As in all leasing contracts the structure of the installments can vary based on whether there is a fixed or variable interest rate, a larger first installment and indexing to a specific foreign currency. The very presence of the final option call means the user can link the leasing transaction to a later financing transaction to cover cash required in the event of purchase at the end.

The "conventional" nature of a leasing transaction makes it particularly effective if the value of the underlying asset is high compared to the financial situation of the intended user, given that in the event of outright purchase this would create a financial strain and also a need to govern risks inherent in the asset's characteristics. This, inasmuch as a leasing transaction can:

- enable contract terms to be modulated in relation to the lessee's cash flow structure;
- generate governable tax benefits, with a different sequence and structure than is achieved by depreciating the asset and covering attendant financial costs arising from the funding policy adopted to purchase the asset;
- finance the possible call option at the end of the leasing transaction.

The three factors mentioned make a leasing contract a very flexible instrument that enables the lessee to position the deal in an optimal manner in relation to cash flow structure, its sustainability over time and the distribution of tax benefits.

While these considerations remain valid whatever the type of underlying asset, it should be emphasized that the assembly of a structured transaction is much more significant when the value of the asset is considerable (whether in absolute terms or relevant to the lessee's potential) and when tax benefits accruing from the leasing transaction are potentially greater and more flexible than those obtained by depreciating the underlying asset and considering financial costs associated with the funding source. This is also consistent with tax-based financing techniques, in which the aim when assembling the transaction is to maximize available financial maneuvering room with reference to tax variables by analyzing the sensitivity of the transaction itself. As a result, assets typically found in structured leasing transactions are, in order of decreasing importance[3]:

– real estate of various types and uses
– industrial plant and equipment
– ships and aircraft.

The decreasing importance of the three asset classes for big-ticket leasing is linked to the versatility and potential represented by utilizing the structured leasing instrument. If in the case of real estate the underlying asset could refer to various types of corporate needs or private purposes (with a view to mobilizing a business person's wealth) in the case of industrial plant and equipment and, above all, ships and aircraft, the need in question tends to be exclusively corporate. Furthermore, real estate can be utilized independent of the asset's use for any given business or personal purpose, in that it represents a form of investment in its own right, utilizable merely as a means to produce value.

The organizational methods and objectives of the transaction can effectively be modeled bearing in mind the nature and role of the asset from the standpoint of:

– its nature in terms of product category
– its nature from a legal standpoint.

In fact, the underlying asset can have different product-category and legal profiles, which condition the financial flows and risk set-up for the leasing transaction. In the first case (product-category profile), real estate can be residential, administrative, commercial, industrial or even infrastructural. In the second case (legal profile), the asset can already exist and be owned by the lessee, can already exist but be owned by a third party or may even not yet be built. By overlapping the two profiles a map of possible structured leasing transactions is obtained that highlights the complexity and level of internal differentiation of such deals (see Fig. 4.2).

In particular it can be observed that while the asset's product category heavily conditions the nature and intensity of financial and equity risks associated with the transaction, the legal profile, if the asset already exists, conditions the financial structure and overall arrangement of the transaction itself.[4] For instance, residential, and to a degree, administrative real estate, ships and aircraft represent, all things being equal, a lower financial and equity risk because they tend to have a better defined market value. This, because there is a much wider secondary market and often, as in the case of real estate, few or even no factors that can generate a negative impact on the asset's value. Vice versa, commercial, industrial and infrastructural real estate, and above all industrial plant and equipment represent an increasing

[3]On this subject see: Leaseurope (2014a).

[4]In this regard the interpretation in terms of legal profile and product-category profile can be extended and broken down further by considering size and whether the deal is of a domestic or international nature. This latter distinction has a significant effect inasmuch as it introduces the aspect of country risk, which interacts with the various other risk aspects of the transaction.

			LEGAL PROFILE OF UNDERLYING ASSET		
			Existing and owned by lessee	Existing and owned by a third party	To be built
PRODUCT-CATEGORY PROFILE OF UNDERLYING ASSET	Real estate	Residential			
		Administrative			
		Commercial			
		Industrial			
		Infrastructural			
	Plant & equipment				
	Ships & aircraft				

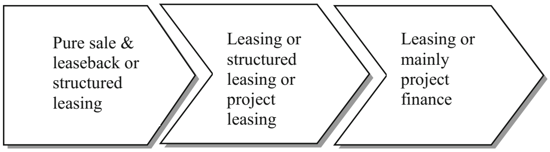

Fig. 4.2 Map of structured leasing transactions based on the nature of the underlying asset. *Source* The author

financial and equity risk because of a progressive reduction of the role played by the secondary market and an increase in factors generating risks of an equity nature. Looking at the transaction from a legal profile standpoint highlights that, given the same asset product-category profile, an already existing asset owned by the lessee technically makes it a sale and leaseback, as the purpose of the transaction has an impact on liquidity of the lessee's assets. The case of an asset owned by a third party is entirely different: if the asset is residential or administrative real estate then in essence it is just like any other financial leasing deal. Instead if it is commercial, industrial or infrastructural real estate, industrial plant and equipment or ships and aircraft, the leasing transaction progressively becomes an integral part of the lessee's business. Hence installments must be structured in such as way as to increasingly take cash flows produced by the asset underlying the leasing transaction very closely into account, making it de facto more like project leasing.[5] In the case, instead, of real estate or plant to be built (so-called "to-be-built leasing"), the leasing transaction can take the shape of project leasing. This, because the weight of the "project" element in the transaction (linked to a project still to be realized over a number of stages) must be seen together with the overall planning, construction and post-construction risks associated with the asset concerned. In these terms, the distinctive element of the transaction is its "project" nature, which encapsulates the equity and financial impacts throughout the various stages of completion.[6] In this sense, the transaction is undoubtedly like a true project financing deal.

[5]On this issue see evaluations made in Caselli et al. (2015).

[6]Particularly as regards project finance transactions, see considerations made in the study of this specific issue.

4.3 Structure of the Structured Leasing Transaction Market

4.3.1 Basic Trends and the International Market Set-up

In Europe leasing transactions in the real estate, plant and equipment and, partially, ship and aircraft sectors are substantial (Table 4.1) and growing strongly both in absolute and relative terms, namely, in relation to overall investment in individual geographical areas (Table 4.2). The reasons for the expansion of big-ticket leasing and growth of its importance depend on certain general and specific factors. The former are linked to big-ticket leasing per se, regardless of the lessee's objective and the specific product-category or legal nature of the real estate or other underlying asset. The second are specifically linked to the asset's legal profile and therefore to the type of leasing transaction—conventional, sale and leaseback, project or structured—it is intended to assemble. Clearly, in addition to the above there are other country-specific, occasional or structural factors that add to the convenience of leasing transactions and fuel their growth.[7]

With reference to general factors, those stimulating demand for big-ticket leasing transactions can be summarized as follows:

- transfer of the risk and risk management of the asset
- exploitation of tax benefits and in-country and extra-country tax loopholes
- overall investment financing, even including a full-service technique
- progressive extension of the average term for leasing transactions, with a consequent expansion of the types of project and parties that can be financed.

Risk transfer is an intrinsic characteristic of leasing transactions and, in cases where the lessee is capable enough, there can also be the further, broader and more complete service of asset risk management. If this is true in general for any type of leasing transaction, the presence of significant risks in the case of real estate, plant and equipment or ships and aircraft means this aspect has added value and so makes it a factor contributing to market growth. This emerges in a significant manner, for instance, in the case of to-be-built leasing, where the lessor shoulders the entire risk structure associated with the transaction, de facto taking on the role of risk manager. In this sense, with an eye to exploiting risk management as an integral service element of the transaction, numerous lessors coordinate their activities with other parties, such as insurance companies or industrial operators, in order to offer a long-term integrated service. Indeed, this is a characteristic feature of project

[7]The above maneuvers have been used widely in the past, above all in Eastern European countries, to attract foreign investors and sustain growth of the investment process at aggregate level. Moreover, similar actions can also be seen in the US market in terms of tax-based leasing transactions that offer strong tax incentives for certain specific real estate leasing transactions. In this regard see Leaseurope and KPMG (2012), Leaseurope (2014b).

Table 4.1 Size of the European leasing market (leased assets at end of 2015; data in billions of Euros)

Country	Real estate leasing	Equipment leasing	Ship and aircraft leasing
Austria	9.7	12.5	1.3
Belgium	1.5	9.2	0.2
Denmark	1.0	14.7	0.2
Finland	0.4	8.3	0.1
France	37.7	49.8	7.6
Germany	37.0	95.7	4.7
Italy	63.3	32.0	2.1
Netherlands	3.0	17.9	1.1
Portugal	4.0	9.0	1.2
UK	38.5	143.0	4.9
Spain	7.4	19.6	2.1

Source Authors' table based on Leaseurope (2014a)

Table 4.2 Penetration level in the European leasing market (leased assets at the end of 2015; data expressed as the percentage of leased assets to total investments at country level)

Country	Real estate leasing (%)	Equipment leasing (%)	Ship and aircraft leasing (%)
Austria	7	16	25
Belgium	2	10	16
Denmark	3	12	16
Finland	2	12	11
France	9	15	29
Germany	15	19	27
Italy	34	16	20
Netherlands	8	13	18
Portugal	11	21	15
UK	19	24	25
Spain	9	14	18

Source Authors' table based on Leaseurope (2014a)

finance transactions, a field with which big-ticket leasing overlaps in the form of project leasing.

Exploitation of tax benefits refers to taking advantage of possible differences in tax treatment between leasing and other sources of financing, with the aim of significantly reducing the net tax liability produced by the deal, or in other words the impact on the lessee's cost of capital. The above tax benefits can be either in-country or extra-country. For instance, in the first case tax legislation in the country where the transaction takes place most likely grants a considerable tax benefit for real estate leasing transactions, so making them attractive. In the second case, independent of the in-country benefit the deal is structured in a way that it (also or in alternative) takes advantage of tax benefits in a country other than that in which the transaction takes place. This is achieved by using a vehicle company domiciled in the other country, which assumes the role of lessee and then proceeds to rent the underlying asset to the effective lessee.

While the above considerations can apply to any type of underlying asset, in the majority of countries at international level three significant factors can be noted that make real estate leasing particularly attractive.[8]

– The minimum duration for real estate leasing is always significantly shorter than the depreciation period prescribed by tax legislation for the underlying asset. As this differential is normally greater than for other asset classes, this means real estate leasing has a significant potential for reducing the net tax effect of the transaction.
– The tax effects of the leasing transaction can be governed based on the duration of the lease, the amount of the one-time first installment and cost of the call option. This enables the lessee to combine benefits deriving from the tax reduction as regards costs with accurate financial planning.
– Often real estate leasing enjoys ad hoc tax benefits created by legislators to enable companies to undertake substantial but risk-free investment programs. This has been and still is the case in Eastern European countries and in other development areas where, moreover, these concessions interact with the development programs of international financial institutions such as, for instance, International Finance Corporation.

Overall financing of the investment itself is also a structural and, in a sense, traditional feature of leasing transactions. However, as already mentioned in the case of risks, the high average value of real estate, plant and equipment or ship and aircraft deals makes this specific aspect particularly attractive and in effect boosts the growth of demand in this market. In addition, more and more frequently proposals offer a full-service solution, in which the lessor provides a series of ancillary services to asset management that supplement and enhance the financing service. Examples of these ancillary services are maintenance, insurance, placement of the redeemed asset in the secondary market or, in the case of real estate, overall management of the asset.[9]

Lastly, extending the average duration of big-ticket leasing deals is a basic trend found typically in the international leasing market and responds to two different requirements to create growth.

The first is a thrust to broaden the market by “type of customer”: in fact, creating long-term leasing contracts makes recourse to leasing transactions more accessible

[8]In this regard see Leaseurope and KPMG (2012).

[9]As regards growth of ancillary services based on the full-service technique for big-ticket deals, see the international market survey and review of possible models presented by Watson (2014) and Rutledge and Raynes (2010).

to a larger number of parties, who appreciate the improved financial sustainability this solution offers.[10]

The second instead concerns the drive to broaden the market “by type of asset”: long-term contracts mean both the lessor and lessee can proceed to finance projects that, given their big-ticket size, could not have been repaid over a “conventional” timeframe. Examples of this are the real estate leasing transactions developed in Germany, which have a duration ranging from 25 to 30 years against a conventional duration of 8–12 years.

With reference to specific factors, aspects that stimulate growth in demand for big-ticket leasing must instead be differentiated according to the legal profile of the underlying asset.

In the case of existing real estate, plant and equipment or ships and aircraft already owned by the lessee, apart from general factors, the specific boost in demand is stimulated by the lessee’s objective to modify the combination of risk and return. In fact, the sale and leaseback technique allows the owner of the asset to sell it, obtain substantial liquid resources available for investment and obtain increased tax benefits—above all in the case of real estate—when compared to depreciation, and then to reacquire the asset at the end of the leasing transaction. The sequence of these effects highlights how a leasing deal competes both by offering securitization alternatives as opposed to a closed real estate fund investment and also enables the lessee to use such a transaction as support for achieving highly differentiated objectives. In this second case the factors stimulating demand can be summarized as follows[11]:

– acceleration of tax benefits by transforming depreciation into leasing installments
– increase in company structural margin and freeing up of financial resources to sustain investment and growth processes
– investment of wealth and net worth in a private banking mode.

In all cases mentioned the sale and leaseback technique can be used in a conventional or unconventional manner. In the former case the owner of the asset becomes the lessee as a result of the leasing transaction; in the latter, the owner can nominate an associated third party as the lessee. This is done when the third party has a more favorable tax profile (either inherent or based on the tax system under which it operates) than the asset’s owner.

[10]See Fabozzi et al. (2006).

[11]The stimulus to demand mentioned represents the “virtuous” factors that encourage users to apply the sale and leaseback framework to big-ticket assets. There are undoubtedly also other stimuli, for instance the need to reduce company debt by using financial resources coming from real estate rather than plant and equipment. In this case the transaction can involve considerable margins of risk as, on the one hand, the company doesn’t acquire resources to sustain an investment and growth process and, on the other, it loses ownership of part of the company’s net worth dedicated to business growth.

Apart from normal factors, in the case of an existing asset owned by a third party or real estate to be built, the specific boost for demand is above all the availability of a flexible instrument that can be modeled (in tax and financial terms) to match the typical requirements of large-scale transactions and, above all, into-be-built projects.

4.3.2 State of the Art as Regards Offer

The many types of deal that can be assembled in the big-ticket leasing field de facto leads to a many-faceted segmentation of the market. This can not only be seen in terms of the breakdown used in Fig. 4.2 based on the characteristics of the underlying asset, but also as regards the geographical positioning of the deal, which can either be domestic or international. This means a framework for analysis comprises the asset product-category, the asset's legal status and the relevant geographical context, which effectively impacts the characteristics of financial intermediaries in terms of choice of positioning and specialization.

The structure of competition found in the international market and within individual domestic markets is complex. In general terms the following profiles can be identified for intermediaries active in developing this market:

- international merchant and investment banks
- leasing companies
- intermediaries specialized in real estate financing, equipment financing and ship and aircraft financing
- private banking boutiques
- family offices
- supranational financial institutions.

The international merchant and investment banks operate on a global scale (JP Morgan, Bank of America and Goldman Sachs) or over large geographical areas (Deutsche Bank, UBS, Credit Suisse, Nomura and Mitsubishi) with an offer that covers the entire spectrum of corporate finance services.[12] Banks always have a dedicated real estate division whose mission as presented to customers is their ability to satisfy all financing and associated management needs in relation to real estate business rather than other big-ticket assets (although in certain cases there is also a plant and equipment or ship and aircraft division). Positioning is both at international and domestic level, given that there are regional banks operating in individual country markets. In the majority of cases the transactions assembled by these banks in the real estate field are big-ticket deals and cover all the various types of transaction examined. In terms of innovation, this is above all their ability to react to changes in the legislative and tax fields within the various countries

[12] In this regard see Caselli et al. (2015).

concerned, where the aim is to prepare contractual solutions with tax profiles that are attractive for customers.

Leasing companies are a highly heterogeneous group inasmuch as it includes large operators competing on a global scale and more marginal operators who do business within specific areas of a single domestic market. In any event leasing companies address the market as universal operators specialized over the entire spectrum of leasing transactions. But in fact only the very largest operators can assemble leasing deals for all "product-category type and legal type" combinations for real estate, plant and equipment or ship and aircraft assets, and so to a degree resemble the international merchant and investment banks. Instead small and medium-sized operators occupy less complex and lower-risk market segments, limiting themselves to leasing deals for already existing residential and administrative real estate. In the latter case the ability to innovate appears to be low.

Specialized real estate financial intermediaries are instead a small, more homogeneous group, inasmuch as the focus on the real estate segment requires a substantial critical mass and normally a coverage at national level. A review of the market reveals that while merchant and investment banks cover the international, mainly big-ticket market, and leasing companies span a wide size range, specialized intermediaries are domestic-sized national operators.[13] Furthermore, there are almost no operators specialized in only the plant and equipment or ship and aircraft segments.

From a company standpoint, specialized financial intermediaries tend to be independents (for example, Depfa Bank) or are part of the real estate construction and sale sector (e.g. KG Allgemeine Leasing GmbH & Co) or yet again, financial group product companies (e.g. Deutsche Immoblien Leasing GmbH). Generally speaking, given their basic strategic focus specialized intermediaries cover the entire spectrum of real estate leasing transactions and are market leaders and innovators. However, in the big-ticket deal area and at international level they meet tough competition from international merchant and investment banks.

Private banking boutiques only entered the real estate leasing scene recently. This, because they wanted to offer customers an all-in management service for that part of their wealth invested in real estate assets. Inevitably the positioning of private banking boutiques is almost exclusively in the segment comprising existing assets owned by the lessee (normally residential or administrative and, in some cases, commercial real estate), where the aim of transactions assembled is rationalization and tax optimization or investment and value creation by utilizing the unconventional sale and leaseback formula.[14]

As regards the mission of family offices, this is completely in line with that of private bankers, given the overlap of business areas in which both operate.

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[13]On this point it should be observed that the presence of specialized intermediaries in the real estate financing segment is particularly strong in Germany. In fact, the League Table prepared by Leaseurope in 2015 covering the main operators in the leasing sector (by asset and by volume) showed that in the top 100 positions at European level there were as many as six German operators specialized in real estate. No other European country shows such high-profile performance.

[14]On this issue see considerations in Caselli et al. (2015).

Supranational financial institutions operate in the real estate leasing sector to provide support for economic growth in developing countries. In this sense transactions mainly fall within the to-be-built, big-ticket area, in particular for infrastructure projects (hospitals, bridges, power stations, etc.). The leading institution in the sector is IFC (International Finance Corporation) The World Bank Group, which continually uses the leasing technique in the real estate and plant and equipment sectors because it can[15]:

- guarantee adequate management of risks throughout the entire lifecycle of a project;
- enable IFC to be in direct charge of the project because of ownership;
- enable the future owner to learn how to realize and manage a project.

So presence of supranational institutions plays a critical role and significantly boosts the big-ticket leasing market and in particular promotes innovation in the risk and legal management fields of transactions, which are typically of a project type.

Specifically as regards the Italian market, there is only a trace of the breakdown reported above, given that the majority of the offer is in the hands of leasing companies owned by the major national banking groups. While in terms of size many of these are substantial at European level, their offer is mainly focused on small and medium-ticket leasing. Looking specifically at structured leasing, the basic traits that emerge compared with trends found in the international market are as follows:

- Asset transfer and risk management is not a factor that the offer exploits as a priority.
- There tends to be exploitation of tax differences and in-country loopholes, but there is almost none at an extra-country level.
- Exploitation of full-service platforms is almost exclusively geared to the retail market, using a technique that is more similar to operating rather than big-ticket leasing.[16]
- Progressive extension of leasing duration in a big-ticket sense is still not underway, although the real estate leasing market is showing strong growth.

4.3.3 Considerations on Certain Critical Aspects Found in the Market

While the big-ticket leasing market is experiencing a substantial growth trend and significant diversification as regards type of user situations and needs, there are a

[15]On this issue see considerations in Ascari and Albisetti (1996) and Della Croce and Sherma (2014).

[16]See Carretta and De Laurentis (1998).

number of factors that pose obstacles to its growth and diffusion. The above factors refer to both elements concerning demand and, above all, to critical management junctures on the offer side. In particular, on both sides factors representing obstacles increase in importance with the size of the transaction and as regards certain more complex types of real estate (industrial, infrastructural) and to-be-built deals. Referring back to the framework proposed in Fig. 4.2, this means the complexity of the transaction and constraints when assembling it increase with a move from the upper left corner (residential real estate—already existing real estate owned by the lessee) to the lower right corner (infrastructural real estate and plant and equipment—asset to be built).

With reference to the demand side, the main factor putting a brake on requests for big-ticket leasing deals stems from two elements.

Firstly, the strong tax acceleration generated by leasing deals and limited duration of the deal itself require a stable level of taxable profits over time and a financial position consistent with the succession of installments. This means that only companies with good liquidity (and the ability to plan and manage it), but above all the ability to generate profits, can organize a big-ticket leasing transaction to good effect. In the absence of these two conditions the leasing transaction may generate a dangerous boomerang effect on the company's overall management equilibrium and even solvency.

Secondly, exploitation of in-country and extra-country tax differences makes tax benefits the center of gravity for the deal. Any changes for the worse in tax legislation over time will not only reduce the transaction's economic justification but, above all, may jeopardize the lessee's ability to sustain the transaction itself. In other words, the tax risk in terms of legislative volatility becomes an obstacle to the growth of these transactions. This is even more probable the more such transactions grow by using the leverage of extra-country tax differences with an eye to assembling frameworks to exploit loopholes and maximize available financial maneuvering room.

With reference instead to the offer side, there are three management constraints that limit or even kill an inclination to participate in the big-ticket leasing market.

First, the substantial size of assets involved in such transactions represents a rigidity factor in the asset side of financial intermediary balance sheets. This, both because of an impact on structural margin and eligible capital ratio, but above all because of risks associated with the asset. From this standpoint, if the financial intermediary doesn't have a sufficient networking capability with other operators, either enabling it to organize offer pools or, above all, contracts for risk sharing and transfer, then the expected remuneration profile would not appear to be consistent with the type and size of the risks taken. This aspect becomes unacceptably critical into-be-built leasing where, in addition to an increase in the number of risks, there is also the issue of the intermediary's civil and criminal liability as regards the construction site for the work concerned.

Secondly, as the real estate asset is owned by the financial intermediary/lessor, in the event of the lessee's insolvency, all things being equal, this increases the loss given default (LGD) value underlying the financing. This, because even if the market value of the asset is substantial, the recovery rate value relevant to real estate leasing is particularly low. This means that in the overall transaction framework, capital absorption for the lessor is significant both because of the size of the deal and its LGD value. The end result is a rationalization impact within the financial intermediary's portfolio that affects the amount or cost of the transaction.

Lastly, in the specific case of sale and leaseback transactions, the lessor's purchase of an asset owned by the lessee can generate dangerous setbacks in the event the lessee should become insolvent. This, in terms of the impact on withdrawal of bankruptcy and also possible liability, issues that the financial intermediary may be summoned to demonstrate before the court from the standpoint of acceleration of the lessee's insolvent status.[17]

4.4 Analysis of the Tax Profile for Leasing Relevant to Value Creation in Structured Operations

4.4.1 Overview of Tax Variables Relevant to Capital Structure

The review of factors behind the growth of the big-ticket leasing market highlighted the centrality of taxation. This is the pivot in transactions that enable a lessee to achieve significant tax savings, in any event much more than by using other financing and investment strategies, above all in real estate. In other words, the tax variable becomes a powerful tool for creating economic maneuvering room to reduce the cost of capital for its users. This means lessees can use the tax benefit this transaction provides in the most appropriate manner to achieve their objectives, bearing in mind their financial and balance sheet profile.

An understanding of this economic maneuvering room can be gained by reviewing the taxation framework in force in the country concerned, examined in terms of tax deductibility of costs for borrowers of financial resources. As regards Italy, the regulatory framework is quite well-defined and stable (Table 4.3), and in the specific case of leasing even tends to reflect regulations found in the majority of other European countries.[18]

With reference to borrowers, tax legislation differentiates between three types of financing source: debt (in whatever form), financial leasing and equity.[19] Debt financing benefits from a tax effect of 27.90%, namely, 24% for IRES (corporate

[17]On this issue see considerations and legislative/regulatory references in Leaseurope and KPMG (2012) and Caselli et al. (2015).

[18]On this subject see again On this issue see considerations and legislative/regulatory references in Leaseurope and KPMG (2012) and Caselli et al. (2015).

[19]For an exhaustive evaluation of tax effects on corporate financing decisions see Eisefeldt and Rampini (2009) and Graham et al (1998).

Table 4.3 Tax profile for corporate borrowers of financial resources based on the Italian system as of 1 January 2017

Source of financing	Type of tax		
	IRES[a]	IRAP	IVA
Debt	Full deductibility of financial charges and expenses, except as regards thin capitalization rules for resources contributed by qualified stockholders	Non-deductibility of financial charges. Full deductibility of the expenses	Outside field of application
Financial leasing	Full deductibility of "average installment" and expenses provided the leasing contract has a duration equal to at least half the asset's depreciation life for tax purposes. The exception is real estate, for which duration of leasing must be at least twelve years	Only full deductibility of the asset value and expenses, provided the leasing contract has a duration equal to at least half the asset's normal depreciation life for tax purposes. The exception is real estate, for which duration of leasing must be at least twelve years	All elements of the contract are subject to IVA
Equity	The tax effect is governed by changes in equity that took place every year, according to the ACE mechanism	No effect	Outside field of application

[a]IRES is the 24% corporate tax calculated on the income before taxes, while IRAP is the regional 3.9% corporate tax calculated on the amount of the EDITDA
Source The author

income tax) net of 3.90% for IRAP (regional tax on productive activities). The tax structure for leasing, instead, cannot be defined a priori but depends on a series of factors, such as the type of underlying asset, duration, the borrower's IVA (i.e., VAT, value added tax) profile and month in which the transaction starts. This means that based on the case in point financial leasing can either have a much better or worse tax profile than debt financing. Lastly, equity financing benefited from the limited, partial effect of the ACE starting from 2014 period whereas the yearly increase of the equity (by cash or by income) is multiplied by a ratio (given by the Government every year and for 2017 is equal to 2.4%) to obtain a number that can be considered as a cost to generate a taxation benefit for IRE. That means if there's a capital increase of 100, the tax benefit for 2017 is equal to 100*24%*2.4%.

Concerning the above, it should be emphasized that while tax variables in the Italian system are highly volatile, rules regarding deductibility of charges relevant to the source of financing and subdivision into the three categories—debt, leasing and equity financing—have essentially remained stable over time. Moreover, introduction by the legislature of a thin capitalization mechanism since 2004 tends, on the one hand, to reduce the convenience of recourse to debt beyond a given level of financial leverage adopted by a company and, on the other, redirects use of stockholder resources away from debt and towards equity. This means the availability of three different categories (and de facto four including thin capitalization) produces interesting room for maneuver and financial planning, above all as regards the comparison between medium-term debt and leasing. Vice versa, there is no short-term maneuvering room, while recourse to equity financing is confirmed as not being structurally convenient given that it doesn't attract any form of tax benefit.

Overall, the tax treatment for financing sources as far as direct taxation is concerned draws on provisions found in the Tax Consolidation Act (TUIR) of 22 December 1986 (Presidential Decree 917) and later modifications, above all the 2004 budget that introduced IRES taxation. In particular, application of rules governing deductibility of interest expense relating to financing contracts means there are effectively only two categories of financing from a tax standpoint: bank financing and leasing. In the case of bank financing, interest expense can be deducted and no specific reference is made to contractual form; as for leasing, specific rules provide for the deduction of installments paid by the lessee to the lessor. Furthermore, treatment for direct taxation purposes must be integrated by provisions governing IRAP tax. This tax was introduced in 1998 by Legislative Decree 466/1997, which indicated a different tax treatment for financing compared to corporate income tax regime in the Consolidation Act. In the specific case of leasing it must be remembered that elements of the contract paid by the lessee to the lessor are subject to IVA, inasmuch as leasing is considered to be a service and, if the call option is exercised, a sale of goods. As there are no specific provisions concerning IVA on leasing, tax treatment is in accordance with general rules that apply to providing services and sale of goods, while the applicable tax rate is determined by the asset underlying the contract (Presidential Decree 633 of 26 October 1972 and later modifications).

The analysis of tax treatment must therefore distinguish between bank financing and leasing, given that the tax implications and impact for the "borrower" are substantially different.

In the case of bank financing, the only significant element from a tax standpoint is interest expense, which the borrower can deduct from taxable income provided profits are sufficient.

In the case of financial leasing, the deductible cost component is the installment paid periodically to the lessor and, if exercised, the asset's call option price, which for all intents and purposes represents the purchase price. The rules determining whether installments can be deducted are different in the case of IRES and IRAP tax. From the IRES tax standpoint, deductibility must be calculated on a day by day basis relevant to the duration of the right. This means that every fiscal year the deductible amount for installments must be calculated according to the accrual accounting method, proportionate to the duration and independent of installments effectively paid or invoiced. In this sense the maxi-installment paid by the lessee at the beginning of the contract is "spread" evenly over all the remaining installments. If the lessee exercises the call option then the price paid has to be depreciated according to the normal rate for the asset concerned, as prescribed by the Ministry of Finance. From the IRAP tax standpoint only the "capital element" of the installments can be deducted. The deductible amount is determined by dividing the lessor's asset cost (calculated as the difference between the asset price and the call option price) by the duration of the contract in terms of days. Then for every tax year the deductible amount is this "daily" value multiplied by the relevant number of days for the year concerned. As regards the depreciation rate used for the asset after the call option has been exercised, this is the same as for IRES tax purposes.

It should be observed that in the case of both IRES and IRAP tax the legislator has subordinated deductibility of the cost elements to a constraint as regards the minimum duration of the leasing contract. For assets other than real estate, installments can be deducted provided the contract duration is not less than half the normal depreciation period established by coefficients stated in Ministerial tables. As for real estate, the minimum duration is twelve years. In this latter case the legislation provides a considerable tax benefit as eight years is considerably less than half the normal depreciation rate for real estate, which on an annual basis is 3%. This represents the competitive edge of a real estate leasing transaction and can be used by the lessee to achieve the various objectives underlying the transaction itself.

Lastly, mention must be made of operating leasing, for which the legislature has applied the same basic approach used for financial leasing. There are, however, two important differences that represent an appreciable tax benefit:

- The constraint concerning minimum duration has been removed with reference to both IRES and IRAP tax.
- The calculation of cost deductibility for IRAP purposes is the same as for IRES tax.

This means that an operating leasing transaction in theory has a greater "tax acceleration" effect than financial leasing, especially when compared to debt financing. Clearly, however, in the case of operating leasing the lessee doesn't have a final call option right and so this transaction is of little interest if the company's business growth depends on owning the underlying asset.[20]

4.4.2 Sensitivity Analysis for Optimizing the Leasing Tax Profile

A review of the regulatory framework for taxation as regards sources of financing highlights two significant factors that must be considered when assembling transactions to take full advantage of tax effects, namely, so that the net tax rate

[20]Moreover, these increased tax benefits afforded by operating leasing have provided a strong stimulus to create "structured" transactions that give operating leasing the same characteristics as true financial leasing. This subject is covered in later sections dealing with synthetic leasing transactions.

reduction is greater than the initial gross interest rate (TAEG—global effective annual interest rate).

Firstly, excluding the case of equity that offers no tax benefits for a company, there is no convenience a priori between debt-based financing as opposed to leasing. The convenience of one as opposed to the other must be verified case by case with reference to the structure of the transaction concerned.

Secondly, a financial leasing transaction's tax profile is more dependent on management and contractual factors and is therefore more flexible than debt financing. The main differences are the following:

- Financial leasing has three tax profiles (IRES, IRAP and IVA tax) and consequently depends on the lessee's overall direct taxation balance, compared to only one tax profile (IRES) for debt financing.
- Financial leasing cost (installment) deductibility is based on a non-financial criterion (spreading the initial maxi-installment over the entire duration of the transaction), as opposed to a financial criterion for deduction of debt financing cost (interest expense). Consequently, given the same TAEG, the net tax rate for a leasing transaction is first and foremost highly dependant on all the temporal factors that modify the deal's structure. Vice versa, in the case of debt financing, again given the same TAEG, temporal factors have no effect on the net tax rate.
- The duration of financial leasing transactions is by necessity tied to the depreciation period for tax purposes. This systematically generates a difference potentially in favor of leasing if financial costs associated with the leasing and debt transaction are the same.
- Real estate leasing transactions potentially have a greater tax benefit than debt financing deals given their shorter, eight-year duration.

Given these elements, a sensitivity analysis must be performed to define the framework of relationships between factors affecting the net tax rate for these financing transactions and the net tax rates themselves. This must be done in order to obtain a framework of functional rules to assemble structured deals that fully exploit the ability of tax leverage to create value from a tax-based finance standpoint.

The sensitivity analysis concerning the net tax rate for debt financing and leasing must be conducted with reference to two macro-factor categories:

- contractual variables
- tax variables.

The first refer to financing contract (debt or leasing) factors and include the individual percentages represented by the maxi-installment, call option price and fees, the duration of the transaction and the start-date in the year.

Instead, the second refer to exogenous tax variables inherent in the transaction and endogenous ones associated with the debtor/lessee. The exogenous tax variable is the depreciation rate for tax purposes applying to the underlying asset, given that this is established by tax legislation. The exogenous variables are instead the

Table 4.4 Sensitivity analysis, given the same TAEG and assuming sufficient taxable profits, to evaluate the impact of contractual variables on the net tax rate for leasing and debt financing

Contractual variables	Impact on net tax rate of leasing	Impact on net tax rate of debt
Maxi-installment	Increases slightly as the maxi-installment percentage increases	–
Call option price	Increases slightly as the call option price percentage increases	–
Fee percentage	Decreases as the fee percentage increases, in the face of a reduction in the maxi-installment percentage	Decreases as the fee percentage increases, in the face of a reduction for financial costs
Duration of transaction	Increases heavily as the duration of the transaction lengthens	No impact
Start of transaction	Increases heavily the more the start-date for the transaction moves towards the end of the year	No impact

Source The author

contracting company's depreciation policy (normal or accelerated), IRES tax profile (in profit or loss), IRAP tax profile (profit or loss[21]) and IVA tax profile (in debit or in credit).

The method for calculating and evaluating the net tax rate for financial leasing and debt financing a model must be built to determine net tax flows for the transactions concerned. For a detailed review of the tables, which in fact are already consolidated in corporate finance, see the referenced works[22] and the review in the example shown in Sect. 4.4.2.1.

The main points emerging from the sensitivity analysis of contractual variables (Table 4.4) and tax variables (Table 4.5) give a sufficiently precise, clear reference framework.

With reference to contractual variables, the joint leasing-debt financing evaluation in cases where there are sufficient profits highlights the strong sensitivity of net tax rate to all independent variables. Vice versa, debt financing is only dependent on the percentage amount of fees, from the standpoint of cross-subsidation compared with financial charges. If these results appear predictable based on the tax regulations in force, then certain basic results must be systematized in an organic reference framework.

The close dependence of the net tax rate for leasing on temporal variables (duration and start-date) is the result of a "non-financial" rule governing deductibility of installments. The results emerging highlight the strong positive tax

[21]The assumption of a pre-tax loss in the case of IRAP is extreme, inasmuch as it would imply a negative difference between the total for items in section A and items considered as deductible in section B of the statutory income statement.

[22]In this regard see the methodology reported by Caselli et al. (2015a, b).

Table 4.5 Sensitivity analysis, given the same TAEG, to evaluate the impact of tax variables on the net tax rate for leasing and debt financing

Tax variables	Impact on net tax rate of leasing	Impact on net tax rate of debt
Underlying asset depreciation rate	Increases strongly with an increase in the underlying asset depreciation rate	No impact
Depreciation policy selected by the company	Decreases strongly with a change from accelerated to normal depreciation	No impact
IRES or IRE profile	Increases depending on if there is a loss for the year	Increases depending on if there is a loss for the year
IRAP profile	Increases depending on if there is a negative taxable base	Increases depending on if there is a negative taxable base
IVA profile	Neutral if the IVA account is usually in debit Increases if the IVA account is usually in credit	No impact

Source The author

impact if leasing exploits the minimum possible duration granted by legislation and the effect of a start-date in the early part of the year. The positive tax impact in these cases can be so considerable that it reduces the net tax rate to lower levels than those achieved using alternative sources of debt financing, even when the TAEG is lower. Vice versa, extending the duration and moving the start-date to later in the year tends to impact the company's cost of capital considerably, de facto meaning that leasing cannot be used for tax-based transactions. In the case of the maxi-installment and option call price, instead, the relationship to net tax rate appears blander, as although there is an increase in net tax rate if either of the two components are higher, the change is marginal. This means these two contractual elements can be modified and governed quite flexibly for financial planning purposes without impacting the company's cost of capital structure.

With reference to tax variables, the overall results seem highly differentiated. The first issue concerns depreciation, seen from the standpoint of the rate and underlying policy. In the case of the rate, there is a very strong direct relationship with the net tax rate inasmuch as assets depreciated over longer periods show a progressively lower cost of capital for leasing. The same effect is essentially achieved by adopting normal as opposed to accelerated depreciation, given that the depreciation period is longer and therefore dilutes the tax benefit associated with ownership of the asset concerned. A second point to evaluate concerns the user's IVA profile. In the event of an IVA creditor position, this favorably affects the net tax cost of leasing as the IVA credit linked to the value of the underlying asset is split up over time. Lastly, a more obvious, general evaluation concerns whether there are profits or not, given that both in the case of leasing and debt financing a loss in one or more years will only serve to increase the cost of capital.

In conclusion, the comparative analysis of leasing and debt financing clearly indicates that financial leasing is much more preferable to debt financing when used for tax-based transactions, provided tax and contractual variables are used correctly to the company's advantage, given their impact on net tax rate. This means that sensitivity analysis can enable formulation of a set of operating rules to devise tax-driven effects when assembling structured leasing deals. These are as follows and, given the tax treatment in force concerning debt financing, can be extended from the Italian case to all other financial systems:

– The tax effect of leasing is maximized every time the minimum duration for the transaction as prescribed by legislation is exploited.
– The above effect is considerably greater if the transaction is organized to start in the early part of a tax year.
– Assets with lower depreciation rates enable leasing to exploit the effect of the minimum duration to a much greater degree. At international level, assets that generally having lower rates are those typically found in structured leasing deals and, in any event, the real estate sector always has the lowest rate.
– Any favorable ad hoc legislative measures concerning minimum duration, such as in the real estate leasing case in Italy, generate particularly attractive synergies for assembling structured operations.
– If the lessee has an IVA balance in credit and a normal depreciation policy profile, then this amplifies the tax benefit produced by leasing transactions.

4.4.2.1 An Example for Purposes of Sensitivity of the Net Rate for Leasing and Debt Financing

This sensitivity analysis is based on leasing and debt transactions for an amount of 900,000 euros, each having a duration of four years and starting 1 February 2017. The assumption in both cases is payment at the time the formal contract is stipulated of 3000 euros (0.33% of the capital sum) to cover relevant costs. The transaction concerns purchase of machinery for which legislation establishes an annual depreciation rate of 12.5%. The purchaser is a joint stock company that is sufficiently profitable, has a monthly IVA debit balance and as a policy has adopted the accelerated depreciation method. The effective annual cost (TAEG) in both cases is 8%. The leasing contract provides for payment of a maxi-installment equal to 10%, followed by 47 monthly installments of 19,800 euros and a final call option price of 1%. On the other hand, the loan contract provides for repayment over 48 months with monthly installments of 21,853 euros. The graphs developed by simulation take into account the Italian tax framework in force as of 1 January 2017. Each simulation is carried out under equal conditions, namely, based on the same input data (Figs. 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 4.10, 4.11, 4.12).

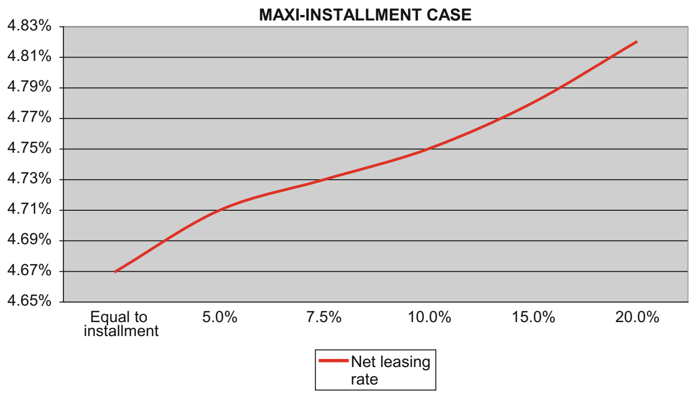

Fig. 4.3 Maxi-installment case. *Source* The author

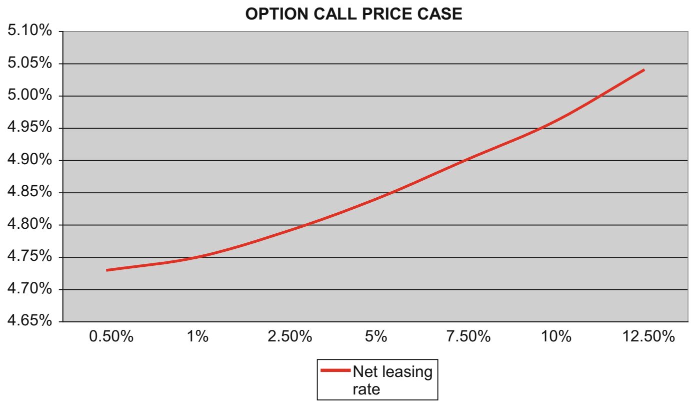

Fig. 4.4 Option call price case. *Source* The author

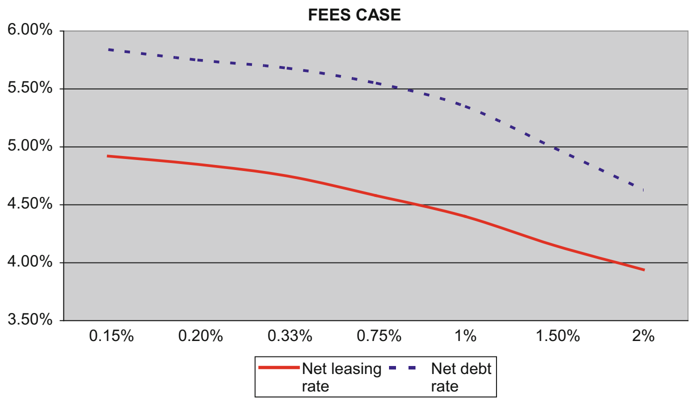

Fig. 4.5 Fees case. *Source* The author

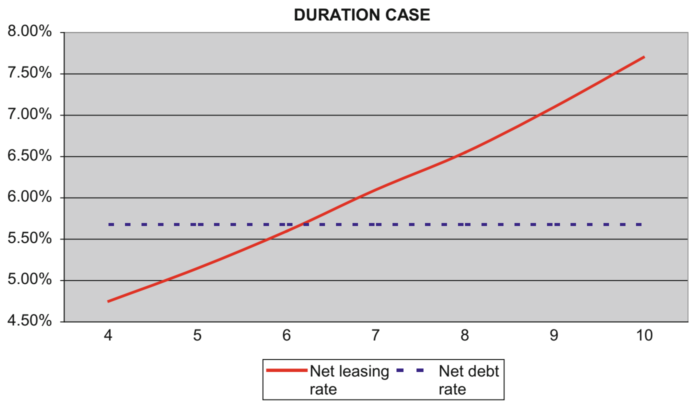

Fig. 4.6 Duration case. *Source* The author

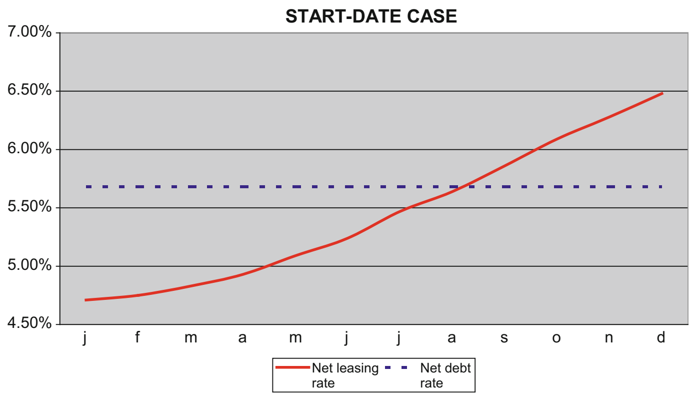

Fig. 4.7 Start-date case. *Source* The author

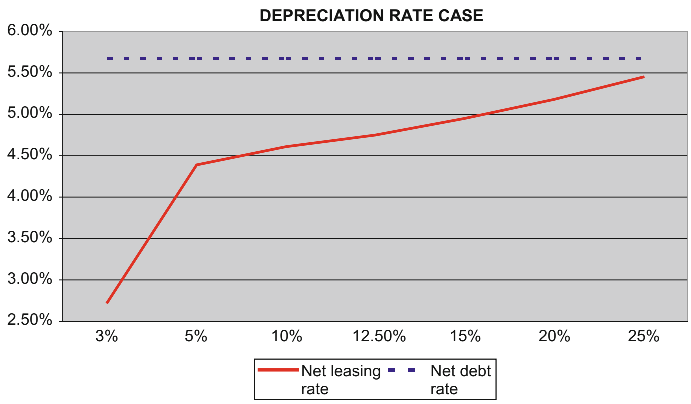

Fig. 4.8 Depreciation rate case. *Source* The author

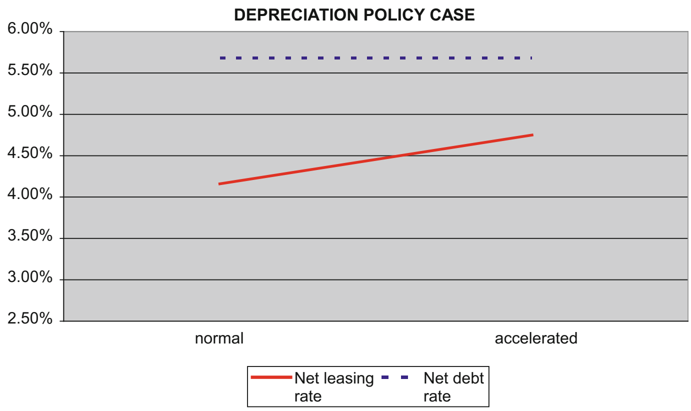

Fig. 4.9 Depreciation policy case. *Source* The author

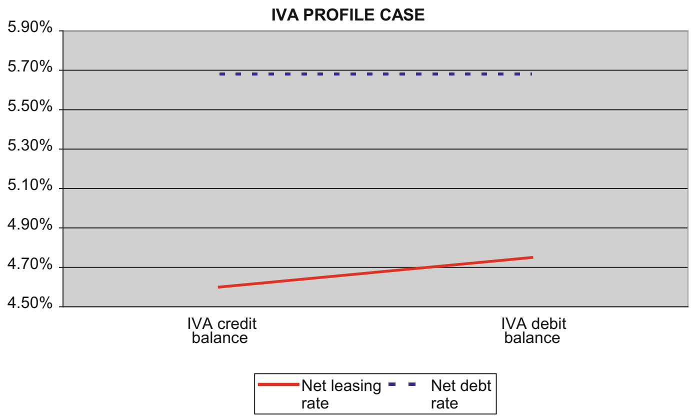

Fig. 4.10 IVA profile case. *Source* The author

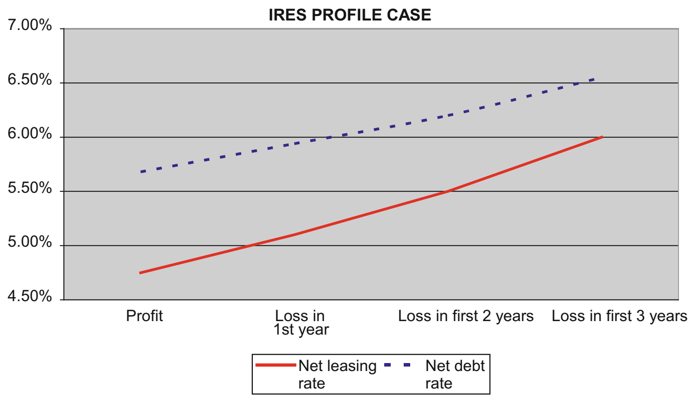

Fig. 4.11 IRES profile case. *Source* The author

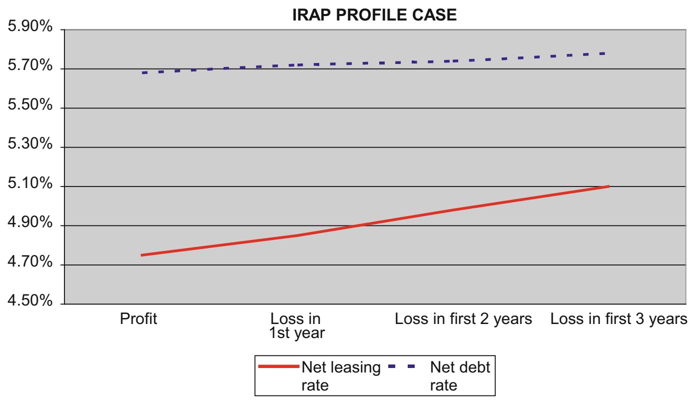

Fig. 4.12 IRAP profile case. *Source* The author

4.5 Assembling Structured Leasing Transactions

4.5.1 A Reference Classification Framework

Assembling structured leasing transactions can involve using the three basic elements, applying different methods to differing degrees, above all in the light of the tax evaluation the various parties to the transaction make in relation to their own objectives in terms of cost and return. For these very reasons, as mentioned at the beginning of this chapter, it is easier to outline the aims for assembling a structured transaction than to offer a comprehensive definition and analytical overview of how it is organized. However, a basic classification framework can be offered that, while not exhaustive, classifies elements recurring most frequently in the international market in certain macro-categories. It also captures the specific nature and objectives of transactions that occur in the market itself.

From this point of view structured leasing transactions can fall within one of three categories (see Fig. 4.13):

– standardized or leveraged transactions
– synthetic transactions (so-called synthetic leasing or synthetic structured leasing)
– asset management transactions.

The first category includes leasing deals basically structured around recourse to financial leverage.[23] This, so the lessor can both exploit the physiological effects multiplying the return as a result of recourse to indebtedness, and expand the number of participants in the transaction from the lessor-lessee axis to comprise a pool of third-party financers. In fact the leveraged technique can be used in any

[23]For an overview of leveraged-type leasing transactions see Caselli et al. (2015).

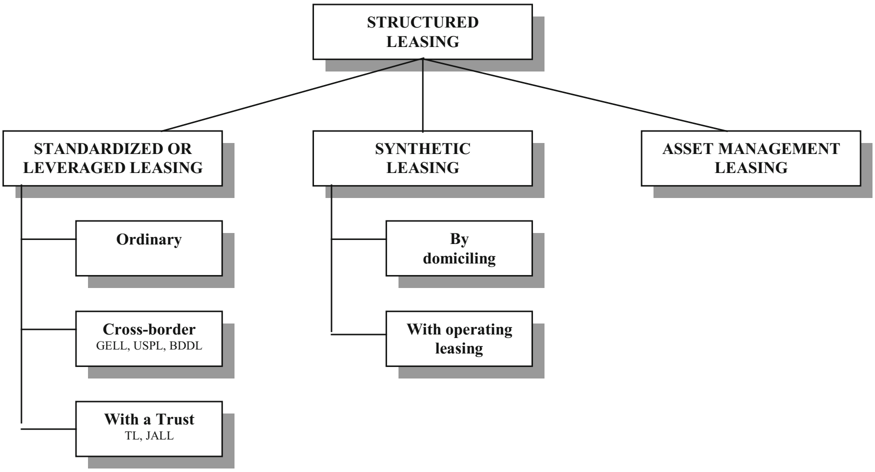

Fig. 4.13 A possible classification of structured leasing transactions. *Source* the author

market situation for any type of transaction, provided the size justifies recourse to a pool of financers external to the original transaction. However, leveraged transactions tend to be used above all in those markets that, given certain conditions, offer specific tax advantages to leveraged leasing transactions.[24] This occurs mainly in two different situations: in international cross-border leasing, and when trusts are used.

The first case refers to the German, US and UK markets where use of cross-border leasing offers specific tax advantages to the lessor, making the transaction more attractive both for the foreign lessee and for potential financers of the leveraged transaction, who can "participate in" the increased tax benefits produced by the deal. At international level the classification used for these deals refers respectively to the terms GELL (German Leveraged Leasing), USPL (US Pickle Leasing) and BDDL (British Double Dip Leasing).

In the second case reference is again made to the US market as well as the Japanese market, where interposing a trust between the lessor and lessee doubles the tax benefits for the deal to the advantage of the parties concerned. In this case the terms used at international level are Trust Leasing (or Pickle Trust Leasing, if a trust is used in conjunction with a cross-border transaction) and JALL (Japanese Leveraged Leasing).

The reference structure for the second category of leasing transactions is instead based on using a dedicated SPV in the transaction.[25] This means the asset underlying the deal is held by the SPV, which therefore becomes the owner. The SPV

[24]From which the term "standardized" classification is derived, given that transactions are based on standard tax frameworks.

[25]On this subject see, in particular, Gregoriou (2008).

then proceeds, on the one hand, to organize the leasing transaction with the lessee and, on the other, to put together the funding to purchase the asset itself. Recourse to synthetic leasing transactions is essentially guided by two factors:

– a search for countries in which to domicile the SPV enabling greater tax benefits to be obtained than in the country of origin;
– the optimzation of tax benefits by transforming financial leasing into operating leasing.

In the first case the SPV's income statement will comprise three basic elements: depreciation of the asset underlying the transaction, financial costs for servicing the debt and installments received from the leasing contract. The degree of net income not only varies based on the spread applied between the leasing and the financing transaction interest rates but above all will depend on criteria governing deductibility of costs and recording of revenues. It will therefore be in the lessee's interest to find the country offering the most favorable relationship between cost deductibility and revenue recording in order to reduce taxation and attract financers for funding the SPV.

In the second case the aim of recourse to the SPV (apart from objectives already mentioned for the first case) is to utilize an operating leasing rather than a financial leasing contract, so increasing the tax benefits from the transaction. To achieve this the SPV stipulates an operating leasing as opposed to financial leasing contract with the lessee, whereas the lessee acquires a call option on the shares of the SPV, de facto obtaining the right to become the owner of the asset as in the case of financial leasing transactions.

Lastly, the third category of structured leasing transactions refers to asset management, namely, the use of leasing as a tool to mobilize wealth, in terms of an asset that can potentially underlie a leasing contract. In this case the aim is value creation by using tax leverage.[26] The concept behind asset management transactions is to use the leaseback technique to accelerate the tax benefit, achieved by financial leasing as opposed to merely owning the asset, and to create available financial wealth. The joint presence of these two effects therefore means leasing transactions, mainly those concerning real estate, can become part of a creative, many-faceted asset management operation in which the overall architecture can also include setting up an SPV as in synthetic-type transactions.

4.5.2 Standardized or Leveraged Structured Leasing Transactions

As the term suggests, by assembling this type of structured leasing transaction the lessor makes considerable use of financial leverage. This basic principle can give

[26]On this subject see Craido and Rixtel (2008) and Cumming (2010).

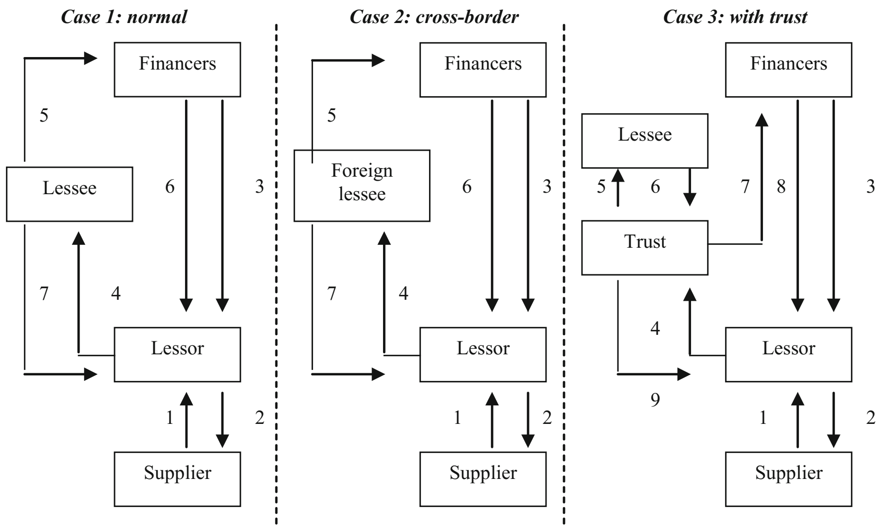

Fig. 4.14 Model for standardized or leveraged structured leasing transactions. *Source* The author

rise to three different solutions with a growing degree of complexity: normal leveraged leasing, cross-border leveraged leasing and leveraged leasing using a trust (see Fig. 4.14).

What these three cases have in common is the reason that induces the lessor to coordinate and manage the leasing transaction together with other financers. The latter can either be banking and financial parties who form a pool as used in medium and long-term financing deals, or the market in a wider sense in cases where the lessor issues bonds linked to a specific deal. The second technique is clearly only possible in the case of big-ticket deals that can justify the syndication and market placement costs. From this standpoint, these are the reasons producing a convergence of interests between the lessor and financers:

- The lessor increases the return on the deal by using greater financial leverage. As the market average for the debt component is 80–85% of the value of the deal, this enables the lessor to reduce utilization of internal sources to the minimum as the non-financed component is covered by the lessee's maxi-installment payment.
- Financers benefit from a dual guarantee of reimbursement given that installments are channeled directly from the lessee to financers and there is a lien on the asset in the event the customer should default, or alternatively a guarantee provided by the lessor.
- The dual guarantee received from the lessor by financiers means the cost of financing can be reduced.
- Financers achieve a more favorable risk-return ratio on the investment made.

Moreover, apart from tax benefits the transaction generates for the lessee, the convergence of financer-lessor advantages benefits the lessee too as the lessor can apply a smaller spread to the funding.

In the normal transaction (see Fig. 4.14, case 1), the lessor purchases the asset from a supplier (1) paying the price (2) utilizing financing provided by the financers (3) in the form of a direct loan or funding obtained from the market. Once the lessee obtains the asset from the lessor (4) payments are channeled to the financers (5), who in turn pay the lessor (6) installments net of the cost due for servicing the debt. At the end of the transaction the lessee (7) can exercise the call option on the asset and so obtain ownership.

The model for the cross-border transaction is identical to the normal deal (see Fig. 4.14, case 2) except that the lessee is domiciled in a different country than the lessor and supplier of the asset. It follows that, apart from general benefits described for the lessor, lessee and financers, tax legislation in certain countries can give the lessor—or even the supplier of the asset—additional tax advantages introduced as support for the country's export business. As mentioned previously, the lessor can currently obtain additional tax advantages in the case of the USA (US Pickle Leasing), Germany (German Leveraged Leasing) and the UK (British Double Dip Leasing). These advantages become an element in negotiations between the parties concerned in order to optimize the lessor, lessee and financers' return-risk-cost profiles.

The model for the trust-type transaction, instead, interposes a trust vehicle between the lessor and lessee (see Fig. 4.14, case 3). This means the trust receives the asset from the lessor (4) and channels payment of installments to the financers (7), who then pay a net installment to the lessor (8), exactly as in the case of the normal model. The difference from the normal model is that the trust subleases the asset to the lessee (5) in exchange for periodic installments (6) that the trust utilizes to pay the channeled leasing contract installments (7). Interposition of the trust is obviously worthwhile if tax legislation in the country in which the deal originates gives the trust a tax advantage utilizable when subleasing, or if the lessee has a greater tax benefit by paying the leasing payments to the trust rather than to the lessor. As of today the most attractive cases for this type of deal are found in the USA (Trust Leveraged Leasing) and Japan (Japanese Leveraged Leasing). Furthermore, in the US market deals can also be assembled that combine the cross-border technique found in Pickle Leasing with the tax benefits generated by using a trust.

4.5.3 Synthetic-Type Structured Leasing Transactions

Assembling a synthetic-type structured leasing transaction requires setting up a dedicated SPV that then becomes the pivot of the entire operation, given that it stipulates the leasing contract directly with the lessee (see Fig. 4.15).

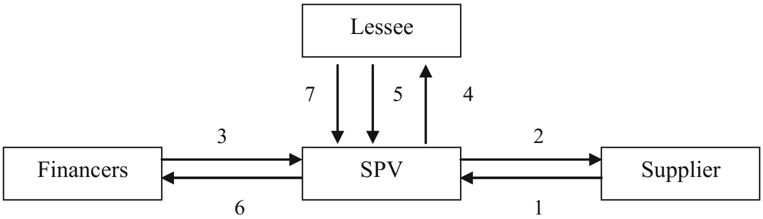

Fig. 4.15 Model for synthetic-type structured leasing transactions. *Source* The author

The overall model for a synthetic deal is based on first setting up an SPV that then purchases the asset (1) from the supplier, paid (2) by utilizing financial resources raised with the financers (3). As occurs in leveraged cases, the financers can fund the debt capital by a direct loan made through a pool or by issuing bonds. Moreover, as opposed to the leveraged case the financers can grant a limited part of the resources in the form of equity, either autonomously or together with the lessee, where the aim is to increase the commitment to and control over the deal. Then the SPV stipulates a leasing contract with the lessee and delivers the asset (4), receiving periodic installments (5) that are used by the SPV itself to repay the financers (6). At the end of the transaction the lessee may, if it wishes, exercise the call option on the asset owned by the SPV (7) or on shares in the SPV, so becoming the owner.

The reasons behind using the SPV are tax-based, given that the lessee can select the most advantageous country from a tax treatment standpoint, where a favorable difference can be created between installments received by the SPV and depreciation of the asset it owns. By creating this differential the lessee can reduce the cost element for the leasing contract while remunerating the investment guaranteed by the SPV's financers.

In addition to the above, an operating leasing contract can also be used and this increases the lessee's tax benefit as a result of a shorter contract duration. This is achieved by stipulating a normal rental as opposed to a financial leasing contract between the SPV and lessee, and the simultaneous stipulation of a call option between the SPV and lessee on the SPV's shares. When the rental contract ends the lessee can exercise the option and so become the owner of the SPV and, therefore, the underlying asset.

This second technique, apart from considerable tax advantages for the lessee, is particularly relevant when the company using the asset wants to set up an off-balance sheet transaction with a view to applying IAS principles or, more simply, to improve its structural margin thanks to a transfer of debt from the company to the SPV. In these terms, creating a tax advantage is accompanied by a significant window-dressing effect as regards the balance sheet, which can have a positive impact on the company's rating and, therefore, cost of money.

4.5.4 Asset Management-Type Structured Leasing Transactions

Use of structured leasing transactions for asset management purposes is based on the sale and leaseback model in which the leasing vehicle is used to create liquidity and greater tax advantages than those deriving from ownership.

This effect is much more appreciable, the greater the tax benefit for leasing and when the underlying asset is part of the investment portfolio of the individual potentially interested in the deal. The result is that most asset management-type structured leasing deals are focused on the real estate sector and for an amount similar to that found in high profile private banking.

The structure of these transactions starts in the private sphere (for instance, a businessman's family) where a real estate asset is producing rental income (see Fig. 4.16).

Rentals are paid by tenants to the owner (1) and the owner pays the highest marginal IRE rate on this income and cannot deduct any costs associated with managing the real estate investment. Consequently there is a very low annual net return on the real estate investment, without considering the risks and outgoings associated with any necessary extraordinary maintenance.

Recourse to a leasing transaction can either be handled through one of the owner's companies that can manage real estate as part of its charter or, more often the case, by setting up an ad hoc SPV as a vehicle specialized in real estate management. The owner therefore sells the real estate to the SPV (2) at market value. This enables the owner to achieve the first objective, namely, availability of liquidity to dedicate to more profitable investments compared to the modest return received previously (3). In order to raise the necessary funding the SPV sells the real estate it has purchased to a bank (4) and simultaneously stipulates a leaseback contract for at least eight years for an amount equal to the market rate. The effect for the SPV is particularly interesting, for the following reasons.

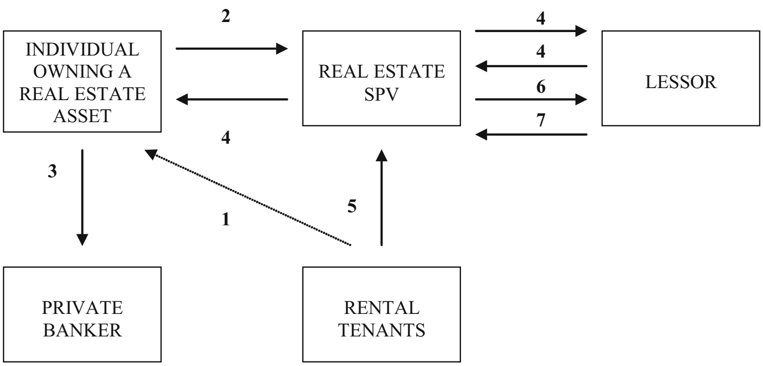

Fig. 4.16 Model for asset management-type structured leasing transactions. *Source* The author

- First, the rentals received by the SPV (5) are channeled to the lessor and therefore used to pay a significant part of the leasing installments (6).
- Secondly, the considerable acceleration provided by the leasing contract generates a significant amount of deductible costs, which heavily reduce the SPV's tax liability.
- Thirdly, management costs for the real estate become deductible items for the SPV.

At the end of the leasing contract the SPV becomes the owner of the real estate (7) and continues to receive the rental payments (5). The original owner therefore achieves a second objective: management costs for the real estate are deductible and the value of the SPV increases, first because the tax liability is reduced and then through purchase of the real estate. An overall financial evaluation of the transaction leads to the observation that over the duration of the leasing contract the original owner's wealth has at least doubled, inasmuch as there is an availability of financial resources equal to the market value of the real estate and, when the leasing transaction ends, ownership of real estate with a nominal value equal to its market value.

4.6 Conclusions and Growth Prospects for the Structured Leasing Market

The analysis in the preceding sections has highlighted the size and structure of the structured leasing deal market from the standpoint of demand and offer profile and, more generally, the competitive scenario at international level. In this sense what has emerged is that while the leasing contract has conventional and consolidated requirements, its use in the case of big-ticket assets, above all real estate, appears to be complex, growing and innovative. A significant aspect in this regard is market segmentation, which has led to the leasing instrument being used in various structured forms, such as leveraged leasing, synthetic leasing or asset management. In more general terms, while different from country to country, it can be said that structured leasing is in the strong growth stage of its lifecycle, even though there are clearly niche segments that must expand (for example, to-be-built leasing), above all in Italy.

Market growth prospects for structured leasing are closely linked to comparative developments in the various markets as regards elements fuelling demand and factors obstructing growth. Depending on which prevails, the structured leasing market will either see further growth or a decline in the use of this instrument.

Three macro-trends can be identified that help define a scenario and map out possible alternatives with a reasonable degree of accuracy.

The first is linked to the tax variable. It is inevitable that growth of both in-country and extra-country structured leasing will depend on changes to tax treatment affecting decisions concerning the structure of company liabilities and investments. This relationship has both a direct and indirect effect. Over the short term, a significant change in tax treatment for real estate financing deals can, as a direct consequence, strongly increase or decrease the size of the market. Over the medium term, innovative financial intermediaries then structure deals in a new way, looking for new market opportunities to replace those that have disappeared. This generates a dialectic exchange between fiscal policies and the creative ability of innovators that progressively modifies the weight and composition of the various structured leasing market segments.

The second, instead, refers to the private banking and family office issue. The considerable weight of real estate assets in the portfolios of high net worth clients and families of business people stimulates the creation of instruments for the management and investment of wealth. If, on the one hand, closed funds and securitization can represent useful tools for this purpose, on the other, real estate leasing has high potential. This is not only because of the simplicity of the technique itself but also the possibility to accumulate tax benefits by exploiting extra-country tax differences. In this sense it is reasonable to assume there will be strong growth in the real estate leasing market over the coming years driven by those operators who manage large real estate investment portfolios.

Lastly, the third macro-trend refers to international leveraged and synthetic-type transactions. The effective match between the leasing instrument and requirements of large real estate deals (in developing countries and others too) is a factor representing a substantial and structural stimulus for market growth. This stimulus, moreover, can only be exploited and sustained by financial intermediaries with a solid reputation in the international market, with a strong networking skill as regards the know-how required to organize complex deals and, above all, the ability to manage and share the asset risk.

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Chapter 5
Leveraged Acquisitions: Technical and Financial Issues

Vincenzo Capizzi

Abstract The main purpose of the chapter is to analyze the technical, financial and fiscal issues associated with leveraged acquisitions. Leveraged buyouts represent a crucial deal class in the area of structured finance. In particular, these types of transaction refer to a class of extraordinary financial operations within the larger "family" of mergers and acquisitions (M&A) transactions, expression qualifying those deals that produce deep and permanent changes in the ownership structure of one or more enterprises. Leveraged acquisitions involve the constitution of a "vehicle company" (also called "special purpose vehicle", "new company" or, simply, "newco") for the transfer of the ownership and, in this case, the acquisition occurs "off balance sheet" for the proponent subject. Leveraged acquisitions have been the subject of extensive discussions, especially with reference to their financial structure and the financial performance of the companies involved. Furthermore, leveraged acquisitions have experienced a number of legal and fiscal challenges strictly connected to the interpretation of their technical and financial structure. In this chapter, the authors want to provide a vision as complete and critical as possible of this kind of operations, in the knowledge that a partial approach to the topic could lead to conclusions too simplistic or, eventually, vitiated from preconceived notions and judgments of value.

The author thanks Paolo Parlamento for technical assistance and for contributing to the empirical analysis.

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S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_5

5.1 Introduction

Leveraged acquisitions are an important deal class in the area of structured finance, namely representing those transactions that result in an acquired company with a debt ratio higher than before the acquisition.

There are a number of studies cited in bibliography about this subject which have been made adopting research perspectives differing in amplitude and within different disciplines. At the level of business and management economics, these transactions are of particular interest for those entrepreneurial contexts characterized by a high number of small and medium sized enterprises (often family-owned), given the possibilities available to this kind of firm for the management of company conflicts and issues related to intergenerational change. Another reason of interest is represented by the capability of leveraged acquisitions to allow managers of the company to change their status by assuming an entrepreneurial role.

From a corporate finance perspective, leveraged acquisitions have been the subject of extensive discussions on the issues of financial structure and financial/economic performance of the companies involved. If, on the one hand, the higher debt that distinguishes such operations allows the exploitation of the effect of financial leverage on the earnings of the firm, on the other hand, these operations increase the financial risks of the company. This, of course, exposes the management to pressures that are difficult to sustain, as it has to define a strategy aimed at generating enough unlevered cash flows to ensure the repayment of the debt and the payment of the financial charges, in accordance with the business plan shared with the lenders. Consequently, the literature is particularly concentrated (i) on the determination of the ideal characteristics that need to be possessed by a firm in order to be considered a good candidate for a leveraged acquisition and, (ii) on estimating the value for the shareholders of such an acquisition.

From a management perspective, these deals are often seen as a way to resolve conflicts of interest between the different stakeholders of the company and, in particular, to realign the objective functions of managers and those of the minority shareholders. By means of a management buy-out, in fact, some managers themselves end up becoming minority shareholders in the acquired business.

The phenomenon of the leveraged acquisitions is, therefore, complex and fits well to be observed and analyzed from different standpoints. In this chapter, the authors want to provide a vision as complete and critical as possible of this kind of operations, in the knowledge that a partial approach to the topic could lead to conclusions too simplistic or, eventually, vitiated from preconceived notions and judgments of value. At the same time, another objective of the authors is to contextualize the instrument of the leveraged acquisitions—born in the USA—to the European reality, evidencing some specific uses as well as applicative issues.

The chapter is structured as follows. Section 5.2 deals with the specificities of leveraged acquisitions compared to more traditional transactions, providing a complete review of the various motivations for the latter. Section 5.3 examines the various types of leveraged acquisitions and analyses the various ways in which it is technically possible to structure this kind of operations. Section 5.4 deals with the financial aspects of a leveraged acquisition, underlining: (a) the valuation methods that are considered the most coherent with the logical assumptions behind similar transactions; (b) the elements that need to be considered during the forecasting of a sustainable capital structure; (c) the various types of capital that can be raised from subjects interested in undertaking these operations. Section 5.5 presents an analysis of the European market for leveraged buy-outs, showing the main figures of the realized transactions with a particular focus on the most recent years. The final section examines the main characteristics of mezzanine finance, given the growing importance of this peculiar kind of financing in the context of leveraged acquisitions.

5.2 Leveraged Acquisitions: Major Aims and Characteristics

Leveraged acquisitions refer to a class of extraordinary financial operations within the larger "family" of mergers and acquisitions (M&A) transactions. The given expression is used to define those operations that produce deep and permanent changes in the ownership structure of one or more enterprises.[1] Among the most common types of transactions that affect the ownership base of the company, it is possible to highlight: mergers, acquisitions, spin-offs, split-offs, equity carve-outs, restructurings of businesses in crisis and listings in officially regulated asset markets (IPO).

Leveraged acquisitions—hereinafter also indicated by the expression leveraged buy-outs, dealing with the most widespread category of such transactions—profoundly affect the ownership base of the involved companies, but, unlike other kinds of M&A activities, have certain unique characteristics. In the first place, these operations require the constitution of a "vehicle company" (also called "special purpose vehicle", "new company" or, simply, "*newco*") for the transfer of the ownership; from a technical point of view, it is this entity that, after being capitalized by the subjects interested in making the acquisition (called the "proponent subjects" or "sponsors") and the industrial and financial partners involved with the proponents, will start the process for the company to be acquired (called the "target firm"), or through a public offer for purchase, in case when the target firm is listed or, otherwise, through private negotiations. Therefore, unlike the traditional M&A activities, in this case the acquisition happens "off balance sheet" for the proponent subject: the only accounting effect for the proponent is represented by the subscription to the capital of the *newco*. The capital raised through debt securities, the associated collaterals and covenants are, instead, pertinent to the *newco*.

Secondly, it needs to be pointed out that the equity raised by the *newco* represents only a minor part of the complex of resources that are necessary for the

[1]For a deeper understanding of M&A services, refer to: Capizzi (2007b), Conca (2011), Confalonieri (2016), Liaw (2011).

operation. The major part of the capital is, in fact, supplied through debt securities by a pool of banks and financial intermediaries. This implies that, being very "pushed" by the recourse to indebtedness (and, therefore, to the exploiting of financial leverage), these operations are characterized by a higher intrinsic structural risk than the one normally observed in the market for corporate ownership and control. It is also possible to define leveraged acquisitions as particular types of M&A activities that leave the acquired firm with a debt ratio (debt capital/risk capital) higher than before the acquisition.

Thirdly, if one reflects upon the circumstance that, de facto, the *newco* specifically created for the acquisition is an "empty box" containing, at least initially, exclusively the risk capital raised by the proponents and by other firms interested in the operation, one would realize that the debt capital needed to allow the exit of the target's previous ownership base is supplied by the banking system as a function of the capacity of the target firm to generate future cash flows. This implies, firstly, that the banks finance the acquirer on the basis of the residual debt capacity of the target firm and its consequent capacity to repay the debt and the servicing charges for the same.[2] In the second place, it is clear that the choice of the target firm has to be carefully considered by the proponents, seeing that they would be responsible for the repayment of the financial obligations of the acquisition. In other words, not all the firms that could be objects of an M&A operation can be considered good candidates for a leveraged acquisition. Without trying to generalize, and above all with reference to the operations that are decidedly personalized, it is still evident that there are certain minimal prerequisites that target firm needs to possess, if an operation of this kind is to be undertaken.[3] In particular:

(1) it is important that the target firm is solid: this means, among other things, (i) that the firm should not have a liability structure that is laden with progressive debt, (ii) that the firm should have a low leverage ratio before the transaction;
(2) the target should be capable of producing, through its own operation, cash flows that are suitable to regularly repay the debt and the associated debt-servicing charges (rule of thumb: EBITDA/Sales > 15x). This implies, as can be guessed, the capacity of the firm to make use of a solid and defensible competitive position, by virtue of the efficacy of its own competitive strategy or the profile of attractiveness of the reference market;
(3) it is important to verify the investment needs of the firm and if the assets of the target (single plants and equipment or corporate divisions not considered strategic) are eventually suitable to be transferred without compromising the

[2]In the section dedicated to the modality of realizing an LBO, it will be seen that, from a technical point of view, the initial financing supplied to the *newco* is obviously conditioned upon the subsequent realization of the acquisition on behalf of the latter. At the end of the operation, it is normally expected that there is an immediate repayment of the financing, this time attributable to the operating company resulting from the merge of the *newco* and the target firm.

[3]*Cfr* Forestieri (2007), Rosenbaum and Pearl (2013).

firm's activity: these assets, moreover, could also be used as financial collaterals in an agreement with the banks;

(4) the target should guarantee the presence of a highly motivated managerial team, possessing professional and operational competences; the management quality, according to a widespread opinion among institutional investors in risk capital, is the most important factor contributing to the success of the entire operation.

It is evident that all three of the specifics just discussed in the frames of the leveraged acquisitions, namely—the role of the vehicle company, the high level of leverage and the centrality of prospective cash flow in order to evaluate the feasibility of the operation—represent the defining elements of operations in project financing, such as the accomplishment of the construction and management of a public work, the securitization of a credit portfolio, even if, obviously, these cases are very different from that of the acquisition of a firm by using a corporate vehicle (*newco*). For these reasons, in accordance with the operative approach characterizing the most important local and global investment banks, leveraged acquisitions are universally placed within the area of structured finance operations.

Before taking a closer look at various types and techniques of implementation of leverage acquisitions, it is considered appropriate to discuss the typical aims and advantages of this kind of operation at a general level and, also, with reference to those contexts (like the Italian market) characterized by a strongly fragmented entrepreneurial structure, in which there is a preponderance of small and medium, family-owned businesses.

In this regard, it is possible to summarize the typical aims of leveraged acquisitions as follows:

(a) to limit the cash commitment of the proponents to the equity capital subscribed in the *newco*, with positive effects in terms of future financing choices available to the proponent themselves;
(b) to accommodate the acquisition of a firm by a team of managers who do not have sufficient financial resources to conduct the transaction autonomously[4];
(c) to provide an effective way-out option to private equity investors and merchant banks;
(d) to resolve existing conflicts among the shareholders of the target firm, most of all among those sharing different opinions about future corporate strategies to be implemented, avoiding at the same time spin-offs and split-offs;
(e) to rationalize the assets of small and medium, family-owned businesses, with a fragmented shareholder base, encouraging a "recompaction" around a subgroup of the proprietary family[5];

[4]In this case, as can be seen in the next paragraph, the type of leveraged acquisition is called "management buy-out" (MBO).

[5]In this case the type of leveraged acquisition is called "family buy-out" (FBO).

(f) to manage problems due to an intergenerational change, giving the possibility to the target company to survive as an independent entity beyond its founders' departure;
(g) to enhance the image and the reputation—and thus, also the creditworthiness—of the target firm through the involvement in the ownership structures of various types of financial intermediaries and institutional investors (banks, merchant banks, financial companies, closed end mutual funds, venture capitalists etc.).

It is clear that, with a specific reference to peculiar contexts (like the Italian market), the most important motivations are the last four, given the specificity of the industrial system, characterized by the centrality of small and medium sized family owned enterprises. These firms, over the years, can experience a widening and diversification of the shareholder base, coherent with the "biological" evolution of the family ownership, so as to generate, at a certain point (i) conflicts between the members and divergence of views regarding the prospects of future development of the firm; (ii) unclear family succession dynamics; (iii) limited contractual power in comparison with some key players (clients, suppliers, financial intermediaries). These problems, together with a somewhat lower quality of management decision making, due to a lack of separation of ownership and control, make the leveraged acquisition an instrument that is, in some cases, crucial to the survival of a firm with similar characteristics.

5.3 Leveraged Acquisitions: Types and Techniques of Realization

As has already been anticipated, there are a number of operations that are technically qualifiable as leveraged acquisitions, differing mainly in relation to the identity of the proponent subjects, or rather in the nature of interests for which these have been planned and brought into being. With regard to the literature, and the professional praxis, the following terminology and acronyms are in use:

- *leveraged buy-out* (LBO): deals with the most generic and typical category of such transactions, in which, on the one hand, there is a significant presence of financial intermediaries and institutional investors in the shareholding of the *newco* and, on the other hand, there is no significant presence of the subjects characterized in the variants of the LBO described as follows;
- *management buy-out* (MBO): the proponents are a group of managers of the target firm, who at the end of the operation, change roles, from managers to manager—entrepreneurs;
- *management buy-in* (MBI): in this case, the managers who are the proponents of the initiative are external to the target firm, often of dimensions more compatible with the limited financing possibilities of the sponsors;

- *buy-in management buy-out* (BIMBO): in this case, the nucleus of the proponent subjects is composed of some managers operating within the target firm, who are in a position to involve in the initiative other managerial resources from the outside;
- *family buy-out* (FBO): this happens when shareholders inside the owning family take over the share of other members who are not interested in the maintenance of their investment in the target. If a part of the owning family decides to "exit" from the firm in order to try to buy another corporate entity using a leveraged buy-out, it is called a *family buy-in* (FBI);
- *workers buy-out* (WBO): this deals with a type widespread in the United States, which envisions that the acquisition is realized by employees of the target, through the intervention of pension funds and the use of employee stock ownership plans[6];
- *corporate leveraged buy-out* (CLBO): in this case, the acquirer is a company wanting to grow through external means without having to enter the acquired company directly into its corporate family. It is clear, on the other hand, that the transaction would be "visible" and connectible with the acquirer at the time when the latter would be considered a part of the group (and, naturally, concentrated upon the analysis at the level of the consolidated financial statements);
- *fiscal buy-out* (FIBO): deals with an operation that comes into being with the precise aim of creating a "fiscal shield" through the reduction in the incidence of direct taxes on the income of the acquirer. In contrast to the other types of LBO operations, in this case a *newco* is almost never created, given that the principal motivation behind the FIBO is the creation of tax savings for the acquirer;
- *owner buy-out* (*OBO*): deals with an operation undertaken by the largest shareholder in the target firm, in order to gain full control over the company;
- *investors buy-out*: in this case, the sponsors of the operation are private equity investors shareholders of the target firm who want to increase the target value in order to benefit from a future exit. If the investors are external to the target firm, this kind of deal is called *investors buy-in*;
- *secondary buy-out*: in this case, the sellers are sponsors of previous buy-out transactions, most of all other private equity funds.[7]

It is appropriate to specify how, in practice, it is difficult to find any of the above-mentioned buy-outs in its "pure" form; more often, a "combination" of these operations takes place. For example, as to the Italian context, it is quite common to find leveraged acquisitions that have elements of an MBO on the one hand, and of an FBO on the other hand. The incentive for the operation could, in fact, be a specific desire to rationalize the ownership base held by a part of the owning family of the firm; moreover, the financial intermediaries could condition their own participation in the initiative requiring the fulfillment of a series of prerequisites, first of

[6]On this point see Bennett Stewart (1999).

[7]For further detail about secondary buyout and their value creation potential, see Capizzi et al. (2009), Bonini (2015), Degeorge et al. (2016).

all the appointment of management (internal or external) of an unquestioned quality and expertise and its involvement in the risks of the initiative through a subscription of a shareholding interest in the *newco*, in order to decrease the probability of opportunistic conduct.

In order to analyze the different techniques of realization of an LBO, not considering anymore the nature of the proponent subject, the most appropriate approach is to discuss in synthesis the different phases that, in chronological order, compose the transaction.

Firstly, the process starts with the proponent subjects identifying and selecting the target company. It is clear that, in case of the MBO, FBO and WBO, this step is practically automatic, considering that, for various reasons, the target firm is the one that the proponent subjects have a high degree of involvement in.

Secondly comes the identification of the financial intermediary—or the team (internal or external) of professionals—that will assist the buyer as advisors. It is the advisor that plans and implements the technical and contractual structure of the transaction, identifying the solution most suitable to the needs and expectations of the proponent subjects.

Thirdly, once the structure of the LBO has been defined, the advisor and the sponsor write a business plan, namely a document that summarizes the considerations about the sustainability of the industrial model related to the new company that will result from the transaction and, also, the financial feasibility of the operation. Naturally, if the proponent subjects are already in possession of a solid and rational business plan, the workload of the advisor—and the corresponding remuneration—would be lower.

Fourthly, we have the identification of a pool of investors—of an industrial or financial nature—to involve with the proponent subjects in the share capital of the *newco*. In this phase, the activities of the advisor assume particular relevance, considering that the latter is usually a subject characterized by a large number of contacts and relations and capable of involving the investors essential to the success of the operation. The advisor's role, as can be guessed, becomes more and more important as a result of possible complications, in terms of capital available for the proponent subjects.

Fifthly comes the identification and capitalization of the *newco* that, as anticipated, represents the vehicle used for the realization of the acquisition; this could involve an already existing company within the group to which the proponent subjects belong to, properly "emptied" of its operating assets, or, more often, a company specifically constituted for the operation.

Next, the sponsor and the advisors proceed with the negotiation of the credit lines needed as a complement for the capital that, united with the equity capital of the *newco*, will allow the due payments for the realization of the transaction. As the reader will see in the following paragraphs, there are many techniques available for the *newco*'s financing, which imply different levels of involvement and different risk-return profiles. Usually, the lenders request that the shares of the target firm, or the real assets of the same company, are considered as collateral. It is not uncommon, moreover, that in this phase the pool of banks grants a bridge loan (short term and unsecured), subordinated, obviously, to the realization of the acquisition

At this point, the operation will proceed on the basis of a multitude of modalities, referring to two major types, which will be discussed hereinafter.

I. Merger Method (or "KKR")

One of the main operative technique of a leverage buy-out is known as "merger" or "KKR" method, from the name of the American bank (Kohlberg, Kravis and Roberts) that first implemented this type of LBO. In this case, the *newco* buys (through a private or public offer) the whole equity base (or the majority equity stake) of the target firm (Fig. 5.1).

More in detail, after the constitution of the *newco*, it is possible to identify 4 additional steps:

(1) the pool of banks affords the debt capital (usually through short-term loans). The (syndicated) loan can be secured or unsecured. In the latter case, the target company's stocks or real assets will be used as collaterals. The loan can temporarily exceed the amount needed, thus giving flexibility to the sponsors and financial investors in the equity capital raising step (bridge financing);
(2) the *newco* buys (through a private or public offer) the whole equity base (or the majority equity stake) of the target company;
(3) the *newco* merges the target company inside itself; more seldom, the *newco* is merged inside the Target company (reverse merger);

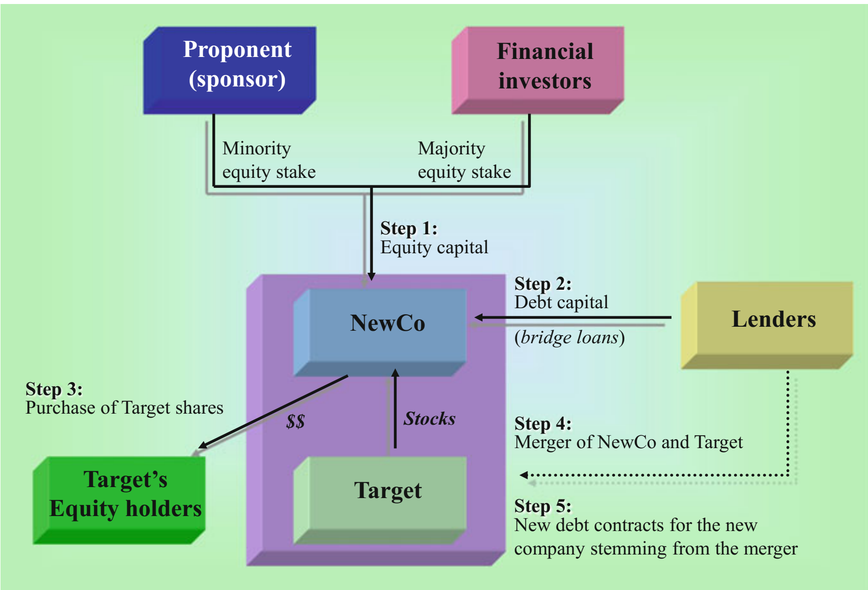

Fig. 5.1 LBO: "Merger" Method

(4) as a final consequence, the *newco*'s indebtedness is transferred downward to the acquired company (the only operating one), which will be the unique source of the financial flows required to remunerate banks and other investors. The short-term debt contracts originally obtained by the *newco* are then substituted with new (long-term and secured) contracts entitled to the new entity stemming from the merger. This new financing, further assisted by collaterals and more ample than the preceding one, is less burdensome on the company, to the advantage of the free cash flows available to the shareholders. It is clear then, that, if there were a need for a further reduction of the total cost of the debt, an opportunity of readdressing the capital markets could be considered by means of issuing debt securities in order to repay the bank debt mentioned earlier.

IIa. Asset sale Method (or "Oppenheimer")—Pure version

Alternatively, the sponsors could use a different approach, called "asset sale" or "Oppenheimer" method, named after the bank that was the first to use this technique. In this case, the *newco* buys only a portfolio of selected assets (or SBUs) from the target company (see Fig. 5.2).

Here, it is possible to identify 3 additional steps following the constitution of the *newco*:

(1) the pool of banks affords the debt capital (usually through short-term loans). The (syndicated) loan can be secured or unsecured. In the latter case, the target company's stocks or real assets will be used as collaterals. The loan can

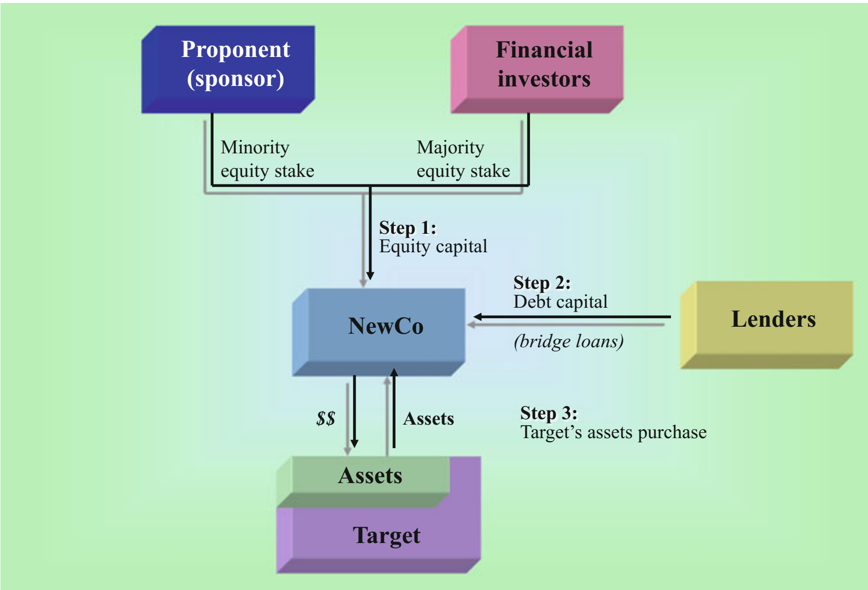

Fig. 5.2 LBO: "Asset sale" Method—Pure version

temporarily exceed the amount needed, thus giving flexibility to the sponsors and financial investors in the equity capital raising step (bridge financing);

(2) the *newco* buys only a portfolio of selected assets (or SBUs) from the target company;

(3) the short-term debt contracts originally obtained by the *newco* are then substituted with new (long-term and secured) contracts entitled now to an operating company (the *newco* itself post-LBO).

IIb. Asset sale Method (or "Oppenheimer")—Modified version

It is possible to identify a modified version of the Oppenheimer method, characterized by the constitution of two *newcos*, in order to avoid legal problems (see Fig. 5.3). Specifically, this method could be implemented in order to avoid the objections raised by those who could see the asset sale technique as an expedient for "emptying" the target firm of its operating assets, undermining the interests of its creditors, as well as of its minority shareholders.

In this particular case, after the constitution of the first *newco*, it is possible to identify 5 additional steps:

(1) The pool of banks affords the debt capital to *newco* (1). The (syndicated) loan can be secured or unsecured. In the latter case, target company's stocks or real assets will be used as collaterals;

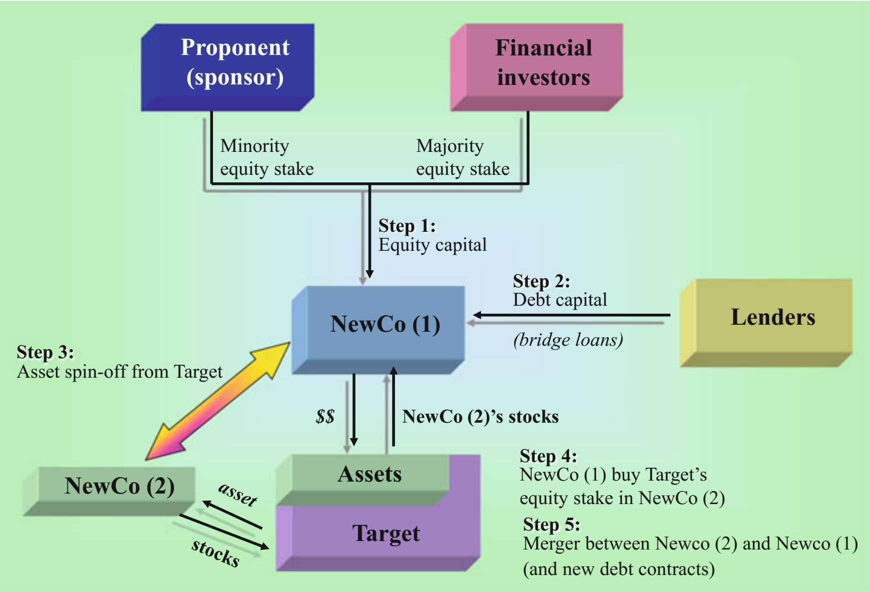

Fig. 5.3 LBO: "Asset sale" Method—Modified version

(2) target company spin-offs its targeted assets (those chosen by the buyers), creating a new company—*newco* (2)—whose stocks will be placed as collaterals, thus improving the credit quality of *newco* (1);
(3) the *newco* (1) buys from the target company its equity stake in the *newco* (2);
(4) the *newco* (2) is merged into the *newco* (1);
(5) as a final consequence, *newco* (1)'s indebtedness is transferred downward to the new (operating) company stemming from the merger, which will be the unique source of the financial flows required to remunerate banks and other investors. The short-term debt contracts originally obtained by the *newco* (1) are then substituted with new (long term and secured) contracts entitled to the new entity.

Obviously, it should be noted that the techniques described above are not necessarily to be followed unquestionably in their original version. Moreover, it is not necessary neither to acquire the total share capital of the target firm, nor to proceed with the merger of the *newco* and the target.

5.4 Leveraged Acquisitions: Financial Issues

As anticipated, the activities of planning and structuring performed by the advisors are critical and require the ability to hypothesize—in a rigorous, rational and realistic manner—a mode of development of the target firm characterized by the creation of a significant value for the new shareholders. The first element that needs to be carefully considered is how to ensure the survival of the target post-LBO, characterized by an increased debt burden and by a financial leverage well above the physiological level—estimated as the weighted average within a sample of firms.

In this regard, it is clear that the capacity for the debt repayment, not to mention the debt remuneration through the payment of financial expenses, is based as on the future unlevered cash flows generated by the target company after management (and strategic plans) change as on the eventual sale of non strategic assets and/or business units considered not essential in order to develop the core business. Thus, there emerges the importance of the business plan, which is the document, prepared by the advisor in consideration of the information and indications provided by the proponent subjects, which has to address clearly the just mentioned aspects, in order to make clear the advantages, in monetary terms, for each of the subjects participating in the operation: the proponents, the financial partners of the *newco*, the credit intermediaries and the providers of debt capital. More in particular, a business plan created for an LBO, must focus on the following areas:

(a) analysis based on the historical trends of the target firm;
(b) analysis of the ownership base and of the internal organization of the target firm;

(c) analysis of the strategic business units;
(d) analysis of the competitive environment;
(e) arrangement of the logical scheme of the transaction (deal structure);
(f) identification of the potential disinvestment areas post-merge;
(g) analysis of the target's potential financial dynamics and estimation of the future unlevered cash flows;
(h) development of a sustainable financial structure;
(i) evaluation of the target company's equity value, in order to show to potential equity owners the IRR and the monetary capital gains expected from the equity investment;
(j) preparation of the final offer memorandum, which is the main reference for the negotiation and closing of the transaction.

In an evaluative context, once understood the logical sequence of a transaction, it becomes clear that the methodologies of analysis and valuation that seem the most consistent with leveraged acquisitions' underlying philosophy are of a financial type, centered upon the present value of the future cash flows, and, first of all, the discounted cash flow analysis (see Fig. 5.4). These methodologies, where the value of the firm (entity value) and economic capital (equity value) depend upon the series of future cash flows (unlevered and levered respectively) generated by the valuated firm, are the most suitable in order to capture the value that can be generated by the companies that, though being heavily indebted, have a favorable market position, an effective business strategy, a solid reputation and, therefore, a

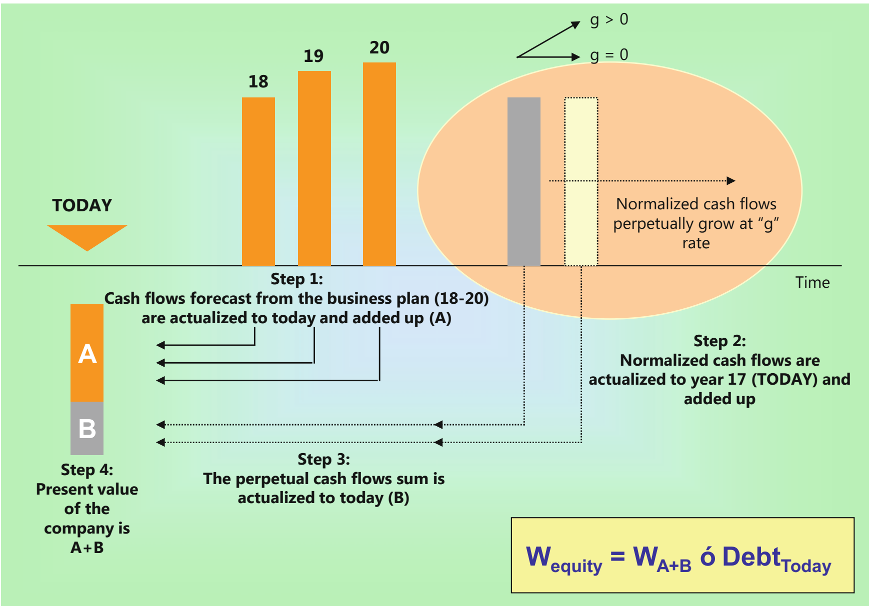

Fig. 5.4 The 'functional logic' of discounted cash flow analysis

high probability of generating positive and increasing cash flows over time. Instead, the value of the target resulting from the application of earnings based, asset based and mixed methodologies could be penalized because of the following points:

(a) they favor non-monetary variables;
(b) they are based on historical or actual data, which result difficult to project;
(c) they treat in an asymmetric manner the payment to the creditors and the one to be offered to the shareholders;
(d) they do not consider the financial value of time.

It is not by accident that leveraged buy-outs originated in contexts—primarily Anglo-Saxon—where it is clear that the value of a business is determined on the basis of its expected future cash flows, discounted by a rate expressing the weighted average cost of capital. If this were not the case, it would be difficult to start a high debt initiative, for a brief period characterized by an absence of monetary remuneration for the shareholders.[8]

Referring back to the vast literature in corporate finance related to a systematic discussion of discounted cash flow analysis,[9] it seems useful, at this point, to make some considerations regarding the construction of the financial structure, which undoubtedly represents one of the most peculiar—and critical—aspects in the realization of an LBO transaction.

In particular, the advisors have to carefully consider the following points:

(a) the sponsors' equity commitment is often a small percentage of the total capital needed for the transaction;
(b) even considering the equity investment afforded by merchant banks, venture capitalists, closed funds and banks, the transaction requires high amount of debt. The latter is, as seen in Sect. 5.2, one of the aspects characterizing leveraged acquisitions when compared to M&A operations in general;
(c) given the nature of the operation, the riskiness of these transactions, the obligations imposed to the *newco* and the "pressure" generated by the debt capital on the unlevered cash flows, it is not convenient to use only "traditional" straight debt (so called "senior debt");
(d) because of the above mentioned reasons, debt capital for the operation should be lent by a pool of banks (credit risk diversification);
(e) in any case, differently from other lending operations, the total debt capital won't be wholly secured through guarantees and collaterals;
(f) moreover, it should be considered that usually there are precise legal limits to the maximum amount of both debt capital obtainable through corporate bonds issues and equity capital obtainable through the issuance of limited voting rights stocks and preferred stocks.

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[8]For an analysis of the methods of business valuation, see: Damodaran (2001, 2009, 2012, 2015), Koller et al. (2015), Guatri and Bini (2009).

[9]*Cfr* Massari and Zanetti (2008), Koller et al. (2015), Damodaran (2001) and Capizzi (2003).

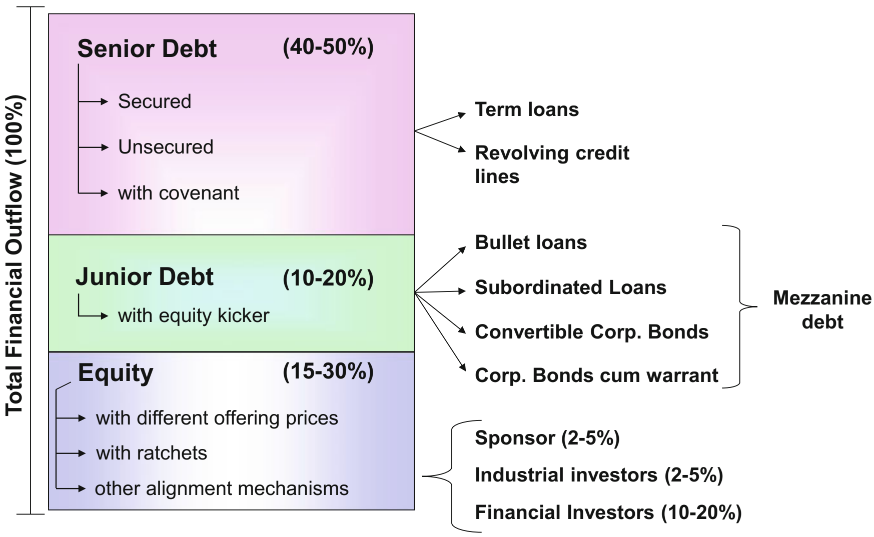

Fig. 5.5 The 'typical' financial structure of a management leveraged buy-out

Therefore, the ability of the advisor to structure a financial plan coherent with the elements and the obstacles just described is considered essential. At the same time, the financial plan must allow the precise formulation of a diversified range of techniques of financing capable of guaranteeing the entire coverage of the required capital.

Figure 5.5, without claiming to generalize and based on the empirical observations with reference to the Italian market for leveraged buy-outs, illustrates the typical financial structure of an LBO operation and, more in particular, a management leveraged buy-out.

First of all, for the reasons stated above, it is necessary to involve a pool of financial investors affording the debt capital needed for the operation. The technical form with which the ordinary finance is supplied will therefore be a syndicated loan, typically a term or standby loan.[10]

Secondly, the advisor must be able to make a balanced distribution of debt capital between senior debt (ordinary bank loans, which has priority among other debts and should be repaid first to creditors in case of insolvency and bankruptcy) and junior debt, in other terms, between technical forms of ordinary debt and "mezzanine debt". This last expression qualifies the so-called "hybrid instruments of fundraising", which include forms of financing (for example subordinated debt, convertible bonds, bonds cum warrant) that share some of the characteristics of debt capital and equity capital.[11] The role of subordinated debt is crucial, incorporating

[10]See Capizzi (2003) for types of financing instruments and risk/cost profiles.

[11]*Ibidem.*

specific options with a well defined market value (warrants, conversion rights, so-called equity kickers meaning options that are incorporated within a bond which allow the bondholders to subscribe to shares of the issuer at a price established ex ante), which is less 'costly' than senior debt with reference to its pure credit component. In other words, the financial charges corresponding to junior debt are determined on the basis of lower than market rates upon which the financial charges corresponding to ordinary debt are parametrized. The benefits produced by the use of these instruments to the unlevered cash flows of the period can be intuited.

Thirdly, Fig. 5.4 clearly shows how the proponent subjects can undertake the operation with relatively little financial expenditure, if considered as a percentage of the total capital necessary for this purpose. Therefore, immediately after the merge their share of ownership will be similarly limited in percentage terms. In order to guarantee to the proponents an adequate level of shareholding in the *newco*, not an unusual thing to do is to fix the differentiated prices of issue attributed to the different categories of participants.[12] Moreover, in order to decrease the probability of opportunistic behavior that could both negatively affect the majority financial stakeholders, and allow the proponents, after a certain period of time, to increase their own share of the equity capital of the firm, it is important to be able to stipulate certain contractual mechanisms which establish a relation between the performance of the company in a pre-defined time horizon and the share quota held by the management.

Fourthly, we should always keep in mind how a financial structure like the one illustrated in Fig. 5.4 could be "unstable" and a source of financial risk; this means that the level of leverage should be decreased to physiological values by the end of the time horizon assumed in the business plan (typically 4–7 years). The reaching of a similar result also allows the company to improve its own activity in the future, realizing new investments and obtaining the capital loans necessary for its growth at a reasonable cost.

In conclusion, it needs to be stressed that the role of the advisor becomes crucial once the financial structure sustainable for a given transaction has been projected. In particular, its capacity to implement what was initially discussed in the business plan: verifying the order of the phases of subscription and return of the capital on part of the shareholders of *newco* (often assisting them in the phase of retrieval of that very capital through new financial operations), obtaining the initial financing for the *newco*, managing the negotiations with the ownership of the target firm (and, eventually, the launch of an open offer, in case of a listed target), realizing the merge, renegotiating/reactivating new financing contracts for the company resulting from the merge, assisting continuously the new ownership and management in order to make them achieve what expected in the industrial plan. Being an advisor to a team of sponsors interested in undertaking a leveraged acquisition is, therefore, a complex activity which requires the possession of competences (professional and relational) of a high degree. This, also, explains the high incomes characterizing the financial intermediaries which are operative in this area of business.

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[12] See Ferrari (2007) for numerical exemplification.

5.5 The European Market for Leveraged Acquisitions[13]

5.5.1 Introduction to the Analysis

The main aim of this section is to offer an in-depth analysis of the evolution of the European market for leveraged buy-outs (LBOs) with a particular focus on the last three years.

Before starting with the analysis, the authors consider appropriate to provide information regarding the database and the sources of the official statistics taken as reference for this specific section. For this purpose, given the complexity of the operations, subject matter of the analysis in the present section, and a still incomplete convergence of opinions in the academic and professional context regarding the exact 'boundaries' of the buy-out transactions compared to a broader market for mergers and acquisitions, the authors have decided to rely upon the 'official' source of periodic statistics on the LBO European market, represented by the Centre for Management Buy-Out Research (CMBOR). Led by Professor Mike Wright, established in March 1986 at the Nottingham University Business School and moved to Imperial College London Business School in 2011, the Centre for Management Buy-Out Research monitors and examines companies buy-outs with a focus on a frequently misunderstood private equity buy-outs, transactions which include the purchase of companies characterized by a large debt. The Centre for Management Buy-Out Research was funded by Equistone Partners Europe Limited, a major European buy-out firm, and is supported by Investec, an international specialist banking and asset management group.

It should be noted that the data provided by the sources includes information about Continental Europe and the UK until the first half of 2016, which, therefore, represents the most recent data available nowadays.

5.5.2 Overview of the European Market

Figure 5.6 shows the evolution of number and value of leveraged buy-outs in Europe starting from 1982. As pictured, the number and the value of these transactions experienced a progressive growth until 2007. In particular, the increase in leveraged buy-out activity experienced in the 2005–2007 period was due to overly favorable terms for debt investors.[14] Starting from 2007, the market was characterized by a negative trend, mainly due to the credit market turmoil.

Considering the most recent period, it is possible to highlight how, in 2015, the aggregate value of the transactions increased despite a decrease in number of deals from 675 in 2014 to 644. The value increase during 2015 is explained by the

[13]Paragraph written by Paolo Parlamento.

[14]*Cfr* Kaplan and Stromberg (2009).

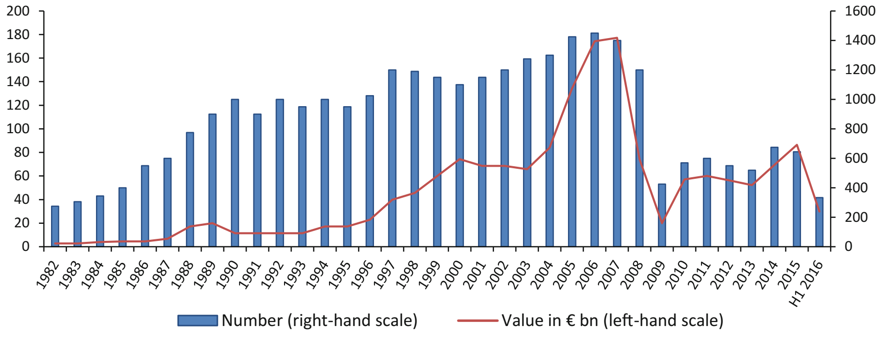

Fig. 5.6 Leveraged buy-outs in Europe since 1982 (number and value). *Source* CMBOR, Equistone Partners Europe, Investec

recorded size of buy-outs. More in particular, in 2014 there were only 13 buy-outs with an enterprise value exceeding € 1 billion, compared to 20 in 2015. The overall deal numbers were impacted by a decline in activity levels at the smaller end of the value range. The sharpest decline is notable in the € 10 million–€ 25 million segment, where the numbers dropped by more than a half (74 in 2015 compared to 155 in the previous year), and in the € 100 million–€ 250 million segment. Similarly to the smaller segment, deals with a value between € 100 million and € 250 million almost halved in number, going from 84 deals in 2014 to 52 in 2015. As far as the value in this particular segment is concerned, it also experienced a sharp fall, from € 13.2 billion in 2014 to € 8.2 billion in 2015.

Before proceeding with the analysis of the figures of the first half of 2016, it is essential to highlight the persistence of the macroeconomic factors that influenced the behavior of investors and buyers in the past months. In particular, we are talking about the slowing down of the Chinese market, the deteriorating credit market liquidity and the pressure on European markets originating from the Brexit and general instability of some of the main banking national systems. Despite these macroeconomic challenges and the declining market sentiment, Continental European buy-out market continued to perform well with the exception of the United Kingdom. In particular, in the UK the concerns over the possible consequences of the Brexit are most pronounced at the larger end of the market, with a steady lower-mid segment.

Following a strong 2015, the first half of 2016 recorded an aggregate value of € 29.9 billion. This value is slightly higher than those recorded at the half year point over the past five years, with the exception of the first half of 2015. Although, it should be noted how the figures of the first half of 2015 (348 deals for an aggregated value of € 40.5 billion) were boosted by 10 "mega-deals", 4 of which exceeding a value of € 2 billion. In the first half of 2016, the smaller mid-market transactions (between € 10 million and € 25 million) kept a strong position in both value and number (respectively € 833 million and 47). The same can be told about the transactions in the € 50 million–€ 1 billion segment. More in particular, in the

European market were recorded (i) 29 transactions for a total of € 2.1 billion in the € 50 million–€ 100 million segment, (ii) 27 transactions for a total of € 3.9 billion in the € 100 million–€ 250 million segment, (iii) 17 transactions for a total of € 5.6 billion in the € 250 million–€ 500 million segment, (iv) 14 transactions for a total of € 9.5 billion in the € 500 million–€ 1 billion segment. Referring to the deals characterized by a value lower than € 10 m, in the first half of 2016 174 deals with an aggregated value of € 701 million had been recorded. In the considered period, only the deals with a value between € 25 million and € 50 million and those exceeding € 1 billion experienced a decrease in the number of deals and in the overall value. Specifically, in the first half of 2016 there were just five buy-outs worth more than € 1 billion.

5.5.3 European Buy-Outs by Country

Table 5.1 shows the trends that characterized the numbers and values of European buy-outs by Country in the 2012–2015 period.

As it is shown, the main three European markets for LBO transactions are the United Kingdom, Germany and France respectively.

In 2015, the UK buy-outs, at € 27.1 billion, experienced a growth rate of 29% compared to 2014. Within the 203 transactions, seven buy-outs exceeded € 1

Table 5.1 European buy-outs by Country in the 2012–2015 period

	2012		2013		2014		2015	
	No.	€ m	No.	€ m	No.	€ m	No.	€ m
Austria	6	342	7	907	5	165	9	2699
Belgium	13	2600	14	2758	18	1016	29	3272
Denmark	21	1662	21	1508	21	2793	27	4877
Finland	13	395	15	948	24	776	15	1040
France	101	6333	99	8656	94	9840	93	11,158
Germany	73	8571	71	12,911	76	12,781	76	12,480
Ireland	5	249	7	69	6	2164	11	1698
Italy	41	1575	30	3944	36	2556	39	4517
Netherlands	34	1450	34	1923	50	4437	47	5296
Norway	19	1461	21	3098	26	6457	9	1002
Portugal	2	22	3	83	3	48	3	59
Spain	27	3553	22	2396	34	2984	45	2867
Sweden	30	3755	23	812	32	1298	28	3929
Switzerland	13	2928	9	1217	14	1320	10	4515
UK	210	20,952	195	18,262	236	21,014	203	27,133
Total	608	55,849	571	59,491	675	69,647	644	86,540

Source CMBOR, Equistone Partners Europe, Investec billion. These transactions include (i) home alarm systems company Securitas Direct Verisure, sold to Hellman & Friedman by Bain Capital for an estimated value of € 2.5 billion; (ii) gym chain Virgin Active sold to Brait (a South African private equity firm) by CVC Capital Partners for € 1.8 billion; (iii) environmental consultancy firm ERM, sold by Charterhouse to Canadian pension fund Omers for € 1.6 billion. Consequently, the UK buy-out market remains ahead of both Germany and France. In the past decade, 2011 was the only year when the French buy-out values exceeded those of the UK market, because of a handful of operations with a value well above the average.

In 2015, the German market value was boosted by four transactions, each with an estimated value exceeding € 1 billion. More in particular, these transactions include (i) Advent International's sale of Douglas Holding to CVC Capital Partners for € 2.8 billion; (ii) Siemens' sale of hearing aid manufacturer Sivantos/Siemens Audiology Solutions to EQT for approximately € 2.2 billion; (iii) BC Partners' sale of laboratory services company Synlab Services to Cinven for an estimated € 1.8 billion; (iv) the sale of the German subsidiary of India's Suzion Energy, the wind turbine manufacturer, to Centerbridge Partners for € 1.1 billion.

In the same year, the French market experienced an increase (+11.5%) in value despite a consistent number of transactions. The value of the French market was boosted by three LBO characterized by a value exceeding € 1 billion. These include (i) French multinational Saint-Gobain's sale to Apollo Management of Verallia, the seller's glass bottle unit, for € 2.95 billion; (ii) the sale, by Astorg Partners, of smart technology company Linxens to CVC Capital Partners for an estimated € 1.5 billion; (iii) Cinven's buy-out from 3i of diagnostics provider Labco for € 1.2 billion.

As anticipated, the first half of 2016 was characterized by a slightly declining market sentiment. This sense of unease was reflected on the German and the UK markets, while France experienced its best first half value in five years. More in particular, French buyouts accounted for 48 transactions with a combined value of approximately € 5.96 billion

In the first half of 2016, the combined value of UK buy-outs dropped dramatically to € 8 billion for 94 transactions. These figures show a slight decrease of the number of deals and a deep fall for the value. The outlook for the UK market highlights the expectations for a state of inertia as the economic realities of Brexit are starting to be realized. Consequently, the 2016 year-end figures are expected to be much lower than those of the past four years.

Considering the German buy-out market, in the first half of 2016, the value of transactions dropped dramatically for no discernible reason. Consequently, it seems like by the end of the year the figures recorded will be still lower than usual.

It is interesting to highlight how, in the first half of 2016, Italy accounted for two of the Europe's three largest deals, with the transactions of TeamSystem and Sisal boosting the Italian total deal value. Specifically, the largest deals recorded in the first half of 2016 were: (i) the sale of UK motoring company RAC by The Carlyle

Group to CVC Capital Partners for €1.9 billion; (ii) the acquisition of Swiss tour operator Kuoni by EQT Partners' for € 1.3 billion; (iii) the sale by HgCapital to Hellman & Friedman for € 1.2 billion of the Italian software applications provider TeamSystem; (iv) the € 1 billion buy-out of the Italian baby and healthcare products company Artsana in which Investindustrial bought a majority stake from the founding Catelli family; (v) the € 1 billion sale by Apax Partners to CVC Capital of gaming and payments Italian operator Sisal (Figs. 5.7 and 5.8).

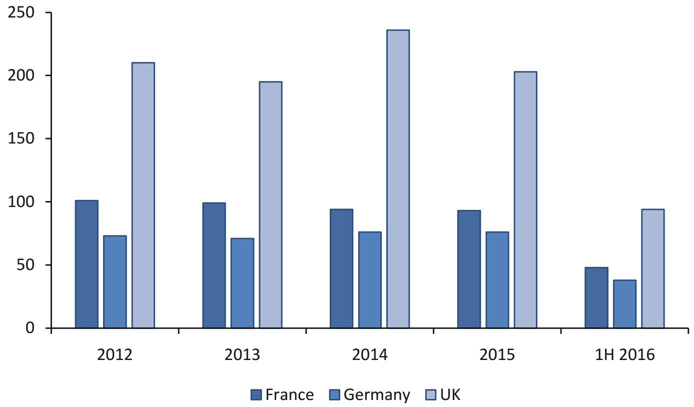

Fig. 5.7 UK, France and Germany number of transactions (2012–1H 2016). *Source* CMBOR, Equistone Partners Europe, Investec

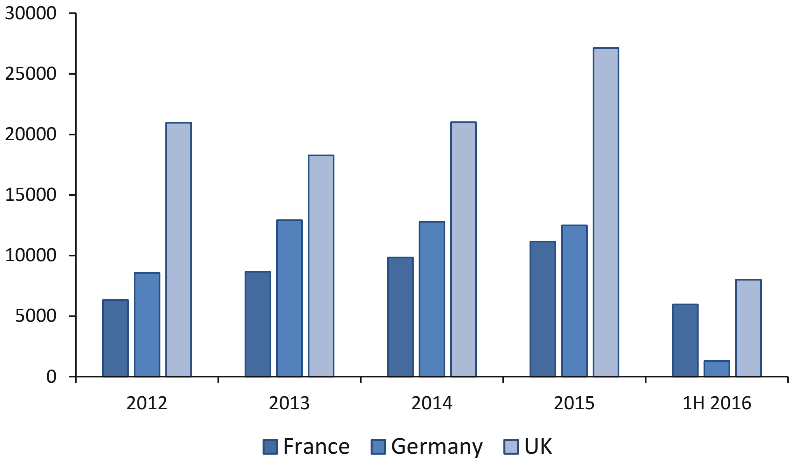

Fig. 5.8 UK, France and Germany buy-out values (2012–1H 2016). *Source* CMBOR, Equistone Partners Europe, Investec

Table 5.2 Financing structures on European buy-outs in the 2012–2015 period

	2012		2013		2014		2015		
	<€ 100 m	>€ 100 m	<€ 100 m	>€ 100 m	<€ 100 m	>€ 100 m	<€ 100 m	>€ 100 m	H1 2016
Equity	65.6%	59.1%	50.8%	41.4%	52.6%	42.1%	43.8%	39.6%	48.9%
Mezzanine	3.3%	2.1%	4.5%	4.0%	0.8%	0.9%	1.8%	1.3%	0.7%
Debt	28.6%	38.4%	41.9%	52.9%	45.4%	56.0%	52.1%	58.7%	50.5%
Loan Note	1.9%	0.2%	1.9%	0.7%	1.2%	1.1%	0.8%	0.3%	0.0%
Other Finance	0.6%	0.1%	0.9%	1.0%	0.0%	0.0%	1.5%	0.0%	0.0%
Total financing (€ m)	31,484	29,734	35,964	34,552	32,041	31,309	45,832	45,311	17,558

Source CMBOR, Equistone Partners Europe, Investec

5.5.4 *Financing Structures on European Buy-Outs*

Table 5.2 shows the financial structures on European buy-outs in the 2012–2015 period, considering separately the segments below and above € 100 million, and the values recorded in the first half of 2016 relating to European deals with a value below € 100 million.

In all the considered years it is possible to see how the transactions characterized by a higher value record a lower component of equity and an increased relevance of debt.

In the considered period, the provided data highlights a steady increase of the debt component. This trend started in 2010, when the market experienced an average value of the debt component of approximately 28.7%.

The ratio of debt to equity experienced a sharp increase in 2013 compared to the previous year. This is because, during the period 2013–2015, the European leveraged finance markets have seen improving liquidity, mainly due to the UK and European quantitative easing programmes and to the improving economic growth rates. Also, the progressive globalization of the European leveraged finance market has determined the development of innovative characteristics, such as the introduction of standardized loan documentation, which was designed in order to increase the level of comfort to North American investors willing to invest in the European market.

The increased liquidity prompted buy-out investors to refinancing in the past years. As demand for refinancing slowed down in 2015, new kinds of buy-out financing spread in the market, determining the highest levels of debt ever seen in buy-out transactions throughout the past decade.

It should be noted how the mezzanine debt component, despite an unsteady trend, represents a constant presence in the European LBO market.

5.5.5 European Buy-Out Exit Values

Figure 5.9 shows the European buy-out exit values and numbers for the 2012–1H 2016 period.

As it is possible to see in Fig. 5.9, buy-out exits reached the highest value in 2015, when 467 deals realized an aggregated value of € 156.6 billion. In this year, there were 42 exits with the value exceeding € 1 billion. 17 of these were trade sales, while 10 were secondary buy-outs and 15 were IPOs. In particular, the IPOs had a high impact on the final value of the exits in 2015 with a combined value of approximately € 48.7 billion for 40 exits. This is the largest combined IPO value ever recorded and breaks the previous highest value, recorded in 2014 when 44 IPOs had a combined value of € 44 billion. These figures show how the size of private equity-backed IPO stock markets are willing to support has increased in the past years. It should be noted that the Centre for Management Buy-Out IPO exits value reflect the full enterprise valuation at IPO, meaning that the provided exit data can be larger than that of other sources. However, this approach reflects the relinquishing of control at the point of IPO and allows CMBOR's annual data to reflect each exit as a singular event. Even allowing for a possible IPO related uplift in the valuation number, the fact remains that average exit values have significantly increased in the past years.

Trade sales exits were consistent in 2015 and by the end of the year presented a combined value of € 64.3 billion, the highest ever recorded.

Secondary buy-out exits value reached € 43.5 billion in 2015, not the highest value considering that in 2006 it was equal approximately to € 52.9 billion and in 2007 it was € 64.6 billion.

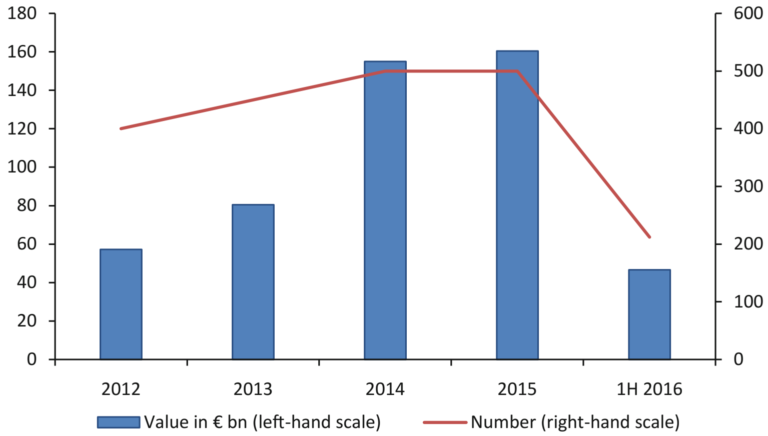

Fig. 5.9 European buy-out values in the 2012–1H 2016 period. *Source* CMBOR, Equistone Partners Europe, Investec

The three main European markets for LBOs, France, Germany and the UK, all saw sharp increase in exit activity during 2015. In particular, UK exits amounted to 199 and reached their highest value (€ 65 billion). This value was 44.5% up in 2014 figure. This trend is mainly attributable to five IPOs that, with a combined value of € 13 billion, represent the majority of the € 19.9 billion uplift on the previous year's total exit value. In addition to those IPOs, there were 10 trade sales and four secondary buy-outs with values exceeding € 1 billion each. Germany, with 47 exits valued at € 22.6 billion, surpassed its previous highest level, recorded in 2007 and equal to € 21.7 billion. France recorded 56 exits worth a total of € 21.2 billion, experiencing its best year since the € 30.1 billion achieved in 2007.

5.6 Mezzanine Finance

Hybrid financial instruments (which can be called "mezzanine debt", "mezzanine finance" or "hybrid debt"), in their extreme variety, are characterized by unique risk-return profiles, different from both ordinary debt and equity.

The spread of mezzanine finance instruments has made it possible to observe concrete examples of financial innovation, meant as the ability to design new financial instruments, characterized by innovative risk return combinations and profiles which can increase the efficiency of the financial system in the allocative sense and its ability to intercept capital from new segments of potential investors, not covered by traditional financial contracts.

These kind of financial instruments are called "hybrid" because they include characteristics of both debt and equity, in particular:

- considering the "Debt" component, mezzanine debt consists in a contribution of financial resources which does not lose the nature of "property with the right of return" in favor of the lender and that is repaid with a fixed or floating interest rate. It should be noted that the debt component of mezzanine finance is subordinated to other forms of debt financing.
- regarding the "Equity" component (Equity kicker), mezzanine debt has embedded equity instruments (usually warrants or call/put options) attached, which increase the value of the subordinated debt for the creditors, who will benefit from any increase in the market value of the firm.

It should be noted that the two components of return (interest rates for the subordinated debt component and the equity kicker) are interdependent values: more interests equal lower equity kicker performance and vice versa.

The specific characteristics of hybrid financial instruments are not clearly delineated from a regulatory point of view: consequently, private autonomy has a crucial role in the modulation and customization of the different features of stocks and bonds, resulting in a range of a wide variety of products.

Consequently, there is a great flexibility that prevents placing this particular kind of financing in a homogeneous category, clearly discernible and that is making the dividing line between debt and equity more and more undistinguishable.

The main advantages of using mezzanine finance can be summarized as shown below:

- hybrid instruments can be used in order to remedy financial shortfalls and to provide capital backing for the implementation of corporate projects.
- mezzanine finance implies a general strengthening of the economic equity capital without the need to dilute equity holdings or surrender ownership rights.
- the use of hybrid instruments improves the balance sheet structure and, consequently, the firm's creditworthiness. This can have a positive effect on the company's rating and can improve the firm's financeability.
- as it was anticipated, this kind of instruments are characterized by high flexibility and level of customization.
- the use of mezzanine finance brings greater entrepreneurial freedom for the company and limited consultation right for the investor compared to other kinds of financing.

Obviously, this kind of financial instruments has some disadvantages that are summarized below:

- hybrid instruments are more expensive than conventional loan financing;
- in case of mezzanine financing, the capital is provided just for a limited term, in contrast to pure equity capital;
- hybrid instruments often have more stringent transparency requirements compared to other kinds of financing.

5.6.1 Main Categories of Mezzanine Finance

In practice, most mezzanine financing takes the form of a preferred stock or a subordinated debt, a medium-long term debt, normally ranked after other debts (senior debts) if a company falls into liquidation or bankruptcy. Between these two main categories, it is possible to find a wide variety of products, very different from each other. The observation of the most frequent mezzanine products makes it possible to create a more systematic categorization based on their different characteristics.

Consequently, the basic product group consists of the following five categories:

(1) Subordinated debt with step-up rates or PIK (payment in kind) interest;
(2) Subordinated debt with warrants;
(3) Subordinated debt with a profit participation scheme (PPS) attached;
(4) Convertible loans;
(5) Preferred shares.

As already anticipated before, it should be noted that, following on from these basic categories, a variety of more "exotic" models has been generated by practice through a gradual customization of these products to the specific needs of businesses in the course of their progressive emergence.[15]

The distinctive features within each product group are summarized further.

Subordinated debt with step-up rates

Subordinated debt with step-up rates is a kind of mezzanine financing which is mainly used in situations where the majority of existing cash flows is consumed by senior debt and plain vanilla subordinated debt. Consequently, in this situation, the company has little room for an additional fixed compensation debt product, especially not on a subordinated level, which is usually characterized by higher interest rates compared to senior debt.

Subordinated debt with step-up rates let the company add an additional layer of subordinated debt, adjustable in consideration of the firm's future ability to service an increasing level of interest charged throughout the product's lifetime. Basically, we assume an increase of the interest payments towards the end of the loan, thereby assuming that, by then, the company will have an increase in both revenues and margins, and have paid off the senior debt (or other subordinated debt the firm has on the financial statements) which will have matured before the considered loan.

Of course, this process implies the need of an estimation of the expected incremental cash flow that the firm will be able to produce over the next years and there is no guarantee that the company will indeed have created additional and sufficient cash flows in order to satisfy the demands of the new lenders over the future years.

An alternative way to proceed is to make the scheduled increase "criteria linked", dependent on the performance of the firm, especially its actual ability to reach the assumed increase in cash flows.

It should be noted that, in practice, some sort of "hybrid" model is often experienced. In this case, a certain level of interest is fixed and has to be paid whatever is the actual performance of the company, combined with additional interest payments which depend on certain quantitative criteria, as agreed between the firm and the lenders.

Subordinated debt with PIK interests

A payment in kind (or PIK) loan is a subordinated debt instrument, which does not yield any interest or principal repayment for the lender between the moment of drawdown and the moment of maturity (or, eventually, the refinance date). Basically, it is an instrument where both principal and interest payments become due in one bullet payment at maturity date.

Given the characteristics of this kind of mezzanine product, the PIK lender takes on a considerable amount of risk and, consequently, will be looking for a certain

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[15] See more extensively: Nijs (2013).

minimum return which can be brought back to three main sources of income: the arrangement and/or commitment fee, the compounded interest and, sometimes, the ticking fee, which is a fee paid by the borrower to a lender in order to compensate for the time lag between the commitment allocation on a loan and the actual funding.

Very often, this kind of financing is accompanied by covenants and warrants, i.e., instruments which include the right to purchase a certain amount of stock, shares or bonds at a given price for a certain period of time. In this case, it then falls within the category of actual equity exposure.

Subordinated debt with warrants

A warrant can be defined as a particular kind of security, which entitles the holder to buy (in most cases) the underlying stock of the issuing firm at a fixed exercise price until the expiry date. The moment when these options are exercised, the firm will issue new common stock, which, at that point in time, will dilute the existing shareholder holdings.

Warrants are frequently attached to bonds/loans or even to a preferred stock as an equity kicker: the investor will potentially be able to get an additional income from the warrant once exercised on top of the interest (or fixed dividend). This implies that the borrower will pay a significantly lower interest rate compared to the subordinated debt of a similar risk profile with no warrants attached.

Because of their characteristics, warrants are mostly used in the Anglo-Saxon countries.

Subordinated debt with profit participation scheme (PPS)

A profit participation scheme (PPS) is a possible mechanism which is often considered in mezzanine financing in order to provide for sufficient downside protection and, also, for an adequate way for the lender to participate in a growth of the financed firm. PPS is used in order to (partially) replace the fixed and/or variable loan mechanism used in straightforward senior and plain vanilla subordinated debt. The profit sharing is often constructed on a percentage basis, and the basis on which it can be applied can be variables such as revenues, EBITDA, EBIT, EBT etc.

This kind of mechanism can also be extended with a cap and a floor potion, resulting in a minimum level of compensation that has to be paid, regardless of the actual performance of the company, and the maximum level, therefore not letting the profit participation scheme have its full effect.

In case of a subordinated debt with profit participation scheme, the lender does not hold a stake or share in the company, resulting in a general inability to influence the firm's business. Although, it should be noted that usually the lenders protect themselves through a set of negative covenants, consequently not participating in the firm's eventual losses.

Other elements (e.g. profit participation scheme cumulative or not and the option for the borrower or lender to withdraw or sell back the instrument) need to be covered in the contract between borrower and lender, resulting in a high degree of customization for this particular kind of mezzanine product.

Convertible loans

Whereas warrants are separate and detachable from the subordinated debt, the convertible loan group has the option of embedding the warrant in the subordinated debt itself. This kind of product can, therefore, change its nature during its lifetime, from a fixed income instrument into common equity. Obviously, at the moment of conversion, this instrument will dilute the existing position of the common shareholders. It should be noted that convertible loans can have other embedded options included like, for example, call or put options.

Therefore, a convertible bond can be considered very similar to a straight bond with an embedded warrant: basically, it is a bond with an option for the holder to exchange it into "new" shares of common stock for the issuing company under specified terms and conditions, such as, for example, the conversion period, the conversion ratio and the conversion price. Because of the conversion privilege, the yield at issuance of convertible bonds is lower than the yield on the same issuer's more senior debt.

Although convertible securities always try to strike a balance between common stock and straight debt attributes, based on their characteristics and their level of customization, some will end up close to common stock in terms of risk and return profile, while others will be more similar to straight debt.

It should be noted that convertible bonds are highly customizable, resulting in a wide variety of alternatives, including more "exotic" models, resulting in being suitable for very different portfolio and situational objectives.

Preferred shares

Preferred shares, which are also known as "preferred equity", are the closest to common equity in a company. Therefore, preferred shares are the most subordinated product there is, resulting more junior compared to all other mezzanine products and positioning right before common equity.

In most cases, preferred shares carry a fixed dividend, resulting not dependent on what the Board of directors of the firm decides yearly regarding the dividend policy in terms of the sum to pay. Anyway, it is the Board of directors which decides whether or not there will be a dividend payment in a particular year: consequently, there is no payout of the preferred shares' fixed dividend if the Boards of directors decides not to pay any dividend in any given year. Therefore, it can be said that the preferred shareholder is a "residual cash flow owner": there is no event of default if the dividend payment is missed.

In case of cumulative preferred shares, the missed dividend payment goes into arrears. Cumulative preferred stock requires that any past, omitted dividends must be paid to the preferred shareholders before the payment of any dividend to the common stockholders.

Usually, failure to make dividend payments to the preferred shareholders may trigger certain restrictions on the issuer's management. For example, in case of cumulative preferred shares, if dividends are in arrears, preferred shareholders may be granted voting rights in order to elect one or more members to the company's Board of directors.

Moreover, it should be noted that this particular kind of mezzanine product can have either no maturity or can be redeemable at a specific point in time (or as of a certain date for a certain period of time).

In general, there are three main types of preferred stock:

(1) *Fixed-rate preferred stock*, as it was discussed before.
(2) *Adjustable-rate preferred stock* (*ARPS*), whereby the dividend rate is reset quarterly based on a pre-determined spread from the highest three points on the Treasury yield curve. The reset spread is added or subtracted from the benchmark rate determined from the yield curve. The three points on the yield curve are (1) 3-month T-bill rate, (2) 10-year T-bond rate and (3) 30-year T-bond rate (or local equivalents). In this case, there are often a cap and/or a floor on the dividend rate.
(3) *Auction rate and remarketed preferred stock*, whereby, like ARPS, the dividend rate is reset periodically, but the dividend rate is established through a Dutch auction process. Participants in the auction are current preference shareholders as well as potential buyers.

5.7 Conclusions

In this chapter we analyzed the main characteristics of leveraged acquisitions and the European market over the most recent period, in order to better understand the relevance of this particular kind of deal.

Leveraged acquisitions are an important deal class in the area of structured finance, namely representing those transactions that result in an acquired company with a debt ratio higher than before the acquisition.

In particular, leveraged acquisitions refer to a class of extraordinary financial operations within the larger "family" of mergers and acquisitions (M&A) transactions, expression qualifying those operations that produce deep and permanent changes in the ownership structure of one or more enterprises.

Leveraged acquisitions have certain unique characteristics. They require the constitution of a "vehicle company" (also called "special purpose vehicle", "new company" or, simply, "*newco*") for the transfer of the ownership and, in this case, the acquisition happens "off balance sheet" for the proponent subject: the only accounting effect for the proponent is represented by the subscription to the capital of the *newco*. The capital raised through debt securities, the associated collaterals and covenants are, instead, pertinent to the *newco*. Also, it should be noted that the equity raised by the *newco* represents only a minor part of the complex of resources that are necessary for the operation. The major part of the capital is, in fact, supplied through debt securities by a pool of banks and financial intermediaries with a high recourse to indebtedness.

In case of leveraged acquisitions, the debt capital needed to allow the exit of the target's previous ownership base is supplied from the banking system as a function of the capacity of the target firm to generate cash flows. It implies, firstly, that the banks finance the acquirer on the basis of the residual debt capacity of the target firm and its consequent capacity to repay the debt and its servicing charges. Therefore, it is clear that the choice of the target firm has to be carefully considered by the proponents, considering that not all the firms that could be objects of an M&A operation can be considered good candidates for a leveraged acquisition.

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Chapter 6
Issues and Trends in Project Finance for Public Infrastructure

Veronica Vecchi, Mark Hellowell and Francesca Casalini

Abstract In the last decades, many governments across the world have sought to use public private partnerships (PPP) as a means of attracting private capital to build economic and social infrastructure. Increasingly, PPPs have strong interest groups support, especially among a new range of investors, such as pension funds, life insurance companies and sovereign wealth funds; however, some policy instruments remain an indispensable element to attract these alternative risk-adverse long-term investors in the infrastructure sector. This chapter reviews the policy actions used by governments to enhance the viability of PPP contracts and shows how economic and financial equilibrium can be established. Crucial to the latter is the expected cost of the equity and the chapter outlines a range of approaches that can be used to estimate this.

6.1 Introduction

In the last thirty years many governments across the world have used private capital to build economic infrastructure (such as roads, railways, airports, broadband, water and sewerage pipes, and electricity and gas pipelines) and social infrastructure (such as hospitals and schools).

Such arrangements were precipitated by the large-scale divestiture of national utilities that took place in the 1980s and 1990s. However, in more recent years, governments have sought to use public private partnerships (PPP) as a means of attracting private capital into sectors in which privatization has been politically or

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© Springer International Publishing AG 2017

S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_6

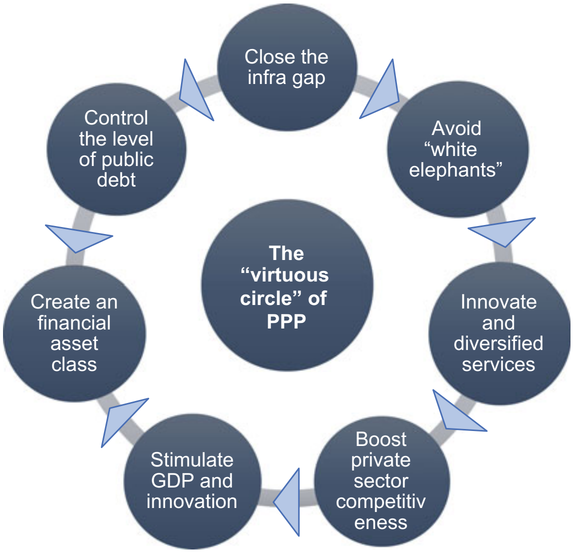

Fig. 6.1 The "virtuous circle" of PPP. *Source* Authors

operationally undesirable. Under some forms of PPP, private consortia are commissioned by governments or agencies to finance and deliver new economic or social infrastructure in return for a stream of payments, to be funded by the public sector and/or service users. This form of public-private engagement is the focus of this chapter.

The PPP model is seen to be capable of generating a range of economic benefits, especially in terms of enhancing the flow of capital to infrastructure (to address an often considerable 'infrastructure gap'[1]), alongside the quality of investment decisions and the effectiveness of project execution.

At the same time, PPPs are attracting considerable interest from a new range of investors, such as pension funds, life insurance companies and sovereign wealth funds (Gatti 2014), which are attracted to the long maturities and stable returns associated with infrastructure-related assets (World Economic Forum 2013). Since the global financial crisis, the institutional investment community has been engaged in a 'search for yield'—i.e. for higher returns than accrue to traditional asset classes such as government bonds—and such securities have been cited as a potentially desirable 'alternative asset class' by scholars (e.g. Gatti and Della Croce 2015). In this context, policies that enable greater private financing of economic and social infrastructure have powerful demand- and supply-side drivers.

[1]The OECD estimates total infrastructure investment requirements until 2030 at approximately USD 82 trillion, i.e. USD 3 trillion a year (OECD 2006, 2007, 2012). There is also evidence that a large gap has emerged between actual investment and the amount required to keep pace with expected income growth and structural change. The World Economic Forum (2014), for example, has estimated the infrastructure 'gap' to be approximately USD 1 trillion per annum.

Figure 6.1 summarizes the main PPP effects, which create a virtuous circle that benefits the whole economic system.

This chapter reviews the policy actions used by governments to enhance the viability of PPP contracts and shows how economic and financial equilibrium can be established. Crucial to the latter is the expected cost of the equity and we outline a range of approaches that can be used to estimate this. The final section draws some concluding remarks by discussing the conditions in which a PPP programme can be placed on a sustainable foundation.

6.2 Project Finance for Public Infrastructure: Contractual Models

As the term PPP can be used to refer to a wide variety of different institutional arrangements (Hodge and Greve 2010; Linder 1999), it is important to be clear about nomenclature. Three are the key elements of a PPP contract (Van Den Hurk and Verhoest 2016). First, it is a long-term contract between a public authority and a private sector company (which is generally a company created to deliver the contract—a special purpose vehicle, or SPV) that is based on the procurement of services, not assets. Second, it involves the transfer of certain project risks to the private sector, notably with regard to designing, building, operating and/or financing the project. Third, it contains a payment structure which is influenced to some degree by service volumes and/or service quality (whether this is paid through user charges (e.g. tolls) or through guaranteed 'availability payments' and/or usage-based 'shadow tolls', which are paid by the public authority).

A PPP project can be analyzed from a number of different perspectives: contractual and financial features; accounting treatment and the judicial forms used to formalize the contract.

From a contractual point of view, the most common PPP forms are:

- Turnkey contracts (design, build and finance, DBF) or finance leases (build, lease and transfer, BLT), in which the private sector partner bears construction risk but is remunerated by the public authority for the whole investment once it is completed—or through leasing installments over a 10–30 year period;
- Operation and maintenance (O&M) of existing/brownfield infrastructure, in which the private sector partner is in charge of maintaining the infrastructure and delivering the service: i.e. managing an outsourced service mainly used by the public authority or delivered to final users;
- Availability and performance based contracts, in which the private sector takes the responsibility to design, finance, build and maintain/operate (DBFM/O) the

infrastructure and delivers non-core services while the public authority retains the responsibility on the core service, generally funded through a mix of taxes and tariffs, and pays the private partner with availability and performance based payments[2];

- Concession contracts (also known as build, operate and transfer, or BOT), where the private sector partner has full responsibility for the design, finance, build and service operation, and generally retains the entirety of the demand risk.

These different models are associated with different risks. In DBF and BLT contracts the private investor bears the construction and financial risk, the latter during the construction phase, when it needs to fund the investment. In O&M contracts, the main risks borne by the private investor are the availability and performance risk when the authority is the main client. When the service is delivered to, and paid by final users, the private operator bears also the demand risk. In bundled contracts, the construction and financial risk are matched alternatively with the performance and availability risk in case of DBFM/O and by the demand risk in case of BOT. Some demand risk can be retained also in the former.

Not all the risks have the same magnitude or impact on the incentive and accountability environment: in particular, the demand risk provides stronger incentives for the private investor to consider the need or demand for its 'product', which may encourage better investment decision-making. However, this is a critical risk that if often difficult to transfer to the private sector, as we discuss later.

Important variations also exist in the approach to financing. In general, project finance (featuring high financial leverage, a non-recourse structure and therefore a credit assessment based exclusively on the project's cash flows) have been applied to DBFM/O and BOT contracts, while DBF and less capital intensive O&M contracts, despite the possible presence of an SPV, are generally funded through a corporate financing approach in which lenders have recourse to corporate assets in cases of default. Table 6.1 shows the main PPP contracts, which are featured by a high degree of flexibility and therefore customizability.

From the points of view of addressing a country's infrastructure investment 'gap', the most relevant contracts are the DBFM/O and the BOT ones, as they imply the funding of an investment.

In recent years, as it will be discussed below, DBFM/O schemes, initially applied in the field of social investments (such as schools and hospitals) have been extended to economic infrastructure, especially transport. DBFM/O schemes are preferred to BOTs because, where the demand risk cannot be identified or mitigated by the private investors, the project is exposed to fluctuations that affect a project bankability (Vecchi et al. 2017).

One of the flagship projects in Europe is the DBFM contract for the A11 motorway in Belgium (see Box 1 below), which is the first greenfield project to

[2] In the UK, DBFO is applied both to hospital and schools, where the core service remains under the competent public authorities responsibility, and to motorways and highways managed through a shadow toll system.

Table 6.1 PPP contracts, funding schemes, services delivered and risks allocation

<table>
<tr><th>PPP schemes</th><th>Investment</th><th>Service management</th><th>Financial structure</th><th>Length</th><th>Main risks retained by the private operator</th></tr>
<tr><td>DBF</td><td>X</td><td></td><td>Corporate finance</td><td>3–5 years</td><td rowspan="2">Construction (on time, on budget) and financial risks, at least during the construction phase</td></tr>
<tr><td>BLT</td><td>X</td><td></td><td>Leasing</td><td>20 years</td></tr>
<tr><td>O&M—outsourcing (the authority pays for the service)</td><td></td><td>X (ancillary services)</td><td>Corporate finance, if some minor investments are included</td><td>3–5 years</td><td>Availability and performance of the service</td></tr>
<tr><td>O&M (the service is delivered to final users, who pay for it)</td><td></td><td>X (core services)</td><td>Corporate Finance, if some minor investments are included</td><td>5–15 years</td><td>Demand risks, which can be mitigated in case the demand is rigid</td></tr>
<tr><td>DBFM/O</td><td>X</td><td>X (ancillary services)</td><td>Project finance</td><td>20–30 years</td><td>Construction (on time, on budget) and financial risks, availability of the facilities built</td></tr>
<tr><td>BOT</td><td>X</td><td>X</td><td>Project finance</td><td>20–30 years</td><td>Construction (on time, on budget), financial and demand risks</td></tr>
</table>

Source Authors

have benefitted from the European Investment Bank (EIB) project bond initiative (Vecchi et al. 2015). Under this contract the authority pays an availability charge to the concessionaire; the payment is subject to adjustments related to the achievement of performance and quality standards and is linked to inflation just for around 16% of its total amount.

In a DBFM/O contract it is fundamental to keep the level of incentives high for the private investor through appropriate risk transfer. Eurostat, in Europe, in its Manual on Government Deficit and Debt—Implementation of ESA 2010–2016 edition, with reference to PPP contracts states that

> application of the penalties should be automatic (i.e. clearly stated in the contract and not subject to bargaining or to a decision by government on whether to apply them or not) and should also have a significant effect on the partner's revenue/profit and, therefore, must not be purely symbolic. Should the asset not be available for a non-negligible period of time, the government payments, as determined by the contractual formula, would be expected to fall to zero for that period, according to a fundamental principle "zero availability – zero payment".

These kinds of hard incentives are rarely welcomed by debt investors, which may decide to fund a project only where it meets a certain level of minimum and average Debt Service Cover Ratio (DSCR) calculated on the net availability payment, i.e. on the part of the availability payment that is largely fixed and unaffected by the application of penalties. This is especially true when the project is funded through bonds, in which case investors are unable to exercise any ex post due diligence on the SPV.

In a DBFM/O contract, authorities may be tempted to link the entire availability payment to inflation, thus ensuring a lower fee in the early years of the concession phase. This decision may generate large exposure of the project to external variables that can't be influenced by the authority or by the private investor. In a DBFM/O project, only management costs (such as the lifecycle maintenance) and dividends change under the effect of the inflation, as the cost of the investment (capital expenditure—capex) and the cost of the debt are fixed. Therefore, when the availability charge is adjusted by inflation, the adjustment is able to neutralize the monetary increases of the operational costs, allowing the project to maintain the expected value across the life of the project.

With reference to traditional applications of DBFMO scheme for social infrastructure, it must be noted that large and often inflexible contracts, including many non-core services (such as, in the case of hospitals, cleaning, catering, laundry etc.), are progressively becoming less inclusive, or 'leaner', and incorporate only those services (such as hard facility management) that are fundamental to the availability of the facilities provided. This trend has been recommended (indeed mandated) by the UK Treasury from 2012, and has been branded by the authors of this chapter as "PPP Light" (Vecchi and Cusumano 2013).

6.3 Public Policies to Attract Private Investors in Public Infrastructure

To sustain the attraction of private finance into the infrastructure sector, especially in the aftermath of the global financial crisis, many governments have introduced policies and financial instruments to mitigate the financial risks associated with infrastructure development, and thereby enhance the availability and/or reduce the cost, of private capital (Hellowell et al. 2015) However, as the traditional debt funding seems to be no longer available for infrastructure financing, as a consequence of the new international regulatory framework for banks (Basel III), these policy instruments remain an indispensable element to attract alternative risk-adverse long term investors in the infrastructure sector.

These measures can be grouped into main five categories: (1) grants, (2) availability-based payments, (3) credit enhancement tools, (4) direct provision of debt and equity capital, and (5) other measures. Each measure can be then articulated into specific instruments.

(1) *Grants*

Grants reduce the capital requirements of the project or integrates revenues; it is generally delivered by contracting authority, even if some dedicated fund at national level may exists. A grant can be of three types:

(1.1) *Lump sum capital grant*, to reduce the need of private capital; it may delivered at the contract signature or during the implementation of the works, usually on a milestone-basis; in the latter case, a performance bond may be required to guarantee the correct allocation of the grant;

(1.2) *Revenue grant*, to increase the revenue volume and stability when the risk of demand is retained by the private player and tariffs are set at social value; it is generally defined at the contract signature and it can be paid by the authority as a periodic fixed amount (with a stronger effect on the mitigation of demand risk) or as revenue integration (it leaves the demand risk on the concessionaire);

(1.3) *Grant on debt interests*, to reduce the amount of interests due to the debt provider, thus mitigating the effect of high interest rates or the volatility of the demand on the debt repayment plan.

(2) *Availability-based payments*

Availability payments neutralize the demand risk, while leaving on the private concessionaire the performance risk and the optimization of the cost/income ratio (Fitch Rating 2015). Though the availability payment is the typical payment mechanism for social infrastructure, where the main user is the public authority (such as in the case of hospitals), increasingly it has been used also for economic infrastructure. In this latter case the service can be delivered free of charge to users or the tariff are retained by the public authority. The availability payment done by the authority to the concessionaire can be reduced by applying penalties that can be linked not only to the "pure" availability, but also to other quality and safety standards (the so called "constructive availability"). Tanks to the positive effects on the stability of the cash flow of the project, the availability payment allows the concessionaire to access lower interest rate debt and more comfortable covenants (such as lower DSCR).

(3) *Credit-enhancement tools*

Credit-enhancement tools are realized directly by a government of by its own controlled agency or development bank and can assume three forms:

(3.1) *Minimum payment* to reduce the demand risk, which is partially retained by the contracting authority, which is committed to guarantee a certain level of revenues, generally those necessary to cover the debt service at some level of the DSCR (debt service cover ratio); Gatti et al. (2013) demonstrated that the presence of revenue guarantees reduces the variability of Cover Ratios and IRRs, since it allows a more accurate and stable estimation of future operating cash flows generated by the project;

(3.2) *Guarantee in case of default*: it pays debt principal and interest in the case of concessionaire's default;

(3.3) *Guarantee in case of refinancing*: it repays lenders if the concessionaire fails to refinance the loan at maturity; actually, in the context of "mini perm" (i.e. a debt structure that can—soft mini perm—or must—hard mini perm—be refinanced after the construction phase) there is a risk that existing debt will not be repaid from new borrowing (risk of refinancing), especially in case of increased interest rates or changed market conditions.

(4) *Direct provision of debt and equity capital.*

Provision realised directly by a government of by its own controlled agency or development bank, can take three main forms:

(4.1) *Subordinated (junior) debt* aimed at enhancing the credit quality of the senior debt through in order to attract investment from insurance companies and pension funds;

(4.2) *Debt, provided at market condition*, to cope just with the liquidity shortage, or at lower interest rate to help the project to meet the expectation of debt capital investors, in term of interest rate, DSCR and maturity. In this latter case it should be considered the crowding out effect that this mechanism can generate;

(4.3) *Equity*, provided at market conditions or at more advantageous conditions: the aim is generally to fill the equity gap; to reduce the financial gearing, therefore reducing the exposure to credit risk and to offer downside protection or upside leverage to private equity holders.

(5) *Other measures*

(5.1) *Officially mandated change to capital structure* (reduced gearing), as introduced in the UK under PFI2, aimed at strengthening the ability to absorb fluctuations in cash-flows and thereby further insulate the lenders' exposure to credit risk;

(5.2) *Favorable taxation schemes*, for the Special Purpose Vehicle to sustain the general viability of the project (the effect is to increase free cash flow to operation) or for the equity investors on "certain qualified dividends" and long-term capital gains.

Figure 6.2 summarizes these forms of public supports, by showing their effects on the main components of the project cash flow. The tools on the left side of the picture are generally introduced to increase and/or stabilize the money inflows of project (i.e. current revenues and long-term funding), while the tools on the right side reduce and/or stabilize the money outflows (i.e. capital costs, operating costs and corporate taxation, interest on debt and dividends).

Table 6.2 shows the main applications of these financial instruments across the world.

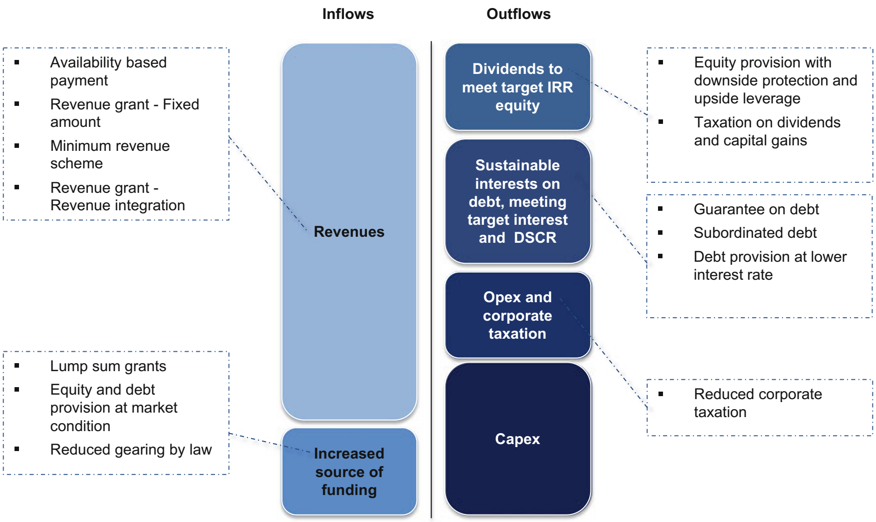

Fig. 6.2 The effect of policy instrument on project cash flows and ratios. *Source* Vecchi et al. (2017)

Box 1—A11 Motorway and the EIB Project Bond Initiative

Under the Project Bond Initiative (PBI), the EIB provides eligible infrastructure projects with project bond credit enhancement (PBCE) in the form of a subordinated instrument to support senior bonds issued by a project company. PBCE is not a guarantee that covers the entire amount of the project bonds, but it is limited in amount from the outset. The maximum size of PBCE available for a single transaction will be the lower of EUR 200 million or 20% of the nominal amount of project bonds issued (EIB 2012).

PBCE facility provides credit enhancement in two different ways:

- Funded PBCE: a loan, subordinated to senior bonds, given to the project company from the beginning;
- Unfunded PBCE: a letter of credit provided upon financial closing for an amount that can be drawn in the event that the cash flows generated by the projects are not sufficient to ensure senior bond debt service or to cover construction costs; in the event that the project runs into difficulties and the credit line is drawn, the EIB will inject funds and create a mezzanine instrument subordinated to senior bonds.

Figure 6.3 graphically describes how PBCE works and highlights differences between funded and unfunded mechanism.

Table 6.2 A summary of international experiences and lessons learned on the use of financial instruments to attract private investors in public infrastructure

Category and experience number	Country/region	Instrument type	Lesson learned
Grants			
Experience 1	EU	Lump sum capital	Within the 2014–2020 cohesion policy cycle, it is recommended to allocate EU structural funds also through PPP. Blended PPP projects have been successful mostly when the private partner was selected prior to the application for the grant and the national public authority was already providing finance for an important portion of the project in addition to any EU grant contribution
Experience 2	Korea	Lump sum capital	"Act on private participation in infrastructure, art. 53" explicitly foresees the awarding of public grants; the situations when the grant may be issued are prescribed by law in art. 37 of the "Enforcement decree"; the amount of subsidy should not exceed the 30% of construction costs for road projects, the 40% for port infrastructures and the 50% for railway projects
Experience 3	India	Lump sum capital	In 2005, the Government of India launched the Viability Gap Funding scheme, to provide up-front capital grants at the construction stage. Grants may not exceed 20% of the project cost and are disbursed only after the private company has made its required equity contribution; the amount of subsidy is determined through a competitive bidding process
Availability-based payments			
Experience 4	US	Availability payment	After that five out twelve toll based concessions projects already operational by 2014 failed, the availability payment system has been introduced by the US Department of Transportation and eight new schemes without traffic risks awarded
Experience 5	Canada	Availability payment	In Canada, almost all the concessions are availability-based since the adoption of PPP and none that reached the operational stage have faced a significant threat of financial failure or termination due to poor contractor performance

(continued)

Table 6.2 (continued)

Category and experience number	Country/region	Instrument type	Lesson learned
Credit enhancement			
Experience 6	Korea	Minimum Payment	From 1995 to 2006, The Minimum Revenue Guarantee guaranteed 70–90% the projected revenues for a period of 15–20 years and it turned out to be a financial burden for the Korean Government. In 2006 the system was revised in 2006 and the Government guaranteed 65–75% of the projected revenue for 10 years only for solicited projects
Experience 7	Brazil	Minimum Payment	This transaction is a good example of demand risk mitigation, where the mechanism used is based on minimum and maximum levels of demand; If passenger traffic fluctuation is within a pre-determine range, the private concessionaire absorbs the upside or downside, while outside this range, the public authority shares the gains or losses
Experience 8	Mexico	Minimum Payment	BANOBRAS, the Development Bank of Mexico, provides a guarantee to cover full and timely payment committed to the private sponsor under a PPP project with the aim to help subnational entities (states and municipalities) attract private investors, and this mechanism is particularly valuable for entities with a lower credit rating
Experience 9	UK	Guarantee in case of default	The UK Guarantee Scheme was introduced by the HM Treasury in 2012 to avoid delays to investment in UK infrastructure projects that may have stalled because of adverse credit conditions making it difficult to secure private finance. It provides, on a commercial basis, a sovereign-backed guarantee, which must cover a financial obligation. However, the contribution of the scheme to the National Infrastructure Plan has been modest to date and it is due to close at the end of December 2016
Experience 10	Mexico	Guarantee in case of default	Always in Mexico, loan guarantees in case of default are offered by the BANOBRAS, the Development Bank, and FONADIN, the Infrastructure Fund of the Mexican Federal Government. However, these schemes have not been as widely used as expected, since they do not assume the construction risk

(continued)

Table 6.2 (continued)

Category and experience number	Country/region	Instrument type	Lesson learned
Experience 11	India	Guarantee in case of default	The Credit Enhancement Scheme, managed by the Indian Infrastructure Finance Company Limited (IIFCL), is available for brownfield projects and provides partial credit guarantee to enhance the credit rating of bonds of infrastructure companies. The scheme is currently under pilot phase
Experience 12	Belgium	Guarantee in case of default	To mitigate the impact of the financial crisis, the Flemish government introduced in April 2009 a refinancing guarantee scheme for projects which had already been tendered or became ready for tender by April 2011. Such a guarantee has been granted to the "Flemish PPP schools" project, where the Government has borne the majority of the financing risk
Direct provision of debt or equity			
Experience 13	EU	Subordinated debt	The Project Bond Initiative (see Box 1), launched in 2012 by the European Commission and the European Investment Bank to facilitate institutional investors financing of infrastructure projects, has been currently applied to 8 projects: Castor energy storage (Spain); Greater Gabbard, Gwynt y Mor offshore transmission link and West of Duddon Sands offshore wind farm (UK); A11 motorway (Belgium); A7 motorway extension (Germany); superfast broadband and expansion of the port of Calais (France). In all these cases, the provision of the subordinated debt by the European Investment Bank enhanced the rating of the bonds, on average, by 3 notches
Experience 14	US	Subordinated debt/debt	The TIFIA program provides direct loans (either junior or senior) to qualified infrastructure projects of regional and national significance, with at least USD 50 million of eligible costs. Loans provided by TIFIA typically have a lower interest with flexible repayment terms
Experience 15	UK	Debt	The first experiment was introduced in 2004, under the Credit Guarantee Finance (CGF) initiative. In March 2009, the UK Treasury announced the establishment of a short-term new private limited company:

(continued)

Table 6.2 (continued)

Category and experience number	Country/region	Instrument type	Lesson learned
			the Treasury Infrastructure Finance Unit (TIFU), which would have provide state loans to projects at prevailing market rates. Both measures were used in few cases
Experience 16	Belgium	Equity	In the Flemish region, the public limited company Via-Invest was established in 2006 with the aim to fill in the missing links in the Flemish road network via PPPs. Via-Invest is a structural joint venture between the Agency for Roads and Traffic, the Department of Transport and Public Works and the investment company PMV. Via-Invest acts as a holding company for various SPV and provides them with risk capital

Source Authors

The A11 road project[3] in Belgium is the first transport TEN-T project and the first greenfield PPP transaction to be financed under the PBI. The project closed with the issuance of almost EUR 580 million PBCE-supported project bonds in March 2014.

Because of the reduced level of risks borne by private investors allowed, the A11 has been assisted by the unfunded PBCE. investors do not bear the demand risk as it is an availability based DBFM contract, where the Authority is also the sponsor of the SPV, with an equity stake of almost 40%.

The EUR 577.9 million senior project bonds were issued by the project company Via A11 NV at par with a maturity of 30 September 2045 and a 4.49% coupon. The bonds are secured by the EIB through the unfunded PBCE facility, consisting in a letter of credit sized at 20% of the senior debt during construction phase, that will be available to provide liquidity in the event that cash flows generated by the project are not sufficient to ensure senior bond debt service or to cover construction costs. After construction phase, the maximum amount secured by the letter of credit will step down to 10% of the bonds.

Moreover, as well as being the PBCE provider, the EIB acted as anchor investor, subscribing approximately 25% of the initial principal amount of bonds.

[3]Information used to develop this section were drawn from the *Deal Prospectus*, available at http://www.iflr.com/pdfs/A11prospectus.pdf.

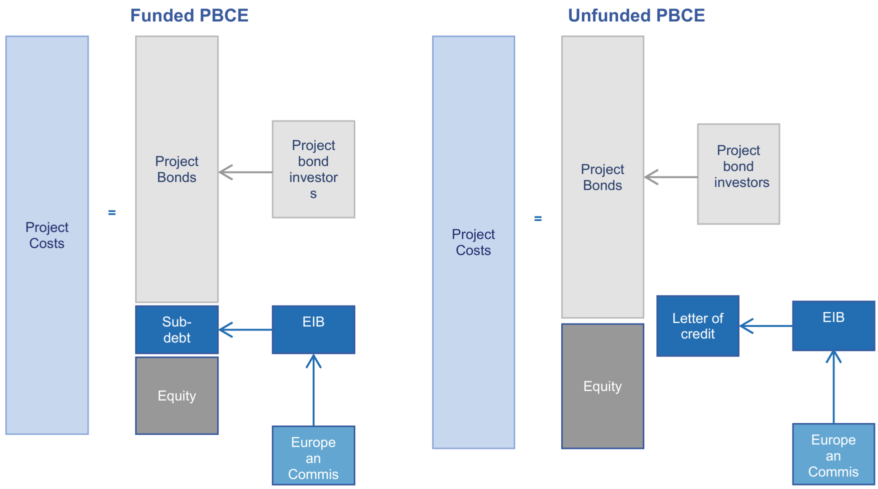

Fig. 6.3 Funded and unfunded PBCE. *Source* Adapted from EIB (2012)

An innovative element of the A11 project financial model is the deferred drawdown structure, that solves the problem of negative carry. The bonds were, indeed, fully issued on the issue date, but only a portion of them was actually subscribed and paid for upon issuance. The remaining part of the bonds will be purchased by bond subscribers following a pre-determined schedule, stated in the Forward Bond Purchasing Agreement, as shown in Fig. 6.3. This structure that allows for deferred capital drawdowns matches construction requirements and eliminates the cost of carry for the SPV.

To mitigate the risk related to the placement of forward purchase bonds at uncertain terms and conditions, on the issue date the SPV issued also EUR 287.5 million of privately placed notes (PP notes). PP notes are unlisted partially-paid senior debt titles with the same subordination, coupon, maturity and pay-up schedule of the project bonds. On pay-up dates, PP notes subscribers may decide to purchase senior project bonds instead of paying up their PP notes.

The pass-through model, i.e. the fact that the SPV is able to allocate to others the main risks, is a fundamental factor for the bankability of the project. Despite the presence of an availability-based payment, the project rating would have been Baa3 in Moody's view.[4] Thanks to the involvement of the PBCE, the rating of the bonds was enhanced by 3 notches, receiving a definitive A3 senior secured rating as shown in Fig. 6.4.

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[4]https://www.moodys.com/research/Moodys-assigns-PA3-rating-to-Via-A11-NVs-senior-secured-PR_294970.

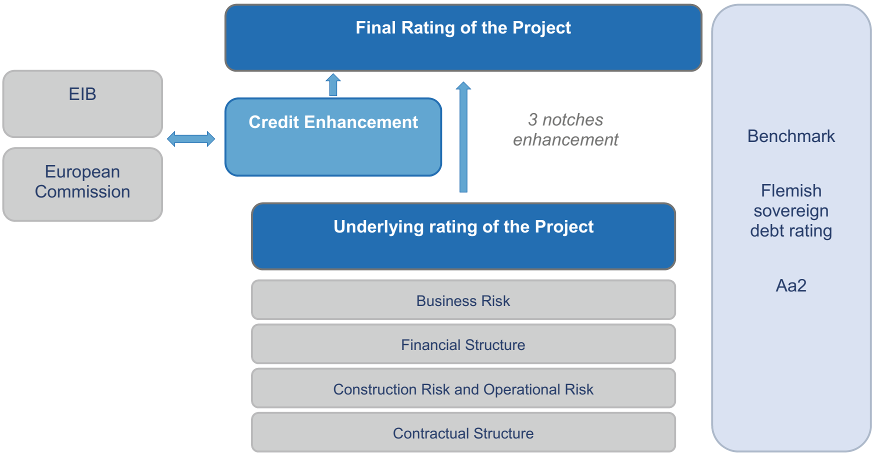

Fig. 6.4 The effect of PBCE on the project rating. *Source* Vecchi et al. (2015)

6.4 The Notion of Economic-Financial Equilibrium in PPP Transactions

When considering an investment, private investors generally use the NPV (net present value) and the IRR (internal rate of return) for assessing the general financial viability of a project, and use the discount rate as a 'hurdle rate', choosing those investments whose IRR is at least equal or above the discount rate, and, consequently, the NPV is positive.

However, PPP projects are based on agreements signed with a public authority, which retains some risks (i.e., part of the environmental risk; the regulation risk; the political risk; part of the force majeure risk) and therefore the selection of this hurdle rate is fundamental. If the project's financial model has been accurately calculated, considering among construction and operation costs also the project risks, the PPP contract, at the moment of its signature, must be in economic and financial equilibrium. It means that the revenues generated by the project, either coming from users or from a fee paid by the authority itself (the availability charge), should be able to cover investment, operational and capital (debt and equity) costs. The hurdle rate must be set to capture the cost of the capital invested in the project. If the contract were signed with a return higher than this hurdle rate, it would mean that the cash flows have been defined in a more conservative way for the SPV, thus creating an extra margin that reduces the exposure to project risks and therefore the incentive to monitor and manage them.

In fact, PPP models must be developed not on the basis of traditional corporate finance rules, according to which an investment is considered only if the IRR is higher than the expected cost of the capital, but pursuing the economic and financial equilibrium, which is reached if the following conditions are met:

- Project NPV = 0
- Equity NPV = 0
- Project IRR = Weighted Average Cost of the Capital (WACC)
- Equity IRR = Cost of equity (Ke).

The return for the equity investors must be calculated by considering the Free Cash Flow to Equity. Often, in practice, bidders refer to the Dividend Discount Model approach and calculate the equity NPV and IRR by considering only the equity invested during the development stage and the dividends paid by the project during the operation. This approach is not appropriate from the point of view of a policymaker or regulator as it embeds subjective considerations—such as the periodicity of dividends' distribution. This is a particularly salient issue as the implied postponement of dividends results in NPV and IRR estimates that are lower than those that estimated using a more accurate definition of cash flow.

Crucial to the definition of the economic and financial equilibrium is therefore the right calculation of the cost of the capital and therefore the hurdle rate. If the selection of the cost of the cost of the debt is rather simple, as it is dictated by the market, more complex is the calculation of the cost of the equity (Ke).

6.4.1 The Estimate of the Cost of the Equity in PPP Transactions

As the most popular method for estimating the cost of equity (and its ubiquitous use among natural monopoly and anti-trust regulators), the Capital Asset Pricing Model (CAPM) forms the basic framework for the analysis of the cost of equity for investors in PPP projects.

It determines that the cost of equity is a function of:

- the rate of return available on risk-free investments; and
- a premium for the amount of systematic risk that is involved in the investment.

In contemporary finance practice, the risk free rate is referenced to the gross redemption yield on the bonds issued by the government in the relevant jurisdiction.

The Equity Risk Premium is found by multiplying:

- the Beta of the investment—the weighted covariance of the forecast excess return on the project with the average excess return on the whole market; by
- the Equity Market Risk Premium (EMRP)—the average premium above the risk-free rate on equities, reflecting the amount of risk in the equity market portfolio.

Hellowell and Vecchi (2012), on which this analysis is based, propose three ways of applying the Capital Asset Pricing Model to estimate the cost of equity for a PPP project:

- the 'orthodox' approach;
- the 'realist' approach; and
- the 'opportunistic' approach.

The "orthodox" CAPM approach

As noted, under CAPM the opportunity cost of equity is equal to the return on risk-free securities plus the product of two variables: the EMRP and Beta.

The risk-free rate is the return on an investment with no variance around the expected return. It is standard practice to use government security rates to estimate risk-free rates, and the selection of the appropriate security rate is a function of the expected holding period for the investment to which the discount rate is to apply. In PPP contracts, by their long-run horizon, the yield on long-term government bond—i.e. 10-year, 20-year or 30-year—is commonly used.

Estimates of the ERPM are not uniform across global equity markets, as they depend on:

- the period over which returns are calculated;
- the method chosen for computing the average rates of return; and
- whether they are set to reflect current or expected market conditions (Damodaran 2015; Vivian 2007).

Nevertheless, the most widely used methodology to estimate ERPM is the so-called historical risk premium approach, where the average return earned on equities over a long time period is estimated and compared to the average return on a risk-free security. The difference, on an annual basis, between the two returns is computed, using the arithmetic or geometric average. This difference represents the historical risk premium.

This is a relatively straightforward process for mature markets, but presents a number of challenges when the analyst is focused on markets with short and/or volatile histories. This is clearly true for emerging markets, in which historical data is either non-existent or unreliable, and where a few large companies (many of them unlisted) may be dominant.

In such cases, Damodaran (1999, 2015) proposes a modified historical risk premium approach, calculating the ERPM as a sum of:

- a base historical risk premium—i.e. the historical risk premium of the US market, in which there is data to estimate the appropriate premium; plus
- a country risk premium—i.e. the additional premium required on investments in a specific country.

The country risk premium is referred to the rating attached to the country's sovereign debt and the country default spread that comes with the rating. Intuitively, we would expect a higher volatility of equity returns in comparison to bond yields. Therefore, this spread is adjusted through the following calculation to derive the country risk premium:

Table 6.3 ERPM calculation for the Russian market

Base risk premium (US ERPM)	5.81%
Country sovereign rating (Moody's)	Ba1
Country default spread (based on rating)	2.50%
Relative volatility (equity/bond)	1.5
Country risk premium	3.75%
Total ERPM	9.56%

Source Damodaran online http://pages.stern.nyu.edu/~adamodar/

$$\text{Country risk premium} = \text{Country default spread} * \left(\frac{\sigma_{\text{equity}}}{\sigma_{\text{country bond}}}\right)$$

We illustrate how to calculate the EMRP using the modified historical risk premium approach in Table 6.3, with reference, as an example, to an emerging market like Russia.

Once the ERPM is obtained, more difficult challenges are presented when deriving the appropriate Beta for a PPP investment.

Equity capital on a PPP project is provided by the owners of the equity in the SPV. This is normally a brand new business that has been established with the sole remit of delivering the contracted infrastructure and related services and earn an income from doing so.

As a result, there are no historical data regarding dividends or share price movements, and therefore no directly observable market data on which to base Beta.

However, it is generally believed that CAPM can be adapted to cope with unlisted businesses (Bowman and Bush 2007; Mitenko and Okleshen 1998; Penman 2006). In such cases, Beta can be derived from industries or firms that undertake activities that are to those undertaken in PPPs and are thus exposed to the similar risks.

Keeping Russia as a relevant example, Table 6.4 shows the relevant sectors to be consider for deriving the Beta for building and managing a motorway in the framework of a PPP contract. This shows average asset Betas for four comparable Russian industries over the period 2005–2015. Our source for this raw data is Bloomberg, but other sources of these data include Thomson Datastream and OneBanker.

It should be noted that the form of a company's Beta that is available on such databases is the Equity Beta. This form of Beta reflects the level of systematic risk that a company's shareholders face in addition to risks relating to the firm's financial leverage (which will be different to the leverage of the specific project under consideration). Therefore, in this table, the Equity Betas of the companies from which the raw data has been extracted are turned into "unleveraged" or Asset Betas through the use of the standard formula:

Table 6.4 Average asset Betas for selected Russian industries, adjusted by Blume

Category	Number of listed companies	Asset beta
Construction	2	0.56
Real estate owners and developer	5	0.57
Telecommunications	7	0.64
Utilities Power generation Networks	8 32	0.59 0.51
Average	54	0.54

Source Bloomberg

$$\text{Asset Beta} = \text{Equity Beta} \div [1 + (1 - \text{tax rate}) \times (\text{amount of debt} \div \text{amount of equity})].$$

Beta are adjusted according to Blume theory (Blume 1971) to reflect the fact that estimated Betas have a tendency to revert to the market mean (i.e. 1) over time.[5]

We have included the regulated utilities sector, as well as the (more intuitive) construction and real estate sectors, because businesses in utilities and PPPs can expect to:

- incur high capital investments;
- experience low volatility in demand (esp. if payments are availability based);
- have relatively predictable costs; and
- have long time horizons.

Damodaran (2015) also suggests that, if Betas are missing for the relevant businesses and sectors in-country, the 'bottom-up' approach can be used for emerging markets by referring to the firms in the same sectors in the United States or other mature economies.

This Beta provides no more than a starting point for estimating the Beta for PPP investors. To turn this into an appropriate Beta for the derivation of ERP, five more steps are needed:

1. The sources of systematic risk accruing to the project must be identified.
2. The relative contribution of each source of risk to the overall risk of the project must be assessed.

[5]The effect of the Blume adjustment is to reduce the difference between the Beta and the market average (i.e. 1). Blume (1971) found that adjusting estimated *Equity Betas* toward unity improved their ability to forecast subsequent period stock returns. The most widely held explanation for this is that unusually low or high *Betas* are subject to measurement error. Blume adjustment is standard in the calculation of *Equity Betas* by regulators in respect of UK, US and Australian utilities in determining the appropriate rate of return to investors, and is recommended in the most prominent corporate finance textbooks (e.g. Brealey et al. 2013). Blume-adjusted Betas are available from most commercial databases, such as Bloomberg and the London Business School Risk Management Service. The formula is: Blume-adjusted *Equity Beta* = (0.67)* βOLS + (0.33) * 1.

3. The allocation of systematic risk between the parties involved in the PPP must be identified to establish an asset Beta for the investors.
4. The Equity Beta for each project is then estimated by re-leveraging[6] the SPV-specific asset Beta according to the leverage of the specific project.

Table 6.5 is a tool that can be applied to calculate the cost of the equity by applying the 'orthodox' CAPM methodology as we have described above.

Column 1 identifies sources of systematic risks to which the project is exposed—they depend on the features of the PPP projects but the main sources are, as noted above: demand, inflation, residual value and downturn;

Column 2 shows the relative contributions of the different types of systematic risk to the beta (based on the qualitative assessment of PPP transactions), and are identified as percentages;

Column 3 contains the overall asset Beta to be applied to the project—we use the figure shown in Table 6.4;

Column 4 outlines the assessment of SPV exposure to risk as a percentage of the total (based on the qualitative assessment of PPP transactions); and

Column 5 shows the total asset Beta accruing to each systematic risk faced by the SPV, and the overall premium to apply to the risk free rate in order to derive the cost of equity.

In row (f), the total asset Beta for the equity investment (share capital and loan stock) is shown; in row (g), the outcome of the re-leveraging process is provided to give the equity Beta (considering a financial leverage of 75%). Row (h) shows the outcome of the equity Beta multiplied by an EMRP of 9.56%, as shown in Table 6.2. In row (i) the free risk rate of the Russian government is summed to the ERP to give the cost of equity in row (j).

Table 6.5 provides a tool that can be used to assess the cost of the equity when the systematic risk is partially retained by the Public Sector. Figures related to the components of systematic risks and the allocation to the SPV represent just an example.

The 'realist' approach

Empirical evidence in mature markets shows that it is often corporate hurdle rates, based on the Betas of an operational investors' core businesses, rather than cost of capital benchmarks appropriate to specific investments, that drive the required return on equity (see National Audit Office 2012). Although this is wrong in theory, it is an established market practice and unlikely to change in the near-term. Therefore, policymakers may wish to be aware of this approach when predicting (or evaluating, e.g. for purposes of regulation) the appropriate cost of equity.

The impact of this approach will vary depending on the payment mechanism adopted on a project. Corporate betas will normally be higher than is appropriate for availability-based projects because the level of systematic risk associated with the

[6]In effect, reversing the process in which the sectoral Asset Betas were derived from the observable *Equity Betas*, thus: *Equity Beta* = *Asset Beta* × 1 + (1 − tax rate) × the amount of debt ÷ the amount of equity.

Table 6.5 Tool to apply ortodox CAPM

	1	2	3	4	5
(a)	Risk type	Contribution to overall risk	Project asset beta adjusted	Allocation of risk to SPV	Beta for SPV capital
(b)	Demand	70%	0.54	90%	0.3
(c)	Inflation	10%	0.54	90%	0.05
(d)	Residual value	0	0.54	0%	0.00
(e)	Downturn	20%	0.54	80%	0.09
(f)	Asset beta for equity				0.44
(g)	Re-leveraged beta according to project-specific gearing (equity beta)				0.70
(h)	Equity risk premium (0.70 × 9.56%)				6.73%
(i)	Risk-free rate (Russia 10 Year bond yield)				9.94%
(j)	Total benchmark cost of equity				16.67%

core business of, say, a construction company is likely to be higher than is appropriate for the project. As revenues on an availability-based contract are more likely to vary according to project-specific factors under the direct control of investors and contractors, such as their ability to deliver contracted assets to time and to budget, rather than market-wide factors, such as changes in demand or inflation, the systematic risk (and thus the Beta) is far lower.

A corporate hurdle rate will therefore produce a target return above that implied by the orthodox CAPM theory for availability-based PPPs. In contrast, for usage-based PPPs, the appropriate Beta for the project may be similar to that faced by the operational investor's business. In this case, the risks to demand, and other market-wide variables such as inflation, for a construction company and the operator of, say, a toll road, may be similar.

In either case, an authority that is concerned not to compromise the supply of equity may wish to reflect the real practices of investors in constructing their corporate hurdle rates.[7]

The (implicit) premise of the corporate hurdle rate approach is that, in a non-recourse project finance transaction, the lowest acceptable equity IRR will be the sponsors' Weighted Average Cost of Capital (WACC). This premise is summarised in Fig. 6.5.

The WACC is a function of the cost of equity for the company and the yield to maturity of the firm's debt, in addition to the ratio (or gearing) between the two sources of finance and the effective tax rate, i.e.:

[7]The exercise be also an illuminating one for policymakers. For, if it is the case that expected returns exceed corporate hurdle rates, this is strong evidence that the cost of equity estimates are inefficiently high. This may indicate a deficiency in the design of the procurement process and/or an unduly risk averse approach by senior debt providers, both of which may indicate the need for a regulatory response.

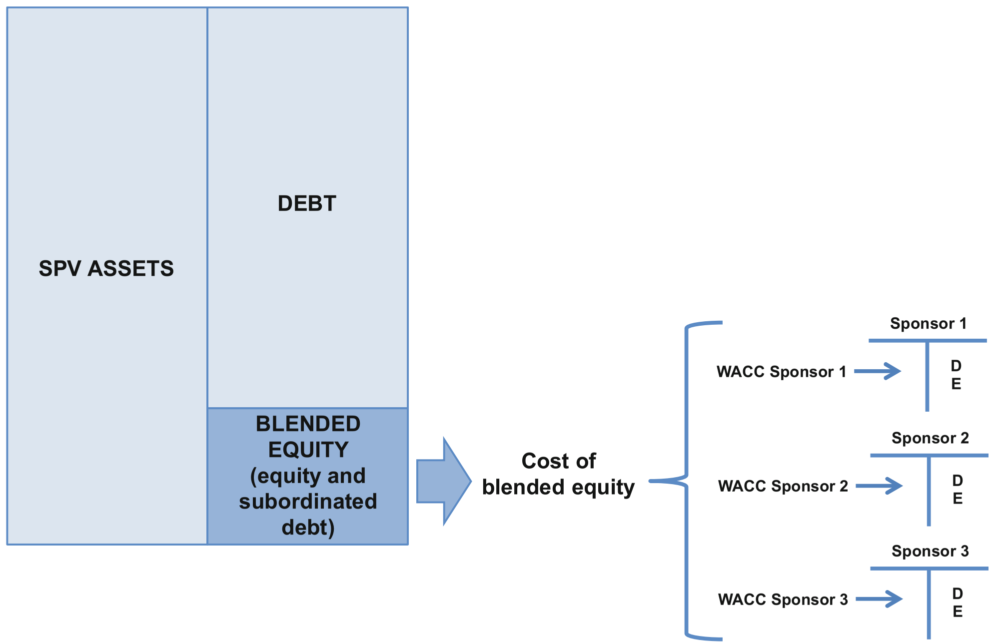

Fig. 6.5 Determinants of the equity invested into an SPV. *Source* Vecchi et al. (2013)

$$\mathrm{WACC} = \left[\frac{E}{E+D} \times K_e\right] + \left[\frac{D}{E+D} \times K_d \times (1 - T\%)\right]$$

where E is the amount of equity; D is the amount of debt; K_e is the cost of capital for equity; K_d is the yield to maturity of the firm's debt and T% is the effective corporation tax rate. The $(1 - T\%)$ function on the right-hand side of the formula reflects the fact that debt interest is often deductible from corporation tax—at least in most jurisdictions.

The components of the cost of equity are then estimated. The Betas of listed sponsors are publicly available (e.g. from providers of commercial project finance data, such as Bloomberg). These then need to be adjusted according to Blume theory, as described above. For unlisted companies, Betas may be calculated as the average Betas of comparable companies, selected according to (i) the nature of the core business, (ii) the reference market and (iii) geographical presence, using data released by each company.

In this case, the Betas are leveraged Betas, reflecting capital structure. These Betas need to be 'de-leveraged' to calculate an average beta to be applied to the group of unlisted firms, as described above. The Betas then need to be re-leveraged, using the capital structure of the project at the point at which the relevant project reached financial close.

The cost of debt may be calculated as the average yield on long-dated corporate bonds, issued by organisations with the same credit rating as that of the equity investor. For rated companies, the rating as quoted in e.g. Bloomberg can be used to determine this. For unrated companies, a proxy creditworthiness merit class between BBB+ and BBB− can be used, reflecting a recent analysis of project finance loan default and recovery rates by Moody's (2012).[8] The cost of debt may then be calculated using appropriate long-dated bonds—e.g. five-to-seven year maturities.

As a final step, data is needed on the capital structure of each investor. For listed companies, the data are publicly available. For unlisted companies, they are not—but analysts may use an average value of comparable companies' capital structure ratios was applied as a proxy for the typical debt/equity ratio in each of the relevant industries.

<u>Opportunistic approach</u>

An alternative approach is provided by Hellowell and Vecchi (2012), in which cost of equity benchmarks are constructed according to the project-specific discount rates actually used by major PPP investors in estimating the Net Present Value of their equity portfolios. This approach can be applied only in the few countries where PPP is a mature market.

[8]This study is based on a data set, which includes 2639 project finance contracts (accounting for 45% of project finance projects globally originated during a period from January 1983 to December 2008).

6.5 Concluding Remarks

Considering the scale of the infrastructure gap, PPP can represent an answer only if contracts are structured appropriately. On the one hand, the public sector is commonly looking for established and efficient private investors that are able and willing to bear project risks and thereby enable better investment decisions to be made, and project delivery to be executed well. On the other hand, the private sector is looking for reliable and stable public sector counterparties, able to create the contracting conditions to allow investors to identify, manage and mitigate project-related risks.

The current evidence is that many PPP transactions and projects have failed to minimize costs for payers (users, governments), and the authorities involved have in many cases found they are required to renegotiate contracts, bail out concessionaires, or even buy back the infrastructure in question. Furthermore, in many countries, such as in the UK and Italy, PPP is increasingly regarded by policy-makers (and in public discussion) as an unaffordable solution, because of supposedly high availability charges, and high rates of return to investors, that impinge on the ability of authorities to meet their socially-defined objectives.

Yet the model remains core to achieving the objectives of governments and supranational institutions across the world.

One of the salient issues, under a regulatory and structured finance perspective, is how to determine the right value for the expected return of equity invested in PPP transactions. In this chapter, we have analyzed the three main approaches used. However, it seems that few investors really apply a specific methodology when assessing the expected return and furthermore it seems that they don't operate any distinction between a traditional private investment and a PPP one. Therefore, much research is needed in order to provide useful feedback to regulating and procuring Authorities and private investors in order to contribute to make PPP a more sustainable and balanced approach to support the closure of the infrastructure gap and to foster even better the economic and social impact the PPP model is undoubtedly capable of.

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Chapter 7
Securitisation of Non-performing Loans

Andrea Fabbri

Abstract The chapter is aimed to understand how the securitisation of non-performing loans works and how to analyse and evaluate it. The first part is dedicated to the analysis of the cash securitisation on performing asset ("standard" cash securitisation). We will not enter into the details of the cash securitisation: we believe though that understanding the fundamentals of a standard cash securitisation is the key to then comprehend better the securitisation of non-performing loans and—above all—the main differences between the two structures. The second part of the chapter is dedicated to an overview of the cash securitisation market in Europe. In the third part of the chapter we will analyse the non-performing loans market in Europe with a focus on Italy where it seems to be a strong interest for this market to reopen and to succeed. The fourth part of the chapter is dedicated to the analysis of the securitisation of non-performing loans. The structure of the deal, the main counter-parties involved, the risks implicit in the securitisation and the valuation criteria will be the main points analysed in the final part of the chapter. The main pricing factors of the securitisation of non-performing loans will be studied from both the Originator Bank and the Investors' standpoint. The perspective which we have adopted in this chapter is the one of securitisation deals originated by banks: we won't refer to securitisation originated by Corporates nor by other type of financial institutions.

7.1 The Fundamentals of Cash Securitisation

Securitisation is a structured financial transaction which involves the sale of a portfolio of assets of a bank ("Originator Bank") with predetermined characteristics to a Special Purpose Vehicle ("SPV") which in turn issues and places to Investors

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S. Caselli and S. Gatti (eds.), *Structured Finance*,
DOI 10.1007/978-3-319-54124-2_7

securities named Asset-Backed Securities ("ABS") whose risk and return are linked to the capacity of the portfolio of assets to generate cash flows during the life of the transaction. This means to segregate specific future cash flows generated by the portfolio of assets sold to the SPV and to transfer them—together with their risks—to the buyers of the Asset-Backed Securities issued by the SPV. It can therefore be defined as a transaction where the future cash flows deriving from a specific asset portfolio are repackaged into a form of liquid securities allocated to Investors.

7.1.1 Standard Structure of a Cash Securitisation

In the structure of a standard cash securitisation the Originator Bank sells a portfolio of assets of its balance sheet to an SPV which issues Asset-Backed Securities in order to finance the acquisition of this portfolio. The transfer of the assets to the SPV is usually at par value ("Portfolio Purchase Price"). The assets exit the balance sheet of the Originator Bank via a True Sale and enter the assets side of the SPV balance sheet: the SPV is the new owner of these securitised assets. The liabilities of the SPV are represented by the Asset-Backed Securities which are usually "tranched" in different rating categories on the basis of the structural protection from which each of these tranches benefit (see Sect. 7.1.5). Typically the total par value of the assets of the SPV corresponds to the total par value of the Asset-Backed Securities issued by the SPV (see also Sect. 7.1.2). The Investors of the Asset-Backed Securities receive interest and capital from the tranches they bought: the capacity of the securitisation deal to pay interest and capital to the Investors is linked to the performance of the underlying portfolio of assets (also known as "collateral" of the deal) together with the different degree of structural protection that the deal could offer. A simplified example of a Cash securitisation transaction is shown in Fig. 7.1.

The Investors benefit from the segregation of the assets from the credit risk of the Originator Bank given the use of the SPV which acts as an intermediary between the Originator Bank and the Investors of the Asset-Backed Securities. The Investors pay the Purchase Price of the tranches to the SPV which is used to finance the acquisition of the portfolio of assets.

Two key elements are important in the legal and structural setup of the securitisation deal:

- *True Sale*: Segregation of the assets from the Originator Bank with portfolio's risk transferred to the SPV usually at Par Value
- *Bankruptcy Remoteness*: the SPV is structured to limit its business scope and, therefore, to isolate the securitisation structure from the bankruptcy risk.

Asset-Backed Securities are structured to obtain for some tranches a higher rating level than the one of the Originator Bank and/or of the underlying asset pool on a standalone basis. The high rating levels of Asset-Backed Securities deals are partially obtained through specific structural methods known as "credit enhancements" which can be both internal and external (i.e. supplied by a third party). Asset-Backed

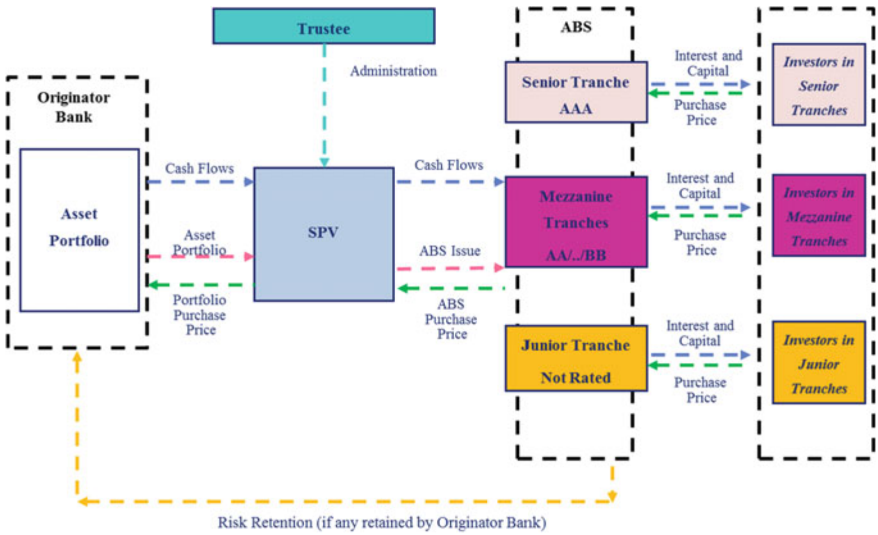

Fig. 7.1 Simplified standard structure of a cash securitisation

Securities issuances can also take advantage of interest rate and currency hedging strategies. Further protection to Investors can be supplied through collateral, or through performance triggers supplied by third parties, which can modify the cash flows "waterfall structure" of the securitisation deal during periods when the structure is under stress. Capital reimbursement and interest distribution can vary from deal to deal, as defined in the waterfall structure. The concept of credit enhancement and waterfall structure will be analysed more in details (see Sect. 7.1.5).

7.1.2 Main Counterparties Involved in the Securitisation

There are different counterparties involved in the securitisation process. The Originator Bank is the bank selling a portfolio of its assets to the SPV. The objectives of the Originator Bank will be analysed in details later on (see Sect. 7.1.4). The Arranger is the counterparty in charge of the structuring of the deal: arrangement fees are paid to the Arranger of the securitisation for the setup of the deal. If the Originator Bank is sophisticated enough it usually undertakes also the Arranger role in the deal. The Servicer collects the flows of payments from the underlying assets (e.g. interest and capital from a mortgage loan) and transfers them to the SPV so that regular cash flows can be transferred to the Investors of the Asset-Backed Securities in form of interest and eventually capital. The Servicer role is usually played by the Originator Bank which is the original recipient of the cash flows of the securitised portfolio of assets: this way it guarantees more effectively the timely flows of payments of the underlying assets to the SPV which then pays the Investors of the Asset-Backed Securities. The Swap Provider (interest rate swap and/or cross currency swap) hedges the possible interest rate and/or currency risk of the deal. This is usually a bank which satisfies minimum criteria in terms of rating. Swaps are needed in certain transaction to guarantee the continuity of the payments to the Investors of the Asset-Backed Securities (they are usually floating rates securities) in case of mismatch with the interest payments from the underlying portfolio (e.g. underlying portfolio represented of fixed rate assets). The Liquidity Provider hedges the possible lack of cash flows to the SPV due to elements which are not related to the deterioration of the credit quality of the portfolio: the Provider offers a Liquidity Line to the SPV to achieve this objective. The Credit Enhancer (if available in the deal) guarantees the possible lack of cash flows from the underlying portfolio: this is usually in favour of the most senior Investors of the Asset-Backed Securities in case there are not enough cash flows to pay them interest and/or capital. The Placement Agent distributes the Asset-Backed Securities to the Investors: it can also be the Underwriter in specific cases bearing the risk of not placing entirely the Asset-Backed Securities issuance to the market. The Investors of the Asset-Backed Securities are usually financial institutions such as banks, insurance companies, asset managers, pension funds and hedge funds. From an Investor's perspective Asset-Backed Securities investments could be considered an interesting investment for the following reasons (amongst others): effective way of diversification by asset type, originator, currency and jurisdiction available and high historical rating for certain asset categories with low price volatility. The difference with comparable alternative fixed income investments is usually linked to the low liquidity of these instruments compared to other bonds (government and corporate bonds).

Other important players are the Trustee of the deal which is appointed by the SPV and guarantees rights and covenants in favour of the Investors: it is also mainly in charge of the administration of the deal and also it guarantees a timely quality information to the Investors about the performance of the collateral of the deal; the Legal firms appointed to draft the main documents of the securitisation (e.g. the Offering Circular also known as Prospectus) and the Rating Agencies. The rating is the instrument through which the market evaluates the quality of the securitisation and therefore plays an important role in the price determination. The main counterparties involved in a securitisation deal are shown in Fig. 7.2.

7.1.3 Phases of the Securitisation

The main phases of the securitisation are the following:

- Selection of the assets to be included in the securitisation
- Preliminary analysis of the structure of the securitisation deal
- Determination of the technical features of the ABS issuance
- True sale and segregation of the assets in the SPV with final placement to the Investors of the Asset-Backed Securities.

In the first phase of the securitisation the Arranger together with the Originator Bank have to select a certain pool of assets to be sold to the SPV. The assets to be

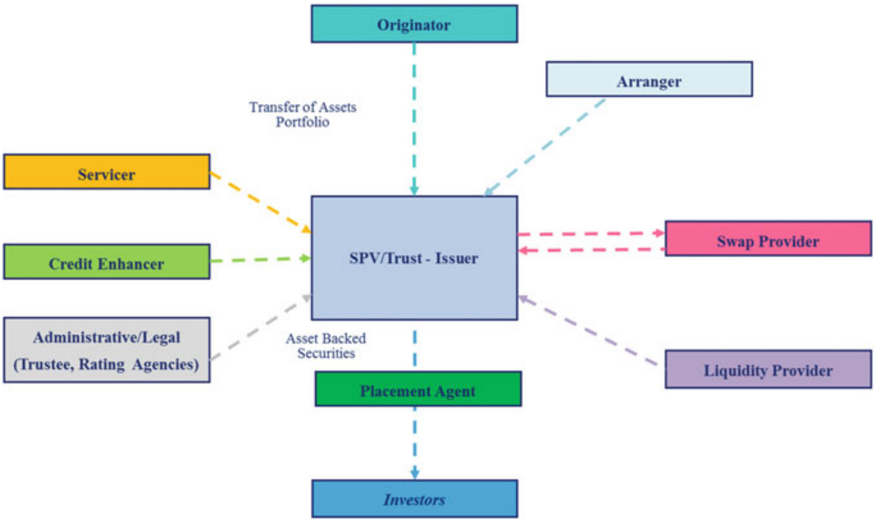

Fig. 7.2 Main counterparties of a securitisation

included in the securitised pool usually have the following characteristics: homogeneity, transparency in terms of risk profile, cash flow stability and frequent payment dates. The main rationale behind the mentioned characteristics is to have a pool of homogeneous assets by nature with predictable, stable and quantifiable cash flows which are measurable in statistical-insurance terms. Based on the economic characteristics of the assets sometimes the price value of the securitised portfolio can differ from its nominal value.

An example of origination criteria set up by the Originator Bank in the selection of the asset pool could be represented by the following:

- No mortgage is more than one payment in arrears as at the closing date
- No residential letting at origination; owner occupied properties
- First-ranking mortgages
- Monthly interest payment
- Maximum "Original Loan Balance" to "Original Foreclosure Value" of 125%
- Types of mortgage loans: linear, annuity, interest-only, savings, investment-based and combinations of the mentioned loans
- Maximum original principal balance of €750,000
- Borrower is over 18 years old and not an Originator Bank employee
- At least one monthly payment under the relevant Mortgage Loan has been made.

During the second phase of the transaction the Rating Agencies evaluate the Asset-Backed Securities issuance and determine the format and the level of credit enhancement of each of the notes: this process is also known as the "tranching" of the securitised portfolio. During this phase an analysis of legal, accounting, regulatory and tax issues is performed together with the evaluation of impact on the risk-return profile of the Originator Bank. A pre-sounding exercise with possible Investors is usually organized to "test" their possible interest in the Asset-Backed Securities issuance.

During the third phase of the deal the potential Asset-Backed Securities issuance is fine-tuned on the basis of technical specifications such as rating, return, cash flows originated by the collateral of the deal, its length and the indicative subordination level for each of the tranche of the Asset-Backed Securities. It is also important in this phase to add structural features to make the notes issuance appealing to Investors and mitigate the risk of the underlying securitised portfolio: as an example the fixed rate deriving from underlying activities could be swapped with a variable rate used to remunerate the notes placed to Investors via an interest rate swap (as mentioned Asset-Backed Securities are usually floating rates notes).

The credit quality and the rating of the swap provider are important factors to evaluate the counterparty risk of the interest rate swap in order to isolate the Investors from the potential interest rate risk of the underlying pool of assets. The SPV issues different classes of notes backed by the collateral bought from the Originator Bank. Each class of notes has a different risk and rating level according to the level of subordination they benefit from.

7.1.4 Objectives of the Securitisation for the Originator Bank

The objectives of the cash securitisation are mainly related to the possibility of reducing the cost of funding compared to other possible alternatives such as Originator Bank's bonds issuances placed to institutional Investors. The reason why this is possible has to do with the segregation of the assets sold to the SPV and with the fact that a "true sale" happens between the Originator Bank and the SPV (i.e. a sale with no recourse to the assets of the Originator Bank).

Through the segregation of assets, the risk-return profiles of the ABS placed to the Investors is strictly related to the credit quality of the assets portfolio sold to the SPV: this allows the Originator Bank to raise funding on the basis of the asset quality of the underlying portfolio sold to the SPV whilst avoiding to bear the cost of funding which is related to the credit quality and the rating of the Originator Bank itself (as in the case of the issuance of senior unsecured bond of the bank).

Usually the securitised portfolio of performing assets is comprised of good quality assets (e.g. prime mortgage loans for Residential Mortgage-Backed Securities): this allows the Originator Bank to get funding at a lower cost compared to the issuance of (senior unsecured) bonds placed to the Investors whose risk-return profile reflects the credit quality of the bank itself.

Other possible applications of the cash securitisations are:

- Improvement in the balance sheet structure also via a diversification of funding sources
- Reduction of illiquid assets or non-performing loans in the balance sheet

- Fund raising also via "intangible assets" (e.g. securitisation of lottery and season tickets etc.)
- In specific cases, reduction of assets with high regulatory capital absorption compared to return (Return on Capital Optimisation).[1]

7.1.5 Risk Evaluation of the Asset-Backed Securities

The two key macro-factors for evaluating the risk of the Asset-Backed Securities by the Investors of the securitisation deals are (see Fig. 7.3).

- Underlying Assets
- Securitisation Structure

7.1.5.1 Underlying Assets

The collateral of a securitisation deal can be represented by different type of financial assets: the most common category in Europe is represented by the Residential Mortgage-Backed Securities. Amongst the criteria to evaluate a given asset class the most relevant ones are: the economic environment; the geography of the assets; the type of origination process of the assets; the servicing/serviceability of the asset; the past performance of the asset and the likely future performance (e.g. historical default rates, historical prepayment rates, historical delinquency rates etc.); the granularity of the portfolio; the past recovery rates of defaulted assets.[2] Each Asset-Backed Security category is determined by the type of asset class underlying the securitisation deal.

One example of a Residential Mortgage-Backed securitisation deal is represented by Hermes XVI, a deal issued by Holland Mortgage Backed Series (Hermes) XIV B.V. For the evaluation of the deal, whose notes are backed by performing residential mortgage loan parts, secured by first ranking mortgages of borrowers based in Netherlands, different factors were taken into consideration by Rating Agencies such as the historical data on Dutch mortgage loans and a comparison of the underwriting criteria across the country.

The following list represents some examples of possible Asset-Backed Securities categories:

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[1]Capital Requirement Regulation (CRR) defines in art. 243–250 the Recognition of Significant Risk Transfer. Under certain conditions, the originator institution of a securitized exposure might exclude this exposure from the calculation of risk weighted amounts and from expected loss amount. In particular, the originator should provide evidence that a significant part of risk associated with the exposure has been transferred to third parties. Also, under certain conditions, a mitigation of credit risk exposure might be recognized.

[2]Rating Agencies reports are available for more information on the criteria used to evaluate Asset Classes. For RMBS criteria, see for example Fitch Rating (2013) EMEA Residential Mortgage Loss Criteria.

- Residential Mortgages (Residential Mortgage-Backed Securities)[3]
- Commercial Mortgages (Commercial Mortgage-Backed Securities)
- SME Loans, Large Corporate Loans & Leverage Loans (Collateralized Loan Obligations)
- Corporate Bonds (Collateralized Bond Obligations)
- Credit derivatives (Collateralized Swap Obligations)
- Trade Receivables
- Credit Cards (Credit card receivables)
- Personal Financing Loans
- Consumers Loans
- Student Loans
- Leasing
- Intangible goods or future income flows (cinema and music rights, stadium tickets, etc.).

Consumer loans are usually divided into Purposes loans which is financing given to consumers for the acquisition of durable goods; Auto loans i.e. financing given to consumers for the purchase of automobiles (can be further divided into “new auto” and “used auto”) and Personal loans: personal loans without specific or declared use.

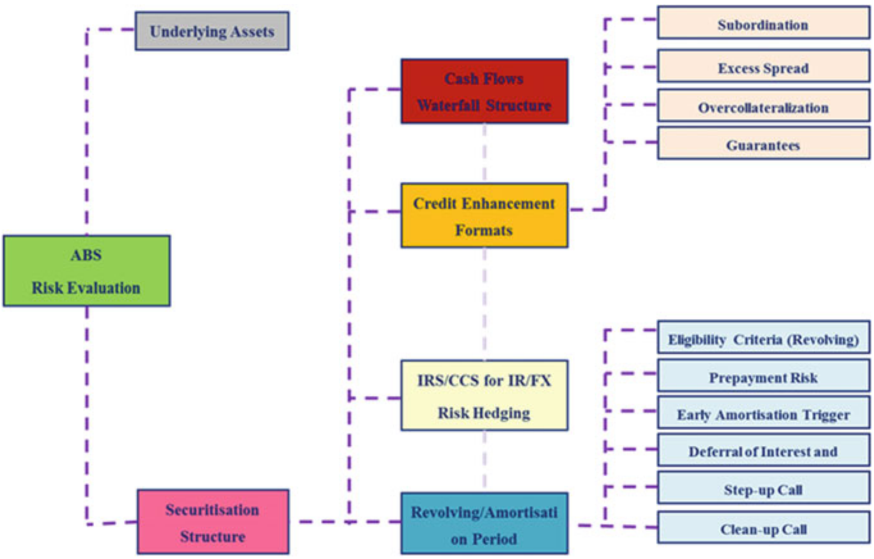


Fig. 7.3 Risk evaluation of the Asset-Backed Securities

[3]If the underlying portfolio is represented by a combination of Residential and Commercial Mortgages the Asset-Backed Securities will be generically named Mortgage-Backed Securities (“MBS”).

7.1.5.2 Securitisation Structure

The main building blocks used to evaluate the soundness of a securitisation structure are the following: credit enhancement, revolving and amortization period; cash flow waterfall structure and hedging agreements included in the securitisation.

Credit Enhancement. Credit enhancement mechanisms can have different formats; the most utilized ones—amongst the others—are subordination of tranches; excess spread; overcollateralization and third parties guarantees (if any). Credit enhancement, while improving the deal structure (making it less risky) has the function of qualitatively transforming the portfolio of assets into ABS tranches with different risk profiles, making them more appealing to Investors who can choose the tranche on the basis of their target risk-return profile. This is an important factor in determining the overall cost of the securitisation: the more efficient is the credit enhancement structure the lower the cost of the deal.

Credit tranching is one of the most common techniques to improve the credit profile of a structured transaction: this technique involves the allocation of the cash flows generated by the underlying collateral pool into different classes of notes issued by the SPV through a multi-layered capital structure ("tranches"). Each class of notes has specific characteristics in terms of subordination level, rating, risk and performance: this creates a cash flow waterfall structure where the most senior tranches have a priority in terms of payments over the mezzanine and junior tranches (subordinated tranches). Investors can then choose a tranche with a different risk-return profile according to its seniority/subordination level. The risk and return of each tranche reflect the different risks of the securitised portfolio of assets which can be ideally divided into three different macro classes of risks: senior, mezzanine and junior risks. Junior risk of the portfolio (also known as equity risk or first loss of the portfolio) represents the riskiest part of the portfolio and could be possibly held by the Originator Bank. It concentrates the most significant amount of credit risk of the securitised portfolio (or, at least, a multiple of its expected loss): the equivalent ABS tranche is not officially rated. Mezzanine risk is represented by an intermediate risk portion of the asset portfolio which is usually further divided into tranches with different risk levels hence different ratings. The senior risk is embedded into the tranche with the higher credit enhancement which also benefits—as a consequence-from the highest rating. Highest credit enhancement means that the Investors of the most senior tranche of ABS are the last ones to be affected by the possible realized net losses of the underlying assets portfolio. Conversely, the Investors of the not rated tranche of ABS (representing the junior risk of the portfolio) are hit by any first loss which could materialize on the securitised portfolio.[4] The creation of subordinated tranches is finalised running the tranching models provided by the Rating Agencies which are aimed to model the distribution of losses which can potentially happen during the life of the deal and consequently

[4]From time to time the tranches are structured on a "reverse enquiry" basis as they are created to reflect particular Investors' needs in terms of interest, subordination, maturity etc.

to provide a rating for each tranche based on its degree of protection from being affected by occurring losses.

The most used tranching approaches are the Probability of Default and the Expected Loss methods. These two approaches are usually used by Rating Agencies to determine the rating of a given tranche. Both models are based on the concept of attachment and detachment points which are used to identify a given tranche in respect to the other ones. The attachment point is the starting point of the tranche and any net losses (i.e. Losses Given Default) below it do not have any impact on the specific tranche. Once the net losses have exceed its attachment point the tranche starts to be impacted by them until it is completely wiped out if the level of net losses has climbed above the detachment point. The difference in the two approaches lays in the different target values which they consider namely the Probability of Default or the Expected Loss. Given a target value the Rating Agencies are able to set attachment and detachment points in order to allocate the tranches within a specific risk class. In the Probability of Default approach if the tranche's default probability (i.e. the probability of default of collateral losses to be above the attachment point) is below a target value (defined as Target Default Probability) then the tranche is considered eligible for the rating associated to that particular target value. In this method the thickness of the tranche (difference between the detachment point and the attachment points) is not taken into consideration so thinner or thicker tranches with equal attachment and detachment points are given the same rating: in a nutshell it is the probability of the first Euro of loss of a given tranche which determines its rating.

The Expected Loss approach is different since the target value is defined through the determination of the expected loss itself: if the net losses are greater than the attachment point a given tranche suffers until it has been completely wiped out if the net losses have exceed the detachment point. Differently from the Probability of Default approach the method of the Expected Loss depends on the thickness of the tranche: net losses that fall between the attachment and the detachment points will represent a different percentage of the total size in a thicker tranche compared to a thinner one hence different rating assigned to that given tranche. The Expected Loss method can be used also to define a tranche by setting a requested Expected Loss target level for the senior tranche (the detachment point will anyway always be 100% for the senior tranche) and reiterating the same process for the subordinated tranches as well.[5]

Excess spread is a credit enhancement technique usually represented by the difference between the return of the securitised portfolio and the overall cost of the securitisation deal (both fixed and variable costs). It is determined net of the realised

[5]As showed by Das and Stein (2013), it is unclear whether PD or EL approach provides higher level of Credit Enhancement. Usually PD models are considered to be more suitable for analysis of default due to their focus on the Probability of De-fault whilst Expected Loss models should be used for managing economic losses. Different target level choices and the absence of any constant LGD assumption on collateral, that can translate a PD model into a EL model, make indeed, these two models difficult to compare.

losses during the life of the deal. During the life of the securitisation the size of the expected excess spread is the key to understand the level of credit enhancement of the overall securitisation deal: this analysis is usually based on historical return rates of the underlying assets and on the evaluation of the quality of the underlying portfolio. Excess spread can be cumulated into a reserve account and periodically liquidated in favor of the junior tranche holder (the Originator Bank if the junior tranche is retained) and/or used to protect the Investors (starting from the holders of the most senior tranches) from the possible lack of interest payments or from potential losses (in this case it works as pure credit enhancement tool).[6]

The potential credit losses related to the underlying portfolio of assets affect first of all the junior tranche holders: if the net losses of the underlying portfolio are larger than the notional of the junior tranche (i.e. larger than it is detachment point) then the next in line are the Investors of the most subordinated tranches "above" the junior tranche and so on and so forth. The potential losses can affect also the Investors of the most senior tranche of ABS if the credit enhancement "below" this tranche is wiped out by the net losses realized during the life of the securitisation deal.

In some securitisation deals it is possible to have also *over-collateralization* which is a credit buffer created when the value of the issued notes (SPV liabilities) is lower than the par value of the assets sold to the Special Purpose Vehicle by the Originator Bank. This is a typical credit enhancement format used to decrease the exposure to default risk of the Investors (it could be possible to find it in the non-performing loans securitisations).

Revolving and Amortization Periods. The securitisation deal has usually two different periods which characterizes its life. During the revolving period the Originator Bank has the right to replenish (and substitute) assets which naturally amortize or are pre-paid in order to keep the notional of the securitisation deal at full capacity (i.e. maximum notional of the deal). The SPV uses most of the cash flows received from the underlying pool of assets to buy new assets according to certain specific criteria which are specified in the Prospectus of the transaction ("eligibility criteria"). The Investors of the Asset-Backed Securities receive only the interest due under the notes during this period: no principal repayment is covered during the revolving phase of the deal. During the amortization period the securitisation starts to amortize and the cash flows coming from the underlying portfolio of assets are used also to pay the capital back to the Investors of the Asset-Backed Securities. No replenishments (nor substitutions) are allowed anymore during this phase. The Asset-Backed Securities are sequentially paid notes: during the amortization period the tranche which benefits from the higher credit enhancement (the most senior) is the first one to be repaid in terms of capital (hence a shorter duration compared to the tranches more subordinated in the capital structure of the deal).

[6]Under this form of credit enhancement it is also possible to have a Cash Advance Facility (or Cash Reserve Fund) provided by the Originator Bank to guarantee the payment of Interest to the Investors of the ABS tranches during a certain period of time (e.g. term of 364 days).

Eligibility criteria play an important role in determining the overall risk of the securitisation since they regulate what kind of replenishments (and substitutions) are allowed to be done by the Originator Bank. Eligibility Criteria are carefully analyzed by the Rating Agencies and by the Investors. The Trustee of the securitisation has to guarantee the fulfilment of the eligibility criteria as stated in the Prospectus of the deal.

Examples of Eligibility Criteria—within a given asset class—are:

- Maximum Outstanding Principal Amount
- Maximum Weighted Average Life
- Geographic Concentration
- Maximum Concentration for each Asset
- Maximum Concentration for each Economic Industry
- Limits related to past performance of the underlying assets
- Limits related to minimum return of the new assets etc.

To understand the importance of the revolving and the amortization periods it is important to notice that the Investors decide to invest in a given ABS tranche also on the basis of its Weighed Average Life ("WAL") which can be defined as the average time to get the entire principal investment repaid: it is weighted for the times when principal repayments are made (the years when many principal repayments are made have a higher weight). To recap the WAL is the average time which takes for every euro of principal to be repaid. The WAL of a securitisation deal is different from the Legal Maturity of the deal itself which corresponds to the maturity of the longest asset within the securitised portfolio. The WAL of the deal is a function of different elements which are typically the assumptions on the prepayment rate of the deal; the existence of step-up calls; the presence of clean-up calls; the estimated default rate of the underlying portfolio and the length of the revolving and amortization periods.

It is important to highlight that the securitisation can include the presence of Early Amortisation Trigger Events which—under certain circumstances—can prevent the Originator Bank from further replenishing (or substituting) assets during the revolving period so that the deal "early amortises" making its WAL shorter. The possibility of replenishments becomes unavailable if certain events occur ("trigger events") related to the performance of the underlying portfolio. In this case the Originator Bank cannot make any portfolio replenishments and the deal "naturally" amortizes ("early amortization").

These circumstances are usually related to the deterioration of the performance of the underlying portfolio of assets such as a given default rate (and/or net loss rate[7]) is reached during the revolving period. Some examples:

[7]Default rate can be defined as the gross loss rate i.e. it doesn't take into account the recovery rate of the defaulted asset. Net loss rate is simply another way of defining the loss given default of the underlying defaulted asset i.e. the severity of the loss multiplied by the notional of the defaulted asset.

- Net available excess spread is below a certain percentage
- The cumulative default rate reaches a certain percentage
- The cumulative net loss rate reaches a certain percentage
- The cumulative delinquency rate reaches a certain percentage etc.

The prepayment rate estimate is an important variable when it comes to measure the overall risk of the deal and its estimated duration. The prepayment rate assumption should be an effective way to quantify the prepayment risk embedded in any securitisation deal. Prepayment risk is related to the possibility for the assets of the underlying securitised portfolio to prepay prior to their scheduled repayment date (e.g. in a RMBS deal, the borrower could decide to prepay the loan since he/she can find a lower refinance rate from a different lender compared to the one offered by the Originator Bank). Prepayments matter since they result in a loss linked to future interest collections which means that the SPV have less cash flows coming from the collateral pool to pay the interest and/or the principal of its liabilities i.e. the Asset-Backed Securities which have been issued and placed to the Investors. Also unexpected and unscheduled prepayment of principal will be distributed amongst the different ABS tranches according to their seniority level reducing the outstanding principal to be repaid making this way effectively shorter the WAL of the deal compared to the investment horizon of the Investors. The standard way to measure the prepayment risk in a securitisation deal is to use as a proxy the Constant Prepayment Rate ("CPR"): this is also known as "conditional prepayment" rate. It measures prepayment as a percentage of the current outstanding loan balance. The level of CPR varies according to the type of asset underlying the Asset-Backed Securities, the overall economic and financial conditions of the geography where the assets are based, the liquidity—in terms of offer—of the underlying pool of assets etc. As an example of RMBS when the underlying mortgage market is in good health (e.g. high growth of domestic product, solid banking system with good level of competition between banks) then it is easier for the borrower to prepay the mortgage and switch to a lender which offers better conditions. This results in a higher CPR hence a lower WAL for the ABS tranches in general. Vice versa when the financial market experiences a "credit crunch" phase together with low level of growth of GDP then it is more difficult for the borrowers to switch their mortgage loans hence this results in a lower CPR and a higher WAL of the ABS tranches. ABS tranches which are evaluated assuming high CPRs at deal's inception (e.g. 15–20%) can later on be revisited using a lower one (e.g. 0–5% CPR) if—for instance—there is a turndown of the economy together with a weakening of the banking system. In these cases WAL turns out to be much higher than the one originally assumed by the Investors in their ABS investment decisions.

Another important feature of the securitisation which can impact the WAL of the deal (by affecting the length respectively of the revolving and amortisation periods) is the potential deferral of payments to certain classes of ABS. The *deferral of interest* is a protection mechanism usually in favor of the Investors of the most senior ABS tranches which allows postponing the payments of interest to the

Investors of more subordinated when certain trigger events verify. This is also true for the repayment of the investment capital when the deal is in its amortization period: we have in this case the *deferral of principal*. As an example the payment of interest to the mezzanine tranches can be postponed when the most senior tranche is "under-collateralised" due to losses on the underlying assets portfolio. This can happen for instance when the outstanding amount of the most senior tranche is larger than the correspondent outstanding of the underlying asset portfolio and of the available funds prior to interest payment. Other measures to protect the Investors of the senior tranches are aimed to accelerate the repayment of the principal hence to shorten the WAL of their ABS tranche: this happens when a given ABS tranche is in breach of the Coverage Tests.[8] Once again these measures are intended to mitigate the risk of payment defaults under the ABS senior tranches in case the performance of the underlying portfolio of assets is much worse than the one expected at the inception of the deal.

Securitisations can feature a mechanism of *step-up call* and/or of *clean-up call*. The step-up call is included when—by contract—the Asset-Backed Securities placed to the Investors pay a coupon which is higher than the original one from the step-up date onwards (this is usually a multiplier factor of the original credit margin). The Originator Bank has the right to unwind ("to call") the deal at the step-up date: if the deal is unwound then the Investors of the Asset-Backed Securities receive the principal repayments at par unless their tranches have been affected by losses related to the defaults on the underlying portfolio of assets. If the deal is not "called" the Originator Bank commits to pay a higher coupon ("step-up coupon") to the Investors of the ABS tranches from the step-up date onwards. Usually the Investors assume that the step-up call date coincides with the WAL of the deal when they are in presence of this feature. Originator Bank can disagree on this assumption though: in fact they will probably evaluate the cost opportunity of issuing a new securitisation deal at the step-up call date compared to the one of paying a higher coupon on the ABS issued in the original deal. In the decision making process it will also be taken into consideration the possible "stigma" attached to the fact of not having called the deal if the Originator Bank needs to tap again the securitisation market in the future. During "credit crunch" phases the Originator Bank would rather pay the step-up coupon than issuing a new deal due to the substantial lack of appetite of the Investors for ABS deals and the general illiquidity of the market.

The *clean-up call* is a standard feature in a securitisation deal. It allows the Originator Bank to buy back the ABS held by the Investors when the notional of the

[8]The most frequent Coverage Tests are the Interest Coverage ("IC") and the Over-Collateralisation test ("OC"). OC test aims at ensuring that the assets are adequate to cover principal and coupon payment. IC test on the other side is a liquidity test, which measure the ability of the assets to generate adequate interest to cover interest payment and obligations (plus a buffer). In both tests each tranche is subject to its own test and the higher the seniority of the tranche, the higher the OC or IC ratio required, in order to guarantee a satisfactory protection of Investors.

underlying portfolio of assets becomes lower than a certain percentage (usually 10%). This allows the Originator Bank to "clean-up" the deal which means to avoid that the deal becomes economically unsustainable since its overall running costs (mainly fixed costs at this stage) are higher than the return of the residual assets pool. Figure 7.4 shows the determinants of the WAL of the securitisation.

Cash Flow Waterfall Structure. The structure related to the payment of interest and the repayment of principal for the Investors of the Asset-Backed Securities varies depending on the expected securitisation structure and on the relevant phase of the deal itself (revolving period vs. amortization period). Figure 7.5 represents the typical cash flow waterfall structure which determines the priority of payments in respect of principal and interest.

The cash flow waterfall structure is based on sequential payments which usually pursue the following hierarchy of payments: (i) costs and commissions due to the main counterparties of the securitisation (e.g. servicing fees, swap fees, legal expenses, also taxes etc.); (ii) interest on senior tranches (principal repayments during amortization period); (iii) interest on mezzanine tranches (principal repayments during amortization period), (iv) cash collateral account (if envisaged in the structure); (v) purchase of new assets (during the revolving period), (vi) interest on equity tranche (principal repayment during amortization period).

Figures 7.6 and 7.7 recall the example of Hermes XVI.

Figure 7.6 shows in details the different rating for each tranche, the WAL and the step up margin that increase with the riskiness of the tranche. Figure 7.7 shows the credit enhancement generated by the waterfall structure, which results in higher excess spread for each tranche. The credit enhancement is provided by the subordination structure and by the excess spread over the swap, which is equal to 0.35%, considering outstanding principle, note interest, servicing fees and senior expenses.

Hedges from Interest Rate and Currency Risks. Hedges from the possible interest rate and currency risks linked to the collateral pool are sometimes provided in the securitisation. The existence of an interest rate swap (fixed rate vs. floating rate) is aimed to cover the Investors from the risk of a reduction in the cash flows in case of increase of the interest rate. This can happen when the collateral pool has a certain portion of fixed rate assets whilst the ABS notes are usually floating rate notes (paying risk free rate and a credit margin).

The swap can be an important element for evaluating the overall risk of the securitisation: in fact the credit rating of the swap counterparty has to satisfy certain minimum rating criteria[9] and the securitisation has to cover possible replacement mechanisms in the case of a downgrade of the swap counterparty below the minimum rating allowed by the Prospectus of the deal. The existence of a cross

[9]Usually the swap counterparty has to be rated minimum single A but this can vary according to the different models of the Rating Agencies.

Fig. 7.4 Determinants of the weighted average life of a securitisation Deal

Revolving Period	**Amortization Period**
Interest amount	*Principal amount*
Cost and Commission	Principal on the senior tranches
Interest on senior tranches	Principal on the mezzanine tranches
Interest on mezzanine tranches	Principal on the equity tranche
Cash collateral amount (if contemplated)	
Purchase of new assets	
Interest on equity tranche	

Fig. 7.5 Priority of payments in respect of principal and interest

Note Class	Amount (€)	Ratings S&P Moody's Fitch	WAL	Payment Window	Expected Maturity	Legal Final Maturity	Coupon	Step-up Margin
Class A1 Euro Senior Notes	500,000,000	[AAA/Aaa/AAA]	[1.47] yrs	[05/07-05/10]	Febr-13	Nov-39	3mE + [•]	3m E + 33bp
Class A2 Euro Senior Notes	1,398,000,000	[AAA/Aaa/AAA]	[5.04] yrs	[05/10-08/12]	Febr-13	Nov-39	3mE + [•]	3m E + 38bp
Class B Euro Mezzanine Notes	16,000,000	[AA-/Aa1/AA]	[5.39] yrs	[08/12-08/12]	Febr-13	Nov-39	3mE + [•]	3m E + 60bp
Class C Euro Mezzanine Notes	54,000,000	[A+/Aa3/A]	[5.39] yrs	[08/12-08/12]	Febr-13	Nov-39	3mE + [•]	3m E + 110bp
Class D Euro Junior Notes	14,000,000	[BBB+/A3/BBB+]	[5.39] yrs	[08/12-08/12]	Febr-13	Nov-39	3mE + [•]	3m E + 180bp
Class E Euro Subordinated Notes	18,000,000	[-/Ba2/BB]	[5.39] yrs	[08/12-08/12]	Febr-13	Nov-39	3mE + [•]	3m E + 325bp

Fig. 7.6 Credit enhancement: excess spread, subordination: Hermes XVI securitisation

Credit Enhancement		
Note Class	Credit Enhancement	Annual excess spread
Class A Euro Senior Notes	5.1%	[0.35%]
Class B Euro Mezzanine Notes	4.3%	[0.35%]
Class C Euro Mezzanine Notes	1.6%	[0.35%]
Class D Euro Junior Notes	0.9%	[0.35%]
Class E Euro Subordinated Notes		[0.35%]

Fig. 7.7 Credit enhancement: excess spread and subordination

currency swaps is meant to protect the investor from currency risks when the collateral pool is denominated in different currencies.[10]

[10]Other considerations on the structure of the securitisation are the legal soundness of the structure: evaluation of the efficiency of mechanisms such as Interest Coverage test and early amortization triggers events for the benefit of the investor; the experience of the Originator Bank and of the Servicer in relation to the specific underlying asset class. If the junior tranche is retained by the Originator Bank this can be a sign of alignment of interest between the Originator Bank and Investors of both the mezzanine and senior tranches.

7.2 Market Overview of the Cash Securitisation

The beginning of modern securitisation is generally identified in 1970, when a Government National Mortgage Association (Ginnie Mae or GNMA) sold the first Mortgage Backed Securities. Before that, Investors used to trade whole loan or un-securitised mortgages, illiquid instruments and risky for Investors, due to the interest rate's exposure and costly for the paper procedures. The strong desire for home ownership, and lack of funding, created a fertile environment for securitisation. With the process of securitisation indeed, Investors had access to more liquid instruments while at the same time lenders could move risks associated with these particular assets off balance sheet. As such securitisation was considered as an effective process to channel investment capital from global Investors to provide access to capital for affordable housing. The process was facilitate with the creation of other government-sponsored entities such as Freddie Mac and Fannie Mae and a supportive regulation environment with REMIC legislation and revised SEC rules. Money market Investors such as pension funds and insurance companies as well as a growing consumer culture, fuelled the expansion of securitisation. The securitisation process started to include assets—with the creation of the first Asset-Backed Security in 1985, such as auto-related receivables, home equity loans, and basically anything with a predictable cash flow stream (American Securitisation Forum 2003). In 2001, Ginnie Mae, Freddie Mac and Fannie Mae securitised 48% of outstanding mortgages, from 1% in 1965, but soon new players started to penetrate the market and in 2007 more than $2.2 trillion of home mortgages were securitised by non-agency entities (Barth et al. 2008). Between 2001 and 2008 the average annual securitisation issuance in European was €374billion (Rützel 2016).

The issuance of securitisation products has dramatically decreased after the financial crisis in 2009 after having reached its historical peak in 2008. After 2007, almost two-thirds of the new transactions have been retained for ECB funding purposes. According to SIFMA data, in the first half of 2016, retained transaction rose to 98billion euro, more than double the 40billion in 1H 2015. UK, Netherlands, Spain and Italy together represent 76% of the market. Notably, Germany, which used to have more than 25% market share in 2015, represents only 6% in the first half of 2016.

Figures 7.8 and 7.9 give a general overview of the securitisation market in terms of collateral and countries.

Figure 7.10 shows the breakdown of MBS issuance in Europe. RMBS has always represented the majority % of the transaction. In 2015, RMBS were 17.11 times CMBS. Since 2012, the "mixed" part of the MBS has disappeared (i.e. the combination of RMBS and CMBS).

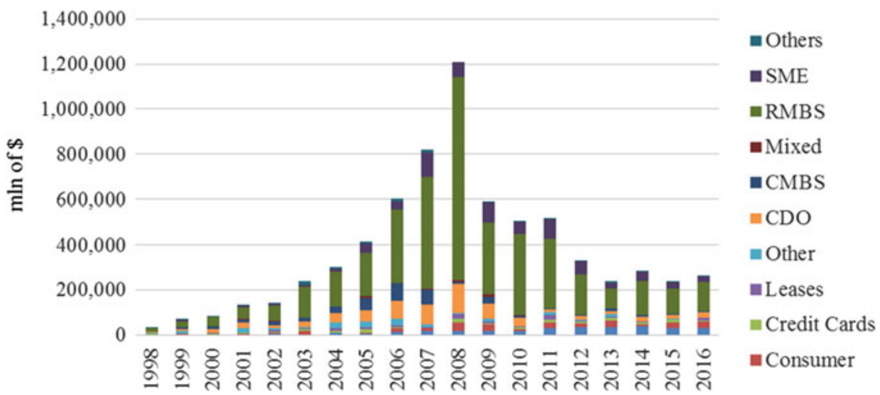


Fig. 7.8 EMEA securitisation Issuance (SIFMA: Europe Structured Finance Issuance and Outstanding 2017)

Asset-Backed Securities volumes have not followed the same path of Mortgage-Backed Securities during the period of time taken in consideration. Although auto and consumer loans seem to be the most active asset classes, there are not clear trends in volumes for the others asset classes. Figure 7.11 identifies the evolution of ABS in terms of underlying.

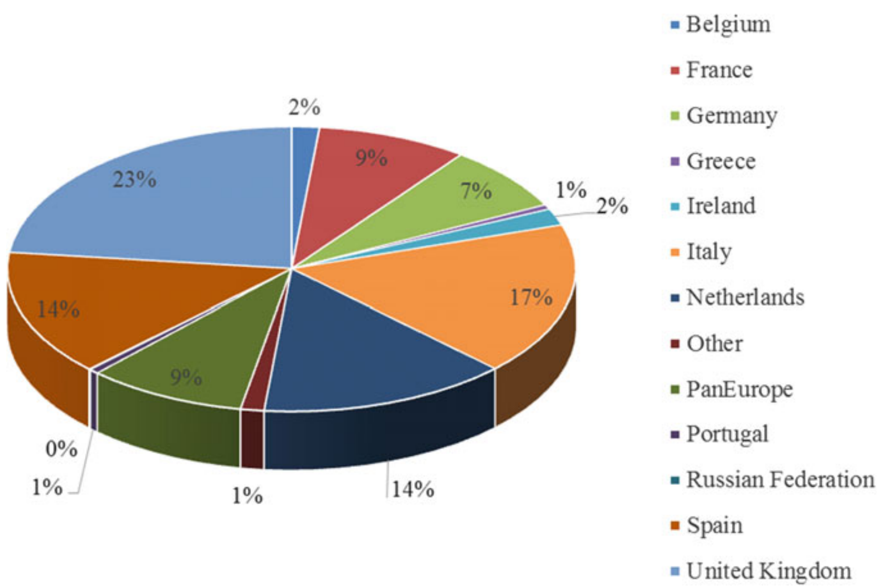


Fig. 7.9 Security transactions by country in % (SIFMA: Europe Structured Finance Issuance and Outstanding 2017)

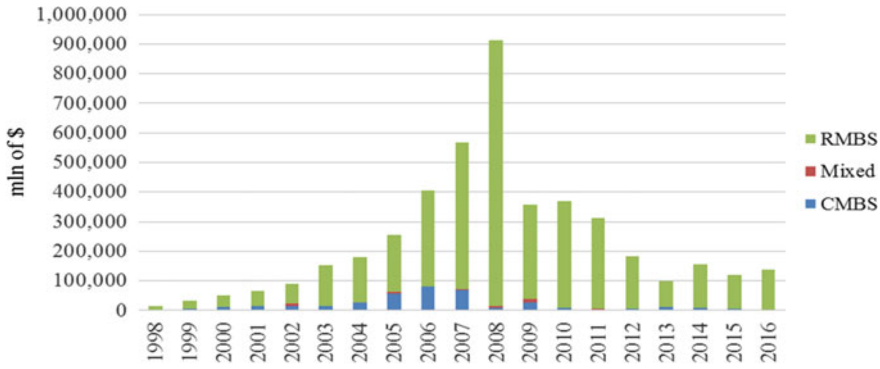


Fig. 7.10 MBS transactions by underlying assets (SIFMA: Europe Structured Finance Issuance and Outstanding 2017)

7.2.1 *The Role of the European Central Bank*

The European Central Bank ("ECB") has become a fundamental player in the securitisation market. As discussed the main important objective of the securitisation is to reduce funding costs.

Originator Banks can refinance themselves through reporting to the ECB highly rated RMBS and some specific categories of Asset-Backed Securities according to given eligibility criteria. This has been one of the measures set up by ECB to provide the banking system with liquidity. Instead of borrowing directly from the Investors by placing ABS to them, the Originator Banks obtain liquidity by entering repo deals with ECB at a less expensive cost compared to other possible financing alternatives. Originator Banks then structure securitisation deals to retain them and then place them to the ECB for financing: these are known as "Retained securitisation deals". Figure 7.12 shows how important the volume of Retained securitisation deals has become after the 2007 "credit crunch" phase.

The ECB provides funding to the banking system through repurchase agreements ("repos")[11] done with European banks under the ECB "Open Market Operations". In recent years, there has been an increase in volumes of Asset-Backed

[11]Repo is an abbreviation of "repurchase agreement", which is a contract where one party (the seller) sells an asset to another party (the buyer) at a specific price and commits to repurchase the assets at a different price (on demand or at a date in the future). The difference between the price paid at the start of the transaction and the price the buyer receives at the end of the transaction is the return (usually quoted in percent percentage per annum) on the cash that the buyer effectively lent to the seller and it is called "repo rate". If the seller defaults, the buyer can use the asset as collateral and offset the losses.

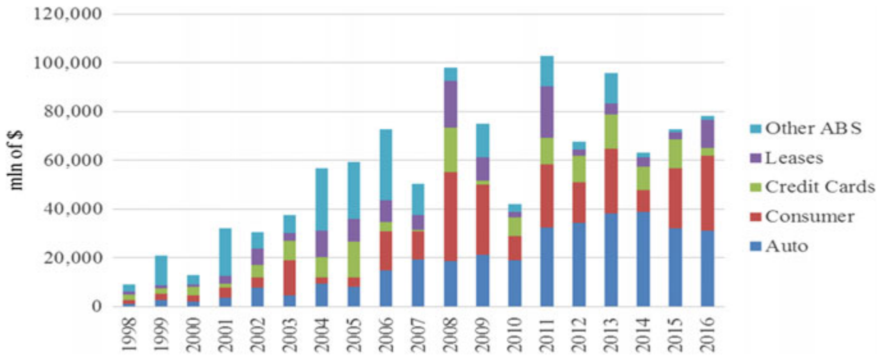


Fig. 7.11 ABS transaction by underlying assets (SIFMA: Europe Structured Finance Issuance and Outstanding 2017)

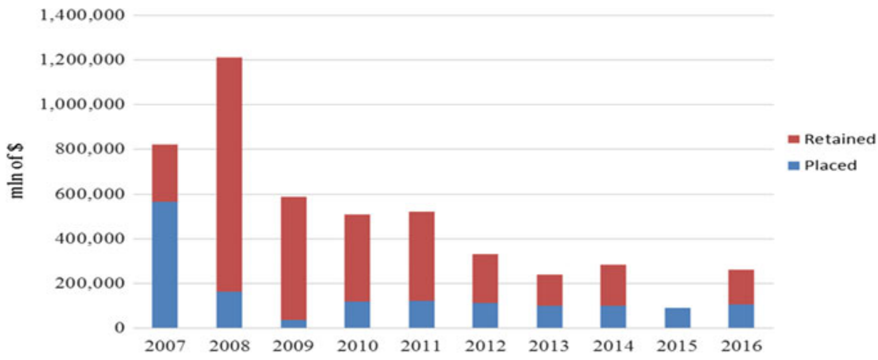


Fig. 7.12 Retained versus issued securitisation deals in Europe (SIFMA: Europe Structured Finance Issuance and Outstanding 2017)

Securities being financed with ECB. Open Market Operations run by ECB are usually undertaken either as “main refinancing operations” or “longer-term refinancing operations”. Refinancing maturities range from 1 week to 3 months. These operations are performed at a one-week horizon (Main refinancing operations) and at a three-month horizon (Longer-term refinancing operations). They are effectively repurchase agreements, based on eligible assets used as collateral. The main goals are to steer short term interest rates, manage liquidity and to signal the monetary policy as well as providing some financing to the financial sector through the Long Term Refinancing Operations (“LTRO”) at a longer term.

Eligibility criteria for assets to be considered eligible under ECB rules are both generic for bonds and specific for Asset-Backed Securities.

Among the generic criteria, valid for all the securities we find:

- Coupon structure (with Guidelines ECB/2016/31, the Eurosystem accepts also those coupon that result in negative cash flows, to reflect the current environment of low/negative interest rates)
- Non subordinations with respect to marketable assets
  - Euro currency, Issued in EEA by Central Banks or Security Settlement System.

Specific criteria for non-financial issuers:

- Settlement: debt instruments shall be transferable in book-entry form through an account with an NCB or with an SSS that respect ECB standards. Instruments should be held and settled in Member States whose currency is euro
- Admitted to trading on regulated market or a certain acceptable non-regulated market.

Specific criteria related to Asset-Backed Securities have been added such as:

- Homogeneity of the cash flows which should be linked to one of the existing loan-level templates as set out by the Eurosystem. Assessment should be needed in case of not homogeneity. No cash flow generating assets originated by the same ABS's SPV
- Non subordination of tranches
- Assets generating cash flows must no consists of tranches of other ABS, Credit-Linked Notes or similar claims, including structured or leveraged loans
- SPV should be established in the EEA as well as the originator of the assets or the intermediary
- Assessment of the claw-back rules
- Loan Level data should be made available, with respect to particular procedures, and should be standardized and comprehensive
- All ABS must have at least 2 single 'A' credit ratings from any accepted external credit assessment institution ("ECAI") for the issue
- Haircuts: under the new directive ECB/2016/32, haircut of ABS (category V) from AAA to A—changes according to the weighted average life. On ECB/2016/33 is outlined the new haircut scale, on the base of the WAL, for ABS below A—(in the previous guideline, the haircut was set at 22%).

The ECB is also carrying out the so-called ABS Purchasing Program ("ABSPP") in the secondary market aimed to the outright purchase of Asset-Backed Securities. On September 4th 2014, the Governing Council of the European Central Bank announced the so-called ABSPP—Asset Backed Securities Purchase Program—which officially started in November 21st 2014.

The official goal is to enhance the transmission of monetary policy, facilitate credit provision to the euro area economy, generate positive spill-overs to other markets and, as a result, ease the ECB's monetary policy stance, and to contribute to a return of inflation rates to levels closer to 2% (Decision ECB/2014/45). Through this program, ECB instructs its agents to purchase eligible ABS, both in the primary and in the secondary market, from eligible counterparties. Since 2016, the ABS purchase is now performed through National Central banks only. The Purchase program is limited in terms of eligible instruments and size. The maximum purchase of ABS for ISIN is limited to 70% while there is no maximum or minimum maturity and issue size of the securities (a different limit has been set for Greece or Cyprus). The official criteria and limits are then followed by a set of guidelines (which are not intended to substitute the eligibility criteria as set out in the Decision ECB/2014/45) which is non-binding and non-exhaustive.

As for December 2016, the ECB reported €22,830 mil ABSPP holdings, more than 60% purchase in the secondary market.

There are a set of criteria that ABS should respect to be considered eligible under the ABSPP:

- Minimum rating as provided by at least two External Credit Assessment Institutions (ECAIs) accepted within the Eurosystem Credit Assessment Framework (ECAF). ABS are required to have a minimum second best rating of single A. However, the "temporary framework" permits the eligibility of some BBB ABS subjected to some condition (ex. performing of the loans at the time of the purchase)
- Only senior and mezzanine tranches. Eligibility of mezzanine tranches has been granted with Guideline (EU) 2015/510 (ECB/2014/60), subjected to different criteria (secured by a guarantee, credit assessment of the guarantor etc.) other than the ones outlined in the general framework. Equity tranches are not eligible for the ABSPP
- Eligible under the collateral framework
- Currency: denominated in euro. Residence of the issuer should be within euro area
- Be secured by claims in the euro area.

7.3 Overview of the Non-performing Loans Market in Europe with a Focus on Italy's Case

It is important to begin this section with some definitions of non-performing loans ("NPLs"). Non Performing Exposures are defined by the European Banking Authority ("EBA") either or both the following two criteria:

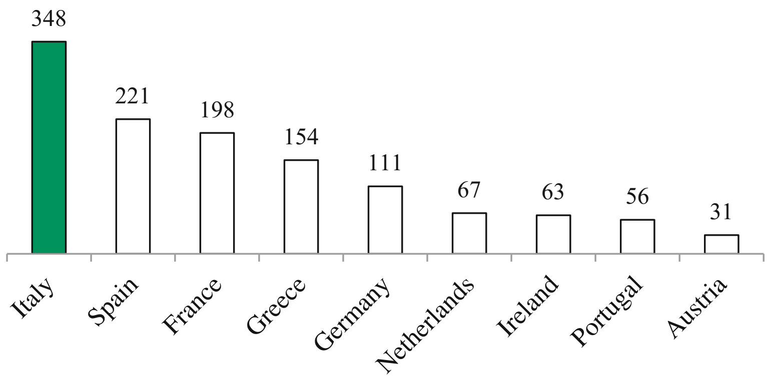


Fig. 7.13 NPLs volumes: european comparison (European Central Bank, Supervisory Banking Statistics, Third quarter 2016. GBV data in EUR bn)

- Material exposures which are more than 90 days past-due ("past-due" criterion)[12];
- The debtor is assessed as unlikely to pay its credit obligations in full without realization of collateral, regardless of the existence of any past-due amount or of the number of days past due ("unlikely to pay" criterion).[13]

Looking at the stock of non-performing loans in Europe (Fig. 7.13) it can be recognized that Italy has the most striking stock compared to the other European countries (Fig. 7.14). The growth of non-performing loans in Italy has been significantly higher compared to any other European country. Italian stock of non-performing loans is equal to around EUR 360billion: it is indeed the largest in Europe and a legacy of the long recession.

Since 2008 gross total non-performing loans increased from EUR 87 billion to EUR 361 billion; in particular gross Bad Loans increased from EUR 42 billion to EUR 207 billion. They are slightly slowing down: in 2015 the loan defaults rate—i.e. the flow of new securitisation of non-performing loans-decreased markedly (3.3% in 4Q15 which is the lowest since 2008). The classification of the non-performing loans into the following different categories helps in understanding better the Italian case: to this extent the next picture shows the comparison between Bad Loans (the most critical ones: "Sofferenze") versus Doubtful loans (for which the borrower's difficulties may be temporary) and Past due loans. Figure 7.15

[12]The past-due criterion is recognized only if the payment was compulsory and there was a legal obligation—as such discretionary payments, such as Additional Tier 1 coupon payments should not be considered NPE in the non-payment event.

[13]The "unlikely to pay" criterion is based on more qualitative criteria. Paragraph 145(b) of Annex V of Commission Implementing Regulation (EU) No 680/2014, set some defined situation that triggers the "unlikely to pay" criterion, such as bankruptcy of debtor. Banks are left some space to internally define some unlikely to pay linked situation, which should regularly be assessed. Further, in the context of an unlikely to pay situation, the exposure should be considered as non-performing even on the presence of full collateral.

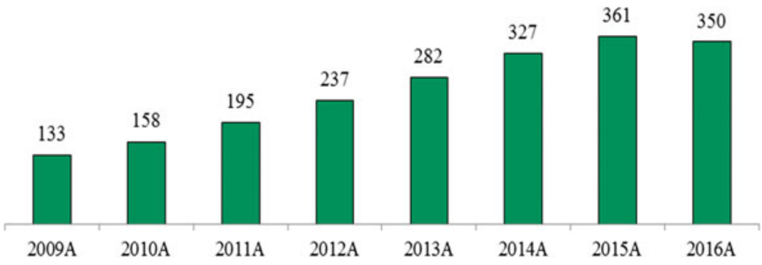


Fig. 7.14 NPLs trend in the Italian banking system (Bank of Italy—Financial Stability Report No. 1/2016. GBV data in EUR bn)

shows the Bad loans rate trend compared to the NPL's one. NPL definition includes also Doubtful loans and Past due loans.

In Italy the total stock of gross NPLs amounts to a Gross Book Value of €361bn a legacy of a long recession phase (corresponding Net Book Value is EUR 197bn): in terms of Bad Loans (the most critical ones) the GBV amounts to EUR 210bn (corresponding to a Net Book Value of EUR 85bn).[14] They represent the 10.4% of the total outstanding exposures in the Italian banking system,[15] while their Net Book Value the 4.3% of total loans. Using the Italian case it is possible to analyse in details the main features of the non-performing loans as an asset class. The impact of the overall situation in Italy is even clearer when the banking system is analysed. Italian banks have the largest amount of non-performing loans in Europe with about EUR 360bn of NPLs of which bad loans represent around EUR 200bn when marked at the average value of 43 cents. Assuming an average market price for bad loans of 27 cents,[16] top 10 Italian banks would need to book an additional EUR 20bn of provisions. Assuming an average market price for other non-performing loans of 60 cents, the top 10 Italian banks would need to book an additional €8bn of provisions (2016, Q2 16 figures, Company reports, see Fig. 7.16).

Figure 7.17 shows the evolution of Italian bank credit default swap ("CDS") spreads starting from December 2015, with the resolution of the small four banks. Monte dei Paschi CDS went as up as 700 bps in February. From then, Italian CDS started to tighten, spiking up with EBA Stress test results in June. The positive comments on a no—impact on Monte dei Paschi senior bondholder droves spread lower again, regardless the failure of the market solution of the world's oldest bank just before end of 2016.

Notwithstanding the importance in terms of volume of the NPL market, the number of NPLs portfolio sales finalised in Italy is significantly lower than the one done out of the rest of Europe: the trend is growing though (see Figs. 7.18 and 7.19).

[14]For definitions of Gross Book Value and Net Book Value see Sect. 7.4.3.

[15]Breakdown by type of debtor is 7.1% for individuals and 17.7% for corporates.

[16]This is just an estimated value based on press news related to Monte dei Paschi possible sale of non-performing loans portfolios.

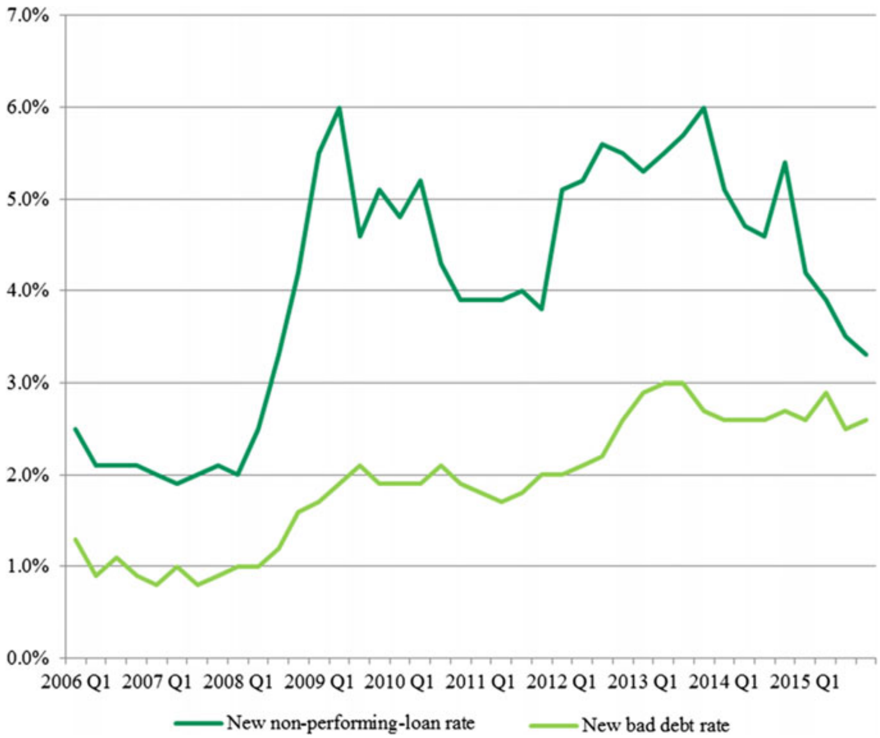


Fig. 7.15 New NPLs rate and new bad loans rate in Italy: a comparison (Bank of Italy, Financial Stability Report No. 1/2016)

Name	Gross NPLs	NPL ratio	Loan loss reserves	Sofferenze (gross)	Sofferenze mark	Other NPLs (gross)	Other NPLs mark	Last CET1r (%)	ADD LOSSES: Sofferenze at 27c	ADD LOSSES: Other NPLs at 60c	TOTAL ADDITIONAL LOSSES	Target CET1r	CAPITAL SHORTFALL	Target CET1r	CAPITAL SHORTFALL
ISPIM (Italy)	58,130	16.8%	-28,902	36,569	39c	21,561	70c	12.80%	-4,294	-2,123	-6,418	10.00%	0	11.50%	-2,699
UCGIM (Italy)	***60,300***	***28.0%***	***-31,200***	***41,300***	***39c***	***17,400***	***67c***	***11.00%***	***-5,149***	***-1,160***	***-6,309***	***10.00%***	***-2,419***	***11.50%***	***-8,283***
UBIIM	13,231	15.2%	-4,898	7,491	52c	5,740	77c	11.68%	-1,890	-976	-2,866	10.00%	-1,854	11.50%	-2,760
BACRED	***2,139***	***4.8%***	***-1,122***	***623***	***41c***	***1,516***	***50c***	***12.10%***	***-87***	***148***	***61***	***10.00%***	***0***	***11.50%***	***0***
MONTE	45,584	35.7%	-23,075	28,230	39c	17,354	67c	11.50%	-3,281	-1,194	-4,475	10.00%	-3,455	11.50%	-4,479
BPIM+PMIIM	***26,585***	***21.7%***	***-10,063***	***14,660***	***55c***	***11,924***	***75c***	***13.27%***	***-4,088***	***-1,760***	***-5,847***	***10.00%***	***-3,331***	***11.50%***	***-4,484***
VICEN	9,394	33.9%	-4,342	4,612	41c	4,783	66c	10.75%	-635	-303	-938	10.00%	-766	11.50%	-1,109
VENBAN	***7,891***	***32.6%***	***-3,033***	***3,867***	***47c***	***4,024***	***75c***	***10.74%***	***-776***	***-624***	***-1,400***	***10.00%***	***-1,242***	***11.50%***	***-1,563***
BANCAR	7,107	20.1%	-3,259	3,685	39c	3,422	71c	12.30%	-437	-362	-800	10.00%	-397	11.50%	-663
TOTAL (excl. ISPIM/UCGIM)	123,146	28.5%	-53,835	77,828	47c	60,687	71c						-14,375		-19,542
TOTAL (incl. ISPIM/UCGIM)	256,946	24.9%	-119,957	155,697	43c	99,648	70c						-16,794		-30,524

Fig. 7.16 Italian banks and NPL

Coverage ratios improved since 2012 from 37.37% in June 2012 to 47% in Q2 2016 for NPLs and from 54.46% in June 2012 to 57% in Q2 2016 for Bad Loans: these ratios are now generally aligned with the European average (see Fig. 7.20).

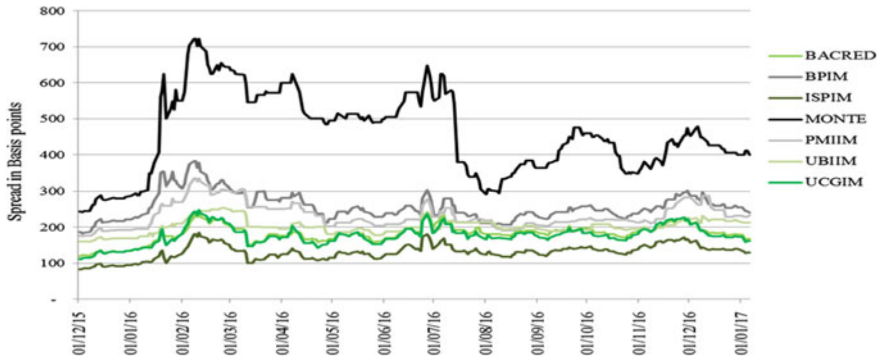


Fig. 7.17 Credit default swap spreads levels for Italian banks

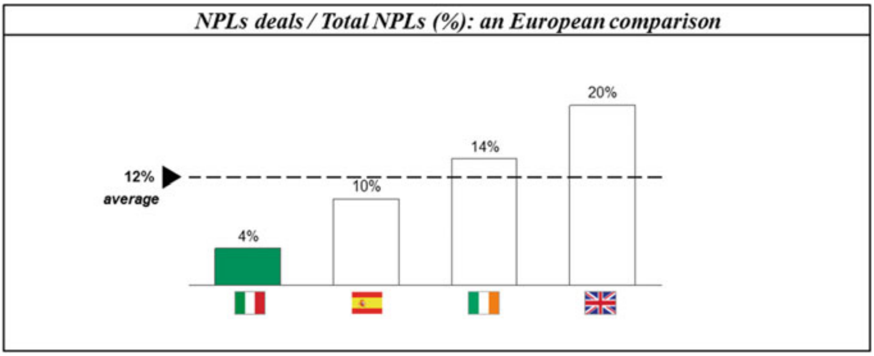


Fig. 7.18 NPLs deals/total NPLs (%): a European comparison

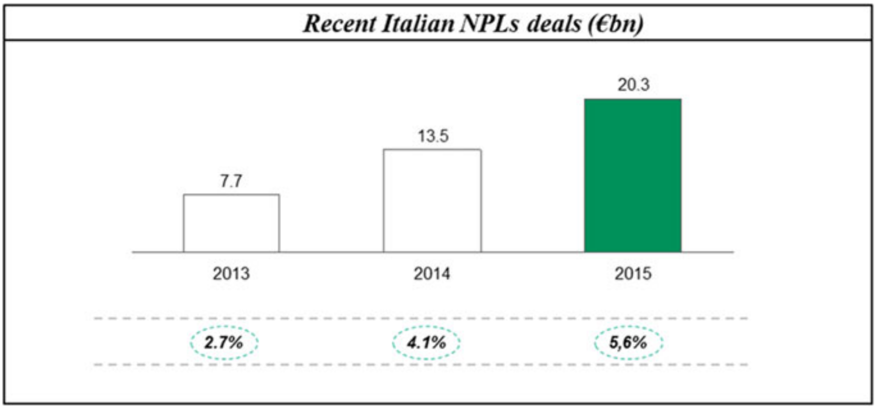


Fig. 7.19 Recent Italian NPLs deals (EUR bn)

Non-performing Loans		
Country (Q2 2016)	Amount	Coverage ratio
Germany	75.19	*37.82%*
Ireland	40.05	*37.42%*
Greece	116.35	*47.78%*
Spain	148.15	*44.31%*
France	164.89	*47.36%*
Netherlands	46.22	*35.50%*
Austria	22.75	*54.84%*
Portugal	40.61	*39.45%*
Total	**654.22**	***43.97%***

Italy	Gross book value	% tot loans	Net Book value	% tot loans	Coverage
NPLs *of which*	**350**	**21.4%**	**164.5**	**10.1%**	**47 c**
Bad loans	**200**	**12.2%**	**114**	**7.0%**	**57 c**
Doubtful and past due	150	9.2%	45	2.8%	30 c

Fig. 7.20 Coverage ratio: a European comparison and Italy breakdown

7.3.1 Breakdown of the NPL Asset Class

Non-performing loans are mainly classified by the type of debtor and by the type of guarantee (collateral) which has been offered to the lender to mitigate the de-fault risk of the debtor. In terms of debtors types the main two macro categories are Re-tail and Corporate. In the Retail category it is possible to further classify the non-performing loans into Consumer and Residential. As to the Corporate category it is possible to analyse further into two sub-categories which are Large Corporates and Small Medium Enterprises ("SME"). The underlying debtor category can be further divide into subsectors such as Manufacturing; Services; Commerce and also into Real Estate (mainly Construction and Services).

Figure 7.21 shows the breakdown for Italian non-performing loans by type of underlying debtor and type of guarantee.

In terms of collateral the main two classifications are related to Unsecured loans (loans with no guarantee attached to them) and Secured loans. It is possible then to further breakdown the Secured non-performing loans based on the type of collateral attached to the loan itself.

Figure 7.22 shows the non-performing loans composition by debtor type.

	GBV	NBV	Mortgage and other guarantee	Personal guarantee	Coverage ratio %	Coverage ratio for unsecured %
			CORPORATES			
Npls	250.0	136.0	119.0	49.0	45.5	57.7
of which Bad Loans	144.0	58.0	62.0	35.0	59.7	74.5
			HOUSEHOLDS			
Npls	54.0	32.0	36.0	2.0	41.3	66.0
of which Bad Loans	34.0	17.0	22.0	1.0	52.0	76.5
			TOTAL			
Npls	317.0	175.0	160.0	52.0	44.7	58.6
of which Bad Loans	184.0	76.0	85.0	37.0	58.5	74.8

Fig. 7.21 Breakdown by Type of debtors and guarantee (Bank of Italy, Statistical bulletin I-2016, 2016a and Financial Stability Report No. 1/2016, 2016a)

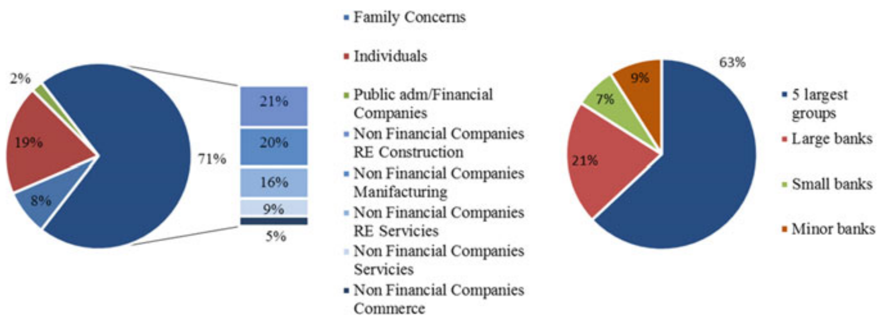


Fig. 7.22 Non-performing loans and guarantees by debtor type (EUR bn) as of December 2015 (data is based on consolidated balance sheet; fair value of the guarantee capped at debt amount. Total includes also public administration, insurance and financial companies and no-profit organizations) (Bank of Italy, Financial Stability Report No. 1/2016, 2016a)

In terms of secured non-performing loans Italian banks have guarantees for EUR 160bn, which represents around 50% of total gross NPLs; for the bad loans the guarantees are equal to Eur 85bn, above the Net Book Value of the total bad loans (Eur 76bn). About 80% of the overall NPLs volume refers to corporate loans. The residual part is composed of retail loans—either consumer or residential mortgages. Retail cluster mainly refers to secured loans: 67% of the total volume consists of residential mortgages. Corporate loans are split among Commercial Real Estate and Unsecured exposures.

7.3.2 ECB Role in the Non-performing Loans Market

In July 2015, a group of high level figures has been mandated by the Supervisory Board of the ECB to outline a consultation paper on NPLs, i.e. a public guidance, addressed to banks, which represents a first step towards a consistent and common supervisory approach to NPLs. This project is the first answer to the needs of more provision for NPLs that started to arise in 2014, when, with the comprehensive test, ECB assessed for the first time the quality of banks' assets using the same methodology. In the paper, which being in the consultation phase, is non-binding, ECB banking supervision has identified some best practices, which banks are expected to follow and to implement, according to the degree of NPL issue. The structure of the paper is defined to cover all the relevant aspects of the life cycle of NPLs management, including the definition of correct strategy, governance, operation and accounting valuation. A clear strategy, for the correct management of the issue, should be defined in accordance to the operating environment, the internal capabilities and the external environment and the ECB proposed different implementation options for medium and long terms. The strategy should also define some quantitative target levels for NPLs, not only on general basis, but also on portfolio one. Management team should monitor and review the strategy and setup sufficient internal controls and procedures. The paper suggests the creation of separated NPL units, in order to eliminate potential conflicts of interest with loan creation units. These units should also be tailored to the different portfolio specificities. The paper, other than focusing on how to manage NPLs, also describes some best practices that can be used to prevent performing borrower to become non-performing, i.e. the forbearance measures. Different measures, short and long term, can be used in combination according to banks' specific needs. Part of the focus of the guideline is also related to accounting valuation of NPLs such as impairment measurement, write offs and collateral valuation for immovable properties. Beside the paper, ECB conducted a qualitative stocktake of the best practices adopted by eight member countries on NPLs. These best practices have been used as a source for some of the guidelines included in the paper and to underline the different legislation policies. According to the stocktake, some countries have actively implemented new strategies and practice in terms of management and reviews but still there is room to enhance internal expertise and supervisory policies for NPLs measurement and recognition. From the legal point of view, some in-scope countries have adopted many reforms, but not always the legal and judicial system are found to be on the same page in terms of amendments. Regarding the information framework, while most of the countries included in the study already have good supervisory tools in place, which represent a valuable source of information, for others the access to this kind of source is still limited.

7.4 Non-performing Loans Securitisation

Non-performing loans securitisation has become one of the major topics in Italy when Bank of Italy agreed with the European Commission a Guaranteed Scheme which facilitates the disposal of NPL via a securitisation process, providing a state guarantee for the senior notes issued. This Scheme has re-opened the door to the NPL securitisation market which still involves only few Investors in an already niche market. The main difference between the NPL securitisation and the standard one, relies on the non-predictability nature of the cash flows of the assets which are indeed unstable and more difficult to predict.

The very first securitisation on NPL was structured in 1989 in USA, by the Resolution Trust Corporation which bought bad commercial mortgages of savings and loans association at deep discount on par, funding the acquisition through the sale of government-guaranteed bonds and treasury. Overall, the Resolution Trust Corporation bought USD 460billion in assets and managed to recover almost 90% through structured operation (Kothari 2006a, b). The securitisation market for these miscellaneous assets started to develop also in Asia, where NPL represented a major issue for banks. Korea Asset Management Company was the first NPL securitisation deal in Korea, a country that was heavily hit by currency crisis in 1997. At the time, when NPL were at 20/30% level, the government was forced to mandate the Korea Asset Management Company to acquire NPL from the banks backed by bond issuances.

In Europe, NPL securitisation has been widely used especially in Italy before 2007 (begin of the “credit crunch”) and marginally in Germany, where the first transaction was finalised in 2006. Recently, the process has gained more momentum also for the moral suasion exercised by the ECB (see Sect. 7.3.2. on ECB role).

7.4.1 Overview of the Structure

There are few key differences between the standard structure of the securitisation on performing assets and the securitisation of non-performing loans. True sale and bankruptcy remoteness remain the pillars of both types of securitisations though.

The structure of the non-performing loans securitisation is similar to the standard cash securitisation one in its main terms (see Fig. 7.23). The main differences between the securitisation of performing loans and the one of non-performing loans are related to the lower degree of homogeneity of the underlying pool of non-performing loans (especially in terms of continuity of cash flows), the counterparties involved in the non-performing loans securitisation, the credit enhancement mechanism (“natural overcollateralisation”) and the importance of external variables such as the state intervention and the judicial system of the country where the underlying pool of non-performing loans is originated.

As discussed, one of the main differences between the two structures is related to the fact that the non-performing loans securitisation always implies a certain degree of overcollateralisation due to the fact the Gross Book Value (“GBV”) of the portfolio which is sold to the SPV via a true sale is always higher than the notional of the liabilities of the SPV i.e. the ABS placed to the Investors.[17]

7.4.2 Main Counterparties of the Securitisation of Non-performing Loans: The Key Role of the Special Servicer

In non-performing loans securitisations, one of the key counterparty is the Special Servicer.[18] The role of the Special Servicer is to assist and support the Originator

[17]See also Sect. 7.4.3.

[18]We focus on the role of the Special Servicer which has mainly an operating function. Master Servicer is the counterparty which is usually involved in the fulfillment of regulatory tasks and undertakes a guarantee role of proper representation and information toward the main stakeholders of the non-performing loans deal and toward the market.

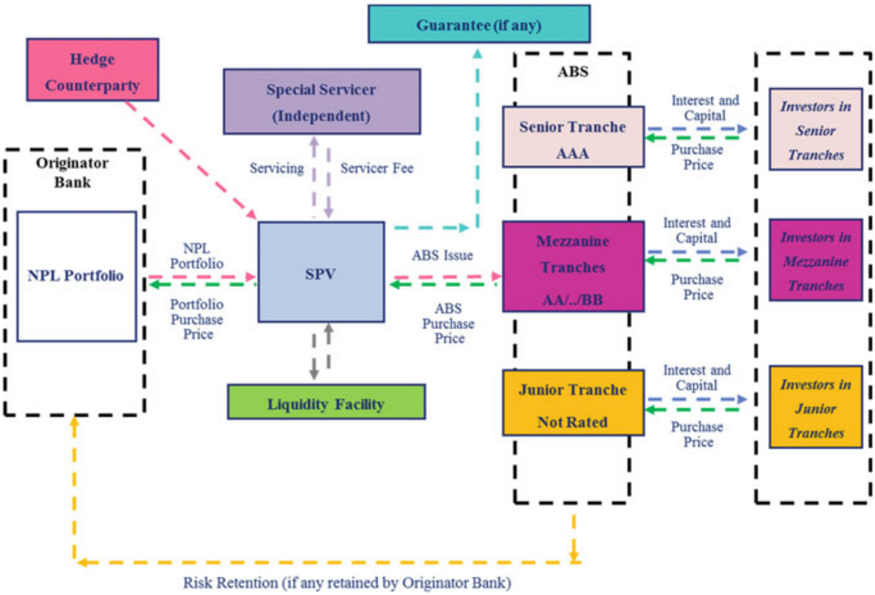
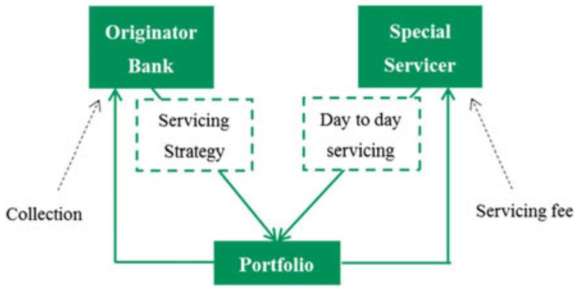


Fig. 7.23 Structure of a non-performing loans securitisation

Fig. 7.24 The servicer role



Bank in the management of the non-performing loans and their cash flows (e.g. collections) and in their recovery process (Fig. 7.24).

An effective servicing platform for the NPL portfolio is essential for the success of the program. Servicers are effective in the recovery process and able to provide high quality reporting on their activity. Having an independent servicer in the non-performing loans securitisation deal is perceived as a strong point for the Investors compared to the case where Originator Bank manages the deal via their specialized servicing unit. The importance of the Special Servicer is related—amongst the others—to the following factors-: experience in working-out the non-performing loans including due diligence; effectiveness of the business plan; recovery strategies; loan management; legal network; reporting and IT (also for data quality purposes). In its loan management role the Special Servicer manages on behalf of the bank (or together with the Originator Bank) the non-performing loans portfolios: the key businesses in this role are the due diligence and the collateral evaluation.

The quality of the Servicer is usually evaluated based on the following variables:

- Management experience and company history: when the company started the activity; experience on the business of the key internal personnel; internal organization including different department specialization; corporate plans achievements
- Financial condition: net earnings during time; source of earnings (e.g. transaction-based; due diligence role; etc.)
- Staffing; number of headcounts; distribution of headcounts (e.g. loan management, legal department, risk management, compliance and auditing departments); training programs
- Specialization according to the type of underlying portfolios (e.g. secured vs. unsecured; granular vs. lumpy portfolios; Small-tickets; Real Estate etc.)
- Procedures and Policies: e.g. existence of committees frequently held for risk management policies; compliance; audit; anti-money laundering
- Loan Administration: new loan set up and loan accounting and cash management
- Reporting: quality of reporting before external distribution; degree of accuracy; timeliness of the distribution; access to reporting system
- Technology: this is one of the key aspects in the Servicing role since it has a strong impact on the degree of quality of the data. Capacity of the servers, possible back-up disaster recovery tests and contingency plans are strictly evaluated. The technology system can also include possibility for clients to run their own portfolio simulations and monitor the assets performances in an user-friendly way.

Servicers which are independent from the Originator Bank can bring up-to-date recovery technology and would be able to treat all portfolios on a similar ground. Since the Originator Bank servicing unit has usually a valuable direct historical knowledge of their non-performing loans, it is also reasonable to have some form of joint ventures which can make more effective the recovery process.

Special Servicers can be effective in managing the non-performing loans directly or on behalf of the Originator Bank. In this loan management role they can offer different services which include the due diligence and collateral evaluation, the set-up of an efficient database to create a watch list and to identify past due loans. Usually a specialized unit is dedicated to the collections of past-due loans (e.g. reminders letters, in-bound phone collection, assistance to the debtor in selling the underlying property, formal final notice to the debtor etc.). One of the most important roles is the set-up of business plans aimed to maximize the recovery value of the non-performing loans (e.g. extra judicial settlements/private workouts, actions on assets etc.). The quality of the business plan and the track record of the Special Servicer in implementing business plans is one of the key elements in the choice of the Servicer. The loan manager can pave the way to the best recovery strategy. It is also responsible for the correct performance of the recovery flows: by doing so it can work effectively with its legal network (usually it is an external one). The breadth of the legal network is an important element to understand the width of the coverage of the courts scattered across the relevant country. The commercial network is relevant as well especially when the underlying is represented by a real estate portfolio they can be effective when the debtor wants to sell the property. The commercial network can assist in both judicial proceedings and out-of-court settlements. As to type of actions for out-of-court settlements some examples are the following: marketing of properties which the debtors could consider to sell; setting conditions for the release of the underlying collateral[19]; facilitating forms of lettings with the view that the income generated will be used to repay the debt; assisting clients during judicial auction sales. Marketing actions can include providing broker opinions, implementation of effective commercial strategies, approaching possible buyers and organizing visits to the properties, strategies to increase the value of the property and all the administrative and legal actions to assist in the property's sale.

The commercial network can cooperate and realize synergies with the legal network to make sure that the recovery activity is properly supported: they can meet with the debtor to check the actual financial situation, can work to try to find an agreement for the repayment plan or, if it is difficult to reach an agreement they can ask for the mandate to sell the underlying asset into the market. These actions are aimed to avoid entering into judicial recovery phase related to the default of the debtor. An important role can be played also during the judicial phase. In case of auctions, the Servicer can help find buyers during the auction process and the same objective can be achieved if a Real Estate Owned Company ("Reoco") is set up. Reoco are real estate companies owned and created by the Investors whose aim is to support the real estate collateral value by participating at auctions and creating a competitive tension between parties. They also try to increase the marketability of the collateral by renting the vacant property. The advantage of setting up a Reoco is that they can support the auction price in order to make it closer to the recovery value defined in the business plan and that can avoid an excessive depreciation of the collateral by accelerating the process of credit recovery via the resale on the real estate market.

The Servicer can play an important role also in assisting the Originator Bank in selling their non-performing loans portfolios providing the seller with data room, due diligence fact sheets and valuations to facilitate the sale process.[20] They can also play a role in setting up the optimal risk and accounting structure to minimize the pricing gap in favor of the Originator Bank (see Sect. 7.4.3).

[19]Examples are sale of full title or of bare interest with the right of usufruct, leaseback etc.

[20]The Servicer can also assist the Investors in the purchase of non-performing loans portfolios by offering advisory services such as structuring and management of the special purpose vehicles, due diligence on the buy-side, managing and servicing the loans also after purchase etc.

The decision about having a Special Servicer involved in the securitisation deal can be also related to the size of the internal specialized units of the Originator Bank. The Originator Bank could need to select external work-out teams from the Servicer to manage the different segments of the non-performing loans portfolios and to speed-up the recovery process also thanks to a more efficient process: the Servicing fee will be paid by the Originator Bank. The non-performing loans portfolio remains into the Bank's balance sheet which is still exposed to the deterioration of non-performing loans value. The servicing strategy is agreed between the special servicer and the Originator Bank which is still the owner of the non-performing loans. In some cases, the Originator Bank could decide to out-source the smallest positions and keep the management of the larger claims in order to reduce the pressure on their servicing units. At the same time, in order to keep complexity and servicing costs of the deal under control, a minimum threshold could be introduced for the on-boarding of the non-performing loans portfolio.

An ideal model of organization of work between the Special Servicer and the special servicing unit of the Originator Bank ("Workout Unit") could be reached by setting up Joint Ventures for the servicing activities dedicated to each specific asset class within the underlying securitised portfolio. Also the joint venture could be in place for the residual non-performing loans portfolio which remains in the balance sheet of the Originator Bank. This scheme could increase the efficiency in the recoveries and in the timing of collections related to the underlying portfolio and—at the same time—could reduce significantly some of the costs of servicing activity due to the synergies which could be realized between the work out group of the Originator Bank and the Special Servicer (see Fig. 7.25).

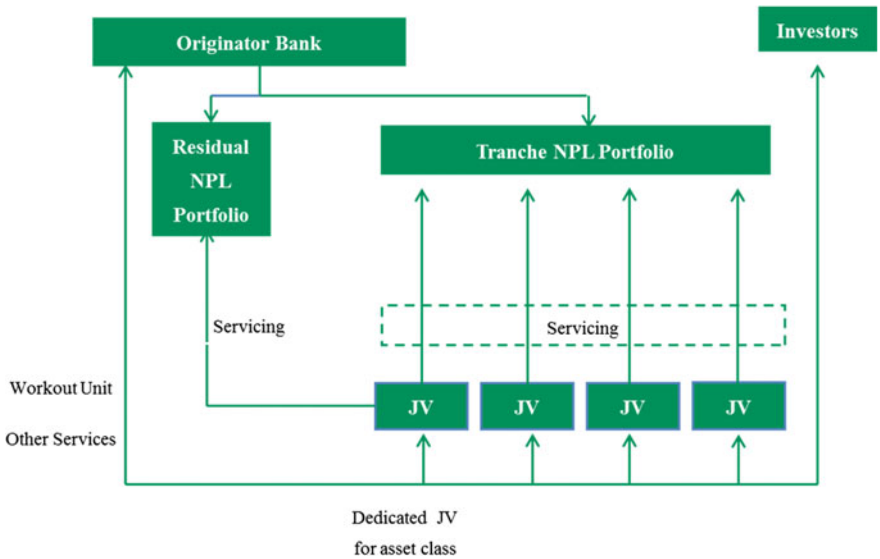


Fig. 7.25 A possible scheme of joint ventures for the servicing activity

7.4.2.1 The Role of Rating Agencies

The analysis of the Rating Agencies in terms of tranching models is not dissimilar from the one of the standard cash securitisation backed by performing assets: the amount of losses are estimated under different stress conditions: asset default probabilities and loss severities are combined into the analysis to estimate the overall loss of the underlying portfolio.[21] This analysis can take care also of the granularity or concentration of a given portfolio and may add the roll rate i.e. the rate at which delinquent loans roll into the default state. Once again underlying asset risk evaluation and soundness of the securitisation structure are the two main pillars in their assessment. The main differences with the cash securitisation backed by performing loans can be recap in the following points:

- Historical Data Analysis, Business Plan and Servicer Assessment: the historical data and the reliability of the business plan are an important factor in the Rating Agencies assessment, the Servicer historical performances, its expertise in the underlying asset class and in the recovery strategies for granular pool and for loans secured by properties, its workout experience and staff expertise and tenure are amongst the most important factor in evaluating the overall quality of the securitisation deal
- Focus on Property Evaluation: when the loans are secured by properties the nature of the property evaluation, the procedures related to this and the underlying assumptions are the main factors for evaluating the recoveries associated to these loans. In a non-performing loans securitisation more than on the loan-to-value the Rating Agencies rely on the notional amount of the recoveries to calculate the loss severities. Also—whilst in a standard securitisation—the focus is more on the market value decline of the property under certain stress scenarios in the non-performing loans securitisation the analysis is based more on the specificities of the given underlying portfolio
- Properties Standard and Recovery Procedures: the non-performing loans secured by properties have mainly lower standard than the ones of the standard securitisation. They are usually located in areas which are not appealing to potential buyers or their conditions are not up to standard (e.g. run-down, battered etc.) hence their recovery rate is lower than those of a portfolio of performing loans. Focus is here more on the stage of the recovery process: the more advanced the better it is since the predictability of the recoveries increases as the process advances
- Macroeconomic and Legal Factors linked to certain Geography. Geography is an important element of the risk assessment of the non-performing loans securitisation by the Rating Agencies. Under the stress scenarios simulations are also based on a slowdown of the economy of the geography where the non-performing loans portfolio is originated. The increase in stock of non-performing loans can be a legacy of long recession phase: if the recession

[21] See Sect. 7.1.5 for tranching models of the Rating Agencies.

persists the property prices can be under more pressure and the time for selling properties can be much longer so that it is harder to recovery collections. Geography matters also for legal factors: inefficiencies in the country judicial system can cause longer time to process bankruptcies and also longer time for the foreclosure process.

7.4.2.2 Other Relevant Counterparties

The non-performing loans securitisation can envisage also the presence of the Hedge Counterparty and of the Monitor Agent in addition to the counterparties which are part of the standard cash securitisation.[22] The Hedge Counterparty could provide hedging against the fluctuations of the interest rate which is usually the index to which the ABS securities are linked to (e.g. Euribor). The swap counterparty typically enters into an interest rate cap as cap guarantor: the hedge is aimed to mitigate the risk that an increase in the interest rate would not be matched by an increase in the collections from the underlying assets pool which are not directly linked to a floating interest rate. Usually the notional of the interest rate cap is at least equal to the notional of the ABS senior notes. The securitisation can also envisage the presence of a Monitoring Agent. The role of the monitoring agent is mainly to control that the actual performances of the business plan are in line with what was planned originally by the Special Servicer. If this is not the case then the deal can include trigger events to mitigate the risk of the ABS senior Investors. As in the standard cash securitisation senior notes can be guaranteed by a third party guarantor: in some cases the Government can play a significant role (see Sect. 7.4.6). Lastly non-performing loans securitisations have usually a Back-up Servicer whose role is to replace the Special Servicer in case of an unexpected and sudden disruption of the activity of the Special Servicer which was originally appointed.

7.4.3 *The Pricing Gap Issue*

As described in Sect. 7.4.1, the standard securitisation process differs from the non-performing loans one for many aspects. One of the main differences is related to the fact that the non-performing loans securitisation always implies a certain degree of overcollateralisation due to the fact the Gross Book Value ("GBV") of the portfolio which is sold to the SPV via a true sale is always higher than the notional of the liabilities of the SPV i.e. the ABS placed to the Investors. The Originator Bank will usually try to sell the collateral portfolio at a value which is close to its accounting value, the so-called Net Book Value ("NBV") and the notional of the ABS issued by the SPV corresponds to the purchase price of the portfolio which is

[22]See Sect. 7.1.2 for the main counterparties in the standard cash securitisation.

usually lower than this value to provide the Investors of the ABS with an extra credit enhancement (overcollateralisation) and—more importantly—their expected target Internal Rate of Return ("IRR"). This is different from a standard cash securitisation where the pool of assets is sold at par by the Originator Bank and usually corresponds to the notional of the Asset-Backed Securities issued by the SPV. In a non-performing loans securitisation the Originator Bank can realise a loss by selling a portfolio of non-performing loans at its Net Book Value: this is the case if the Portfolio Purchase Price of the non-performing loans portfolio paid to the Originator Bank is lower than its Net Book Value. The amount of the potential loss is linked to the Coverage Ratio of such portfolio at the time of the true sale to the securitisation SPV. The Coverage Ratio plays a key role in the decision of the Originator Bank to finalise the securitisation deal. The determinants of the Coverage Ratio are:

- Gross Book Value: the GBV is the sum of the discounted cash flow originated from the underlying pool of assets. It is calculated as the sum of outstanding principal, due and unpaid instalments, accrued default interest and recoverable legal costs
- Net Book Value: the NBV is the revised book value of the assets due to the fact that the original debtor has difficulties in repaying its debt. It is calculated by the Originator Bank as the present value of all the future recoveries from the NPL discounted at the same rate of the underlying contract. The revised value of the assets can be a function of factors such as—amongst the others—the probability of the debtor of not repaying the loan, the recoverable amount (which is related to the type of collateral given under the loan—if any) and the expected timing of the possible recovery of the underlying assets (see Sect. 7.4.5)
- Value Adjustment: this is the difference between the Gross Book Value and the Net Book Value. It can vary from time to time based on the amounts of Recoveries and/or further write downs during a given period. This value can be considered as the cumulated amount of write downs done by the Originator Bank
- The Coverage Ratio, which is the ratio between the Value Adjustments and the GBV of the impaired loans, determines the potential amount of loss undertaken by the Originator Bank by selling the non-performing loans portfolio: the lower the Coverage Ratio, the higher will be the probability of suffering a loss on the sale of the non-performing loans portfolio (see Fig. 7.26).

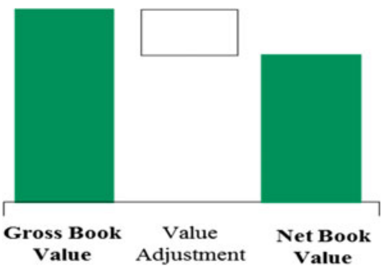


Fig. 7.26 From gross book value to net book value

The difference between the Net Book Value and the Market Value of the portfolio is derived by several factors including: Originator Bank and Investors may have different expectations (i) on the total amount and on the timing of recoveries, (ii) on the discount rate and (iii) on the impact of servicing costs. The discount rate is particularly important in the valuation of the non-performing loans portfolios: the Originator Banks are required by the current accounting rules to discount their expected cash flows at the same rate of the underlying loan contract i.e. they are required to use as discount rate the same rate which was applied at origination[23] while the Investors will typically include in their expectations a much higher Internal Rate of Return. In fact the main determinants of the Internal Rate of Return from an Investors perspective are the country risk; the type of underlying asset class and income producing assets; the volatility of the time to recovery; the servicing capabilities; the associated costs and—last but not least—the data quality.

Servicing costs play an important role: Originator Banks—when calculating the net present value of their non-performing loans portfolios—are not required to deduct servicing costs from their expected cash flows[24] while Investors, when assessing the value of an NPL portfolio, will discount all expected cash flows, including the deduction of servicing fees. In addition to the accounting criteria the Value Adjustment can also be related to capital and regulatory requests and constraints.[25] As a consequence, even if the Originator Bank and the Investors agree on the same recovery assumptions on amounts and timing, there could be still a gap between the net book value and the market value of the non-performing loans portfolios. As showed in Fig. 7.27 in addition to the Coverage Ratio which determines the difference between the GBV and the NBV the differences in discount factors and the recovery costs (mainly servicing costs) play an important role in the pricing gap.

To evaluate non-performing loans portfolio means to estimate the value of expected future cash flows both on the Originator Bank and on the Investors side.

Main determinants are:

- Discount rate: “historical” discount rate at origination versus target Investors IRR
- Amount of collections expected to be recovered

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[23]If there is objective evidence that an impairment loss on loans and receivables or held-to-maturity investments carried at amortised cost has been incurred, the amount of the loss is measured as the difference between the asset’s carrying amount and the present value of estimated future cash flows (excluding future credit losses that have not been incurred) discounted at the financial asset’s original effective interest rate (i.e. the effective interest rate computed at initial recognition). The carrying amount of the asset shall be reduced either directly or through use of an allowance account. The amount of the loss shall be recognised in profit or loss (Commission Regulation (EC) No 1126/2008).

[24]Servicing costs are accounted on an annual accrued basis on the P&L of the Originator Bank.

[25]Regulators such ECB can ask to for a certain amount of Coverage Ratio to make homogenous the Coverage Ratios across countries in Europe.

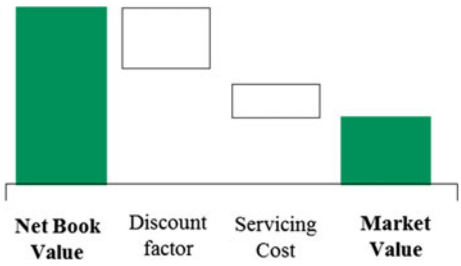

Fig. 7.27 Main Determinants of the Pricing Gap

Fig. 7.28 Originator banks versus investors perspective

	Originator Banks perspective	Investors perspective
IRR	Low rate (e.g. single digit)	High rate (e.g. double digit)
Recovery Rate	Expected Future Cash Flows	
Time to recovery	5 – 7 years (above the European average)	
Costs	▪ Servicing (internal and independent) ▪ Legal	▪ Due diligence ▪ Servicing ▪ Legal

- Time to recovery
- Recovery costs (mainly servicing and legal fees)

Figure 7.28 represents the building blocks of the analysis performed by both Investors and Originator Banks.

As to costs related to non-performing loans portfolios they are not typically embedded into the Net Book Value of the portfolio by Originator Banks: at the same time they are deducted from the net present value of the future expected cash flows of the non-performing loans portfolio by the Investors.

While most banks would be happy to sell their NPL portfolio and many Investors would be interested in buying them quite often the significant Pricing Gap hinders the possibility of non-performing loans transactions.

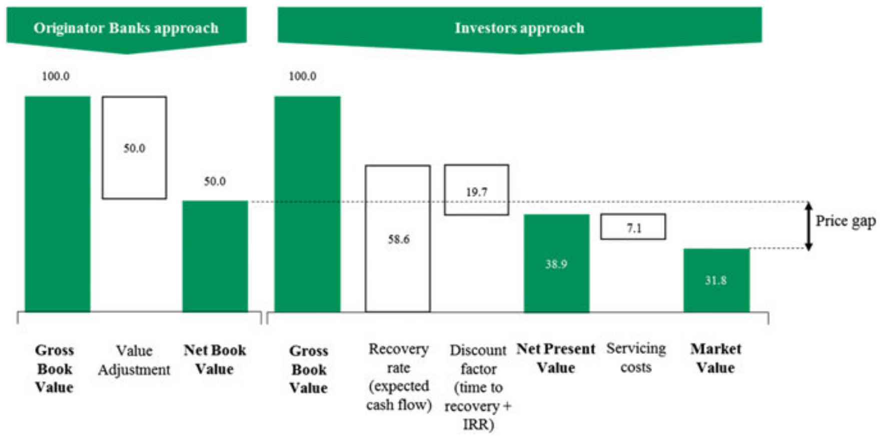

Fig. 7.29 From gross book value to market value (EUR MM)

A simplified numerical example can help to recap the path from the Gross Book Value of the non-performing loans portfolios to their Market Value (see Fig. 7.29).

Let's assume to have a non-performing loans portfolio whose Gross Book Value is EUR 100MM. Using the internal accounting approach of the Originator the Net Book Value of the same portfolio is EUR 50MM: this is determined by reviewing the expected cash flows of the loans by taking into consideration the revised expected repayments (i.e. the recovery rate) due to the financial stress situation related to the debtor.[26]

The Investors perform a similar exercise based as well on the expected cash flows of the non-performing loans portfolio, i.e. the recovery rate, applying their own discount factor which is a function of their view on the timing of these cash flows and of the target return for this kind of asset class i.e. their Internal Rate of Return. Once they have determined the Net Present Value of the portfolio (EUR 38.9MM in our example), the Investors deduct the costs related to the deal (mainly servicing costs) to determine the Market Value of the portfolio (EUR 31.8MM). The Price Gap is then equal to EUR 18.2MM which is the difference between the Net Book Value of the portfolio as determined by the Originator Bank and the Market Value estimated by the Investors.

The selection of the initial portfolio and the securitisation technique can be an effective way to reduce the pricing gap between Originator Bank and Seller.

[26]According to IAS 39, the discount rate remains the same though as per the one adopted in the original loan concession (Commission Regulation (EC) No 1126/2008).

7.4.4 Phases of the Securitisation

The selection of the underlying portfolio by the Originator Bank is one of the key elements to try tightening the pricing gap between its target transfer price (which is the Net Book Value) and the Market Value i.e. the Investors price as per Fig. 7.29.

The potential criteria in the choice of the securitised portfolio can include the notional of the loans line by line: this can have an implication both on the operational aspects and on the capital aspects. The minimum and maximum size of the single non-performing loan matter: from an operational point of view a sizing under EUR 200M could represent more than 90% of total numbers of loans and only 10% of total GBV whilst from a capital point of view a sizing over EUR 500M could transfer less than 5% of total loans and about 60–70% of total GBV loans.[27] Another important element is represented by the presence of collateralised loans within the underlying portfolio i.e. secured versus unsecured loans: the current pricing gap affects mostly the mortgage backed portfolios. Asset class is another important determinant of the underlying portfolio. Loans to Small Medium Enterprise versus collateralized loans to private individuals: a certain minimum percentage of mortgage-backed loans and/or loans to private individuals could make the portfolio cash-flows more predictable and more interesting for the Investors by decreasing the cash flow volatility and by increasing the timely repayment of the underlying loans. Seasoning of the underlying non-performing loans is also important. Loans beyond initial stage of foreclosure and bankruptcy procedures could be more predictable both in terms of recovery amount and residual time to recovery; at the same time a long-lasting procedure should be carefully analyzed, as it could imply serious recoverability issues. Last but not least the geographic distribution of the underlying non-performing loans is an important aspect to analyze: a wide geographic distribution of the portfolio might reduce portfolio concentration risk. Geographic distribution matters also for the judicial length of the recovery process. Section 7.4.5 will further analyze this aspect.

Figure 7.30 shows how different classes reflect a different market appetite. A good market appetite generally characterizes asset classes such as single large corporates, consumer finance and leasing. On the other side SME corporate and residential mortgages are considered asset classes for medium to low market appetite. Usually the IRR required by Investors in each class is driven by factors such as the quality of the collateral, the market and the volatility of the recovery. Overall the IRR range for these underlying might is between 10 and 20%.

[27]These hypothetical numbers could refer to a typical non-performing loans portfolio of a medium sized Italian bank.

Asset class	Market appetite	IRR Driver
Single Name Large Corporate	**HIGH** On single file/ industry basis (more on Restructuring than NPL)	Volatility of recovery expectation
Consumer Finance	**HIGH** An active and growing market with large investor and servicer universe	Market dynamic/ competitive tension
Leasing	**HIGH** Real Estate, luxury boats and Equipment	Quality/ information on collateral
SME Corporate	**MEDIUM** Limited universe of investor / servicer	Volatility of recovery expectation
Residential Mortgages	**LOW** Long foreclosure process and concern about collateral	Quality of data/ information on collateral

IRR range 10% - 20%

Fig. 7.30 Different asset classes and different IRR targets

7.4.5 *Risk Evaluation of the Asset-Backed Securities*

The building blocks in analyzing a securitisation of non-performing loans are the same as the ones for the cash securitisation of performing assets namely the valuation of the underlying collateral pool and of the structural features of the deal. The objective is to estimate future expected cash flows i.e. the amount collected and the timing of collections. We assume during the course of the analysis that the underlying portfolio is represented by non performing mortgage loans.

Analysis of the underlying portfolio of non-performing loans In terms of underlying assets portfolio the following variables should be considered:

- Underlying Type of Debtor: the breakdown between individual versus companies is the first step of the analysis of the underlying collateral pool. Recovery rates associated to individuals tend to be higher than the ones related to companies which are typically “shielded” by the limited liability legal structure. Individuals also tend to make more efforts to take responsibility of their liability situation more than companies when it comes to mortgages (especially on their first house)
- Underlying Type of Debt: the key distinction here is linked to the nature of the loan namely if the loan is unsecured or it is a secured one. Further important analysis is based on the nature of the mortgage: first lien has higher recovery rate than second lien and the same applies to residential compared to non-residential properties
- Geographical Distribution: geography matters for both the ability of repayments based on where the property is located (e.g. macroeconomic factors) and also in

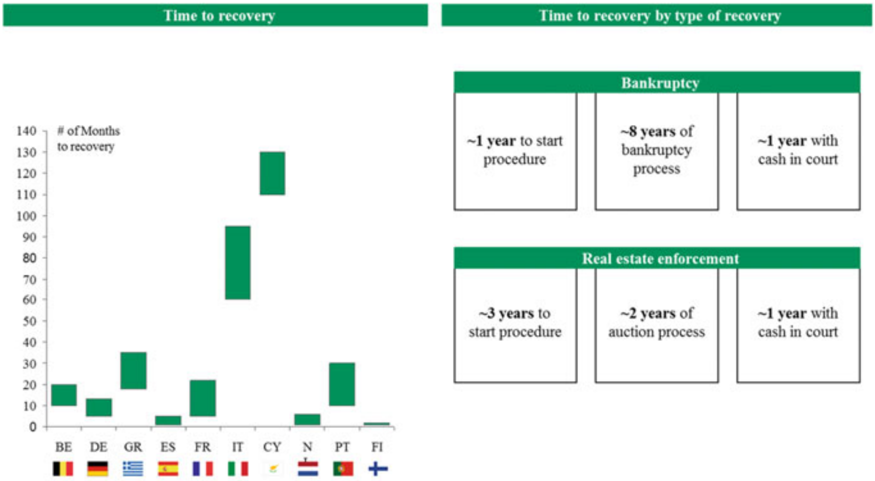

Fig. 7.31 Time to recovery by country and by type of recovery

relation to where the court is based. If we take the Italy example, North of Italy has recovery rates much higher than Centre/South of Italy and also the speed of the judicial process is much faster than the one of the tribunals of the center and south of Italy. This difference arises also from the point of view of the amount of Bad Loans. In 2014, the bad debt ratio for central and southern region for instance increase above 20%, compared to level below 10% in all the regions in 2009 (Garrido et al. 2016).

The timing of recovery is significantly affected by the efficiency of the judicial system. Compared to many countries in Western Europe foreclosure and bankruptcy procedures in Italy are long lasting causing significant impacts on the amount of non-performing loans and timing of recoveries and hence on their market value (see Fig. 7.31).

Given the unusually long duration of recovery procedures in Italy a difference in the discount rate generates a large valuation gap between the Originator Bank's Net Book Value and the potential Investor's Market Price.

- Step of the Recovery Process: the breakdown of non-performing loans between the ones which have already entered a specified stage of the legal process and the ones which have not yet entered is also an important element in the risk evaluation. If—for instance—the majority of the non-performing loans have not yet reached the initial legal stage this means that their timing of recovery could be negatively affected and hence their market value
- Type of Legal Process: if the underlying portfolio is represented mostly by bankruptcies more than foreclosures then the timing of recoveries will be negatively impacted. For a typical bankruptcy process it will take on average one year to start the process, eight years for the average duration of the legal

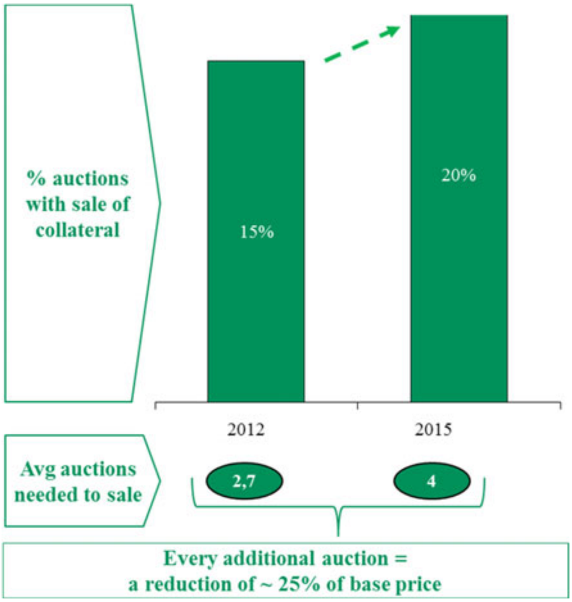

Fig. 7.32 Auctions process in Italy (Bank of Italy)

process and finally one year with cash in court. In case of a non-performing loan collateralized by Real Estate the foreclosure procedure could take three years to start, two years is the average duration of such a process and finally one year with cash in court (Bank of Italy 2016b)

- Auction Phase: the final stage of the legal process i.e. the auction phase is also important to consider. The relevant breakdown is non-performing loans at the first auction compared to the ones already into additional auctions. This impacts the liquidity of the collateral pool, the timing of recovery and the amount of the collections. In fact each additional auction is estimated to push down the base price of 25% and also delays further the timing of recovery (see Fig. 7.32). For the Italy case on average four auctions are needed to sell the collateral and twenty percent of the auctions end up in an effective sale of the collateral. The higher is the number of auction the less liquid becomes the collateral to sell (more difficult to sell)
- Historical Data: the role of the Special Servicer is an important one when it comes to historical data: the independent Servicer is important to align the interest of the Originator Bank with the ones of the Investors compared to the case where the specialized servicing unit of the Originator Bank is solely in charge of this task. Structural features can also help to achieve this objective.[28] First of all the Servicer provides the historical data on the recovery rating and on

[28]See Sect. 7.4.1 for the structure of NPL securitisation.

the timing of the collections both for unsecured and secured loans. The more reliable are the database of the Servicer and its experience in the non-performing loans market the better are the chances to maximize the expected future cash flows. The Servicer can work jointly with the specialized servicing unit of the Originator Bank or its work-out group in order to maximize the quality of the information to be provided to the potential Investors of the deal

- Business Plan: the strength of the business plan is another important determinant in evaluating the amount of collections and the timing of recovery of the underlying non-performing loans portfolio. Typically the business plan is adjusted from time to time (e.g. on an annual basis) to reflect the actual performances of the portfolio
- Data Quality and Market Valuation: the quality of the data related to the underlying collateral pool is an important factor for the reliability of the assessment of the investment by the potential buyers. In the case of mortgage loans the reliability of the evaluation derives also from the frequency of the market valuations updates. The market evaluation can be made by a third party expert or by an expert appointed by the court if the legal phase started already (or by both parties). Historically market evaluation done by independent third parties are the ones closer to the actual market value of the collateral of the secured loans whilst when performed by experts appointed by tribunals have on average overestimated the actual market price of the properties.

7.4.6 Analysis of the Structure

Main features of the cash securitisation can apply to the non-performing loans securitisation deals.

Credit Enhancement. The creation of subordinated notes has the same purpose which we analyzed for the standard cash securitisation. The senior notes holders will have priority in the payment of interest and capital compared to the holders of the mezzanine and junior notes according to the cash waterfall structure of the securitisation. A numerical example could be of help (see Fig. 7.33).

Let's assume that the Originator Bank has a portfolio of non-performing loans whose Gross Book Value is EUR 1bn. The securitised portfolio is sold to the Special Purpose Vehicle at EUR 300MM (Portfolio Purchase Price). The credit

Fig. 7.33 Example of credit enhancement for a portfolio of non-performing loans

	EUR (MM)
Gross Book Value	1,000
Portfolio Purchase Price	300
Senior Tranche Purchase Price	255
Mezzanine Tranche Purchase Price	30
Junior Tranche Purchase Price	15

enhancement structure contemplates the creations of three ABS tranches: the senior Tranche is bought by senior Investors at EUR 255MM (Senior Tranche Purchase Price), the mezzanine tranche at EUR 30MM (Mezzanine Tranche Purchase Price) and the junior tranche at EUR 15MM (Junior Tranche Purchase Price).

The senior tranche represents the 85% of the notional of the ABS issued, whilst the mezzanine and the junior tranches represent 10 and 5% respectively of the securitisation deal notional: 15% is the subordination level which the senior tranche holders enjoy i.e. the underlying non-performing loans portfolio has to be impacted by 15% of net losses during the life of the deal before any loss affect their initial investment. Official rating considerations matter as well: Rating Agencies will assign a rating to the senior and mezzanine tranches base also on the subordination level they benefit from[29]: senior tranches are typically investment grade rated in order to be placed to a wider public of Investors (banks and real money Investors such as insurances, pension funds, asset managers etc.).

The pricing gap hits the Originator Bank as seller of the underlying portfolio if the Purchase Price of the portfolio is lower than the Net Book Value at which the portfolio is accounted in the balance sheet of the Originator Bank. As described previously this can happen for instance if the target Internal Rate of Return of the Investors (prominently the junior Investors) is higher than the one used by the Originator Bank to determine the Net Book Value.

Liquidity Facility. The securitisation can envisage a liquidity facility (or cash reserve). Liquidity risk is much more relevant in non-performing loans securitisations compared to the standard ones on performing assets. The underlying non-performing loans portfolio may not generate any cash flow for a certain period of time: in this case the SPV doesn't receive cash flows to cover senior expenses (according to the securitisation waterfall structure) and to make payment to the ABS holders beginning with the most senior ones. The liquidity facility could enable the SPV to meets its obligations toward the ABS holders and toward the securitisation main counterparties. Collections on non-performing loans are by nature both lumpy and uncertain so the securitisation deal can envisage a liquidity facility to counterbalance temporary lack of cash flows due to the above connotations of the underlying portfolio. The commitment of the Liquidity Provider can be valid only during a certain period of the deal (e.g. first year or first two years) and it is usually aimed to payment of interest of the most senior notes only. Counterparty risk is an important element of this feature: for instance Liquidity Facility Providers are subject to a minimum rating to be considered eligible by Rating Agencies in their rating assessments. In the non-performing loans securitisation the risk exposure to the Liquidity Facility Provider is greater than in the one backed by performing loans: hence the credit quality of the Provider should be assessed and monitored with great attention.

[29]See Sect. 7.1.5 for tranching models.

Trigger Events. The Special Servicer plays a key role in the non-performing loans securitisation deal. This type of deals are much more servicing intensive than the standard ones backed by performing loans due to the extreme importance of the collection process in general including the timing of the collections. Any disruption to the servicing activity can jeopardize the whole deal's structure. To mitigate the risk related to the important role played by the Special Servicer—including its potential conflicts of interest and its possible misalignment of interest—the structure can contemplate trigger events of the following type:

- Subordination of the Servicing Fees: if the actual performance of the business plan is diverging from the one presented by the Special Servicer at the inception of the deal the Monitoring Agent can ask for changing the priorities of the payments under the waterfall structure and for diverting the payments of the servicing fees after the interest of the most senior notes has been paid. This mechanism is aimed to incentive the Special Servicer to perform in line or better than the business plan when it comes to timing of collections and expected recoveries
- Servicing Termination Event: if the actual performance of the business plan is worse than the one agreed at deal's inception, the Monitoring Agent—together with a committee composed of junior and mezzanine Investors-can serve a Termination Event to the originally appointed Special Servicer.[30] In this case the Back-up Servicer enters the deal or it can be decided to appoint a new external Special Servicer.
- Triggers related to Servicing Decisions: Special Servicer could be interested in making decision outside the collection policy which could harm the Asset-Backed Securities holders. As examples, the Special Servicer could decide to sell non-performing loans with additional discounts so below their target and agreed price or to grant certain waivers to the debtors.[31] To reduce the potential conflict of interest of the Special Servicer the Monitor Agent can trigger a call of a Committee composed of representative of the mezzanine and junior noteholders to decide on this kind of matters and to give permission to the Special Servicer to proceed or not to proceed.

Other "traditional" trigger events can be included in the non-performing loans securitisation deal such as payment of interest to mezzanine Investors conditional to the repayment of capital to the senior notes Investors' etc.[32]

Swap Hedge. The tranches of ABS placed to the Investors (at least the rated ones) are usually linked to a money market index i.e. they are usually floating rate

[30]Due to their subordination in the cash flow waterfall structure the junior and mezzanine Investors are the ones which promptly suffer from a negative performance of the underlying portfolio compared to the expected one at inception.

[31]Servicing fees are usually amongst the most senior fees in the cash flow waterfall structure: namely they are more senior than the interest payment to the senior noteholders.

[32]See Sect. 7.1.5.

notes which typically pay Euribor 6 months and a credit spread which is based on the risk of the given tranche. Due to the fact that the underlying assets are typically not linked to any money market index the existence of the swap counterparty in the deal is key to guarantee ongoing payments of the liabilities of the SPV linked to the money market indexation. A swap deal is then usually featured in the securitisation whereby the cap guarantor (i.e. the swap counterparty) provides an interest rate cap to hedge the interest rate risk for the Investors of the rated ABS: the notional of the swap is equal to the outstanding balance of the rated notes. Again the counterparty risk is a relevant one: the cap guarantor has to have a minimum rating to be eligible to enter such deal.

Guarantee from Third Parties The non-performing loans can benefit from third parties guarantees which usually affect positively the holders of the most senior tranche. An important example is the one related to the *Garanzia Cartolarizzazione Sofferenze* ("GACS") i.e. the guarantee granted by the Italian government on the most senior tranches of non-performing loans securitisation. The GACS is a guarantee provided by the State on the senior tranche of securitisations of NPLs. Key elements of the GACS are the following: the originators must be banks or financial intermediaries with legal head office in Italy; the issuer is an Italian securitisation SPV; the tranche must be senior in principal repayments; interest on mezzanine tranches could be paid senior to payment of principal on the senior tranche; no payment on the junior tranche before the full redemption of the guaranteed tranche; the servicer must be external and independent from the seller; the guaranteed tranche must be rated at least "investment grade" by one rating agency; the senior notes must be floating rate and liquidity and hedging agreements are allowed and—lastly-the Originator Bank has to place at least 51% of the junior Notes to private Investors in order to try to deconsolidate—also from an accounting perspective—the non-performing loans portfolio from its balance sheet. The cost of the guarantee is linked to the risk of a basket of Italian issuers.[33]

The GACS is an unconditional, irrevocable and first-demand State guarantee which covers interest and principal due on the senior notes and it is released upon request of the Originator Bank. This guarantee is an example of Public initiative aimed to reach two objectives: (i) make it easier for the Originator Bank to decrease the amount of non-performing loans on their balance sheet via (ii) reducing the pricing gap compared to a straight non-performing loans sale to specialized Investors. securitisation of non-performing loans by providing leverage through tranching to Investors may reduce the overall costs for the Originator Bank and so may improve the portfolio pricing: this is precisely the purpose of the GACS which has been introduced by the Italian State as a way to reduce the cost placement of the senior tranches and—by doing so—reducing in general the pricing gap.

[33]The guarantee fee paid by the Originator Bank to the State is determined as the average of daily mid-prices of CDS for baskets of Italian issuers as quoted on Bloomberg in the six months preceding the request for guarantee (Gazzetta Ufficiale 2016, Decreto Legge 14 February 2016, n. 18).

7.4.7 Securitisation of Non-performing Loans: Pricing Factors and Impact on Pricing Gap

This paragraph focuses on how Originator Banks could use the securitisation of non-performing loans as a way to reduce exposure to non-performing loans on their balance sheet. The main objective of the Originator Bank when arranging a securitisation of non-performing loans is to reduce the exposure of outstanding non-performing loans in its balance sheet. The securitisation can play an important role in achieving this by providing a certain degree of leverage (where by leverage we define the percentage of the senior tranche compared to the overall tranches issued by the Special Purpose Vehicle). This can be an effective tool for reaching this objective compared to the outright sale of non-performing loans portfolios which has never worked properly especially in countries where the non-performing loans issue is prominent as showed by the scarce number of deal finalised.[34] The main reason of the lack of success of a secondary market of non-performing loans portfolios is mainly due to the pricing gap between the Net Book Value determined by the Originator Bank and the different expectations in terms of yield by the specialized players in this market.

As we described previously the main determinants of the pricing of the non-performing loans portfolios are:

- Amount of collections expected to be recovered
- Time to recovery
- Discount rate
- Recovery costs: most important servicing and legal fees.

The following examples (Figs. 7.34, 7.35 and 7.36) relate to the analysis of the pricing factor for three sample hypothetical non-performing loans portfolios originated in Italy.

The three above figures show the impact of each component of the pricing of the non-performing loans portfolios in the pricing gap. Internal Rate of Return assumptions are clearly one of the most important driver of the magnitude of the pricing gap: each type of portfolio will have a different IRR assumption based on the investor perception of the risk of the specific asset class even if both Originator Bank and the Investors agree on the amount of recoveries, their timing and the overall recoveries expenses (see Sect. 7.4.4).

It is important to notice that the more granular and data intensive is the underlying portfolio (such as in Residential Mortgages portfolios) the wider is the pricing gap since the IRR expected by the prospective Investors is much higher than the one used by banks to determine the Net Book Value of the non-performing loans.

[34]See Sect. 7.3.

Assumptions

LOAN INFO		
GBV	100	
REAL ESTATE ASSUMPTIONS		
Original LTV	65.00%	
Auction Discount	-35.00%	
RECOVERY ASSUMPTIONS		
	No/Ce	**South**
Time to recovery (yrs)	4.5	7.0
Recovery Expenses (% of gross rec.)	16.00%	
Servicing Fees (% of recoveries)	7.50%	
IRR ASSUMPTIONS		
IRR Bank	4.00%	
IRR Investor	12.00%	

Output

RECOVERIES		
Real Estate Original Value	153.8	
Judicial Sale Value	100.0	
Recovery Expenses	16.0	
Net Recoveries	**84.0**	
VALUATION		
	No/Ce	**South**
Bank PV	70.4	63.8
Investor PV	46.7	35.1
Pricing Gap	23.7	28.7
PRICING GAP FACTORS		
	No/Ce	**South**
Servicing Fee	3.8	2.8
IRR	20.0	25.9

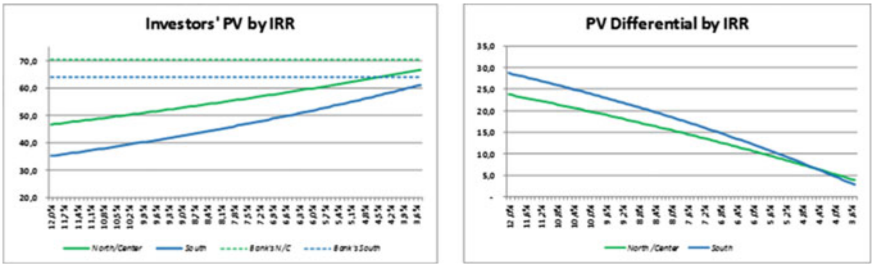

Fig. 7.34 Pricing Factors for a portfolio of residential mortgage loans (No/Ce stands for loans originated in the North and Centre of Italy. South refers to loans originated in the South of Italy)

Assumptions

LOAN INFO		
GBV	100	
RECOVERY ASSUMPTIONS		
Gross Recovery rate	31%	
	No/Ce	**South**
Time to recovery (yrs)	5.0	7.5
Recovery Expenses (% of gross rec.)	20.00%	
Servicing Fees (% of recoveries)	9.00%	
IRR ASSUMPTIONS		
IRR Bank	6.50%	
IRR Investor	20.00%	

Output

RECOVERIES		
Gross Recoveries	31.0	
Recovery Expenses	6.2	
Net Recoveries	**24.8**	
VALUATION		
	No/Ce	**South**
Bank PV	18.1	15.5
Investor PV	9.1	5.7
Pricing Gap	9	9.7
PRICING GAP FACTORS		
	No/Ce	**South**
Servicing Fee	0.9	0.6
IRR	8.1	9.2

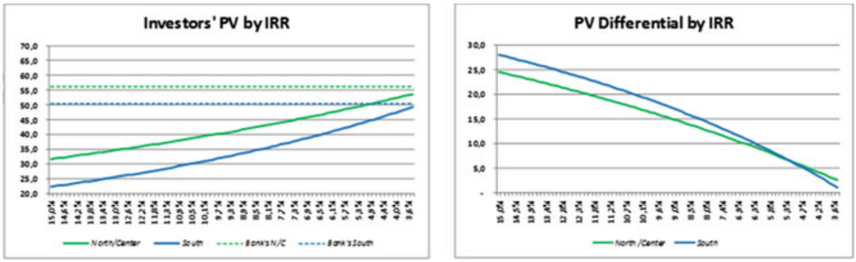

Fig. 7.35 Pricing factors for a portfolio of small medium enterprises secured mortgage loans (commercial/industrial)

Assumptions

LOAN INFO		
GBV	100	
REAL ESTATE ASSUMPTIONS		
Original LTV	60.00%	
Auction Discount	-50.00%	
RECOVERY ASSUMPTIONS		
	No/Ce	**South**
Time to recovery (yrs)	5.0	7.5
Recovery Expenses (% of gross rec.)	16.00%	
Servicing Fees (% of recoveries)	9.00%	
IRR ASSUMPTIONS		
IRR Bank	4.50%	
IRR Investor	15.00%	

Output

RECOVERIES		
Real Estate Original Value	166.7	
Judicial Sale Value	83.3	
Recovery Expenses	13.3	
Net Recoveries	**70.0**	
VALUATION		
	No/Ce	**South**
Bank PV	56.2	50.3
Investor PV	31.7	22.3
Pricing Gap	24.5	28.0
PRICING GAP FACTORS		
	No/Ce	**South**
Servicing Fee	3.1	2.2
IRR	21.4	25.8

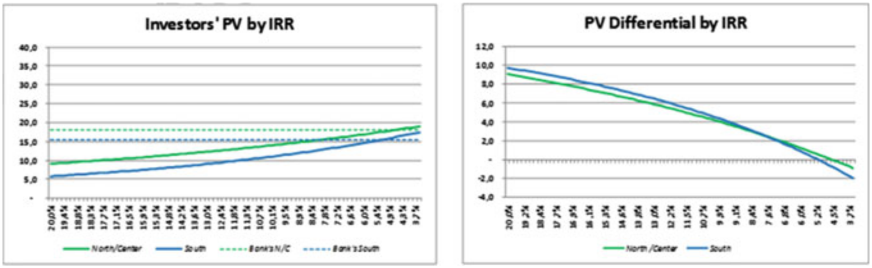

Fig. 7.36 Pricing factors for a portfolio of small medium enterprises unsecured loans

We now combine the three above portfolios into one which contemplates different weights for each of the three categories. We assume that the amount of recoveries is net of legal costs and gross of servicing fees paid to independent Special Servicers (Figs. 7.37 and 7.38).

Figure 7.39 shows the theoretical valuation of such portfolio by the Originator Bank and by the Investors under the assumptions previously displayed in particular the Originator Bank and the Investors share the same recovery assumptions and the servicing fees are calculated at 7% of total recoveries. Outrights portfolio sale to Investors by Originator Bank would cause a loss of around EUR 1.8bn which is equivalent to around 18% difference in the valuation in terms of notional value.

We analyse now the case of a non-performing loans securitisation deal done on the same portfolio. We assume that the senior notes are priced via senior financing deals[35] or placement to the market at a level typical of investment grade senior ABS

[35]An example can be a third party bank interested in financing the senior tranche of non-performing loans securitisation deal based on their internal models after a thorough analysis of the specific of the underlying portfolio and of the soundness of the securitisation structure. In this case an implied internal rating would be assigned to the senior tranche without the need of an official rating.

	%	GBV	Total Recoveries	% of GBV	WAL of Recoveries (yrs)
Residential Mortgages	20.00%	2,000,000,000	1,680,000,000	84.00%	6.0
Comm/Ind Mortgages	40.00%	4,000,000,000	2,800,000,000	70.00%	6.5
Unsecured SME	40.00%	4,000,000,000	1,000,000,000	25.00%	6.2
Total	**100.00%**	**10,000,000,000**	**5,480,000,000**	**55.00%**	**6.3**

Fig. 7.37 Structure of a theoretical EUR 10bn securitisation portfolio

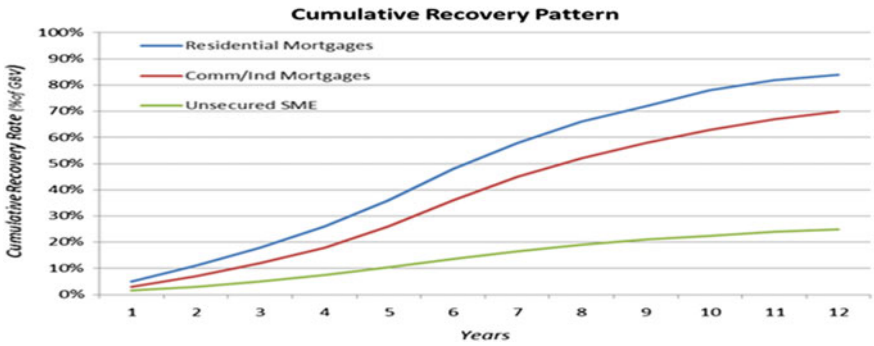

Fig. 7.38 Assumptions on cumulative recovery patterns

notes.[36] Figure 7.40 shows the outcome of a non-performing loans securitisation in terms of pricing gap. The assumptions are the ones showed previously: we also assume that the securitisation follows the standard cash flow waterfall structure discussed in this paragraph including junior notes capital repayment only after payment of interest and capital to the senior noteholders and the existence of triggers event at junior note level. As we noticed the pricing gap is significantly reduced compared to the case of the outright sale of the non-performing loans portfolio.

The sensitivity of the potential purchase price to different assumptions on leverage and yield of the senior tranche is showed in Fig. 7.41 with the assumption of an IRR equal to 14% for the junior Investors.

The final impact on an actual NPL portfolio will depend on both quantitative and qualitative elements such as pricing/return requested by Investors on both senior and junior tranches, Asset-Backed Securities terms (maturity, size, ECB eligibility, ratings, etc.), guarantee terms, portfolio features as well as legal/regulatory measures implementations and servicing arrangements. The underlying asset class has a significant impact as well as previously discussed (see Fig. 7.30).

[36]The use of third parties guarantees could help to achieve this outcome when senior notes are placed into the market. We assume that the senior notes return includes the price of the guarantee.

	Bank Discount Rate	Investors' IRR	Bank Valuation		Investos' Valuation	
Residential Mortgages	4.00%	12.00%	133,612,120	66.70%	829,210,497	41.50%
Comm/Ind Mortgages	4.50%	15.00%	2,122,870,143	53.10%	1,141,511,720	28.50%
Unsecured SME	6.50%	20.00%	686,819,350	17.20%	343,267,050	8.60%
Total	**4.70%**	**15.00%**	**4,143,301,613**	**41.40%**	**2,313,989,267**	**23.10%**

Fig. 7.39 Pricing Gap for the outright sale of the non-performing loans portfolio (EUR)

Tranches	Yield	Size	%	WAL (yrs)
Senior	2.50%	3,536,922,285	90.00%	6.1
Junior	14.00%	392,991,365	10.00%	11.4
Total	**4.48%**	**3,929,913,650**	= Portfolio Purchase Price	

Fig. 7.40 Pricing Gap after the securitisation of the non-performing loans portfolio (EUR)

Fig. 7.41 Sensitivity of the purchase price under different senior IRR and leverage assumptions

		Senior Leverage		
		95.00%	90.00%	85.00%
Senior Yield	1.50%	4.35	4.11	3.91
	2.00%	4.24	4.02	3.83
	2.50%	4.13	$\boxed{3.93}$	3.76
	3.00%	4.03	3.84	3.68
	3.50%	3.93	3.76	3.61

7.5 Conclusions

The new market for the securitisation of non-performing loans is still at its early steps compared to the potentiality which has been already experienced in the securitisation market for performing loans. Many factors have to be in the right place for reducing the pricing gap and then revamp this market especially in countries such as Italy where this market is most needed. To mention a few: judicial system reform to shorten the time to recovery; public initiatives such as government guarantees; high standard of quality data; improvement in the Special Servicers offerings and in their range of services and—last but not least—a recovery in property prices especially in the peripheral countries. At the same time some of the Originator Banks need to be more disciplined and determined to set up a reliable business plan and a strong sale strategy (market appetite understanding, portfolio cluster identification, etc.), to improve non-performing loans data quality, to develop specific methodologies for comprehensive assessment of the costs associated (not yet embedded in their Net Book Value calculation), to setup a transparent and straightforward process of non-performing loans dismissal and to assess options such as financing structures at least for the senior portion of the non-performing loans portfolio. Time is of the essence: Originator Banks needs be equipped to promptly catch the potentially strong opportunities offered by the market of securitisation of non-performing loans.

Acknowledgements I would like to thank my colleagues Antonella Tagliavini, Francesco Magliocchetti and Marco Germani for their important contributions to the non-performing loans section of this chapter. Special thanks to my colleague Vittoria Perocco who assisted me in drafting this chapter. Last but not least I would like to thank also Geoffroy De Pellegars and Marco Busin for their kind collaboration.

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